

Price Policy for Rabi Crops

Marketing Season 2026-27



Commission for Agricultural Costs & Prices

Department of Agriculture & Farmers Welfare

Ministry of Agriculture & Farmers Welfare

Government of India, New Delhi

July 2025

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Preface & Acknowledgments

I have the privilege and honour to submit the report of “**Price Policy for Rabi Crops: The Marketing Season 2026-27**”. The report contains the recommendations on Minimum Support Prices (MSPs) for the mandated crops, namely, **wheat, barley, gram, lentil, rapeseed & mustard and safflower**, and a set of non-price measures. I hope that these recommendations will serve the interest of both producers and consumers, incentivise farmers to adopt new technologies and develop production pattern in the light of emerging demand patterns and improve competitiveness of Indian agriculture.

Summary of recommendations is followed by an overview of Indian agriculture in Chapter 1. Chapter 2 of the report discusses issues related to demand-supply, prices trends and procurement operations. Trends in productivity of *rabi* crops and drivers of productivity are analysed in Chapter 3. Broad trade patterns, performance and outlook, comparison of domestic and world prices and recent trade policy changes are presented in Chapter 4. Costs and returns and cost projections for *Rabi* Marketing Season 2026-27 including inter-crop parity are discussed in Chapter 5. Major considerations and non-price and price policy recommendations are given in the Chapter 6.

Many people have assisted in the preparation of this report. First and foremost, I am pleased to acknowledge the constructive contributions of farmers, farmers’ associations, officers from Central and State Governments, research organizations, industry associations and various other stakeholders during the meetings of the Commission. I would like to express my special appreciation and thanks to Government of Assam, Government of Bihar, Government of Gujarat, Government of Haryana, Government of Madhya Pradesh, Government of Maharashtra, and Government of Telangana for organizing meetings of the Commission with farmers, State Government Departments and other stakeholders. Special thanks to the Economics, Statistics & Evaluation Division, Ministry of Agriculture & Farmers Welfare for providing key data on cost estimates for this report.

I would like to express my appreciation to Sh. Ratan Lal Daga, Member (Non-Official) and Dr. Prem Chand, Member (Official) for their excellent support in preparation of the report. Sincere gratitude goes to Sh. Cyriac George, Member Secretary, who not only contributed to the report but managed timely completion of the report. The report would not have been possible without scholarly contributions of Advisors, Dr. R. Sathish, Dr. Seema and Sh. Vivek Shukla and I would like to thank them all. My sincere gratitude to Ms. Divya



Sharma, Ms. Sonal Ramesh, Sh. Baidyanath Guru, Smt. Shivani, Ms. Shraddha Pal, Sh. Deepak Peeceeyen S, Sh. Vipin Singh and Ms. Parul Singh for their valuable contributions in preparation of report. I gratefully acknowledge analytical support provided by Sh. Md. Abdul Aleem, Sh. Mintu Kumar, Sh. Chandra Kumar, Sh. B.K. Barnawal, Sh. Ravi Kumar Sharma, Sh. Ashok Kumar, Smt. Savitri Ramani, Sh. Mahender Kumar Gupta and Sh. Mohd. Shoeb Malik. I would like to make a special mention of the dedicated efforts put in by officers and staff members of the Commission for providing administrative and other support.

14th July 2025

(Vijay Paul Sharma)



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Acronyms

Acronym	Description
A_2	Paid-out cost
A_2+FL	Paid-out Cost plus Imputed Value of Family Labour
ABARES	Australian Bureau of Agricultural Economics and Sciences
AGRTot	Terms of Trade for Agriculture
AIBP	Accelerated Irrigation Benefit Programme
AIDC	Agriculture Infrastructure and Development Cess
AIDE	App for Intermediary Enrolment
AP	Andhra Pradesh
APMC	Agricultural Produce Market Committee
AS	Assam
ATMA	Agricultural Technology Management Agency
BCD	Basic Customs Duty
BR	Bihar
C_2	Paid out cost plus imputed value of family labour plus rental value of owned land (net of land revenue) and interest on fixed capital excluding land
CACP	Commission for Agricultural Costs & Prices
CAGR	Compound Annual Growth Rate
CCEs	Crop Cutting Experiments
CFPI	Consumer Food Price Index
CG	Chhattisgarh
CGS-NPF	Credit Guarantee Scheme for e-NWR based Pledge Financing
CIPI	Composite Input Price Index
CoC	Cost of Cultivation
CoP	Cost of Production
CPI	Consumer Price Index
CPO	Crude Palm Oil
CRM	Crop Residue Management
CROPIC	Collection of Real-Time Observations and Photographs of Crops
CS	Comprehensive Scheme



Acronym	Description
DCP	Decentralized Procurement
DCS	Digital Crop Survey
DGCES	Digital General Crop Estimation Survey
DGCIS	Directorate General of Commercial Intelligence and Statistics
DGFT	Directorate General of Foreign Trade
DPI	Digital Public Infrastructure
DPIIT	Department for Promotion of Industry and Internal Trade
e-NWR	e-Negotiable Warehouse Receipts
EC	Economic Cost
EDI	Electronic Data Interchange
ES&E	Economics, Statistics and Evaluation
EU	European Union
FAO	Food and Agriculture Organisation
FAQ	Fair Average Quality
FCI	Food Corporation of India
FFPI	FAO Food Price Index
FLD	Front Line Demonstration
FPMA	Food Price Monitoring and Analysis
FPOs	Farmers Producing Organisations
FToT	Farmers' Terms of Trade
GAP	Good Agricultural Practices
GCA	Gross Cropped Area
GCF	Gross Capital Formation
GIA	Gross Irrigated Area
GJ	Gujarat
GOBARDhan	Galvanizing Organic Bio-Agro Resources <i>Dhan</i>
GST	Goods and Services Tax
GVA	Gross Value Added
GVO	Gross Value of Output
HCES	Household Consumption Expenditure Survey



Acronym	Description
HKKP	<i>Har Khet Ko Pani</i>
HP	Himachal Pradesh
HR	Haryana
HRW	Hard Red Winter
HSD	High Speed Diesel
ICT	Information and Communication Technology
IGC	International Grains Council
IND	India
INM	Integrated Nutrient Management
JH	Jharkhand
JVCs	Joint Venture Companies
K	Potassium
KA	Karnataka
KCC	<i>Kisan</i> Credit Card
LoC	Letter of Credit
MEP	Minimum Export Price
MH	Maharashtra
MIF	Micro Irrigation Fund
MISS	Modified Interest Subvention Scheme
MP	Madhya Pradesh
MRP	Maximum Retail Price
MSP	Minimum Support Price
MSR	Marketed Surplus Ratio
Mt	Million Metric Tonne
N	Nitrogen
NABARD	National Bank for Agriculture and Rural Development
NAFED	National Agricultural Cooperative Marketing Federation of India Ltd.
NCCF	National Co-operative Consumers' Federation of India Ltd.
NCEL	National Cooperative Exports Limited
NFSA	National Food Security Act



Acronym	Description
NFSM	National Food Security Mission
NFSNM	National Food Security & Nutrition Mission
NIP	New Investment Policy
NMEO-OP	National Mission on Edible Oils-Oil Palm
NMEO-OS	National Mission on Edible Oils-Oilseeds
NSSF	National Small Saving Fund
NWR	Negotiable Warehouse Receipts
OECD	Organisation for Economic Cooperation and Development
OGL	Open General License
OMSS(D)	Open Market Sales Scheme (Domestic)
OWS	Other Welfare Schemes
P	Phosphorous
PACS	Primary Agricultural Credit Societies
PB	Punjab
PDMC	Per Drop More Crop
PI	Procurement Incidentals
PMFBY	<i>Pradhan Mantri Fasal Bima Yojana</i>
PMGKAY	<i>Pradhan Mantri Garib Kalyan Anna Yojana</i>
PMKSY	<i>Pradhan Mantri Krishi Sinchayee Yojana</i>
PM-KUSUM	<i>Pradhan Mantri Kisan Urja Suraksha evam Utthaan Mahabhiyan</i>
PM-PRANAM	PM Programme for Revival, Awareness, Nutrition and Improvement of Mother Earth
PSF	Price Stabilization Fund
PSS	Price Support Scheme
QRs	Quantitative Restrictions
R&D	Research and Development
R&M	Rapeseed and Mustard
RBD	Refined, Bleached and Deodorized
RFI	Rural Financial Institution
RJ	Rajasthan
RKVY	<i>Rashtriya Krishi Vikas Yojana</i>



Acronym	Description
RMS	<i>Rabi</i> Marketing Season
RWBCIS	Restructured Weather based Crop Insurance Scheme
SATHI	Seed Traceability, Authentication and Holistic Inventory
SLUSI	Soil and Land Use Survey of India
SMAE	Sub-Mission on Agricultural Extension
SMAM	Sub-Mission on Agricultural Mechanization
SRR	Seed Replacement Rate
SUR	Stock to Use Ratio
TE	Triennium Ending
TG	Telangana
ToT	Terms of Trade
TPDS	Targeted Public Distribution System
UAE	United Arab Emirates
UAV	Unmanned Aerial Vehicle
UK	United Kingdom
UN	United Nations
UP	Uttar Pradesh
UPAg	Unified Portal for Agricultural Statistics
USA	United States of America
USDA	United States Department of Agriculture
UTs	Union Territories
VRR	Varietal Replacement Rate
WB	West Bengal
WDC	Watershed Development Component
WDRA	Warehousing Development and Regulatory Authority
WINDS	Weather Information and Network Data System
WPI	Wholesale Price Index
WTO	World Trade Organization
WTSR	World Trade Statistical Review



Summary of Recommendations

- S.1 The Commission, as per its mandate, has considered cost of production, demand and supply situation of various crops and their prices in domestic and global markets, overall terms of trade between agriculture and non-agriculture sector, likely effect of the price policy on the rest of the economy and competitiveness of the sector and a minimum margin of 50 percent over the cost of production. These factors have been discussed in detail in various chapters of this report. In addition, suggestions and inputs received from various stakeholders including farmers and farmers associations, State and Central Government Ministries/Departments, industry associations, research organizations, etc. during the Commission's meetings and field visits have been considered. Based on the above-mentioned factors and stakeholder consultations, the Commission makes the following price policy and non-price recommendations for *rabi* crops for the marketing season 2026-27.

Minimum Support Price (MSP) Recommendations

- S.2 After due deliberations, the Commission recommends the following MSPs for the mandated *rabi* crops for *Rabi* Marketing Season (RMS) 2026-27 as given in Table S.1.

Table S.1: MSPs Recommended for *Rabi* Marketing Season 2026-27

(₹/qtl)

Crop	Projected A ₂ +FL Cost for RMS2026-27	MSP for RMS2025-26	Recommended MSP for RMS2026-27	MSP as percent of A ₂ +FL
Wheat	1239	2425	2585 (6.6)	209
Barley	1361	1980	2150 (8.6)	158
Gram	3699	5650	5875 (4.0)	159
Lentil (<i>Masur</i>)	3705	6700	7000 (4.5)	189
Rapeseed & Mustard	3210	5950	6200 (4.2)	193
Safflower	4360	5940	6540 (10.1)	150

Note: Figures in parenthesis represent an increase in MSP (%) over the previous year

Non-Price Recommendations

Emphasis on Productivity Improvement

- S.3 Over the last 10 years there has been remarkable growth in yield in most crops but annual growth rates of productivity of all mandated crops, except lentil and rapeseed & mustard, have decelerated during



the recent years. There are also considerable yield gaps in most crops across States indicating large scope for increasing yields. In case of Punjab and Haryana, crop yields are nearing their potential and need a major technological breakthrough for pushing productivity frontiers. Therefore, the Commission recommends strengthening technological interventions through Research and Development (R&D), availability of quality inputs and services, expansion of irrigation facilities and better farm practices to improve productivity. In case of oilseeds, special emphasis should be given on improving both seed yield and oil content to increase production and profitability.

Aatmanirbharta in Pulses: Role of Technology, Trade Policy and Incentives

- S.4 The spectacular growth of India's pulses production between 2015-16 and 2021-22 was primarily driven by three factors; technology, remunerative prices and conducive trade policy. However, pulses production has stagnated in the recent years mainly due to technology fatigue resulting in deceleration in area and yield, and unrestricted cheap imports of pulses, particularly yellow peas, gram, and lentil. The Commission recommends a two-pronged strategy for achieving *aatmanirbharta* in pulses: (i) concerted efforts to develop high-yielding climate-resilient varieties and ensure availability of quality seed to farmers, and (ii) restrict cheap imports of gram and lentil through higher tariffs and a complete ban on imports of yellow peas.

Strategies for Reducing Edible Oils Import Dependence

- S.5 Despite significant increase in oilseeds production in the country during the last 10 years, dependence on imports has remained high due to fast rising demand for edible oils. The performance of rapeseed and mustard (R&M) has been remarkable, but safflower, a premium oil crop in Maharashtra and Karnataka, has seen a steep decline in both area and production. The Commission recommends a special programme for promotion of safflower cultivation in Maharashtra, Karnataka and parts of Telangana and Andhra Pradesh under the National Mission on Edible Oils (NMEO) and developing high-yielding varieties/hybrids of oilseeds with high oil content using conventional and biotechnological approaches. The Commission believes that effective implementation of the National Mission on Edible Oils - Oil Palm (NMEO-OP) would help in bridging the gap between demand and supply of edible oils.
- S.6 Since global vegetable oil markets are highly volatile, the Commission recommends a dynamic duty structure linking import tariffs to international prices and in case of sudden fall in global prices, trigger points can be identified for quick action to raise import duties.

Link MSP of Oilseeds to Oil Content

- S.7 To incentivise high oil content in oilseeds and ensure better returns to farmers, MSP of rapeseed & mustard and safflower should be linked to the oil content. The Commission recommends that for every 0.25 percentage point increase in oil content above the basic level, premium should be paid to farmers.

Address Low Fertilizer Use Efficiency and Pricing Issues

- S.8 The urea sector in India, which was highly dependent on imports, is now moving towards achieving self-sufficiency due to Government initiatives. However, the current fertilizer subsidy structure has led



to excessive use of urea resulting in imbalanced use of nutrients. The Commission recommends that urea price should be increased in a phased manner and subsidy savings on account of increase in urea prices should be used for providing higher subsidy to farmers on P & K fertilizers to address the problem of nutrient imbalance. As problems of micronutrient deficiencies and low organic carbon in Indian soils are widespread, a nationwide intensive awareness campaign on balanced nutrition with special focus on use of micronutrients and bio-fertilizers should be launched.

- S.9 Since balanced fertilizer application based on soil test recommendations is crucial for cost reduction and improving nutrient efficiency and long-term soil fertility, the Commission recommends strengthening soil testing laboratories, installing modern equipments, and providing hands-on training to staff for real-time soil analysis and advisory services to farmers.

Strengthen PSS Procurement Operations at State Level

- S.10 Procurement of pulses and oilseeds under Price Support Scheme (PSS), which is implemented on the request of the State Governments/UTs, faces several challenges such as delays in procurement operations, inadequate scientific storage and warehousing infrastructure, jute bag availability, timely payment to farmers, etc. In order to make PSS more effective, the Commission recommends that States should make adequate and timely arrangements for procurement including registration of farmers, details of purchase centers, quality specifications, scheduling of date and time for procuring produce at purchase center, storage and warehousing, gunny bags, QR tags, timely payment, etc. well in advance and communicated to farmers.

Strategies for Reducing Costs

- S.11 The Government is promoting use of renewable energy, farm mechanization, nano-fertilizers, drones and various other initiatives to reduce costs and promote efficient resource use. In order to enhance adoption of such interventions by the farming community, awareness campaigns and demonstrations should be organized. Besides, the Commission recommends significant increase in allocation under Sub-Mission on Agricultural Mechanization (SMAM) and the issue of high GST rates on farm machinery and implements requires favourable consideration to encourage mechanization. In addition, farmers should be encouraged to shift from less profitable crops to more suitable and profitable crops.

Commodity Markets and Price Outlook

- S.12 Commodity markets and price outlook reports provide comprehensive assessments and short-term forecasts of market movements and policy developments in various countries. The Commission recommends that a comprehensive project on commodity markets and price outlook should be launched. This would enable farmers to make informed decisions on production and marketing as well as help Government to manage price volatility and take evidence-based policy decisions.

Improve Irrigation Water Use Efficiency

- S.13 *Pradhan Mantri Krishi Sinchayee Yojana* (PMKSY) has played a significant role in increasing irrigation potential and micro-irrigation coverage, but there are large disparities in irrigation intensity across States and area under micro-irrigation is concentrated in a few States. Therefore, future efforts



should focus on reducing regional disparities in irrigation coverage and improving water use efficiency and productivity. The Commission recommends that awareness campaigns and training programmes should be organized to encourage farmers to adopt micro-irrigation and special emphasis should be given on repair, renovation, and restoration of traditional water bodies with active involvement of district administration and local bodies including gram *panchayats*.

Make Agricultural Credit More Inclusive

- S.14 Policy initiatives taken by the Government of India for increasing flow of credit to agriculture have resulted in significant increase in credit flow and also improved small and marginal farmer's access to institutional credit. However, large disparities across States and regions remain, necessitating concerted efforts to reduce these disparities by strengthening Rural Financial Institutions (RFIs) and improving their presence in Central, Eastern, and North-eastern regions.

Remove Market Distortions

- S.15 It has been observed that high market fees, taxes and other incidental charges on procurement restrict efficient and competitive functioning of markets as it drives out private trade. Therefore, to facilitate price discovery, enhance efficiency of market outcomes and participation of private sector, the Commission recommends that States should rationalise market fee and other charges on procurement.
- S.16 The Government has imposed various restrictions including stock limits and ban on exports of wheat. However, the Commission is of the view that such regulatory measures should be temporary in nature and efforts should be made to improve wheat supply in the country.

Measures to Address Crop Losses/Damages by Wild/Stray Animals

- S.17 The incidences of crop damages, particularly pulses, by wild/stray animals have significantly increased in recent years. Despite provision for coverage of crop losses due to wild animals under *Pradhan Mantri Fasal Bima Yojana* (PMFBY), awareness among farmers is very low. Therefore, there is a need to increase awareness about such provisions and feasibility of providing financial assistance to farmers for solar fencing of fields on community/group basis could be explored.

Awareness about MSP and FAQ Specifications

- S.18 Various initiatives taken by the Central and State Governments to promote public awareness about MSP and procurement operations have resulted in increased awareness and participation of farmers. However, lack of knowledge about FAQ specifications, in particular moisture content in mustard, and non-availability of basic infrastructure facilities for cleaning and drying of produce are major constraints faced by farmers. Therefore, the Commission strongly recommends that basic infrastructure facilities for cleaning and drying of produce should be provided at farm/village level and special awareness campaigns using different media platforms of publicity including print, electronic/digital, audio/video, etc. should be organized to spread awareness about MSP in areas of low procurement.



Sample Size and Coverage of States

- S.19 It has been observed that in some cases sample size under the Comprehensive Scheme (CS) is small and some States are not covered under the CS. Therefore, in order to ensure reliability and robustness of the cost estimates used for MSP determination, the Commission recommends that sample size of gram in Haryana, Jharkhand and Telangana, rapeseed & mustard in Chhattisgarh, and safflower in Karnataka and Maharashtra should be increased and cost data should be collected on continuous basis for gram in West Bengal, lentil in Jharkhand, rapeseed & mustard in Chhattisgarh and safflower in Maharashtra. In addition, the Comprehensive Scheme should be implemented in Jammu & Kashmir for wheat and rapeseed & mustard, and Uttarakhand for wheat, barley, lentil and rapeseed & mustard.



CHAPTER 1



Overview

Introduction

- 1.1 This chapter provides an overview of the performance of Indian agriculture and various Government initiatives in the sector. With a brief discussion on the trends and growth rate of area, production and yield of the mandated crops in the recent period, the chapter analyses trade performance, procurement trends, stock position, inflation, capital formation in agriculture, irrigation development, issues related to fertilizer, seed, credit, etc. It includes a brief review of the demand-supply situation of pulses and edible oils as well as the changing composition and major drivers of growth in agricultural sector. The chapter concludes with an outlook of demand-supply in domestic and world markets.
- 1.2 Indian agricultural sector witnessed a remarkable performance in 2024-25 with a record production of 353.9 million tonnes of foodgrains, up by 6.5 percent from 2023-24, as per the 3rd Advance Estimates of Production of Foodgrains and Oilseeds, Ministry of Agriculture and Farmers Welfare. Wheat production is estimated to increase by 3.7 percent to 117.5 million tonnes in 2024-25 as compared to record 113.3 million tonnes in 2023-24. An appreciable increase of 4.1 percent in total pulses production is estimated in 2024-25 at 25.2 million tonnes but still lower than the record production of 27.3 million tonnes in 2021-22. The production of *rabi* pulses is estimated to increase by 1.9 percent during 2024-25, while gram production is estimated to reach 11.3 million tonnes, an increase of 2.7 percent over last year. Lentil production, which reached a record level (1.8 million tonnes) in 2023-24, marginally declined (-1.1%) in 2024-25. The production of total oilseeds in 2024-25 is also estimated to significantly rise by 7.4 percent at 42.6 million tonnes but production of *rabi* oilseeds is estimated to fall by 4.9 percent. Rapeseed & Mustard (R&M) production is expected to fall by 4.9 percent at 12.6 million tonnes in 2024-25, while production of safflower, which has witnessed continuous decline during the last few decades, is estimated to fall by a whopping 14 percent. The increase in total pulses and oilseeds during 2024-25 is largely driven by an increase in production of *kharif* pulses (7.5%) and *kharif* oilseeds (15%) and this increase augurs well for a much-needed shift towards these crops to achieve *atmanirbharta*. The real Gross Value Added (GVA) from agriculture and allied sectors is estimated to grow at 4.6 percent in 2024-25 as compared to 2.7 percent last year, primarily attributed to record production and favourable monsoons.



- 1.3 During last 10 years, there has been a significant increase in production of all major crops/crop groups in the country (Table 1.1). The average foodgrains production increased from 278.9 million tonnes during 2015-16 to 2019-20 (Period I) to 328.5 million tonnes during 2020-21 to 2024-25 (Period II), while production of total pulses increased from 22 million tonnes to 25.7 million tonnes during the period. Both area expansion and yield improvement contributed to higher production. However, average annual growth rate in production of wheat, gram and R&M decelerated over the last five years. Improvement in yield, which was a major contributor to higher production in all crops except safflower during Period I, became less important and production growth was primarily driven by area expansion during the recent period. While, average yield of all crops/crop groups has increased, growth of all crops/crop groups except lentil was lower during 2020-21 to 2024-25 compared to 2015-16 to 2019-20. The declining rate of growth in crop productivity during the recent period contributed to the slower output growth rate, therefore, deceleration in growth rate of crop yields is a serious concern and needs to be addressed.
- 1.4 All-India estimates of area, production, and yield of principal crops are given in Annex Tables 1.1 to 1.3, while share of major States in total production of mandated *rabi* crops during the Triennium Ending (TE) 2024-25 is given in Annex Table 1.4.

Table 1.1: Trends and Growth Rate of Area, Production and Yield of Mandated Rabi Crops in India: 2015-16 to 2024-25

(Area in million ha, production in million tonnes, yield in qtl/ha)

Crop	2015-16 to 2019-20	2020-21 to 2024-25	2015-16 to 2019-20	2020-21 to 2024-25
	Area		Growth Rate (%)	
Total Foodgrains	126.3	132.3	0.5	1.6
Wheat	30.3	31.5	0.0	0.9
Barley	0.6	0.6	-3.0	2.6
Total Pulses	28.3	28.7	3.8	-0.1
Gram	9.6	10.1	3.6	0.0
Lentil	1.4	1.6	-1.8	5.8
Total Oilseeds	25.7	29.7	1.3	2.2
Rapeseed & Mustard	6.2	8.3	3.5	5.1
Safflower	0.1	0.1	-17.6	6.8



	Production		Growth Rate (%)	
Total Foodgrains	278.9	328.5	3.4	3.6
Wheat	100.4	111.7	4.5	1.7
Barley	1.7	1.7	1.9	3.2
Total Pulses	22.0	25.7	7.6	2.1
Gram	9.8	12.0	9.9	0.9
Lentil	1.2	1.6	3.6	11.4
Total Oilseeds	30.5	39.5	4.4	5.2
Rapeseed & Mustard	8.3	12.1	7.9	6.9
Safflower	0.1	0.1	3.2	8.1
	Yield		Growth Rate (%)	
Total Foodgrains	22.1	24.8	2.9	1.9
Wheat	33.1	35.5	4.7	0.8
Barley	27.1	30.0	5.1	0.9
Total Pulses	7.8	8.9	3.2	2.2
Gram	10.2	11.9	5.5	0.8
Lentil	8.9	9.9	4.6	4.7
Total Oilseeds	11.9	13.3	3.2	2.9
Rapeseed & Mustard	13.5	14.7	4.6	2.1
Safflower	6.0	7.7	15.5	-1.0

Source: Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare

Pulses and Oilseeds: High Import Dependency

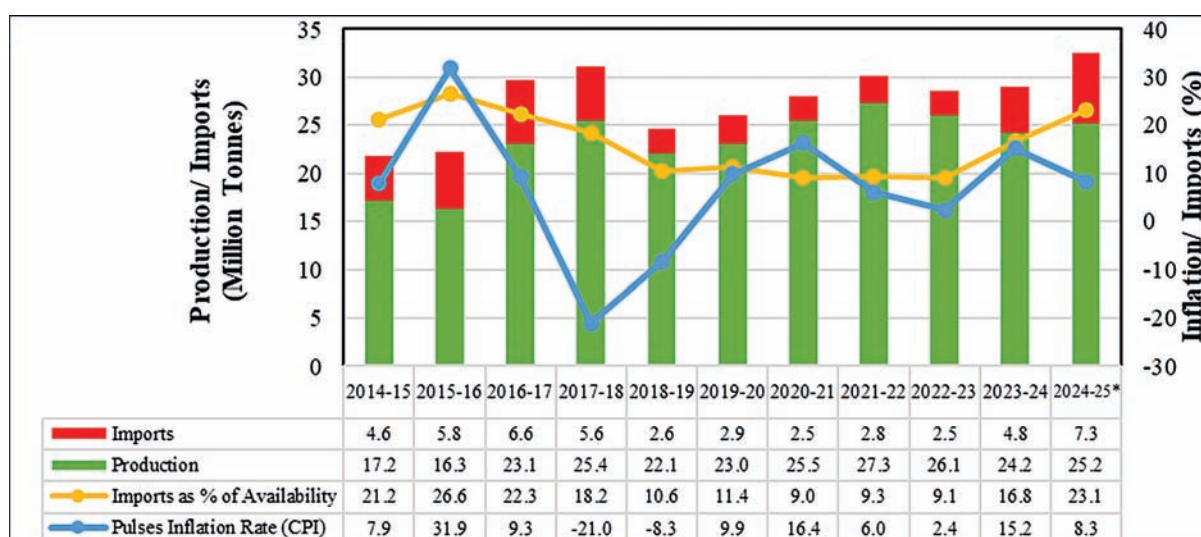
Pulses

- 1.5 India's pulses production, which increased significantly during the last 10 years and reached a record level of 27.3 million tonnes in 2021-22, witnessed a declining trend during 2022-23 and 2023-24 (Chart 1.1). However, trend reversed in 2024-25 and total pulses production is estimated to increase by 4.1 percent at 25.2 million tonnes. Production of gram, the most dominant crop that accounts for over 45 percent of total pulses production in the country, declined during 2022-23 and 2023-24 and is estimated to improve by 2.7 percent during 2024-25 but still much lower than the record 13.5 million tonnes in 2021-22. Lentil production, which improved significantly during 2022-23 and 2023-24, is estimated to marginally decline by 1.1 percent during 2024-25.



- 1.6 Decline in production and fast rising demand for pulses have resulted in steep rise in imports of pulses during the last two years. Imports, which average around 2.7 million tonnes between 2018-19 and 2022-23, sharply rose to 4.8 million tonnes in 2023-24 and hit a new record of 7.3 million tonnes in 2024-25. Between 2020-21 and 2024-25, import dependency in pulses increased from nine percent to 23.1 percent. However improved availability of pulses due to higher imports has softened inflation in pulses in 2024-25. To address the issue of high import dependency and promote self-sufficiency in pulses, Government of India has launched various initiatives to enhance domestic production of pulses, including Mission for *Aatmanirbharta* (self-reliance) in Pulses', significant increase in Minimum Support Prices (MSP), 100 percent procurement of *tur/arhar*, *urad* and *lentil/masur* till 2028-29, etc.

Chart 1.1: Trends in Pulses Production, Imports and Inflation



Note: *Production as per 3rd Advance Estimates, Availability = Production + Imports - Exports

Source: 1. Economics, Statistics and Evaluation Division, Ministry of Agriculture & Farmers Welfare

2. Directorate General of Commercial Intelligence and Statistics

3. Ministry of Statistics and Programme Implementation

- 1.7 India is the largest producer of pulses in the world but yield is significantly lower compared to other major producers (Table 1.2). To meet growing demand for pulses, India must scale up production by targeting both horizontal and vertical expansion. Horizontal expansion (an increase in area under cultivation) could be achieved by targeting rice fallow areas, promotion of short duration varieties and intercropping with other crops. A vertical expansion (improvement in productivity) could be achieved by adoption of new seed varieties, which are drought/disease resistant, better access to farm machinery, protective irrigation in rainfed areas and bridging yield gaps. Pulses are mostly grown on rainfed marginal lands and are affected by drought and heat stress, thus, major focus should be on development of drought-tolerant varieties.



Table 1.2: Global Scenario by Top Ten Pulses Producing Countries (2018-2023)

Countries	Area (Mha)	Production (MT)	Yield (qtl/ha)	Contribution (%)		Ranking (Yield)
				Area	Production	
India	33.6	24.9	7.4	35.6	27.3	9
Canada	3.3	6.2	18.3	3.5	6.8	2
China	2.5	4.9	18.3	2.6	5.3	3
Nigeria	5.1	4.2	8.3	5.4	4.6	8
Russian Federation	2.0	3.9	17.6	2.1	4.3	4
Myanmar	3.8	3.6	9.5	4.1	4.0	7
Australia	2.6	3.3	14.4	2.8	3.7	5
Ethiopia	1.6	3.0	18.9	1.7	3.3	1
Brazil	2.8	2.9	11.0	2.9	3.2	6
Niger	6.0	2.5	4.2	6.3	2.8	10
World	94.5	91.2	9.6	-	-	-

Note: Pulses include imputed value of dry beans and top-10 countries are based on production share in 2023
Source: Computed from FAOSTAT

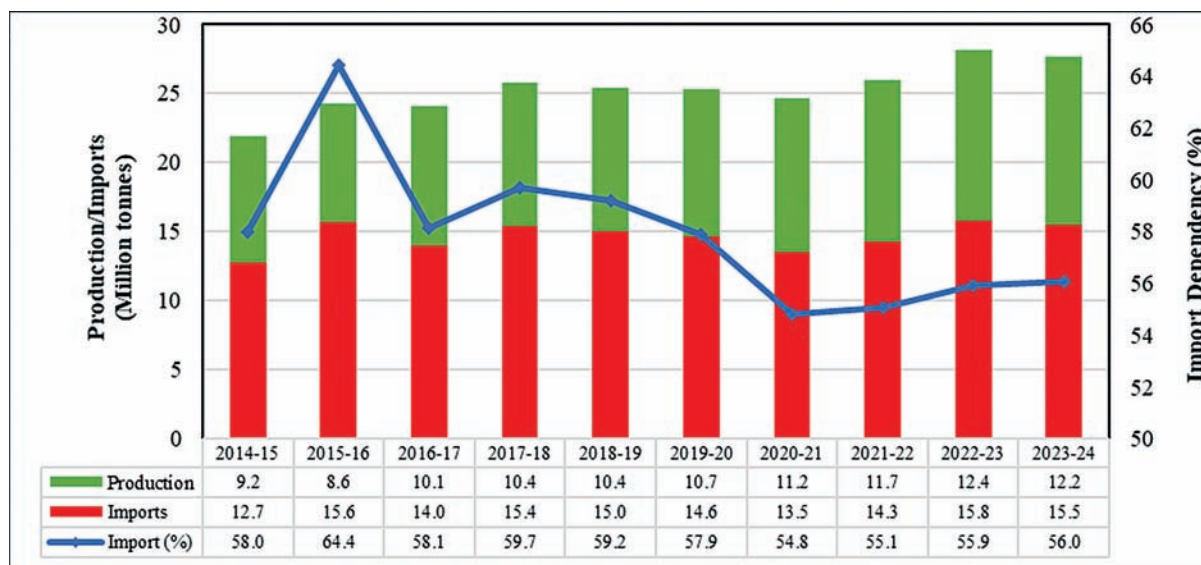
Oilseeds

- 1.8 Despite impressive growth in domestic production of oilseeds, the country has not been able to keep pace with growing consumption of edible oils. The domestic availability of edible oils has increased considerably from 8.6 million tonnes in 2015-16 to 12.2 million tonnes in 2023-24, while consumption has increased from about 24 million tonnes to about 28 million tonnes during this period (Chart 1.2). In 2023-24, domestic production of edible oils, including secondary sources, was estimated at 12.2 million tonnes, while consumption was estimated at about 27.7 million tonnes. Due to the persistent large gap between demand and supply, the country has remained dependent on imports over the past two to three decades.
- 1.9 India is the largest importer of vegetable oils in the world and imported about 16.4 million tonnes costing around ₹1.47 lakh crore in 2024-25. Import dependence, which declined from 59.7 percent in 2017-18 to 54.8 percent in 2020-21, increased during the last three years. Due to high volatility in global prices, edible oil imports (in value terms) showed large fluctuations. For example, between 2020-21 and 2022-23, volume of imports increased by 16.3 percent, from 13.5 million tonnes to 15.8 million tonnes but value of imports more than doubled from ₹82 thousand crore to ₹1.67 lakh crore. As per NITI Aayog's estimates, based on the static/household demand approach, the demand for edible oils is projected to reach 29.8 million tonnes by 2030 with a deficit of 14.1 million tonnes. In order to bridge this gap and boost domestic production, the Government has launched several initiatives



including National Mission on Edible Oils - Oil Palm (NMEO-OP) and National Mission on Edible Oils - Oilseeds (NMEO-Oilseeds).

Chart 1.2: Trends in Domestic Production and Imports of Edible Oils in India



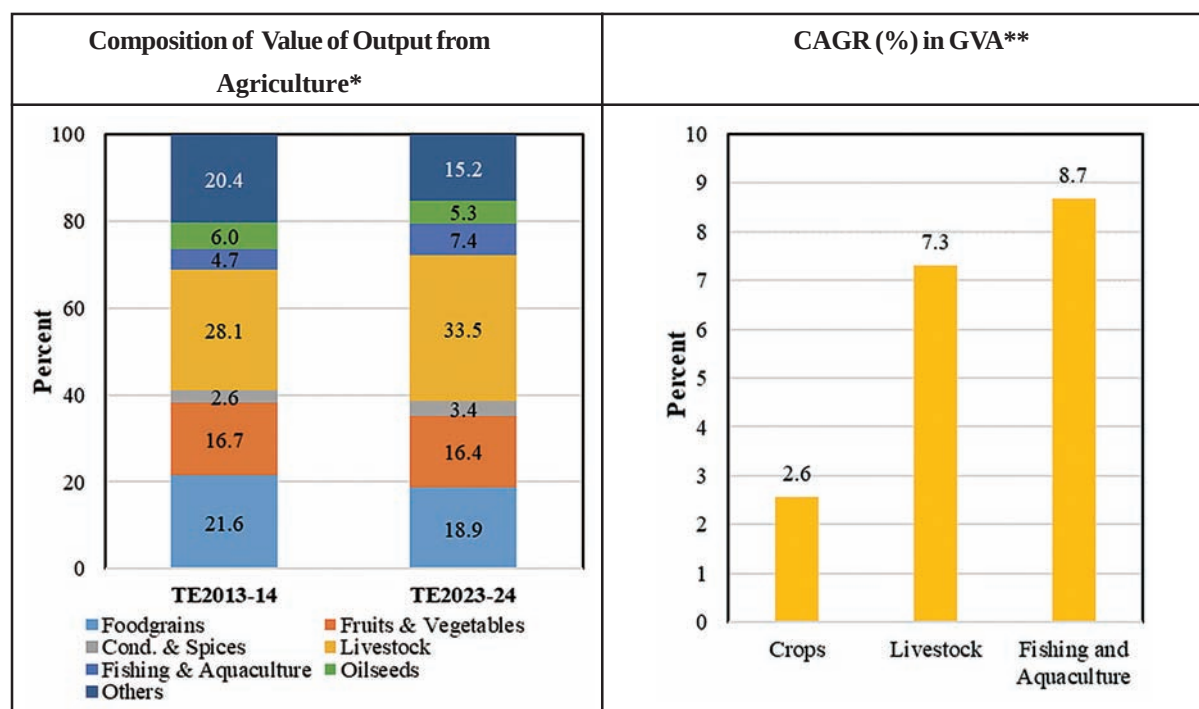
Source: 1. Department of Food and Public Distribution
2. Directorate General of Commercial Intelligence and Statistics

Changing Composition and Drivers of Agricultural Growth

- 1.10 The composition of agricultural output in India has changed, from a traditional cereal-based staples to high-value crops and allied sectors. This shift is driven by changing consumption pattern, rising income, market dynamics, demographic factors, etc. The Household Consumption Expenditure Survey (HCES) 2023-24 clearly shows a shift in consumption pattern from cereal & substitutes to beverages & processed food, milk and dairy products, and meat, eggs & fish during the last two decades. The change in composition of agricultural output is reflected in the changing shares of agricultural commodities and allied sectors in total value of output (Chart 1.3). The share of foodgrains in total value of output has declined from 21.6 percent in TE2013-14 to 18.9 percent in TE2023-24, while share of livestock products has increased from 28.1 percent to 33.5 percent and fishing & aquaculture from 4.7 percent to 7.4 percent during the period. The GVA from crop sector has grown at Compound Annual Growth Rate (CAGR) of 2.5 percent during 2011-12 and 2023-24 but fishing & aquaculture (8.7%) and livestock sector (7.3%) have grown at a much faster rate. However, this composition and growth have differed across regions and States, given their heterogeneous agro-climatic and geographic endowments and policy initiatives.



Chart 1.3: Share and Growth Rate of Different Components of Agricultural Output and Gross Value Added (GVA)



Note: *excluding forestry and logging; **2011-12 to 2023-24 period

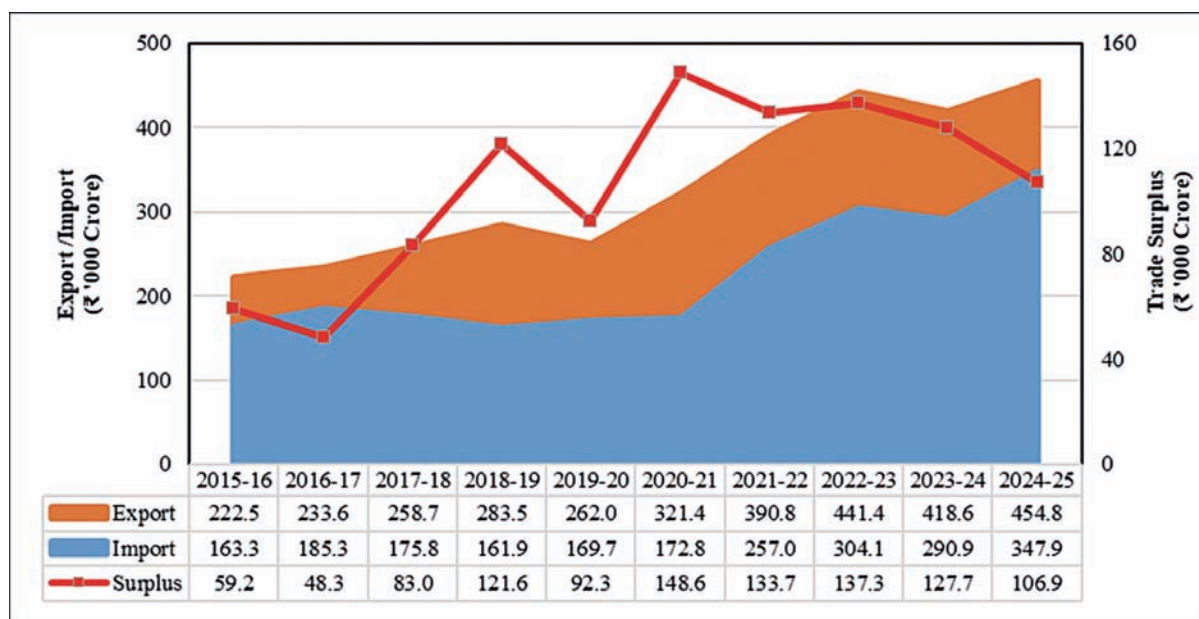
Source: Ministry of Statistics and Programme Implementation

Agricultural Trade Performance

- 1.11 During the last 10 years, both agricultural exports and imports have more than doubled, exports increased by 104.4 percent and imports by 113 percent between 2015-16 to 2024-25. India's agri-exports during 2024-25 were ₹4.55 lakh crore against an import of agri-commodities worth ₹3.48 lakh crore with agricultural sector having a large trade surplus (Chart 1.4). While both exports and imports registered an impressive increase in 2024-25, increase in imports (19.6%) was much higher vis-a-vis increase in exports (8.6%), leading to a fall in trade surplus to ₹1.07 lakh crore, a decline of 16.3 percent.
- 1.12 India has emerged as the world's largest exporter of rice and the largest importer of vegetable oils (with imports of 16.4 million tonnes worth ₹1.47 lakh crore) and pulses (with imports exceeding 7.3 million tonnes worth ₹47 thousand crore) during 2024-25. Rice had the largest share (23.2%) in the agri-exports, followed by marine products (13.8%), spices (8.3%) and buffalo meat (7.6%), while import basket primarily comprised of vegetable oils (42.1%), pulses (13.3%) and fresh fruits (7.4%). India is net exporter of agricultural commodities but exports were adversely affected due to various restrictions on agri-exports during the last two to three years. Therefore, to realize full export potential, an open, stable and rational agri-trade policy is needed.



Chart 1.4: India's Exports and Imports of Agricultural Commodities (2015-16 to 2024-25)



Source: Directorate General of Commercial Intelligence and Statistics

Foodgrains Management: Procurement, Stocks and Distribution

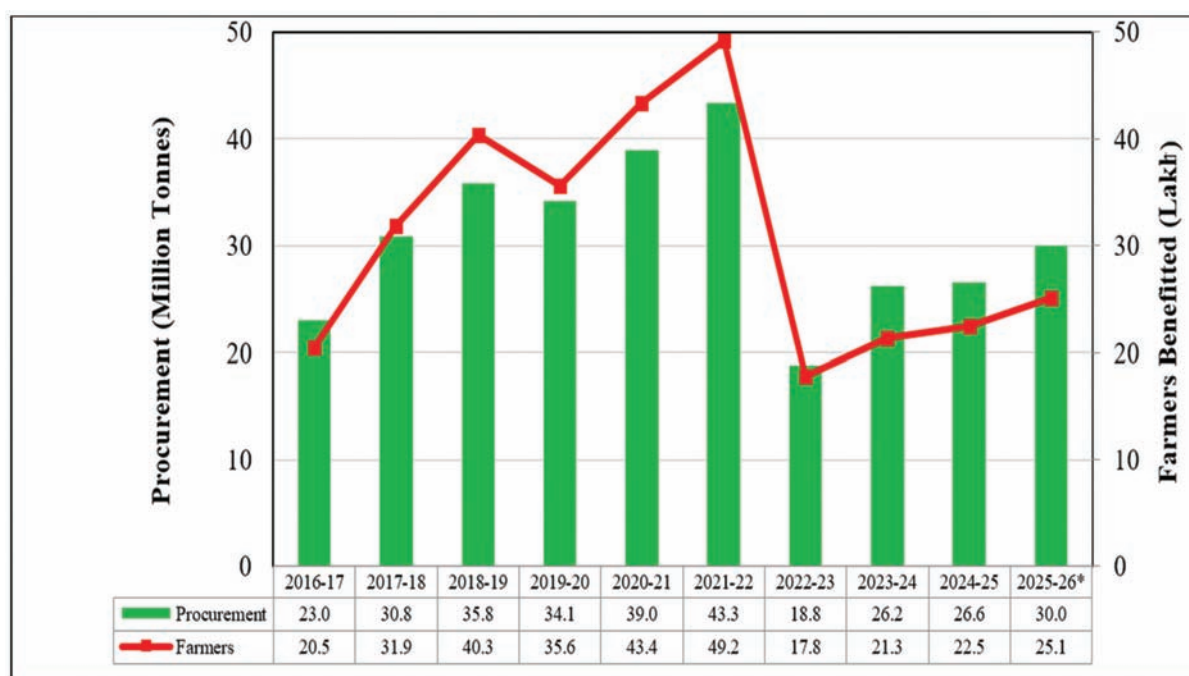
Cereals

- 1.13 Government of India's foodgrains management policy involves procurement, storage, and distribution of foodgrains, mainly rice and wheat, under Targeted Public Distribution System (TPDS) and Other Welfare Schemes (OWS) and maintenance of buffer stocks. However, policy is confronted with a dilemma of record production of rice and wheat and at the same time, high inflation for these commodities.
- 1.14 Wheat procurement, which showed a rising trend during 2016-17 to 2021-22 and reached a record high of 43.3 million tonnes in 2021-22, recorded a steep fall to 18.8 million tonnes in RMS2022-23 but improved during the last three years and is estimated at 30 million tonnes during 2025-26 (Chart 1.5). The number of farmers benefitted from procurement operations significantly declined during RMS2022-23 but increased during the last three seasons and 25.1 lakh farmers have benefitted from wheat procurement operations during RMS2025-26.
- 1.15 The total stocks of rice and wheat held by FCI and State agencies as on May 31, 2025 (closing stock) were 74.9 million tonnes, 36.9 million tonnes of rice and 38 million tonnes of wheat. Wheat stocks, which fell to the lowest level of 7.5 million tonnes in April 2024, showed an improvement during 2024-25 and remained higher than stocking norms. However, lower procurement and stocks of wheat during the last few years have resulted in reduced allocation of wheat under the National Food Security Act (NFSA) and Other Welfare Schemes (OWS). Allocation of wheat declined from 33.8 million tonnes in 2021-22 to 28.9 million tonnes in 2023-24 and was 24.1 million tonnes in 2024-25 (Chart 1.6). Total wheat allocation for 2025-26 is 18.4 million tonnes. The average offtake of wheat has



been over 92 percent of the allocation. However, excess stocks of rice in Central Pool, about three times the stocking norms, despite record offtake under NFSA, OMSS (D), and by distilleries for ethanol production, have implications for storage space for coming season and rising food subsidy.

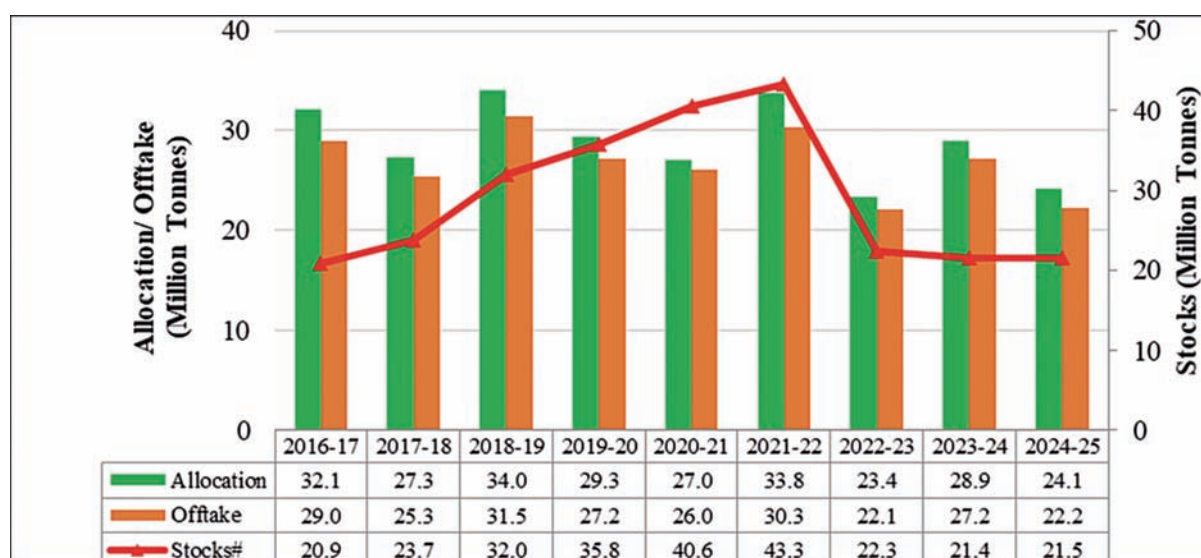
Chart 1.5: Trends in Procurement of Wheat and Beneficiary Farmers



Note: *As on June 30, 2025

Source: Food Corporation of India

Chart 1.6: Trends in Stocks, Allocation and Offtake of Wheat: 2016-17 to 2024-25



Note: #Monthly average of each financial year

Source: Department of Food and Public Distribution, Ministry of Consumer Affairs, Food and Public Distribution

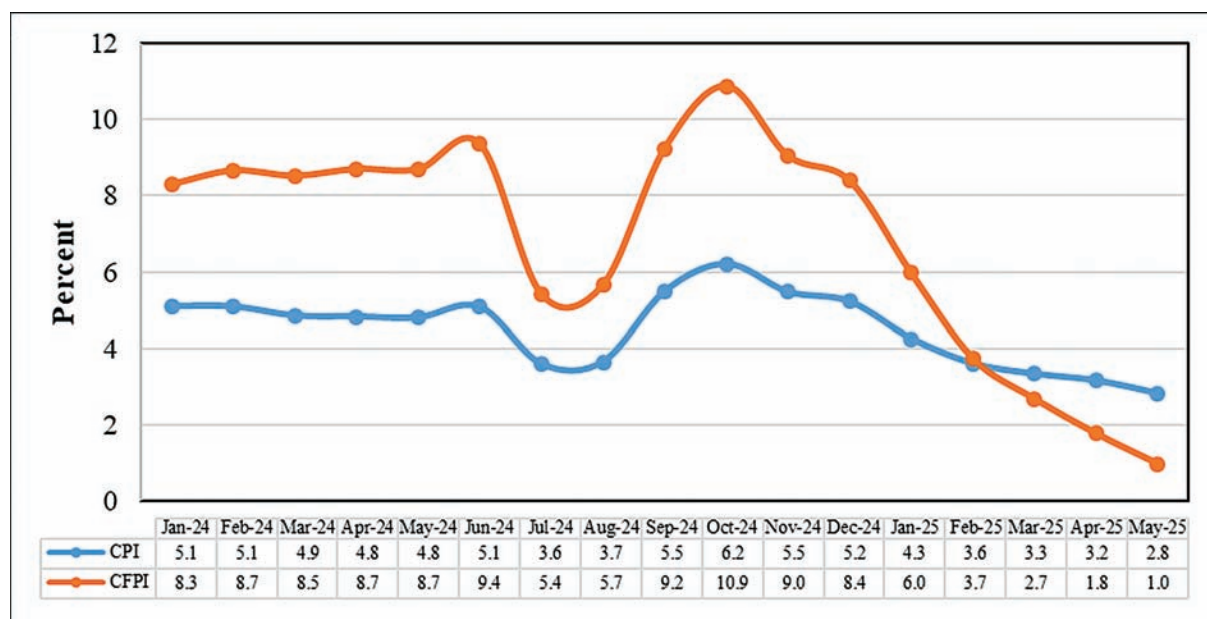


Food Inflation

- 1.16 High food inflation has been a major concern over the last few years despite record production. Food inflation in India remained high during the last three years but has moderated during January to May 2025. Year-on-year inflation rate based on All India Consumer Price Index (CPI) during 2024 was more than five percent for seven months, with a high of 6.2 percent in October 2024, after which it has constantly been declining and stood at 2.8 percent in May 2025 (Chart 1.7). All-India Consumer Food Price Index (CFPI) based inflation remained above eight percent during 10 months in 2024 with a high of 10.9 percent during October 2024, thereafter it has been continuously falling and reached the lowest level (0.99%) in May 2025. The CPI (General) inflation rate, which remained much below the CFPI inflation between January 2024 and February 2025, increased and was above the CFPI inflation during March-May 2025. The disaggregated analysis of food inflation indicates that pulses & products, vegetables, spices, cereals and products, milk and products, eggs, fruits and sugar and confectionary were major drivers of food inflation during 2023-24, while during 2024-25, food inflation was mainly driven by vegetables, pulses and products, cereals and products, fruits and oils & fats (Chart 1.8).
- 1.17 Inflation rate based on CFPI averaged seven percent during 2023-24, which marginally declined to 6.8 percent in 2024-25 (Chart 1.9). The *El Niño* induced adverse impact on domestic production of various food commodities and increase in world edible oil prices led to higher food inflation. Consequently, food inflation remained relatively higher during the last two years, with vegetables, cereals and products, pulses and products, fruits, livestock products and oils and fats being key drivers. However, inflation rate for cereals and products, pulses and products, milk and products declined during 2024-25 compared with 2023-24, while inflation for vegetables, meat and fish and oils and fats increased. These trends clearly indicate that in addition to supply-side constraints, changing consumption pattern towards high-value commodities including fruits & vegetables, milk and dairy products, meat & fish also influence food inflation dynamics. In this changing scenario, the need for location specific crop diversification strategy with considerations on environment and soil health aspects and overall food security, emerges.

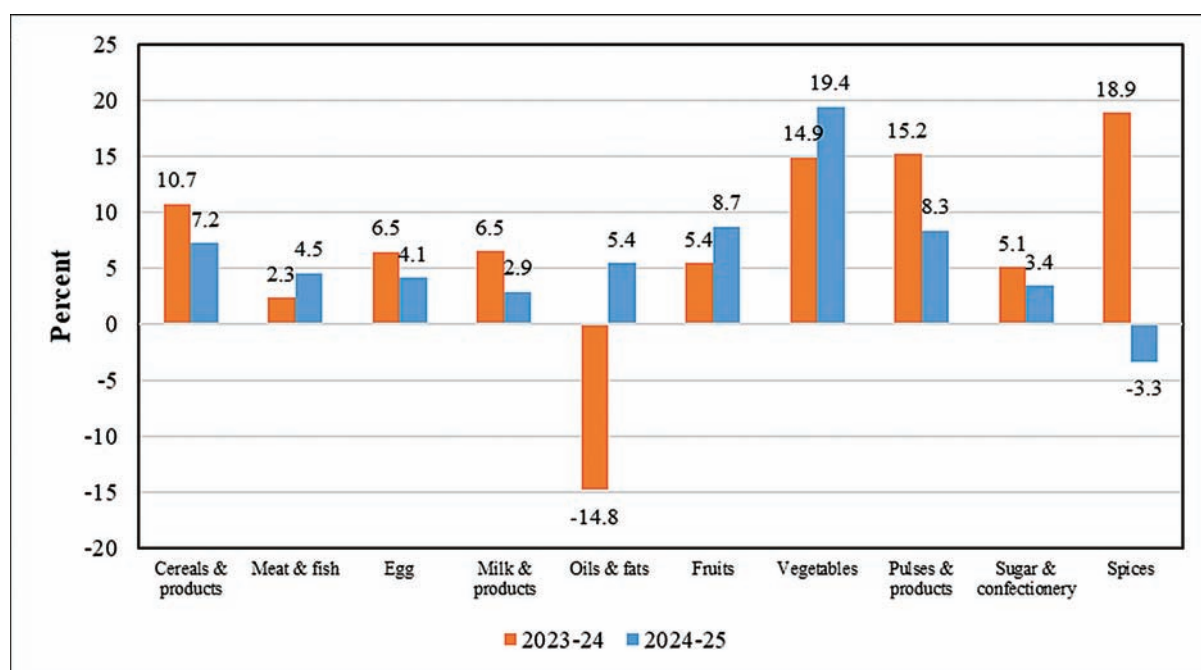


Chart 1.7: All-India Inflation Rates for CPI (General) and CFPI



Source: Ministry of Statistics and Programme Implementation

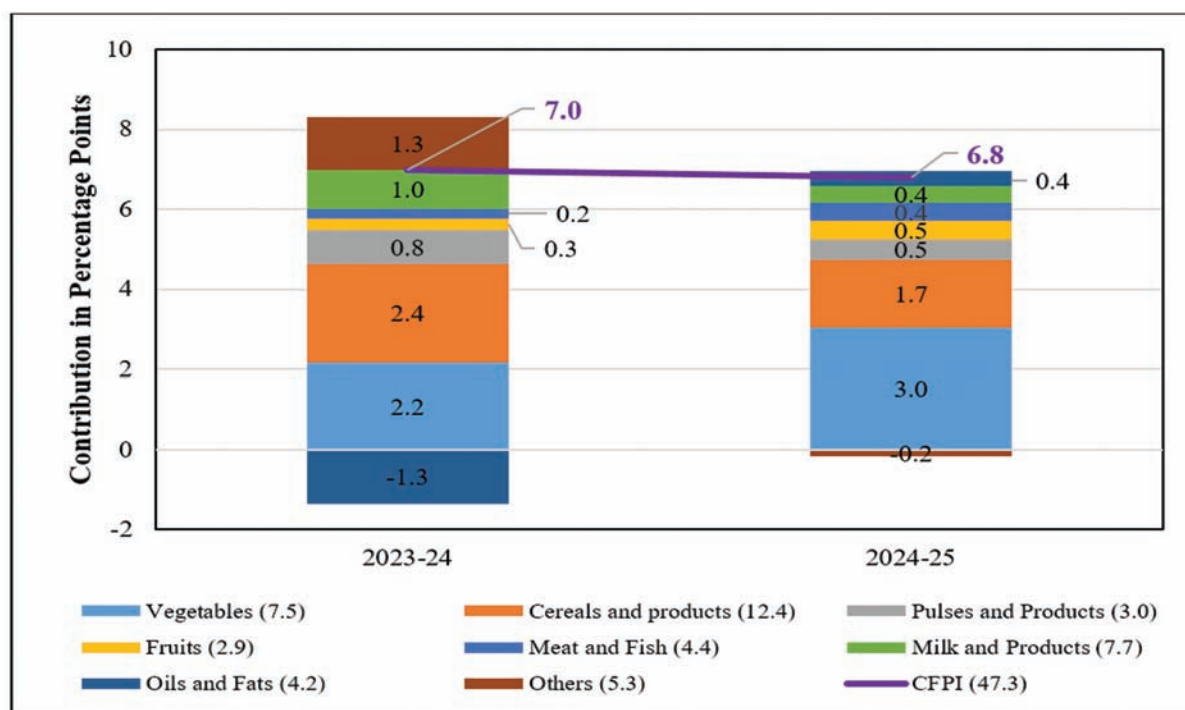
Chart 1.8: Trends in CPI based Food Inflation: 2023-24 and 2024-25



Source: Ministry of Statistics and Programme Implementation



Chart 1.9: Drivers of CFPI Inflation (Y-o-Y)



Note: Figures in parenthesis denote the weight of respective item in CFPI basket

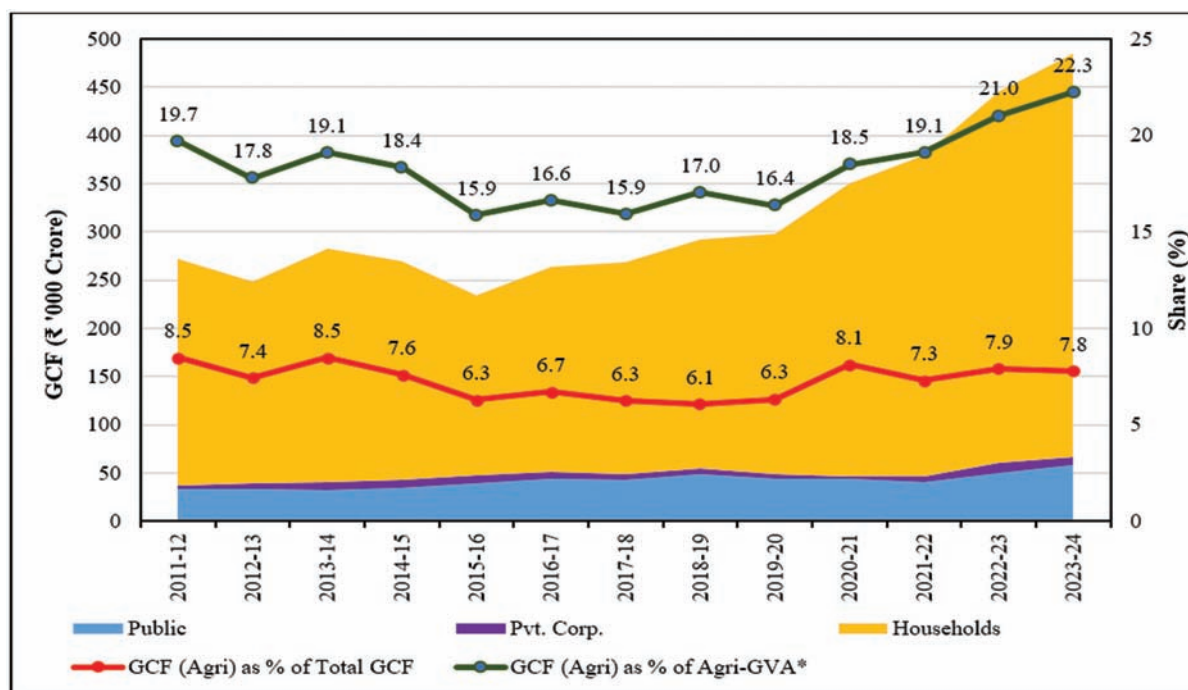
Source: Ministry of Statistics and Programme Implementation

Capital Formation in Agriculture

- 1.18 Capital formation in agriculture is crucial for improving productivity and sustainable growth of the sector. It is encouraging to note that total Gross Capital Formation (GCF) in agriculture has witnessed a continuous increase during 2016-17 to 2023-24, with dominant share (86.7%) of household sector, followed by public (11.4%) and private corporate sector (1.9%) during the TE2023-24. The GCF in agriculture as a percentage of the Gross Value Added (GVA) from agriculture excluding forestry and logging has shown an increasing trend since 2019-20 and reached its highest level of 22.3 percent in 2023-24 (Chart 1.10). The GCF in agriculture as a percentage of total GCF, which declined from 8.5 percent in 2011-12 to 6.1 percent in 2018-19, showed a steady improvement and was 7.8 percent in 2023-24.
- 1.19 The private corporate sector investment is essential for improving economic efficiency and boosting productivity in agriculture but its share in total investment has remained meagre (<2%). This indicates, the scope for enhancing participation of private corporate sector in the agriculture through appropriate policy interventions to boost productivity, improve farmers' incomes and promote sustainable development.



Chart 1.10: Trends in Gross Capital Formation in Indian Agriculture



Source: Ministry of Statistics and Programme Implementation

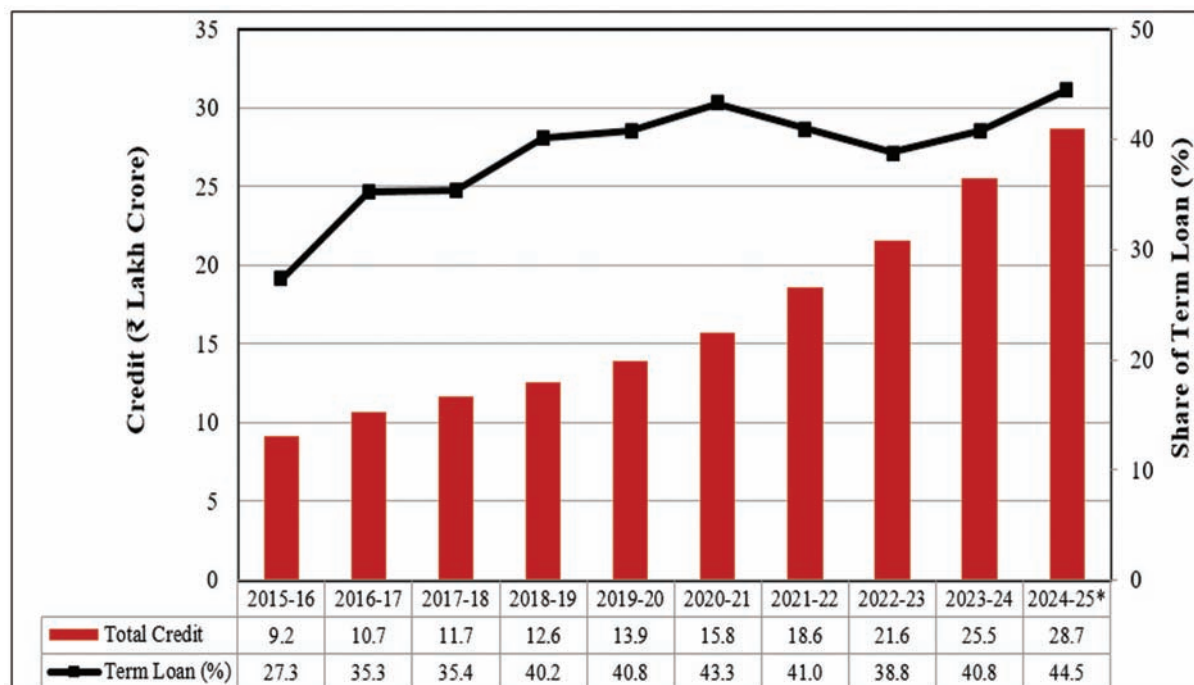
Flow of Credit to Agriculture

- 1.20 Access to agricultural credit is an important policy instrument for agricultural development, therefore, Government has taken several initiatives including *Kisan Credit Card* (KCC), Interest Subvention Scheme, priority sector lending, collateral-free loans, etc. to improve access to affordable institutional credit. These efforts have resulted in significant increase in flow of institutional credit to agricultural sector. During the last 10 years, total credit flow to agriculture has more than tripled, from ₹9.2 lakh crore in 2015-16 to ₹28.7 lakh crore in 2024-25 and share of investment credit, essential for capital formation, has increased from 27.3 percent to 44.5 percent during the period (Chart 1.11).
- 1.21 It is worth noting that share of small and marginal farmers in the total agricultural credit disbursed has increased from 44.1 percent in 2013-14 to 56.5 percent during 2023-24, which is much higher than their share (47%) in total operated area as per Agriculture Census 2015-16.
- 1.22 The share of cooperative banks in agricultural credit has witnessed a steep decline during the last three decades, while share of commercial banks has risen sharply and was more than 79 percent in 2023-24. The decline in share of cooperative banks in agricultural credit is a cause for concern because cooperatives play a critical role in providing financial services to agricultural and rural sector. However, Government's initiatives for strengthening Primary Agricultural Credit Societies (PACS) would help in reversal of this trend and improve farmers' access to institutional credit.
- 1.23 The KCC scheme, a single window facility for providing working capital as well as investment credit to farmers, has made significant progress and amount outstanding in operating accounts has increased from ₹5.02 lakh crore in 2014-15 to more than ₹10 lakh crore in 2024-25 and number of operative KCC



accounts has increased from 7.36 crore to 7.72 crore during the period. The Reserve Bank of India (RBI) has enhanced the collateral-free loan limit for farmers from ₹1.6 lakh to ₹2 lakh with effect from January 1, 2025 and Government of India increased the KCC loan limit from ₹3 lakh to ₹5 lakh under the Modified Interest Subvention Scheme (MISS) in the recent budget to improve access to institutional credit. However, regional disparity in distribution of credit, needs to be addressed for more balanced distribution of agri-credit and growth of agriculture in the country.

Chart 1.11: Trends in Institutional Credit to Agriculture



Note: *Provisional

Source: NABARD

Availability of Quality Seeds

- 1.24 Access to quality seeds is critical for food and nutritional security but lack of access to good quality seeds of new varieties is one of the major constraints faced by farmers. Despite large number of improved varieties released by Indian Council of Agricultural Research (ICAR) and State Agricultural Universities and potential benefits of these varieties, widespread adoption remains limited due to various socio-economic, institutional and infrastructure constraints as well as less developed seed value chains. Seed Replacement Rate (SRR) and Varietal Replacement Rate (VRR), instrumental in improving crop productivity, have improved over the years, but are still quite low in some crops, making introduction of new varieties difficult.
- 1.25 In August 2024, 109 varieties of 61 crops comprising 34 field crops and 27 horticultural crops were released and 'National Mission on High Yielding Variety/Hybrid Seeds' with an allocation of ₹100 crore aimed at strengthening the research ecosystem, targeted development and propagation of seeds with high yield, pest resistance and climate resilience was announced in the Union Budget 2025-26.



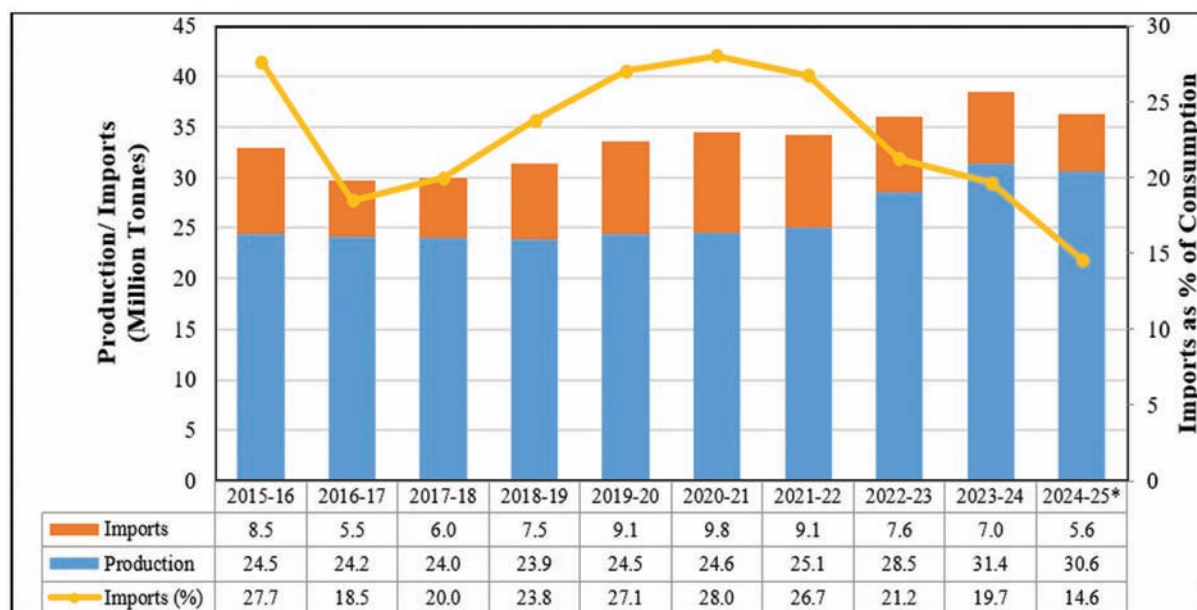
But supportive policy environment must also be in place to strengthen seed value chain and to create an enabling mechanism to make available certified/quality seeds to farmers. This would help in improving SRR and VRR and enhancing productivity of crops. In order to ensure availability of high-quality seeds to farmers, Government has launched SATHI (Seed Traceability, Authentication and Holistic Inventory) portal to track seeds from production to distribution, for inventory management, traceability and prevent spurious seeds from entering the market.

Improved Fertilizer Supply and Efficiency

- 1.26 India is heavily dependent on imports of both raw materials and finished fertilizers and about 30 percent of demand is met through imports. In 2023-24, the gap between consumption and production of N was 3.35 million tonnes, and P_2O_5 was 3.43 million tonnes, while entire K_2O requirement is met through imports. However, various initiatives taken by the Government have resulted in significant expansion in urea production and progress towards achieving *atamnirbharta*. Between 2015-16 and 2023-24, India's domestic urea production increased from 24.5 million tonnes to 31.4 million tonnes but marginally declined to 30.6 million tonnes in 2024-25. On the other hand, imports declined from 8.5 million tonnes in 2015-16 to seven million tonnes in 2023-24, after reaching a peak of 9.8 million tonnes in 2020-21 (Chart 1.12). The share of imports in total urea consumption has declined from 27.7 percent in 2015-16 to 14.6 percent in 2024-25. However, Indian agriculture is also grappling with issues stemming from imbalanced use of nutrients, excessive use of fertilizers in some regions, deficiency of micro-nutrients, depletion of soil organic carbon and rising fertilizer subsidy. Excessive use of highly subsidised urea has resulted in distorted NPK ratio and much higher consumption than recommended levels, more than 500 kilogram per hectare in some districts of Andhra Pradesh and Tamil Nadu. Therefore, it is critical to take corrective actions and mitigate adverse impacts on crop yield, soil health and environment. Government of India has launched various initiatives including Prime Minister's Programme for Restoration, Awareness Building, Nourishment, and Amelioration of Mother Earth (PM-PRANAM), Galvanizing Organic Bio-Agro Resources *Dhan* (GOBARDhan), promotion of nano-fertilizers and bio-fertilizers, etc. to promote sustainable and balanced use of fertilizers.



Chart 1.12: Trends in Production and Imports of Urea in India



Note: *Provisional

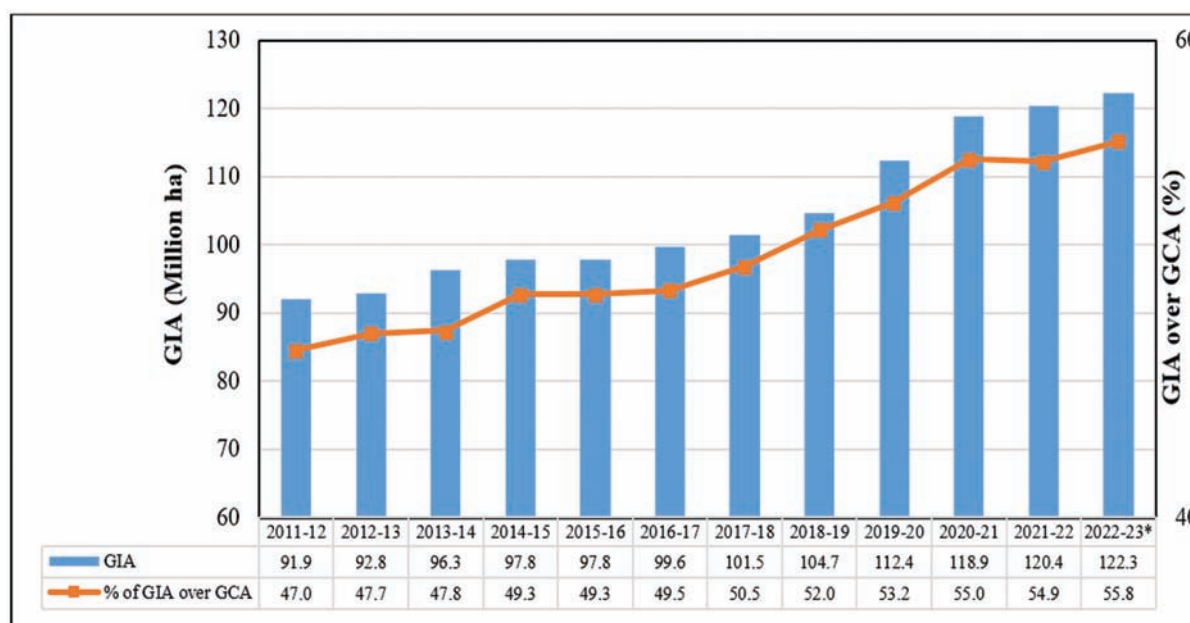
Source: Fertilizer Association of India

Enhanced Irrigation Coverage and Water-use Efficiency

- 1.27 Irrigation will continue to play a key role in agricultural production growth but shrinking availability and inefficient use of irrigation water is becoming a growing concern. Therefore, in order to expand irrigation coverage and promote efficient irrigation systems and practices, Government of India launched *Pradhan Mantri Krishi Sinchayee Yojana* (PMKSY) in July 2015. In addition, Micro Irrigation Fund (MIF) was operationalized by NABARD in 2019-20 to facilitate State Governments for expanding coverage under micro-irrigation and incentivizing its adoption.
- 1.28 Government's efforts have resulted in significant expansion in irrigation and coverage under micro-irrigation during the last few years. Gross Irrigated Area (GIA) has increased from 97.8 million hectares in 2015-16 to 122.3 million hectares in 2022-23 and percentage of GIA over Gross Cropped Area (GCA) has increased from 49.3 percent to 55.8 percent during the period (Chart 1.13). During the last 10 years, GIA rose by 27 percent compared to about nine percent increase in GCA. However, there are large disparities in irrigation intensity across States, ranging from as high as 98 percent in Punjab to less than 25 percent in States like Jharkhand, Assam, Kerala and Maharashtra. Micro-irrigation under PMKSY has made good progress and more than 95 lakh hectare area has been covered under micro-irrigation, but nearly 75 percent of which is concentrated in Karnataka, Rajasthan, Maharashtra, Gujarat, Tamil Nadu and Andhra Pradesh, which indicates a need for wider adoption of micro-irrigation in the country.



Chart 1.13: Trends in Gross Irrigated Area in India



Note: *Provisional

Source: Economics, Statistics and Evaluation Division, Ministry of Agriculture and Farmers Welfare

Risk Management

- 1.29 It is crucial to manage risks both production and market risks to improve growth and competitiveness of agriculture sector and provide farmers with a safety net. Protecting farmers from various risks ensures a stable income and encourage farmers to innovate and adopt new agricultural technologies and practices. *Pradhan Mantri Fasal Bima Yojana* (PMFBY) was launched from *kharif* 2016 to provide an affordable crop insurance product to ensure comprehensive risk cover for crops against all non-preventable natural risks. The Scheme has made significant progress in terms of coverage of farmers and participation of non-loanee farmers since inception. In order to improve efficiency and effectiveness of the Scheme, Government has taken several steps including technological interventions such as YES-Tech, Weather information and Network Data System (WINDS), App for Intermediary Enrolment (AIDE App) etc. A centralized platform National Crop Insurance Portal (NCIP), Digi claim Module to monitor the claim disbursement process and *Krishi Rakshak* Portal helpline have been setup for the grievance redressal of farmers.
- 1.30 During the Commission's interactions with various stakeholders, problem of damage to agricultural crops, particularly pulses, caused by wild/stray animals was raised by farmers. Under the revamped operational guidelines of *Pradhan Mantri Fasal Bima Yojana* (PMFBY), States have been given liberty to consider providing add-on coverage for crop loss due to attack by notified wild animals, wherever the risk is perceived to be substantial and identifiable, but awareness among farmers about such provision under the Scheme is very low. Some State Governments provide subsidy to farmers to install fencing including solar fencing, but a comprehensive plan to address crop losses due to wild and stray animals needs to be prepared.



Digitalization in Agriculture

- 1.31 The Digital Agriculture Mission, approved on September 2, 2024 with a budget of ₹2,817 crore, is a key initiative to enable a robust digital ecosystem for driving innovative farmer-centric digital solutions and making available timely and reliable information to farmers. The Mission envisages creation of Digital Public Infrastructure (DPI) for agriculture such as Agri Stack, *Krishi* Decision Support System (*Krishi*-DSS), Comprehensive Soil Fertility & Profile Map and other IT initiatives undertaken by Central/State Governments.

Box 1.1: Initiatives under Digital Agriculture Mission

- *Agri Stack* is envisaged as a digital public infrastructure for efficient and effective scheme/service delivery to the farmers. It consists of three foundational registries, namely, Farmer's Registry, Geo-referenced Village Maps and Crop Sown Registry through Digital Crop Survey.
- *Krishi-DSS* integrates and standardizes geospatial and non-geospatial data, including satellite, weather, soil, crop signatures, reservoir, groundwater data, and Government schemes information. *Krishi-DSS* offers crop maps, soil maps, automated yield estimation models, drought/flood monitoring systems, etc., which support the Government's evidence-based decision-making and facilitate innovative solutions by research institutions and the agri-tech industry. This *Krishi-DSS* platform is aligned with the Government's National Geo-Spatial policy.
- *Nationwide Soil Resource Mapping project* has been initiated by the Soil and Land Use Survey of India (SLUSI), which is inventorying soils at a village level on a finer and more detailed 1:10,000 scale using high-resolution satellite and ground data. Under this project, a detailed soil profile study will be carried out to create standardized soil maps for rational land use and crop planning, thus promoting sustainable agriculture.
- *Digital General Crop Estimation Survey (DGCES)* is designed to streamline and automate the entire General Crop Estimation Survey process, encompassing State-level planning, recording the results of Crop Cutting Experiments (CCEs) conducted on the field, and report generation. By digitizing and enhancing the efficiency of CCEs, DGCES seeks to provide States/UTs with crop yield data that is not only accurate but also accessible in a timely manner.

- 1.32 These initiatives would help in creating robust farmer-centric digital ecosystem to improve efficiency in implementation of government initiatives and help in making evidence based policy decisions. Significant progress has been made in creating digital identity of farmers and Digital Crop Survey (DCS). Over 6.4 crore farmers have been issued digital IDs and DCS has been conducted in 17 States covering more than 25.2 crore plots in the *rabi* season of 2024-25.

Cost Reduction Initiatives

- 1.33 Rising costs in agriculture is a major concern, which adversely affect farm profitability and food security. Increase in input prices, high wages, scarcity of labour and non-availability of quality inputs and services



significantly contribute to rising costs. Government has taken several initiatives to reduce cost and promote efficient use of resources. To promote use of renewable energy sources like solar and wind, financial assistance is given to farmers for installing solar-powered irrigation systems, solar pumps and grid-connected solar power plants under *Pradhan Mantri Kisan Urja Suraksha evam Utthaan Mahabhiyan* (PM-KUSUM), which would reduce dependence on conventional energy sources leading to lower cost to farmers. In order to address the problem of shortage of labour and rising wages, Government is promoting farm mechanization under Sub-Mission on Agricultural Mechanization (SMAM). Use of drones in agriculture would help in improving efficiency and reducing cost of fertilizer and pesticide application. In addition, other interventions such as use of nano-fertilizers, Soil Health Cards (SHCs), National Pest Surveillance System (NPSS) to enhance surveillance and management of pests and diseases by using latest digital technologies such as Artificial Intelligence (AI) and Machine Learning (ML) would help in reducing costs and improve profitability.

Outlook for Indian Agriculture

- 1.34 According to the Indian Meteorological Department (IMD) Long Range Forecast for the Southwest Monsoon Seasonal Rainfall during June-September 2025 (May 27, 2025), the seasonal rainfall across the country as a whole is likely to be 106 percent of the Long Period Average (LPA), with a model error of $\pm 4\%$, indicating a high probability of above-normal rainfall during the monsoon season. Regionally, rainfall is most likely to be above normal over Central India and the South Peninsular region, normal rainfall (92-108% of LPA) in Northwest India and below-normal rainfall ($< 94\%$ of LPA) in Northeast India. The latest Monsoon Mission Climate Forecast System (MMCFS) as well as other climate model forecasts indicate that the Neutral *El Nino*-Southern Oscillation (ENSO) condition are likely to continue during the monsoon season.
- 1.35 As per Central Water Commission bulletin (26th June 2025), total live storage capacity of 161 reservoirs was 182.4 Billion Cubic Meter, about 70.77 percent of the live storage capacity of 257.8 Billion Cubic Meter in the country, while the live storage available in these reservoirs was 175.8 percent of the live storage of corresponding period of last year and 152.8 percent of normal storage. The above normal rainfall during the monsoon season and comfortable water level in reservoirs would lead to a better output during *rabi* season.
- 1.36 Area coverage under *kharif* crops as on July 04, 2025 has increased compared to previous season for almost all the crop groups. Area coverage under rice has been reported as 69.3 lakh hectare in 2025, an increase of 7.4 percent compared to the corresponding period of 2024. The area coverage under pulses has increased by 35.2 percent and oilseeds by 14 percent during the same period. Total *kharif* area has increased from 39.4 million hectare to 43.7 million hectare, registering an increase in area sown by 11.1 percent.

World Outlook

- 1.37 According to FAO Food Outlook, June 2025, outlook for food commodity markets appears relatively optimistic, with production of most commodities anticipated to increase. However, global production remains vulnerable to weather conditions, geopolitical tensions, trade policy uncertainties, etc.



- 1.38 Food and Agriculture Organization (FAO), the United States Department of Agriculture (USDA) and the International Grains Council (IGC) forecast a marginal increase in global production, supply and utilization of wheat in 2025-26. Global wheat trade is forecast to partially recover from the previous season's decline largely driven by an anticipated rebound in import demand from Asia. Global wheat stocks and Stock-to-Use Ratio (SUR) are forecast to decline specially in major exporting countries.
- 1.39 International wheat prices remained generally steady in 2024-25 season with some fluctuations. In May 2025, world prices of wheat were 10.5 percent below their May 2024 level and 27.3 percent below their five-year average value. However, initial forecasts for 2025-26 season indicate higher prices due to higher global utilization, stronger import demand and reduction in stocks in major exporting countries as per FAO's latest assessment (June 2025).
- 1.40 Global barley production is estimated to increase in 2025 due to higher outputs in European Union and Russian Federation, while world trade in barley is forecast to increase at a faster rate due to higher import demand from China and Turkey.
- 1.41 Production of pulses in Canada, the largest exporter, is forecast to decline in 2025-26 but total supply, domestic use and ending stocks are forecast to increase.
- 1.42 In 2025-26, global oilseeds production is expected to increase, primarily due to higher output of soybean, groundnut and cottonseed. Global trade, supply and utilization of oilseeds are also expected to increase during 2025-26. World output of oilmeals is forecast to increase due to higher oilseeds supplies. Global consumption and trade in oilmeals is also forecast to expand in 2025-26. FAO's early and tentative forecasts for 2025-26 suggest increase in global output of oils and fats, albeit at a slower pace.
- 1.43 The FAO Food Price Index (FFPI) marginally declined (-0.8%) to 127.7 points in May 2025 from 128.7 points in April 2025 but was six percent higher than its level in May 2024. Cereal Price Index was 109 points in May 2025, registering a decline of 1.8 percent from April 2025. The FAO Vegetable Oil Price Index averaged 152.2 points in May 2025 a decline of 3.7 percent from April 2025, but 19.1 percent higher than May 2024 level. The decline in Vegetable Oil Price Index was driven by decline in world prices of palm, rapeseed, soy and sunflower oils.

Structure of the Report

- 1.44 The Report comprises of six chapters. Chapter 1 provides an overview of performance of Indian agriculture with special focus on mandated *rabi* crops. Chapter 2 discusses demand-supply outlook in domestic and world markets, price trends and procurement operations. Trends in crop productivity at all-India and disaggregated level, yield gap analysis and major drivers of productivity are analyzed in Chapter 3. Chapter 4 examines trade performance and patterns, trends in domestic and world prices, recent trade policy changes and trade outlook. Chapter 5 analyses cost of cultivation, returns and cost projections of the mandated crops. Major considerations and recommendations of MSP and non-price measures are presented in Chapter 6.



CHAPTER 2



Supply-Demand Scenario, Price Trends and Procurement Operations

- 2.1 As per the *OECD-FAO Agricultural Outlook 2024-2033*, emerging economies will be pivotal in shaping the global agricultural landscape, with India expected to overtake China as the leading player. Well-functioning international agricultural commodity markets are imperative for global food security and rural livelihoods since effective global markets help countries overcome hardships, e.g., Covid-19 wherein trade in bulk agricultural commodities was relatively resilient, and other such recent uncertainties. Over the last few decades, global markets have also helped ensure producers can meet rising demand for food and farm goods in different regions and trade in food and agricultural products has increased five-fold between 2000 and 2022, in line with growing population and rising average income. FAO's latest assessments reflect a relatively optimistic outlook for food commodity markets, indicating production of all commodities (except sugar) anticipated to increase, but with varying effect of growth on stock recovery depending on the delicate balance between supply and demand. As per IMF's World Economic Outlook (WEO) Update, April 2025, forecast for global growth has been revised markedly down to 2.8 percent in 2025 and 3 percent in 2026 due to trade tension escalations and considerable degree of policy uncertainty.
- 2.2 Against the backdrop of weather vulnerabilities, geopolitical uncertainties, protectionist policies and evolving market complexities, which lead to volatilities in trade and disruption of supply chain, an assessment of consistent and precise demand-supply forecasts of agricultural commodities is as much challenging as it is crucial to ensure food security. Further, availability of credible and robust estimates of the demand-supply outlook and price trends are important for an effective analysis and informed policy making and implementation. This chapter provides an overview of the supply-demand situation in global and domestic markets, price trends and procurement operations of major *rabi* crops.

Global Outlook

Wheat

- 2.3 Table 2.1 presents the world wheat supply and demand outlook as estimated by the Food and Agriculture Organization (FAO), the United States Department of Agriculture (USDA) and the International Grains Council (IGC). As per the most recent estimates of FAO, global wheat production is forecast to increase marginally by 0.3 percent in 2025-26. USDA and IGC also predict marginal increase in global wheat production in 2025-26. Similarly, global wheat supplies are projected to increase marginally



in 2025-26. Global wheat trade is predicted to increase by 3.8 percent as per FAO, by 6.1 percent as per USDA and by 5.3 percent as per IGC in 2025-26 driven mostly by an expected rebound in import demand from China and Türkiye, along with larger sales from Argentina, the EU, and the Russian Federation. Global utilisation of wheat is expected to increase by about 1.3 percent in 2025-26 as per FAO, USDA and IGC estimates. As per FAO and IGC, ending stocks are forecast to decline by 2.1 percent and 2.7 percent in 2025-26, respectively, while USDA predicts a marginal increase in ending stocks in 2025-26. As per these organizations' estimate, the stock-to-use ratio (SUR) are projected to decline in 2025-26 due to large increase in utilisation compared with supply. The ending stocks and SUR in 2025-26 are also forecast to be lower than TE2023-24.

Table 2.1: Global Supply and Demand Outlook for Wheat

(Million Tonnes)

Particulars	FAO*			USDA ^{\$}			IGC [#]		
	TE 2023-24	2024- 25 ^e	2025- 26 ^f	TE 2023-24	2024- 25 ^e	2025- 26 ^f	TE 2023-24	2024- 25 ^e	2025- 26 ^f
Production	791.1	797.7	800.1	787.1	799.7	808.5	792.4	799.5	805.6
Supply	1095.3	1113.1	1116.9	1064.4	1068.8	1073.7	1070.0	1071.6	1075.0
Trade ^a	203.3	193.3	200.6	215.7	201.9	214.2	206.6	193.2	203.4
Utilisation	782.8	795.4	805.4	790.7	794.9	804.7	795.0	802.3	813.0
Ending Stocks	312.2	316.8	310.0	270.9	265.2	265.7	275.0	269.4	262.0
SUR ^{\$\$}	39.9	39.8	38.5	34.3	33.4	33.0	34.6	33.6	32.2

Note: 1. *: as on 6th June 2025, \$: as on 12th May 2025; #: as on 22nd May 2025

2. a: Trade data refer to export estimate

3. \$\$: Stock-To-Use Ratio (SUR) = Stocks/Utilisation*100

4. e- estimated, f- forecast

Source: 1. FAO AMIS website for TE 2023-24

2. FAO AMIS Market Monitor June 2025 for 2024-25^e and 2025-26^f

Pulses

- 2.4 Canada is one of the world's leading exporters of pulses and the demand supply situation of pulses in Canada is given in Table 2.2. As may be seen from the Table, production of total pulses in 2025-26 is predicted to decrease by 3.4 percent to 6.3 million tonnes, while imports are forecast to decline by 21.4 percent. Total supply is forecast to increase by 4.8 percent in 2025-26 on account of higher ending stocks in 2024-25. Exports are predicted to decrease to 4.2 million tonnes in 2025-26, a decline of 16.5 percent. Domestic use is forecast to increase significantly in 2025-26. For lentils, production is expected to fall by 4.4 percent to 2.3 million tonnes in 2025-26, while exports and ending stocks are forecast to remain unchanged in 2025-26. The average price of lentils is forecast to be lower by 10.4 percent at \$730/tonnes and world supply is lower in 2025-26. In case of dry peas, production is forecast to increase to 3.1 million tonnes in 2025-26 and exports are predicted to significantly decline by about 38 percent to 1.3 million tonnes over the previous year. The stocks are forecast to be 2.5 times higher in 2025-26, while the average price is predicted to decline from \$415/tonnes to \$365/tonnes due to expectations for larger domestic carry-out stocks, higher production and lower exports in 2025-26.



Table 2.2: Pulses Supply-Demand Situation in Canada

('000 Metric Tonnes)

Particulars	Total Pulses			Lentils			Dry Peas		
	TE 2023-24	2024- 25 ^e	2025- 26 ^f	TE 2023-24	2024- 25 ^e	2025- 26 ^f	TE 2023-24	2024- 25 ^e	2025- 26 ^f
Production	5470	6568	6346	1899	2431	2325	2759	2997	3125
Imports	297	304	239	77	120	75	64	40	20
Total Supply	7051	7693	8060	2266	2716	2705	3305	3337	3770
Exports	4940	5030	4200	1828	2100	2100	2326	2100	1300
Domestic Use	1151	1188	1340	259	311	300	633	612	770
Ending Stocks	959	1475	2520	178	305	305	379	625	1700
SUR	-	-	-	9.0	13.0	13.0	13.0	23.0	82.0
Average Price (\$/t)	-	-	-	930	815	730	497	415	365

Note: e: estimated, f: forecast

Source: Outlook for Principal Field Crops – 21st May 2025 released by Agriculture and Agri-Food Canada

Oilseeds

- 2.5 Global demand-supply trends and outlook for oilseeds and oilseeds complex as estimated by USDA and Australian Bureau of Agricultural and Resource Economics and Sciences (ABARES) have been given in Table 2.3. According to USDA, global production of oilseeds is projected to increase by 2.2 percent and global supply is projected to rise by 2.6 percent in 2025-26 over 2024-25. Global utilization and trade in oilseeds are expected to increase by 3.4 percent and 2.8 percent, respectively in 2025-26. Ending stocks are forecast to increase by 1.4 percent in 2025-26, while stocks-to-use ratio (SUR) is forecast to marginally decrease from 25.3 in 2024-25 to 24.8 in 2025-26.
- 2.6 As per ABARES, production, supply, utilization and trade in oilseeds are forecast to increase in 2025-26. Production of total oilseeds is forecast to increase by 2.2 percent, while global supply and utilization are forecast to increase by 2.5 percent and 2.9 percent, respectively. Global trade is expected to increase by 3.3 percent in 2025-26.
- 2.7 With respect to oilmeals, global production and supply in 2025-26 are estimated to increase by 3.4 percent and 4 percent, respectively, as per USDA. Global utilisation is predicted to increase to about 395.1 million tonnes in 2025-26, an increase of 4 percent over 2024-25, while trade is estimated to increase by 1.5 percent. As regards vegetable oils, global production is forecast to increase by 2.8 percent in 2025-26 over 2024-25 and global supply is forecast to increase by 2.1 percent in 2025-26. Global utilization and global trade are estimated to increase by 2.4 percent and 2.3 percent, respectively, in 2025-26.



- 2.8 As per ABARES, global oilmeals production is estimated to grow marginally by 0.6 percent in 2025-26 while supply is expected to increase by 1.1 percent. Global utilisation and trade are forecast to rise by 1.4 percent and 4.1 percent, respectively. For vegetable oils, global production is predicted to increase by 1.8 percent, while supply is forecast to rise marginally (0.6%). Global utilization is estimated to increase by 1.4 percent, while global trade is forecast to significantly rise by 10.1 percent in 2025-26. The USDA projections indicate lower SUR in vegetable oils in 2025-26, while ABARES forecasts show a significant rise in SUR of vegetable oils.

Table 2.3: World Oilseeds Complex Market Outlook

(Million Tonnes)

Particulars	USDA			ABARES		
	2023-24	2024-25 ^e	2025-26 ^{p*}	2023-24	2024-25 ^e	2025-26 ^f
Oilseeds						
Production	658.0	677.3	692.3	658.1	677.2	692.1
Supply	781.0	813.4	834.4	780.4	813.3	833.4
Utilization	543.4	561.5	580.8	644.2	668.4	687.7
Trade	205.1	209.2	215.0	205.1	208.3	215.1
Ending Stocks	136.2	142.2	144.1	136.1	141.3	143.2
SUR	25.1	25.3	24.8	21.1	21.1	20.8
Oilmeals						
Production	371.0	387.3	400.4	372.6	385.7	388.0
Supply	389.3	406.5	422.6	390.0	404.6	409.1
Utilization	366.0	379.7	395.1	366.2	379.4	384.8
Trade	105.7	111.2	112.9	106.2	108.1	112.5
Ending Stocks	19.3	22.2	23.1	18.9	21.1	24.3
SUR	5.3	5.8	5.8	5.2	5.6	6.3
Vegetable Oils						
Production	222.1	228.5	235.0	221.7	227.4	231.4
Supply	254.6	259.3	264.8	252.9	258.2	259.8
Utilization	217.9	224.0	229.4	217.5	224.7	227.9
Trade	86.1	85.7	87.7	86.0	86.2	94.9
Ending Stocks	30.8	29.8	30.0	30.8	28.3	31.9
SUR	14.2	13.3	13.1	14.2	12.6	14.0

Note: 1. e: estimated, p: projected, *: As on 12th June 2025, f: forecast
 2. Supply for ABARES of current year is the addition of production of current year and stocks of previous year

Source: 1. USDA Oilseeds: WASDE Report, June 2025
 2. Australian Bureau of Agricultural and Resource Economics and Sciences Agricultural Commodities, June quarter 2025 for Oilseeds and March quarter 2025 for Oilmeals/Vegetable Oils.



Domestic Market Scenario

Wheat

- 2.9 As per the 3rd Advance Estimates of the Ministry of Agriculture & Farmers Welfare, wheat production is estimated at a record high of 117.5 million tonnes in 2024-25, which is 3.7 percent increase over 2023-24 (Table 2.4). As per Department of Food & Public Distribution (DFPD), wheat stocks in Central Pool were about 38 million tonnes as on 1st June 2025 (Opening Stock), significantly higher compared with the corresponding period of last year (30 million tonnes) and hit the highest level since January 2022. Exports of wheat declined significantly during the last two years due to ban on wheat exports since 2022 to boost domestic supply and moderate domestic price of wheat. Higher production, larger marketed surplus and high stocks in Central Pool point towards a fairly comfortable wheat supply in domestic market during 2025-26.

Table 2.4: Domestic Supply Situation of Wheat in India

(Million Tonnes)

Particulars	2020-21	2021-22	2022-23	2023-24	2024-25
Production	109.6	107.7	110.6	113.3	117.5 [^]
Marketed Surplus	80.9	82.5	84.9	82.3	85.4 ^{\$}
Exports	2.2	7.2	4.7	0.2	0.0
Imports	0.0	0.0	0.0	0.1	0.1
Stocks in Central Pool*	60.4	28.5	30.1	28.3	38.0 ^{**}

Note: 1. *: As on 1st July 2021, 2022 and 2023 and 2024; **: As on 1st June 2025

2. \$: as per 2023-24 Market Surplus Ratio (MSR)

3. ^: Production as per 3rd Advance Estimates 2024-25 released on 28th May 2025

Sources: 1. Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare

2. Directorate General of Commercial Intelligence and Statistics, Ministry of Commerce and Industry

3. DFPD Foodgrains Bulletin, May 2025

Pulses

- 2.10 Table 2.5 gives the supply situation of pulses in India for the past five years. Production of pulses rose during 2021-22 and recorded at a peak of 27.3 million tonnes, which declined to 26.1 million tonnes in 2022-23 and 24.2 million tonnes in 2023-24. However, production is estimated to increase by 4.1 percent at 25.2 million tonnes in 2024-25. India is a net importer of pulses, and a small quantity is exported. Import of pulses recorded an increase in 2021-22, followed by a decline in 2022-23 and recorded a steep increase during the last two years, reaching a record level of 7.3 million tonnes in 2024-25. Stocks of pulses doubled in 2021-22, further increased in 2022-23 followed by a sharp drop in 2023-24 to one-third of the level in 2022-23. Stocks are estimated at 1.5 million tonnes in 2024-25.

**Table 2.5: Domestic Supply Situation of Pulses in India**

(Million Tonnes)

Particulars	2020-21	2021-22	2022-23	2023-24	2024-25
Production	25.5	27.3	26.1	24.2	25.2 [^]
Exports	0.3	0.4	0.8	0.6	0.7
Imports	2.5	2.8	2.5	4.8	7.3
Stocks*	1.8	3.6	4.2	1.4	1.5**

Note: 1. *: As on 1st June 2021, 2022, 2023, 2024 and **: As on 27th May 2025

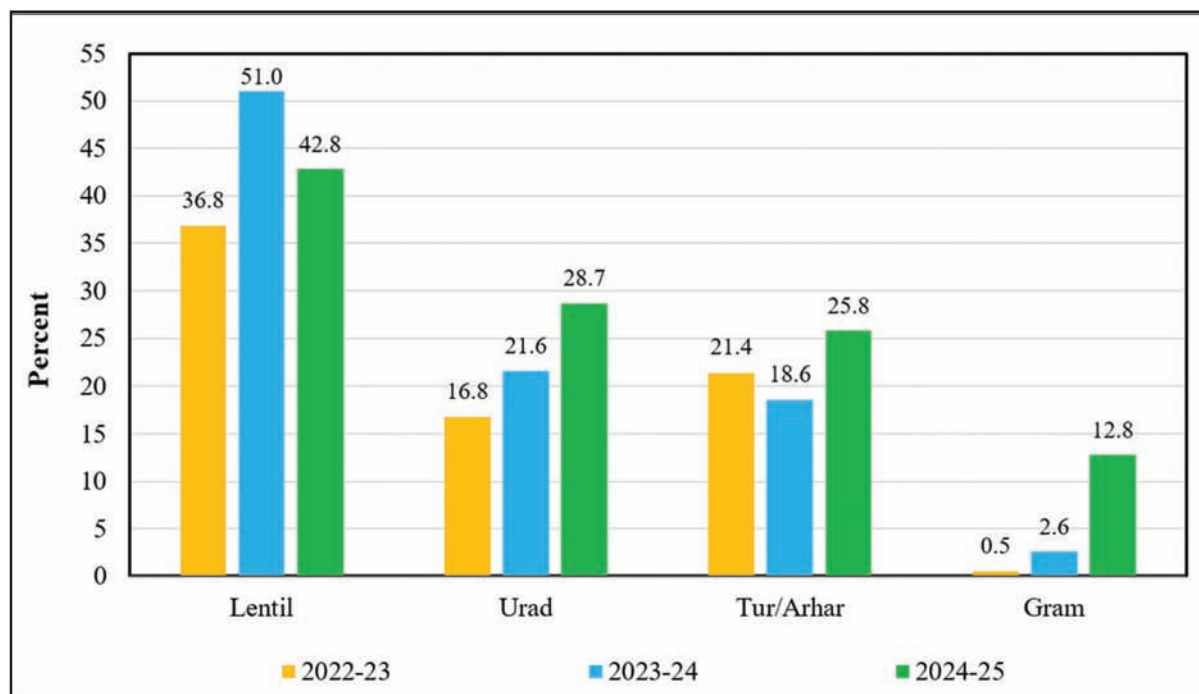
2. [^]: Production as per 3rd Advance Estimates 2024-25 released on 28th May 2025

Source: 1. DGCIS for data on exports and imports

2. Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare

3. NAFED for data on stocks

- 2.11 Chart 2.1 depicts India's import dependency for pulses. As may be inferred from the Chart, import dependency is considerable with respect to lentil, *urad* and *tur/arhar* for meeting the domestic demand. The share of imports in total consumption of lentil increased from 36.8 percent in 2022-23 to 51 percent in 2023-24 and declined to 42.8 percent in 2024-25 due to increase in domestic production. For *urad*, share of imports increased steadily from 16.8 percent in 2022-23 to 28.7 percent in 2024-25 due to a steady decline in production during last three years. In case of *tur/arhar*, share of imports was 21.4 percent in 2022-23, which declined to 18.6 percent in 2023-24 but rose to 25.8 percent in 2024-25. Thus, this calls for increase in production of pulses through both area expansion and productivity improvement to reduce import dependency.

Chart 2.1: Import Dependency in Major Pulses in India

Source: 1. DGCIS for data on imports and exports

2. Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare



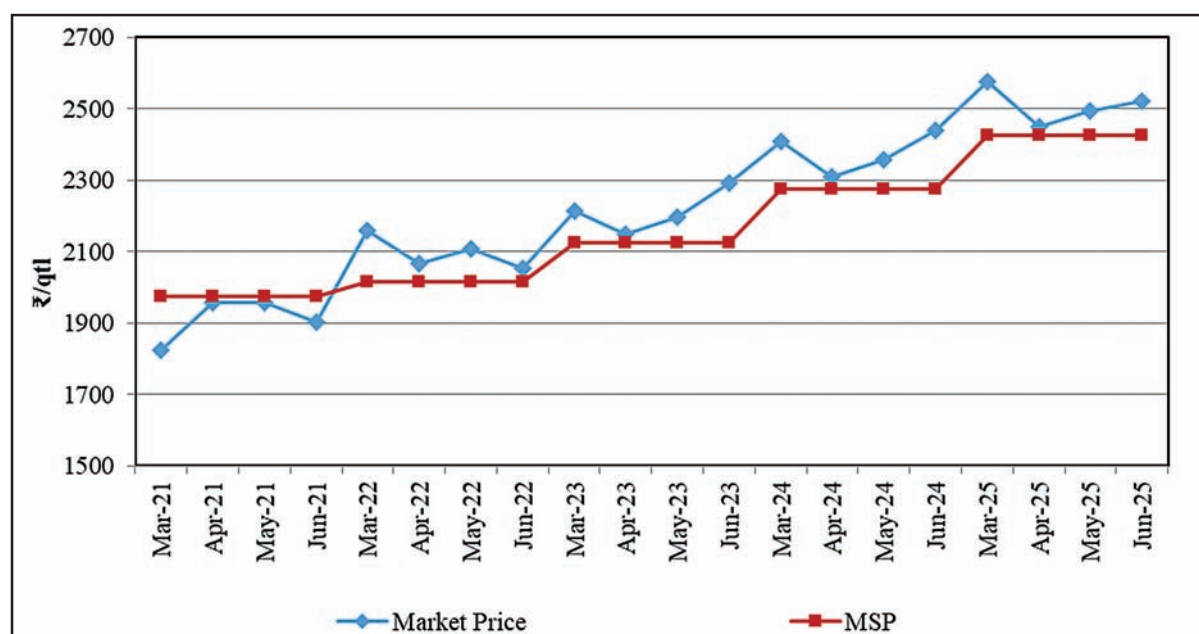
Trends in Domestic Market Prices

- 2.12 This section presents a comparison of minimum support price (MSP) and domestic market prices of mandated *rabi* crops. Market price data compiled from Unified Portal for Agricultural Statistics (UPAg), Ministry of Agriculture and Farmers Welfare has been used for carrying out the analysis. However, it is pertinent to mention that MSP is for Fair Average Quality (FAQ) specification while market prices may or may not conform to the FAQ. Thus, though not precisely comparable but to examine broad trends, comparison of domestic market prices and MSP for mandated *rabi* crops has been done.

Wheat

- 2.13 Chart 2.2 presents the trends in domestic market prices vis-à-vis MSP of wheat during last five marketing seasons. The average market price of wheat was below MSP for RMS2021-22 but market prices showed a rising trend and stayed above the MSP during RMS2022-23, RMS2023-24 and RMS2024-25. In RMS2025-26, average market price recorded a decline during April 2025, but stayed above the MSP.

Chart 2.2: Trends in Domestic Market Prices vis-à-vis MSP of Wheat



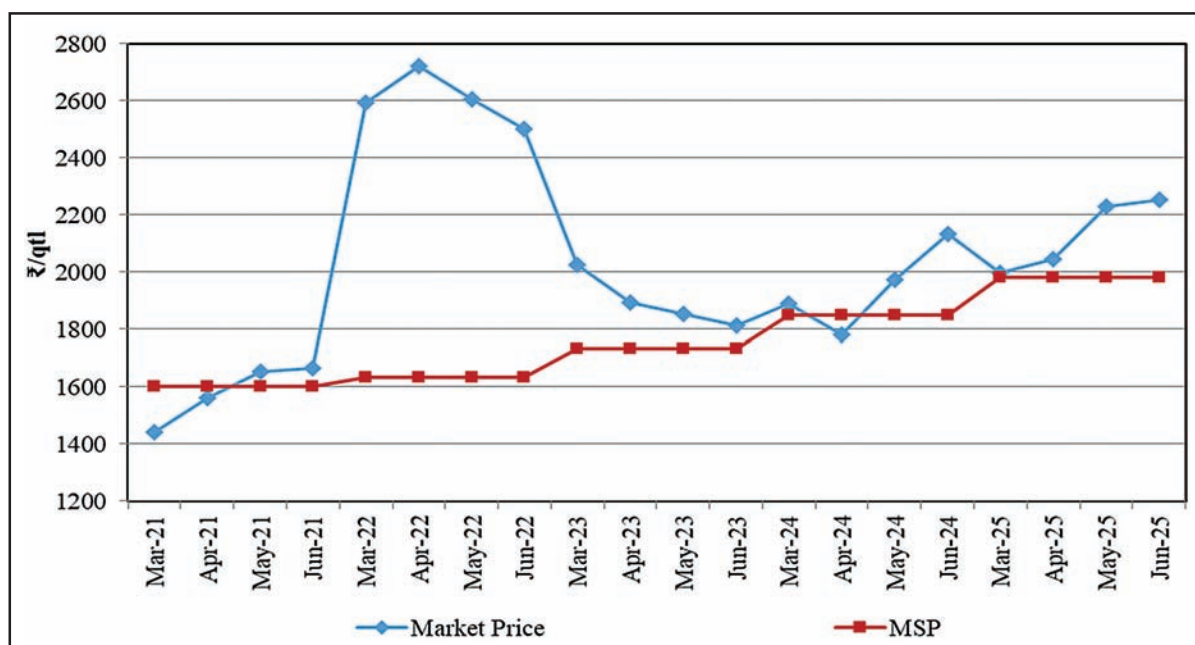
Source: UPAg Portal, Ministry of Agriculture & Farmers Welfare (as on 17th June 2025)

Barley

- 2.14 Chart 2.3 portrays average market prices and MSP of barley. The average market price was significantly higher during RMS2022-23 than RMS2021-22 and the MSP. Average market price declined significantly during RMS2023-24 but stayed above the MSP. Market prices continued a rising trend during RMS2024-25 and RMS2025-26 and were higher than MSP except April 2024. Market prices rose sharply during April-June 2025.



Chart 2.3: Trends in Domestic Market Prices vis-à-vis MSP of Barley



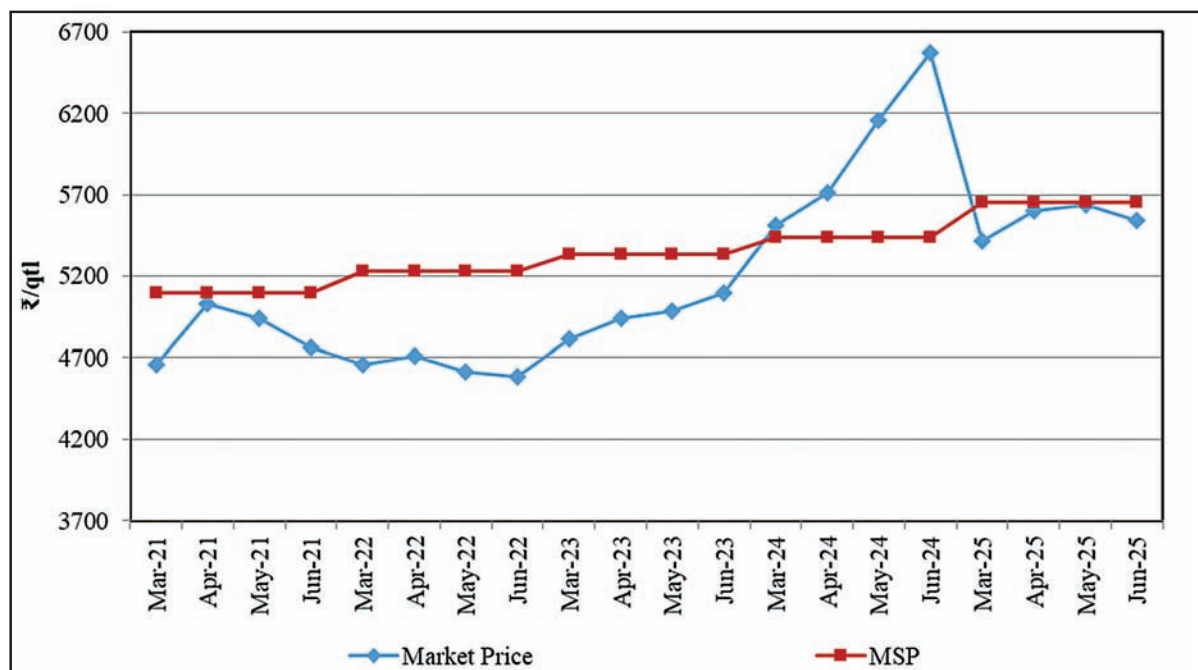
Source: UPAg Portal, Ministry of Agriculture & Farmers Welfare (as on 17th June 2025)

Gram

- 2.15 The average market prices of gram during the last five marketing seasons is shown in Chart 2.4. The average market price of gram remained consistently below the MSP during RMS2021-22 to RMS2023-24, but prices exhibited an increasing trend during RMS2023-24. The market price witnessed a steep increase in RMS2024-25 and surpassed the MSP on account of decreased production in 2022-23 and 2023-24. However, market prices of gram remained below the MSP during RMS2025-26 due to cheaper imports of gram and yellow peas. However, market prices improved during April and May 2025 after Government imposed a 10 percent import duty on gram effective from April 1, 2025. Market prices improved during April-June 2025 as a consequence of the import duty but stayed below the MSP.



Chart 2.4: Trends in Domestic Market Prices vis-à-vis MSP of Gram



Source: UPAg Portal, Ministry of Agriculture & Farmers Welfare (as on 17th June 2025)

Lentil

- 2.16 Trends in domestic market prices vis-à-vis MSP of lentil during the last five *Rabi* Marketing Seasons have been depicted in Chart 2.5. The average market prices of lentil witnessed an increasing trend and stayed above the MSP in RMS2021-22 and RMS2022-23. However, the market prices during RMS2023-24 fell sharply as compared to RMS2022-23 and moved below the MSP. The market prices witnessed an increasing trend during the last three marketing seasons but stayed below the MSP due to large imports of lentils and higher domestic production.



Chart 2.5: Trends in Domestic Market Prices vis-à-vis MSP of Lentil

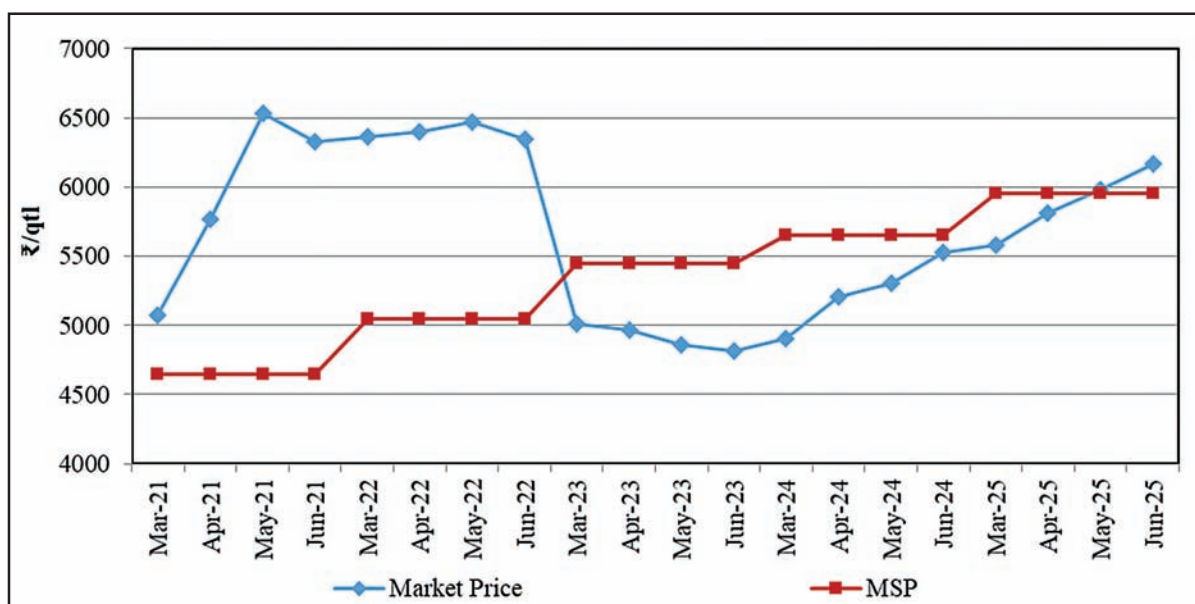


Source: UPAG Portal, Ministry of Agriculture & Farmers Welfare (as on 17th June 2025)

Rapeseed & Mustard

- 2.17 The average market prices of R&M were substantially higher than the MSP during RMS2021-22 and RMS2022-23 (Chart 2.6). However, prices exhibited a decreasing trend during RMS2023-24 and were below the MSP. Market price of R&M gradually increased in RMS2024-25 and RMS2025-26 but were still below the MSP. The gap between market prices and MSP narrowed and average market price was higher than MSP in May 2025 and June 2025.

Chart 2.6: Trends in Domestic Market Prices vis-à-vis MSP of Rapeseed & Mustard



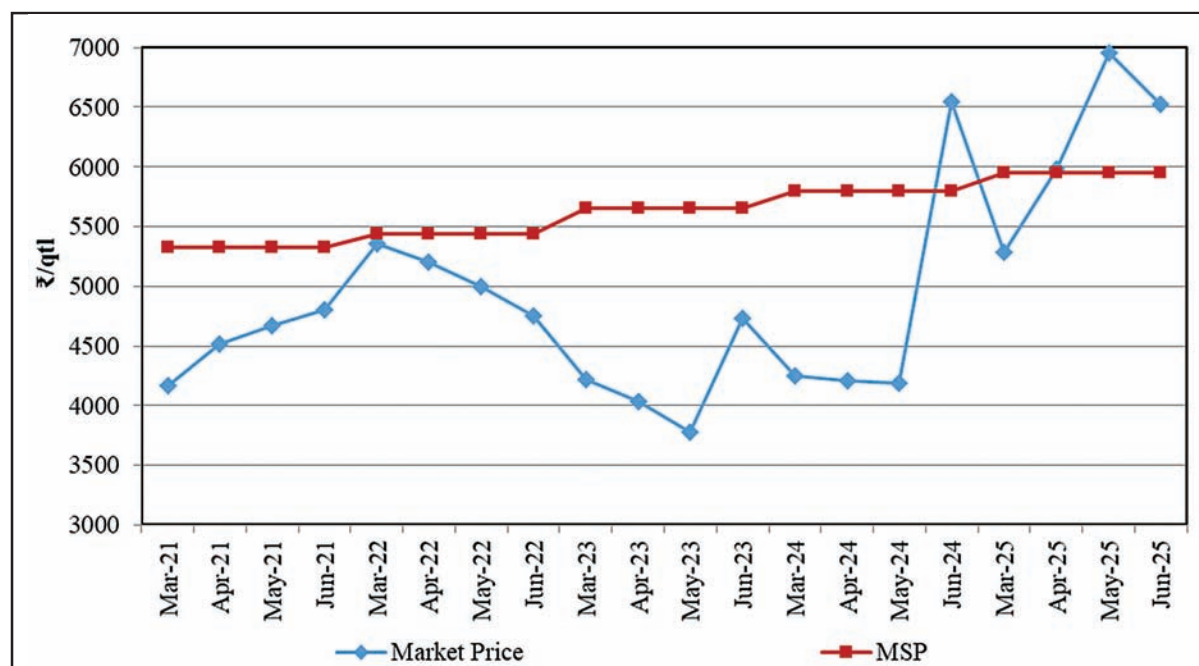
Source: UPAG Portal, Ministry of Agriculture & Farmers Welfare (as on 17th June 2025)



Safflower

- 2.18 As depicted in Chart 2.7, market prices of safflower stayed below the MSP during the last four marketing seasons, RMS2021-22 to RMS2024-25, with large fluctuations. Market prices, which observed an increasing trend during RMS2021-22, recorded a declining trend in RMS2022-23 and RMS2023-24, except for June 2023 which saw a jump in prices. During RMS2025-26, market prices exhibited a steep rise during April-June 2025 and moved above the MSP.

Chart 2.7: Trends in Domestic Market Prices vis-à-vis MSP of Safflower



Source: UPAg Portal, Ministry of Agriculture & Farmers Welfare (as on 17th June 2025)

Food Inflation: Drivers and Dynamics

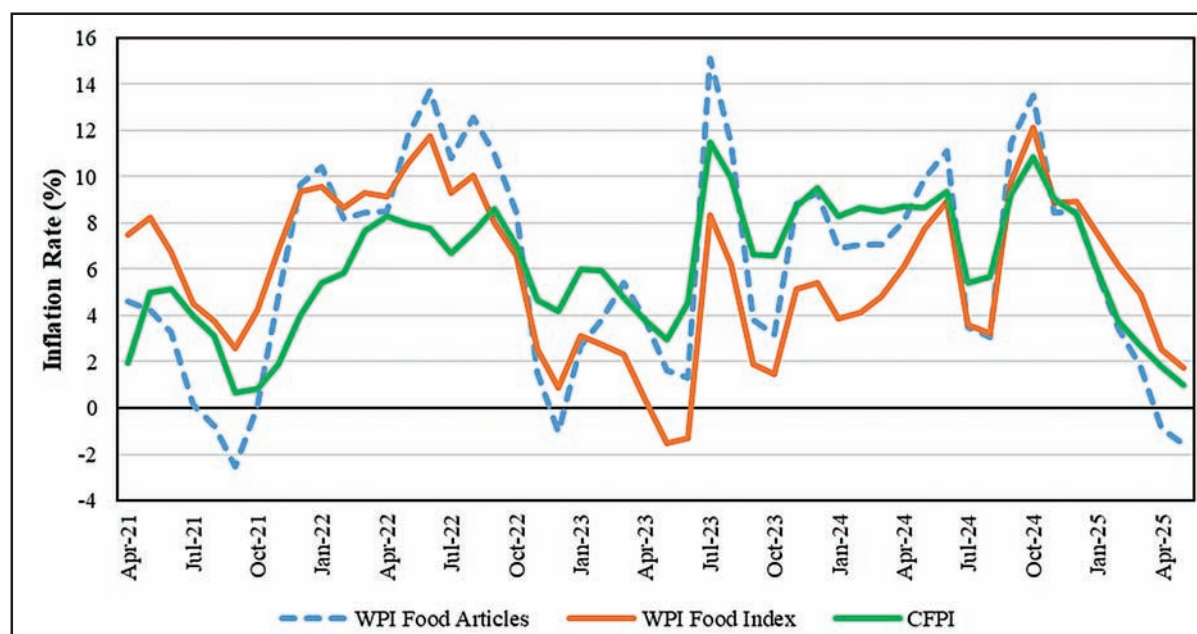
- 2.19 The average annual rate of inflation of food articles (based on WPI base year 2011-12) was at an elevated level of 8.4 percent in 2019-20 owing to supply chain disruptions as a consequence of the global pandemic. However, the effect softened with the eventual unlocking of the economy and inflation decreased to 3.1 percent in 2020-21 but increased to 4.1 percent in 2021-22 and further climbed to 7.3 percent in 2022-23 (Annex Table 2.1). The rate of inflation declined to 6.6 percent in 2023-24, but again rose to 7.3 percent in 2024-25.
- 2.20 The annual rate of inflation of cereals (based on WPI) more or less followed the trajectory of food articles wherein from a high of 7.5 percent in 2019-20, it turned negative at (-) 2.6 percent in 2020-21 but moved upwards thereafter and reached 11.3 percent in 2022-23 before falling to the 2019-20 level in 2023-24 and was 7.9 percent in 2024-25. For pulses, the rate of inflation was 11.6 percent in 2020-21, and declined to 0.6 percent in 2022-23. However, in 2023-24 the inflation rose to 14.4 percent owing to fall in the domestic production in 2022-23 and 2023-24 as a consequence of adverse climatic conditions but declined to 10.7 percent in 2024-25. The rate of inflation for oilseeds was 6.8



percent in 2020-21, which rose to 32.6 percent in 2021-22, but witnessed a sharp decline to (-) 4.3 percent in 2022-23, (-) 9.8 percent in 2023-24, and (-) 1.9 percent in 2024-25.

- 2.21 Consumer Food Price Index (CFPI) has a significant weightage of 39.06 percent in the overall Consumer Price Index. The CFPI inflation increased from 3.8 percent in 2021-22, to 6.6 percent in 2022-23 and to 7.5 percent in 2023-24, which marginally declined to 7.3 percent in 2024-25 (Annex Table 2.2). The inflation in cereal and products (7.2%) and pulses and products (8.3%) was significantly higher than oilseeds inflation (0.9%) in 2024-25. As is observed from Chart 2.8, CFPI declined from 9.5 percent in December 2023 to 8.3 percent in January 2024 and hovered around 8.7 percent during February-May 2024. The CFPI fluctuated in the range of 5.4 percent to 10.9 percent during June-December 2024, followed by a declining trend from January 2025 onwards. Annual rate of inflation of food articles (based on WPI) and WPI Food Index observed somewhat similar movement as CFPI.

Chart 2.8: Trends in Food Inflation

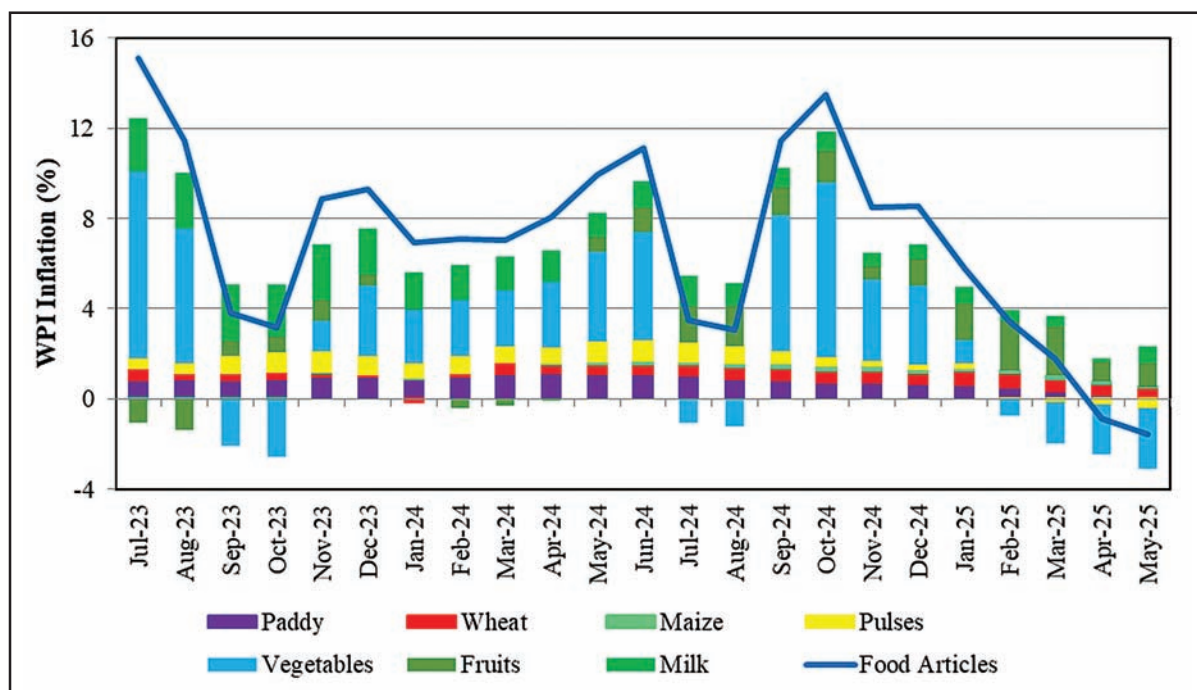


Source: 1. Ministry of Statistics and Programme Implementation for CPI data
2. Office of the Economic Advisor, Department for Promotion of Industry and Internal Trade (DPIIT) for WPI data

- 2.22 Chart 2.9 shows month-wise movement of WPI food articles inflation and its drivers. It is evident from the Chart that vegetables, milk, paddy, wheat, pulses and fruits have been driving inflation during second half of 2024. Fruits, wheat, milk and paddy have been major drivers of inflation during the last three months. From the peak of 15.1 percent in July 2023, inflation based on WPI food articles observed a declining trend till October 2023 (3.2%), which was followed by a rise during November-December 2023. The WPI food articles inflation fluctuated in the range of 3.1 percent in August 2024 to 13.5 percent in October 2024 during 2024. The WPI food articles inflation witnessed a steady decline since January 2025 and reached negative zone (-0.9%) in April and May (-1.6%) driven by declining inflation in vegetables, pulses, fruits, wheat and paddy.



Chart 2.9 Contribution of Major Food Articles to Inflation (Y-o-Y) based on WPI Food Index (Base Year 2011-12)



Source: Office of the Economic Advisor, Department for Promotion of Industry and Internal Trade (DPIIT)

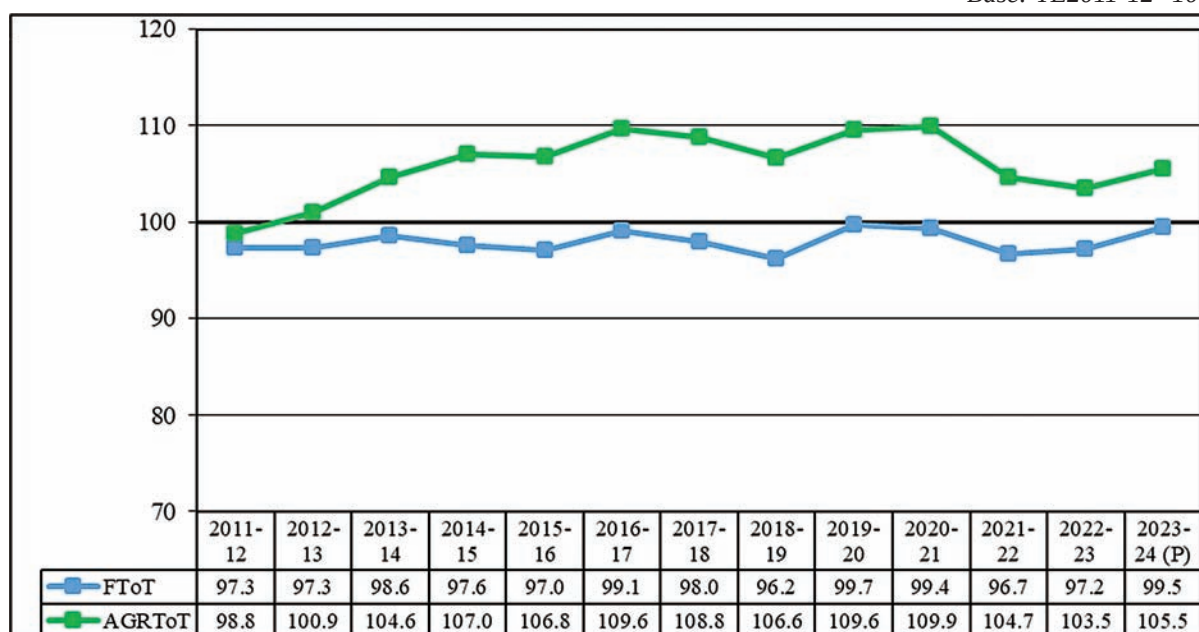
Trends in Terms of Trade

- 2.23 Terms of Trade (ToT) is an important factor considered by the Commission while formulating price policy recommendations. The index is considered as a measure of relative performance of sectors and crucial for growth, income distribution and consequently demand, savings and investment patterns.
- 2.24 Chart 2.10 depicts farmers' terms of trade (FToT) and agricultural terms of trade (AGRToT) during 2011-12 to 2023-24(P). The FToT measures index of prices that farmers receive for their products and the prices paid by the farmers. The FToT index, which witnessed stagnation since 2011-12, showed some improvement during the last 3 years but remained below 100.
- 2.25 The agricultural terms of trade (AGRToT), which includes both farmers and agricultural labourers, index of AGRToT remained above 100 since 2011-12 and reached the highest level (109.9) in 2020-21. The AGRToT rose much faster than FToT and remained higher than the FToT. Improved Terms of Trade along with improvement in productivity are essential for raising farmer's income. In order to improve the FToT and AGRToT and contribute to the economic well-being of farmers, appropriate policy interventions are needed to ensure better prices to farmers for their produce and reduce unit cost of production. The effective extension services, adoption of new innovative technologies and improved practices can help in reducing unit cost of production, thereby, improving farmers' income. In addition, improved infrastructure, efficient and competitive markets, policy reforms, and improved bargaining power of farmers by organizing them into groups like Farmer Producer Organizations (FPOs), cooperatives, SHGs etc. will help in addressing the challenges faced by agricultural sector.



Chart 2.10: Trends in Index of Farmers' Terms of Trade (FToT) and Agricultural Terms of Trade (AGRTToT)

Base: TE2011-12=100



Source: Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare

Foodgrains Procurement Policy and Operations

- 2.26 The broad objectives of Government policy of foodgrains procurement are (i) to ensure remunerative prices to farmers for their produce, (ii) availability of foodgrains at affordable prices to weaker sections by supplying subsidized foodgrains under National Food Security Act (NFSA) and other welfare schemes (OWS), and (iii) to build up buffer stocks to ensure food security and keep prices under control by timely and effective market intervention. The Food Corporation of India (FCI) is the central nodal agency of the Government of India, which undertakes procurement of paddy and wheat along with other State agencies. Procurement of nutri-cereals is undertaken through State Governments/ State agencies as per the directions issued by the Government of India from time to time. The National Agricultural Cooperative Marketing Federation of India Ltd (NAFED) and National Cooperative Consumers' Federation of India Ltd (NCCF) undertake procurement of oilseeds and pulses under the Price Support Scheme (PSS), if the market price of a commodity falls below the MSP. In addition, a corpus, namely, Price Stabilization Fund (PSF) was set up in 2014-15 to regulate price volatility through procurement by State/UT Government and Central agencies/Central PSUs/Cooperative organisations as decided by respective Ministries of selected produce, maintenance of buffer stocks and regulated release into the market and ensuring availability of these food commodities at affordable prices to the consumers.
- 2.27 The Decentralized Procurement Policy (DCP) was introduced in 1997 to improve efficiency in procurement operations, expand its coverage in non-traditional States and crops, and to reduce transit losses and transport costs. Under the DCP, the State Government, or its agencies procure, store and distribute (against allocation for NFSA/TPDS and OWS) within the State and excess stocks are

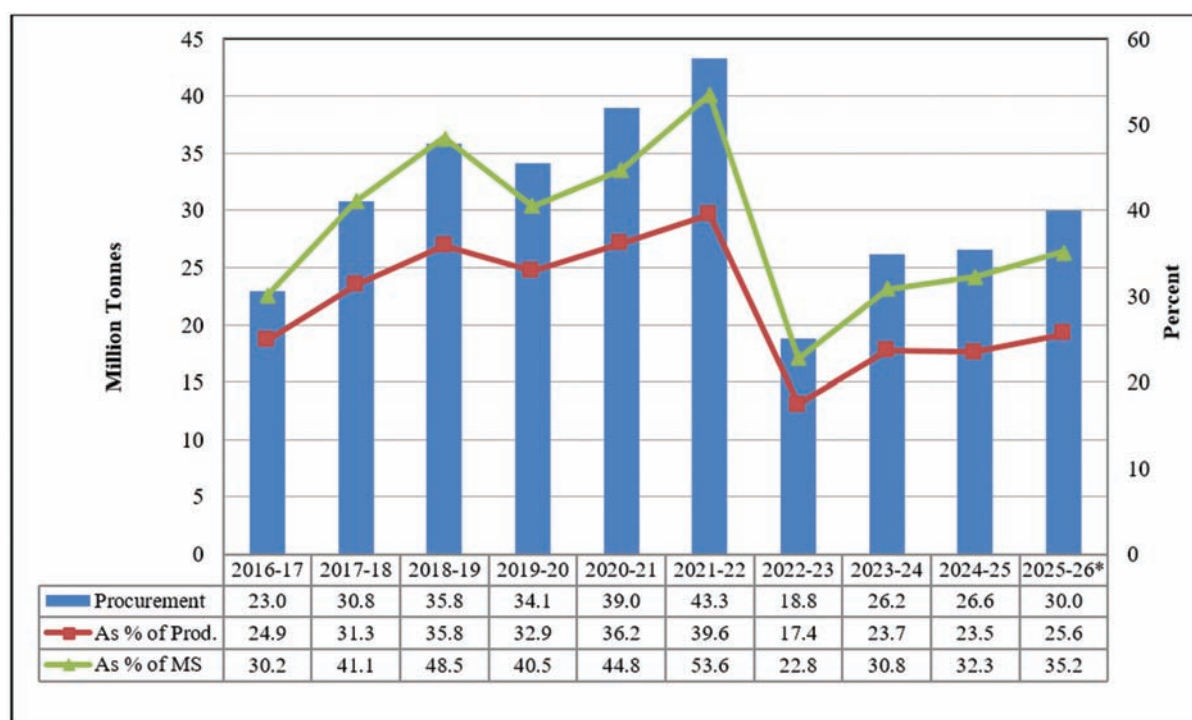


handed over to FCI for the Central Pool. The expenditure incurred by the State Governments on procurement, storage and distribution of DCP stocks is reimbursed by the Central Government on the laid down principles. At present, 17 States for rice and 9 States for wheat are under the DCP system (Annex Table 2.3).

Wheat Procurement

- 2.28 Chart 2.11 gives procurement of wheat during RMS2016-17 to RMS2025-26. Wheat procurement witnessed an increasing trend from RMS2016-17 to RMS2021-22, except for a minor decline in RMS2019-20. Wheat procurement reached a record level of 43.3 million tonnes in RMS2021-22, which sharply fell to 18.8 million tonnes in RMS2022-23 but increased to 26.2 million tonnes in RMS2023-24 and 26.6 million tonnes in RMS2024-25. Wheat procurement was 30 million tonnes during RMS2025-26 (as on 30th June 2025).
- 2.29 Wheat procurement as a percent of production showed an increasing trend from RMS2016-17 to RMS2021-22, with minor decline during RMS2019-20. This was followed by a sudden drop in procurement as a percent of production in RMS2022-23 but increased in RMS2023-24, and was 25.6 in RMS2025-26. A similar pattern was witnessed in procurement as a share of marketed surplus and it was 35.2 in RMS2025-26. It is to be noted that during the last 10 years, there have been wide variations in procurement as a percent of marketed surplus (22.8% to 53.6%) and production (17.4% to 39.6%).

Chart 2.11: Trends in Wheat Procurement: RMS2016-17 to RMS2025-26



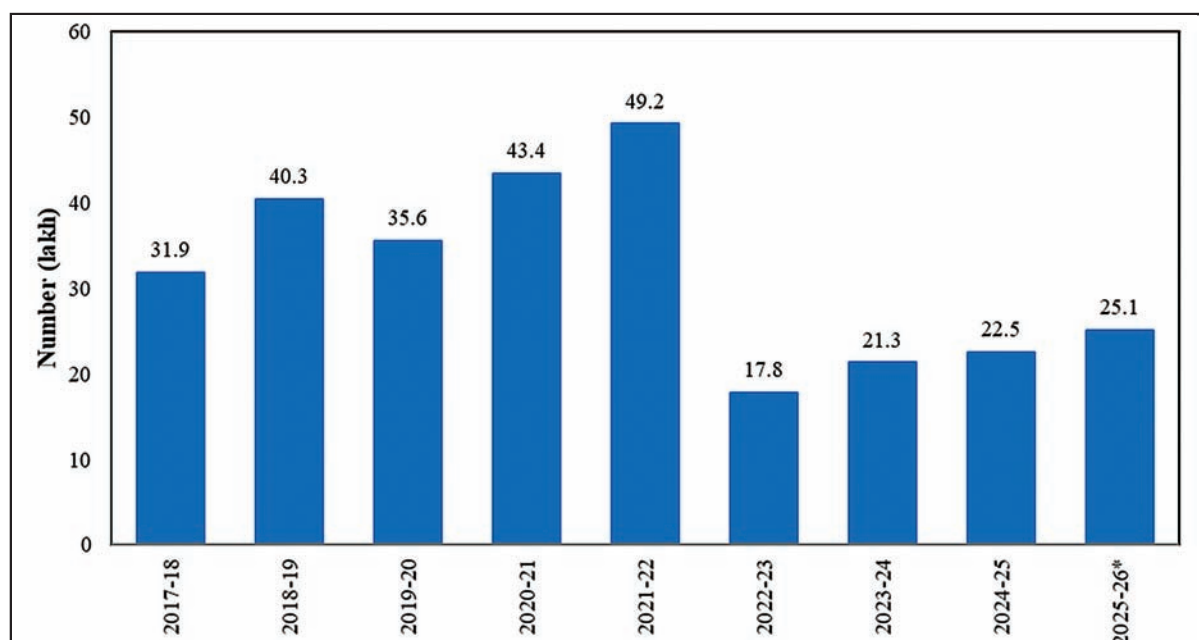
Note: 1. *: As on 30th June 2025; MS: Marketed Surplus
2. Marketed Surplus Ratio for RMS2024-25 is taken as proxy for RMS2025-26

Source: 1. Food Corporation of India
2. Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare



- 2.30 The total number of farmers benefitted from wheat procurement witnessed a general increase from about 32 lakhs in 2017-18 to a record 49.2 lakhs in 2021-22 (Chart 2.12). This was followed by a considerable decline of about 64 percent in the number of beneficiary farmers due to market prices of wheat much higher than MSP in RMS2022-23. The number of farmers benefitted increased by 19.4 percent to 21.3 lakhs in RMS2023-24, about 6 percent to 22.5 lakhs in RMS2024-25 and to 25.1 lakhs in RMS2025-26 (upto 30th June 2025). Except for Rajasthan, Bihar and Madhya Pradesh, number of farmers benefitted from procurement in RMS2025-26 declined in other major wheat producing States like Uttar Pradesh, Punjab and Haryana (Annex Table 2.4). Madhya Pradesh accounted for 35.8 percent of total farmers benefitted from procurement operations, followed by Punjab (33.4%), Haryana (17.9%), Rajasthan (6.9%) and Uttar Pradesh (5.7%).

Chart 2.12: Number of Farmers Benefitted from Procurement Operations from RMS2017-18 to RMS2025-26



Note: *: As on 30th June 2025

Source: Food Corporation of India

- 2.31 Procurement of wheat from DCP States increased by 13.9 percent, from 17.3 million tonnes in RMS2024-25 to 19.7 million tonnes in RMS2025-26 (as on 30th June 2025) due to higher procurement in Bihar and Madhya Pradesh, while wheat procurement from non-DCP States increased by 11 percent driven by higher procurement in Rajasthan and Uttar Pradesh.

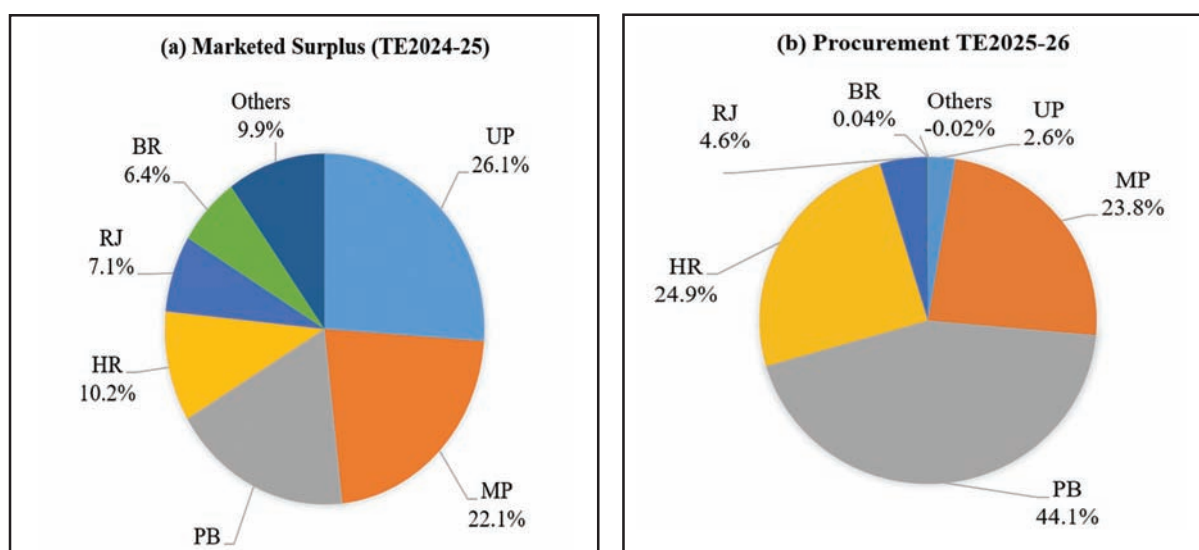
Inequity in Procurement Operations

- 2.32 As may be seen from Chart 2.13, Uttar Pradesh has the highest share (26.1%) in marketed surplus in TE2024-25, followed by Madhya Pradesh (22.1%), Punjab (18.2%), Haryana (10.2%) and Rajasthan (7.1%). It is interesting to note that even though Uttar Pradesh has the largest share in both wheat production and marketed surplus in the country, the State had a mere 2.6 percent share in wheat procurement in TE2025-26. For the second largest wheat producing State, Madhya Pradesh, the



share in marketed surplus aligns with the share in procurement of wheat. Punjab's share in wheat procurement was significantly higher (about 2.4 times) than the share in marketed surplus. The share of Haryana in total wheat procurement was also higher (24.9%) as compared with its share in marketed surplus (10.6%). In case of Bihar and Rajasthan, important wheat producing States, the share in procurement was lower than their share in marketed surplus. However, procurement in Rajasthan has increased significantly during the last three seasons. The low share of Uttar Pradesh, Rajasthan and Bihar in procurement of wheat warrants concerted efforts to strengthen procurement operations in these States.

Chart 2.13: Share of Major States in Marketed Surplus and Procurement of Wheat



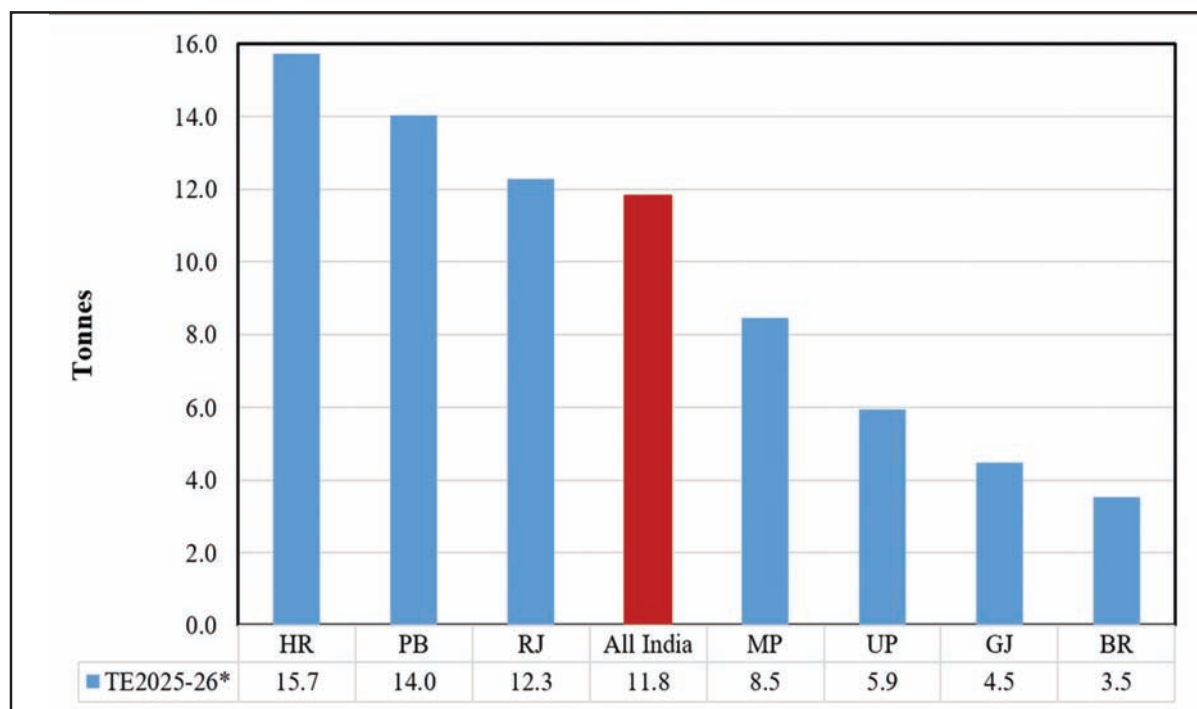
Source: 1. Food Corporation of India

2. Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare

- 2.33 Wheat procurement per farmer exhibits inequity, with Haryana and Punjab having significantly higher procurement per farmer, while other States, namely, Bihar, Gujarat and Uttar Pradesh with much lower procurement per farmer (Chart 2.14). Haryana has the highest average procurement per farmer (15.7 tonnes), followed by Punjab (14 tonnes), and Rajasthan during TE2025-26. However, Madhya Pradesh, Uttar Pradesh, Gujarat and Bihar, other important wheat producing States, have lower levels of procurement, even less than all-India average.
- 2.34 This concentration in total as well as per farmer procurement of wheat in some States leads to an uneven distribution of economic benefits of procurement operations. This inequity in procurement operations calls for stepping up efforts to strengthen procurement operations in States like Bihar and Uttar Pradesh.



Chart 2.14: Procurement of Wheat per Farmer (TE2025-26)



Note: *: as on 30th June 2025

Source: Food Corporation of India

Procurement Trends in Major States

- 2.35 Table 2.6 gives trends in procurement of wheat in leading wheat producing States. Total wheat procurement increased from a historic low of 18.8 million tonnes in RMS2022-23 to 26.2 million tonnes in RMS2023-24 and 26.6 million tonnes in RMS2024-25. Wheat procurement further improved during RMS2025-26 and was 30 million tonnes. In case of Punjab, wheat procurement increased by about 26 percent in RMS2023-24 over RMS2022-23, followed by an increase of 2.8 percent during RMS2024-25 but declined by 4.3 percent in RMS2025-26 at 119.2 lakh tonnes due to higher purchase by private traders. Haryana observed a similar trend, an increase of about 51 percent in RMS2023-24 over RMS2022-23, over 13 percent rise in RMS2024-25 and a very marginal decline in RMS2025-26. In Madhya Pradesh, wheat procurement increased by 54.2 percent in RMS2023-24, followed by a significant decline of about 32 percent in RMS2024-25 but increased by almost 61 percent in RMS2025-26. Rajasthan recorded an increase in wheat procurement from 0.1 lakh tonnes in RMS2022-23 to 4.4 lakh tonnes in RMS2023-24, 12.1 lakh tonnes in RMS2024-25 and reached 21.4 lakh tonnes in RMS2025-26. Additional bonus in Madhya Pradesh and Rajasthan contributed to a significant increase in wheat procurement during RMS2025-26. Wheat procurement also increased in Uttar Pradesh during RMS2024-25 and RMS2025-26 but was significantly lower than the target. In Bihar, wheat procurement has remained very low during all seasons.
- 2.36 Procurement as a percent of production observed an increase during RMS2023-24 over RMS2022-23, followed by a marginal decline during RMS2024-25 and an increase of about 9 percent at 25.6



percent during RMS2025-26. An increase in procurement as a percent of production was witnessed for Madhya Pradesh, Rajasthan, Uttar Pradesh and Bihar during RMS2025-26. Almost a similar trend was observed for procurement as a percent of marketed surplus. More than 25 percent of production and one-third of marketed surplus was procured during RMS2025-26.

Table 2.6: Trends in Procurement of Wheat in Leading Wheat Producing States

(Lakh Tonnes)

State	Procurement				Procurement as % to Production				Procurement as % to Marketed Surplus			
	2022-23	2023-24	2024-25	2025-26*	2022-23	2023-24	2024-25	2025-26*	2022-23	2023-24	2024-25	2025-26*
Punjab	96.5	121.2	124.6	119.2	64.9	72.2	70.2	66.6	74.4	80.2	78.6	74.5
Haryana	41.9	63.2	71.5	71.4	40.1	57.8	63.9	62.2	49.4	67.8	85.8	83.6
Madhya Pradesh	46.0	71.0	48.4	77.7	20.0	31.2	21.4	33.3	22.9	35.8	26.5	41.1
Rajasthan	0.1	4.4	12.1	21.4	0.1	4.1	12.4	18.8	0.2	6.9	22.2	33.5
Uttar Pradesh	3.4	2.2	9.3	10.3	1.0	0.7	2.6	2.8	1.4	1.0	4.2	4.5
Bihar	0.04	0.01	0.1	0.2	0.1	0.02	0.1	0.3	0.1	0.02	0.2	0.3
All India	187.9	262.0	266.1	300.3	17.4	23.7	23.5	25.6	22.8	30.8	32.3	35.2

Note: 1. *: As on 30th June 2025
 2. Production as per 3rd Advance Estimates of 2024-25 released on 28th May, 2025
 3. MSR for RMS2024-25 is taken as proxy for RMS2025-26

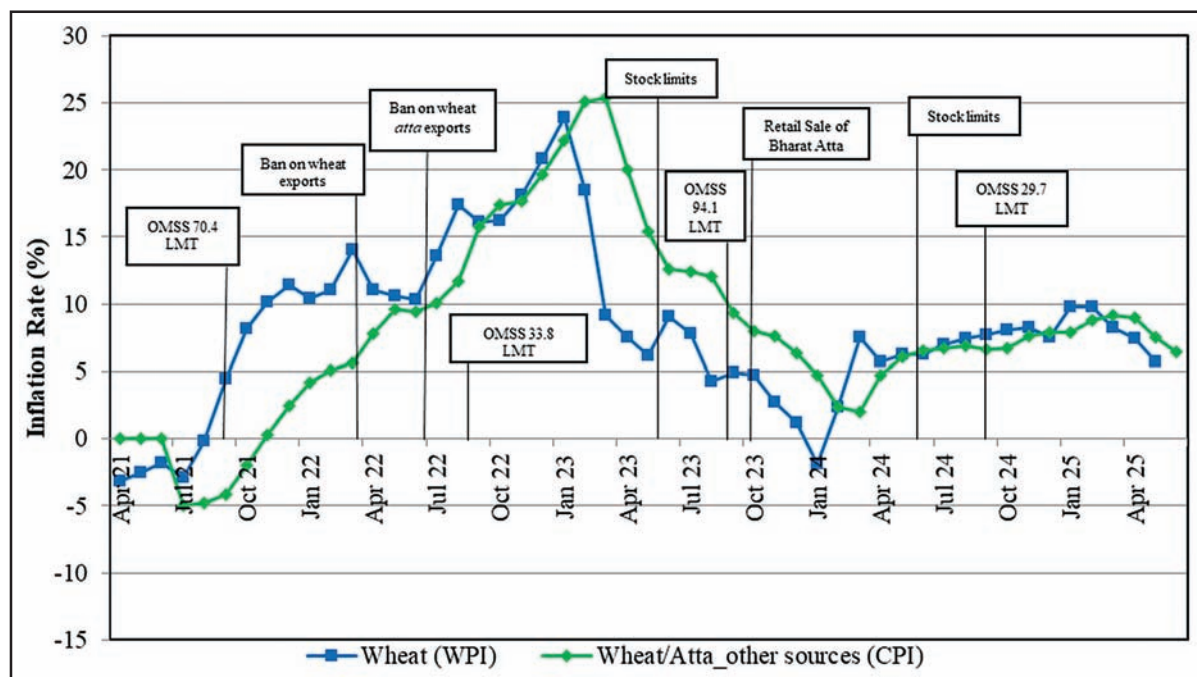
Source: 1. Food Corporation of India
 2. Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare

High Inflation in Wheat

- 2.37 As may be seen from Chart 2.15, WPI based inflation (base year 2011-12) and inflation based on CPI for wheat and wheat based products witnessed generally a rising trend during June 2021 and January/February 2023. The average annual inflation (based on WPI) of wheat increased from 4.7 percent in 2021-22 to 15.5 percent in 2022-23, while retail inflation of wheat sharply rose from 0.4 percent in 2021-22 to 17.1 percent in 2022-23. In order to contain inflation, Government banned wheat exports in May 2022 and wheat flour products in August 2022. Despite these restrictions inflation reached a peak of 23.9% (WPI) and 25.4% (CPI) in January 2023 and February 2023, respectively. The Government commenced sale of wheat under Open Market Sales Scheme (Domestic) [OMSS(D)] in February 2023 to increase the availability of wheat and wheat-based products in market and check rising inflation. In addition, Government imposed wheat stocking limits on wholesalers, retailers and big chain retailers in June 2023. Government offloaded 3.4 million tonnes of wheat in 2022-23 and 9.4 million tonnes in 2023-24. The additional supply of wheat in open market helped in containing inflation of wheat in the country and consequently, inflation of wheat (based on WPI) came down to 4.5 percent in 2023-24 and retail inflation of wheat/atta_{other} sources declined to 7.8 percent in 2023-24. However, wheat inflation witnessed an increasing trend during 2024-25.



Chart 2.15: Trends in All-India Inflation Rates for Wheat



Source: 1. Ministry of Statistics and Programme Implementation for CPI data
2. Office of the Economic Advisor, Department for Promotion of Industry and Internal Trade for WPI data

- 2.38 On 29th March 2024, Government of India ordered traders, wholesalers, retailers and processors across all States and Union Territories to disclose their wheat stock positions on a designated portal every Friday, starting 1st April 2024 to prevent hoarding and moderate prices. Further, as part of continuous efforts to moderate prices of wheat and prevent hoarding and unscrupulous speculation, Government revised stock limits on wheat on 9th September 2024, December 2024 and May 2025 and applicable until 31st March 2026 for all States and Union Territories. The Commission is of the view that such regulatory steps should be temporary in nature and efforts should be made to increase wheat production in the country.

Pulses

- 2.39 The trends in procurement of gram and lentil for the last three marketing seasons are presented in Table 2.7. The total procurement of gram at all-India was about 23.5 lakh tonnes in RMS2023-24 which declined to 43.3 thousand tonnes during RMS2024-25 due to high market prices and increased to about 1.6 lakh tonnes in RMS2025-26. During RMS2023-24, Madhya Pradesh and Maharashtra accounted for two-third of total procurement of gram. However, during RMS2025-26, Gujarat accounted for more than half of the procurement of gram, followed by Madhya Pradesh (29.5%) and Rajasthan (13.4%). Procurement of gram as a percent of total production declined from 19.2 percent in RMS2023-24 to 1.4 percent in RMS2025-26, much lower than procurement ceiling of 25 percent under PSS. In case of lentil, procurement more than tripled in RMS2024-25 over RMS2023-24 but declined by around 36 percent in RMS2025-26 at 1.6 lakh tonnes. Madhya Pradesh accounted for around 83 percent of total procurement, followed by Uttar Pradesh (around 17%) during RMS2024-



25 and RMS2025-26. Procurement of lentil accounted for 13.9 percent of total production in RMS2024-25 and 9 percent in RMS2025-26 due to improvement of market prices of lentil during the last two seasons.

Table 2.7: Procurement of Gram and Lentil in Major Producing States under PSS and PSF Schemes

Crop	State	Procurement (in '000 Tonnes)			Procurement as % of Total Production		
		RMS 2023-24	RMS 2024-25	RMS 2025-26*	RMS 2023-24	RMS 2024-25	RMS 2025-26*
Gram	Madhya Pradesh	796.0	41.4	46.9	25.7	1.4	1.9
	Maharashtra	772.9	0.0	1.9	24.3	0.0	0.1
	Gujarat	328.4	0.1	82.7	25.3	0.0	5.6
	Rajasthan	247.3	1.3	21.2	13.7	0.1	1.1
	Karnataka	79.5	0.0	0.0	14.2	0.0	0.0
	Andhra Pradesh	63.1	0.0	0.2	13.8	0.0	0.1
	Telangana	50.2	0.6	5.8	21.6	0.3	5.2
	Uttar Pradesh	15.6	0.0	0.0	1.7	0.0	0.0
	Haryana		0.0	0.01	0.0	0.0	0.0
	All India	2353.1	43.3	158.7	19.2	0.4	1.4
Lentil	Madhya Pradesh	64.5	205.7	131.6	10.1	25.0	18.1
	Uttar Pradesh	12.0	42.8	27.6	2.3	7.4	4.4
	All India	76.5	248.6	159.2	4.9	13.9	9.0

Note: 1. *As on 5th June 2025

2. Production as per 3rd Advance Estimates of 2024-25

Source: 1. National Agricultural Cooperative Marketing Federation of India

2. Economics, Statistics & Evaluation Division, Ministry of Agriculture & Farmers Welfare

- 2.40 The number of farmers benefitting from procurement of gram at all-India level declined from 10.3 lakh in RMS2023-24 to 15,485 in RMS2024-25, but increased to 80,555 in RMS2025-26 (as on 5th June 2025). In RMS2025-26, Gujarat had the highest number of beneficiary farmers, followed by Madhya Pradesh and Rajasthan. With respect to lentil, number of farmers benefitting from procurement recorded a spike of about 2.6 times in RMS2024-25 over RMS2023-24, followed by a decline of over 38 percent to 72,049 in RMS2025-26. The State-wise break-up of number of farmers benefitted from pulses procurement operations during the last three marketing seasons is given in Annex Table 2.5.



- 2.41 Table 2.8 gives the quantity sanctioned and procurement of pulses under Price Support Scheme (PSS). It is evident from the Table that there is a huge gap between the sanctioned quantity and actual procurement of both *kharif* and *rabi* pulses. Though, procurement of gram, *tur/arhar* and *moong* increased during the current season compared with last season but was significantly lower than sanctioned quantity despite the fact that market prices of lentil were below the MSP during RMS2025-26. Hence, it calls for efforts to strengthen procurement operations and create scientific storage and warehousing facilities to ensure remunerative prices to farmers, which would help in increasing acreage and production of pulses in the country and achieve *atmanirbharta* in pulses.

Table 2.8: Procurement of Pulses under Price Support Scheme (PSS)

(Lakh Tonnes)

Crop	(KMS 2023-24 & RMS 2024-25)		(KMS 2024-25 & RMS 2025-26*)	
	Quantity Sanctioned	Procurement	Quantity Sanctioned	Procurement
Gram	27.7	0.4	27.8	1.5
Lentil	12.8	2.4	10.6	1.5
Tur	16.6	-	13.2	4.2
Moong	3.3	0.5	3.4	1.7
Urad	5.2	-	7.6	0.0
Total	65.6	3.4	62.6	9.0

Note: *:As on 27th May 2025

Source: National Agricultural Cooperative Marketing Federation of India

Oilseeds

- 2.42 As market prices of R&M were below MSP during RMS2023-24 and RMS2024-25, about 11.1 lakh tonnes of R&M was procured during 2023-24 and 12.1 lakh tonnes in 2024-25. However, as market prices rose and were marginally lower than MSP, R&M procurement declined significantly and was 2.9 lakh tonnes in 2025-26 (as on 5th June 2025). Rajasthan, Haryana and Madhya Pradesh accounted for more than 85 percent of total procurement during RMS2023-24 and RMS2024-25, while Haryana's share during RMS2025-26 was nearly 55 percent. The number of farmers benefitting from procurement of R&M increased by about 15 percent in RMS2024-25 over RMS2023-24 but witnessed a decline of about 72 percent in RMS2025-26. Haryana had the largest number of farmers benefitting from procurement, followed by Gujarat and Rajasthan. The state-wise break up of farmers benefitted from oilseeds procurement during the last three seasons is given in Annex Table 2.6. Some of the issues relating to the procurement of pulses and oilseeds are given in Box 2.1.



Box 2.1: Need to Strengthen PSS Procurement Operations at State Level

Procurement of pulses and oilseeds is done under PSS, which is implemented on the request of the State Governments/UTs, after sanction granted by the Government of India for procurement of notified crops. State Government make arrangements for setting up procurement centres, storage facilities, arrangement of gunny bags, and other logistics. However, the Scheme faces several challenges including delay in procurement operations, inadequate infrastructure and logistics issues. There is significant delay in issuing notification by States after approval by the Central Government (Annex Table 2.7). For example, there was significant delay in State government notification in Karnataka (30 days) and Gujarat (41 days) in case of gram, Madhya Pradesh (22 days) and Uttar Pradesh (21 days) in case of lentil/masoor and all major R&M producing States. In addition, there is significant delay in actual procurement after State Government's notification. For example, the time gap between State notification and actual procurement was 73 days in Karnataka and 38 days in Haryana for gram, 35 days in Madhya Pradesh for gram and lentil/masoor, and 28 days in Uttar Pradesh for lentil/masoor. Similarly, for mustard seed, the delay was more than 30 days in Madhya Pradesh and Uttar Pradesh.

Unavailability of scientific warehousing facility and Warehousing Development and Regulatory Authority (WDRA) accredited warehouses for storage of the commodity procured under PSS and availability of jute bags are other infrastructural and logistics constraints. Since procurement of oilseeds and pulses is not a regular activity, State agencies are generally reluctant to procure gunny bags in advance due to the financial burden associated with interest costs and warehousing charges. Delay in allocation of districts between NAFED and NCCF is another issue that needs to be addressed.

The arrangements for procurement including registration of farmers, opening of centers, storage and warehousing, gunny bags, QR tags, etc. should be made well in advance and communicated to farmers.

Source: National Agricultural Cooperative Marketing Federation of India

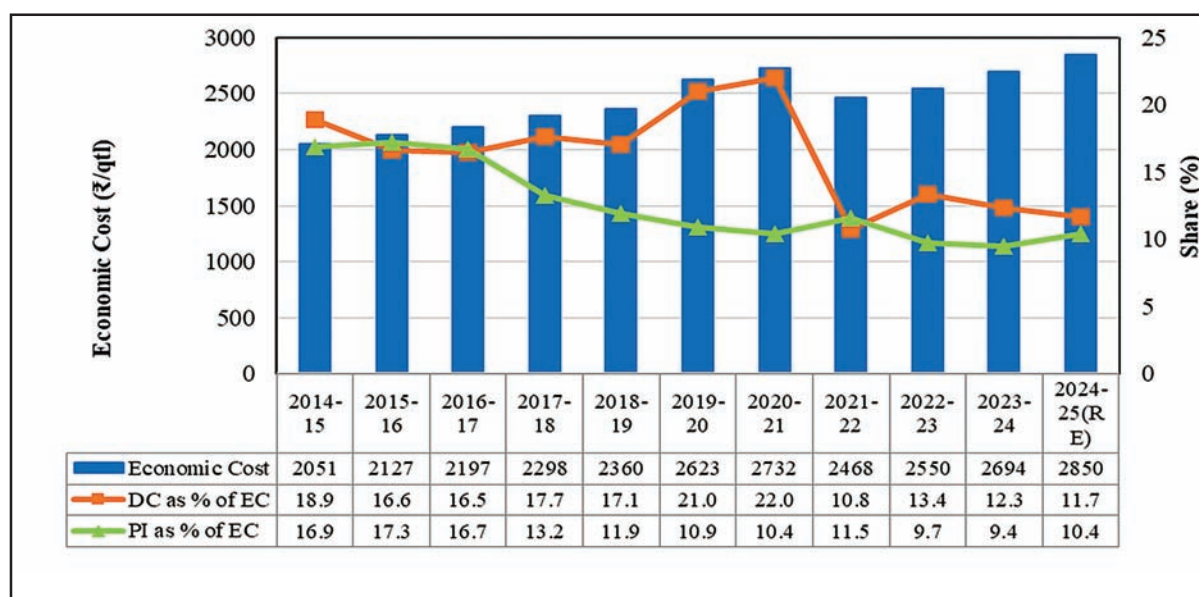
Food Subsidy and Economic Cost of Wheat

- 2.43 Food subsidy serves various objectives of providing price support to farmers, ensuring availability of foodgrains to poor consumers at affordable prices and maintain strategic stocks of grain for national food security and price stabilisation. However, rising subsidy is a major concern due to substantial impact on government budget and efficiency issues. Food subsidy is the largest component of subsidy bill in the country and has increased 1.6 times over the last decade. The food subsidy consists of three major components: (i) Consumer subsidy, difference between the economic cost and Central Issue Price, (ii) Buffer subsidy, operational cost of holding buffer stocks for food security purpose and market intervention operation and excess stocks, (iii) Other subsidies for coarse grain operation by State Government and any other specific distribution of foodgrains by the Government of India etc.



- 2.44 The economic cost (EC) of foodgrains consists of pooled cost of grains, procurement incidentals (PI) such as statutory charges, gunny cost, labour, transportation, storage and interest charges, administrative expenses, etc. and distribution costs (DC) comprising of freight, handling, storage and interest, etc. The economic cost of wheat exhibited an increasing trend during 2014-15 and 2020-21, rose from ₹2,051 per quintal in 2014-15 to ₹2,723 per quintal in 2020-21 per quintal but declined to ₹2,468 per quintal in 2021-22 on account of about 56 percent decline in distribution cost in general and interest cost in particular. The economic cost increased by 3.3 percent in 2022-23 and 5.6 percent in 2023-24 at ₹2,694 per quintal in 2023-24. As regards the components of economic cost, share of procurement incidentals exhibited a declining trend from 2015-16 onwards, which continued till 2023-24 except for 2021-22. The share of distribution cost observed a decline during 2014-15 to 2016-17 but increased in 2017-18, 2019-20 and peaked at 22 percent in 2020-21. A steep decline was observed in 2021-22 due to repayment of National Small Saving Fund (NSSF) loan to FCI. The distribution cost in total economic cost witnessed a declining trend during last 3 years, while procurement incidentals share declined during 2021-22 to 2023-24 but rose in 2024-25. The pooled cost of grains increased significantly during last two years.

Chart 2.16: Economic Cost of Wheat and Share of Different Components of Economic Cost



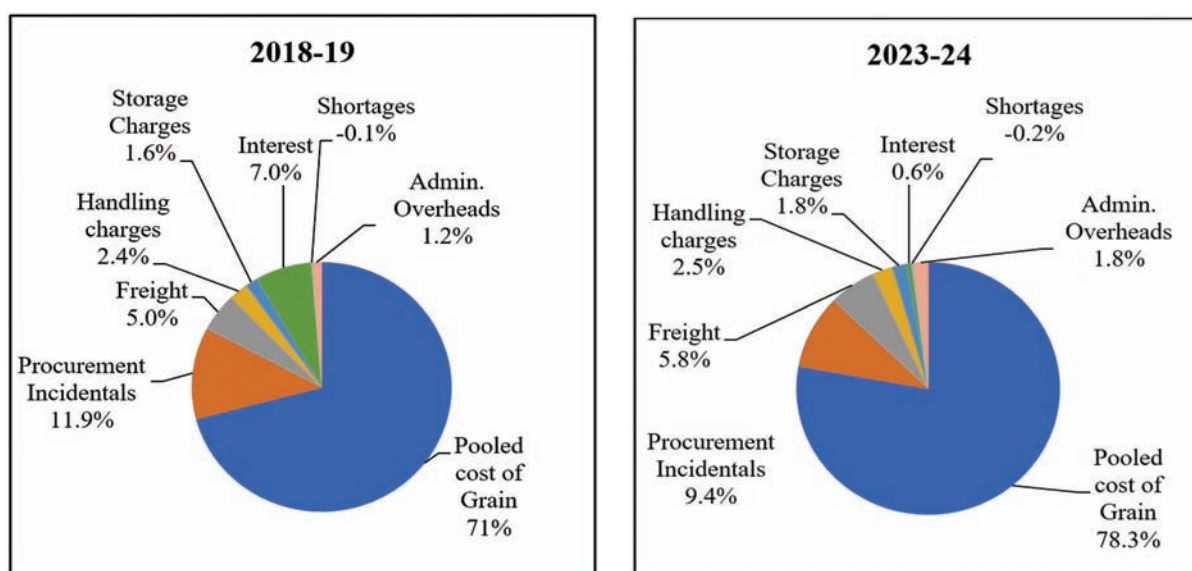
Note: EC: Economic Cost; DC: Distribution Cost; PI: Procurement Incidentals

Source: Food Corporation of India

- 2.45 A comparison of major components of economic cost during 2018-19 and 2023-24 is presented in Chart 2.17. As may be observed from the Chart, pooled cost of grains accounted for 71 percent of the total economic cost in 2018-19, while share of procurement incidentals was 11.9 percent and distribution cost was 17.1 percent in the total economic cost. In 2023-24, share of pooled cost of grains increased to 78.3 percent of economic cost, while share of procurement incidentals declined to 9.4 percent and distribution cost to 12.3 percent. The decrease in procurement incidentals is primarily on account of storage & interest charges and administrative charges paid to State agencies while decline in distribution cost was driven by decline in interest charges.



Chart 2.17: Share of different components of Economic Costs of Wheat Procurement during 2018-19 and 2023-24



Source: Food Corporation of India

- 2.46 During the last decade, food subsidy as a percentage of GDP (at current prices) fluctuated in the range of 0.6 percent in 2024-25(RE) to 1.2 percent in 2021-22, except for the year 2020-21 when it was 2.7 percent. Due to Central Government's provision of ₹1,18,712 crore to FCI for repayment of National Small Saving Fund (NSSF) loan and clear part dues of the FCI, food subsidy declined in 2021-22 and 2022-23. Food subsidy was ₹1,97,240 crore in 2024-25(RE) and is budgeted at ₹2,03,420 crore in 2025-26.

Distortions in Agricultural Markets

- 2.47 Agricultural markets in India face a number of restrictions and distortions pertaining to trade, markets and policies, which impinge on smooth functioning of markets. These, inter alia, include regulatory restrictions, high incidence of statutory taxes and other charges, trade restrictions, bonus over and above the MSP, etc. As a result of such distortions, there is excess production of certain agricultural commodities and underproduction of some crops like oilseeds, pulses, nutri-cereals, etc. in the country.
- 2.48 Several States impose *mandi* fee and other charges on sale/purchase of agricultural produce for development of market infrastructure such as market yards, storage facilities, rural roads etc. so as to make marketing of agricultural produce more efficient. On account of these charges being very high in some States, in addition to large variations across States, inter-State trade is adversely affected thereby creating an inefficient marketing system in the country. As may be seen in Table 2.9, Punjab levies 2 percent of MSP as market/*mandi* fee, 3 percent as rural development fee and 2.5 percent as commission for *ahratiyas*/agents, totalling to 7.5 percent of MSP, which is the highest in the country. Haryana charges 2 percent as market/*mandi* fee, 2 percent as rural development fee and 2.5 percent as commission for *ahratiyas*/agents. In Rajasthan, 1.6 percent as market/*mandi* fee, 0.4 percent as *Krishak Kalyan Cess* and 2.25 percent commission for *ahratiyas*/agents is levied. Uttar Pradesh levies 1



percent as market/*mandi* fee, 0.5 percent as Development Cess and 2.5 percent as the commission for *ahratias*/agents. Madhya Pradesh has the lowest market fee and other levies consisting of 1 percent as market/*mandi* fee and 0.2 percent as *Nirashrit Shulk*. Punjab and Haryana have relatively very high level of market fee and other charges, which drives out private sector trade.

Table 2.9: Market Fees/Taxes/Charges levied on Procurement of Rabi Crops during RMS2024-25

(in percent)

State	Market/ <i>Mandi</i> fee	Rural Development Fee	Infrastructure Development Cess and others	Commission for <i>Ahratias</i> / Agents	Total
Punjab	2	3	-	2.5	7.5
Haryana	2	2	-	2.5	6.5
Rajasthan	1.6	-	0.4*	2.25	4.25
Uttar Pradesh	1	-	0.5	2.5	4
Madhya Pradesh	1	-	0.2 [#]		1.2

Note: *: *Krishak Kalyan Cess*, [#]: *Nirashrit Shulk*; Rural Development Fee and *Nirashrit Shulk* are claimed by the States but not allowed by Department of Food and Public Distribution

Source: 1. Department of Food and Public Distribution, Ministry of Consumer Affairs, Food and Public Distribution

2. State Governments replies

Awareness about MSP and Procurement Operations

- 2.49 Lack of awareness about MSP, procurement operations, FAQ specifications etc. due to lack of adequate and timely publicity among farmers has been highlighted by various studies. Therefore, efforts in awareness creation among farmers about MSP, procurement and FAQ norms would help in making procurement operations more effective. In addition, strengthening of infrastructure for grading, drying, etc. of produce to meet FAQ standards can help in reduction in the instances of rejection of farmers' produce at APMC *mandis*/procurement centres.
- 2.50 Some State Governments have taken several initiatives to create awareness about procurement under MSP, registration of farmers, information about procurement centers and timings, etc. but more efforts are needed. State Governments should make efforts towards awareness creation about the MSP, quality specification, procurements centres, period of procurement, registration procedures, document requirements etc. and actively engage with the farmers and farmers groups in this regard. Information should be disseminated in local languages and well in advance using various print and electronic media like TV, radio, newspapers etc. and social media. Further, institutions like panchayats, co-operative societies, FPOs, etc. can play an important role in disseminating information among farmers.



Recapitulation

- 2.51 As per FAO, USDA and IGC estimates, global wheat production and supplies are forecast to increase marginally in 2025-26. Global wheat trade and utilisation is also expected to increase in 2025-26. As per FAO and IGC, ending stocks are forecast to decline in 2025-26, while USDA predicts a marginal increase in ending stocks in 2025-26. However, the stock-to-use ratio (SUR) is projected to decline in 2025-26.
- 2.52 Wheat production in India is estimated to reach a record level of 117.5 million tonnes in 2024-25, 3.7 percent higher than 2023-24. Wheat stocks in Central Pool were about 38 million tonnes as on 1st June 2025, the highest level since January 2022. Domestic situation points towards a comfortable wheat supply during 2025-26.
- 2.53 The average market prices of wheat and barley in RMS2025-26 remained above the MSP, while market price of gram remained below MSP in RMS2025-26. In case of lentil, market prices witnessed an increasing trend during RMS2025-26 but stayed below the MSP. The R&M prices remained below the MSP in RMS2025-26 with a narrowing gap between market prices and MSP. During RMS2025-26, market prices of safflower exhibited an increasing trend and moved above MSP in April-May 2025.
- 2.54 Wheat procurement increased to 30 million tonnes in RMS2025-26 benefitting about 25.1 lakh farmers. Procurement of gram at all-India level increased in RMS2025-26, but was significantly less than RMS2023-24 due to improvement in market prices of gram during April-June 2025. Procurement of lentil, which more than tripled in RMS2024-25 over RMS2023-24, declined by about 36 percent in RMS2025-26, despite market prices being lower than the MSP, which points towards infrastructural, logistics and administrative constraints in procurement operations.
- 2.55 During the last decade, food subsidy has significantly increased due to rising economic cost, constant Central Issue Price and free foodgrains under *Pradhan Mantri Garib Kalyan Anna Yojana* (PMGKY).
- 2.56 High market/*mandi* fee and other charges on sale/purchase of agricultural produce, which distort markets and drive out private trade, should be rationalized.
- 2.57 State Governments and other related organizations/institutions should make coordinated efforts towards awareness creation about MSP, FAQ specifications, procurement process and procedures to make procurement operations more effective.



CHAPTER 3



Crop Yield: Trends and Drivers

- 3.1 In the current scenario of natural resource depletion and the climate change induced extreme weather events, it is essential to promote agriculture that ensures food security, enhances farmers' income and also promotes sustainable use of resources. Therefore, increasing crop yield is imperative as there is limited scope for area expansion. This Chapter discusses productivity trends of major *rabi* crops at all-India level and compares these with the world average yield and top-yielding countries. Further, productivity trends across major States and districts have also been analyzed to identify yield gaps at disaggregated level for targeted interventions to enhance crop yield. The Chapter also sheds light on crucial factors of yield and policy initiatives for improvement in the yield.

Yield Growth Trends

- 3.2 Table 3.1 presents growth trends in area, production and yield of six mandated *rabi* crops, namely, wheat, barley, gram, lentil, rapeseed & mustard (R&M), and safflower during the last five years.

Cereals

- 3.3 Wheat production registered a positive growth during the last five years, except 2021-22 when production declined significantly due to decline in area under cultivation (-2.1%) and only a marginal increase in yield attributed to heat wave during the grain development stage. During the last three years, positive growth was recorded in wheat production, which was primarily driven by growth in acreage, while yield remained almost stagnant. Low growth or stagnation in wheat yield is a matter of concern. Barley, cultivated in less than one million hectare, showed high volatility in production caused mainly by fluctuations in area. The growth in yield of barley was also not encouraging except 2021-22 when yield increased by 8.2 percent due to low base effect.
- 3.4 The highest growth in area (4.8%) and production (6.7%) of total cereals was recorded in 2024-25, while the lowest were in 2021-22. Yield growth during the last five years remained less than 2 percent except 2021-22 when it was 2.7 percent.



Table 3.1: Trends in Annual Growth Rate of Area, Production and Yield of Major Rabi Crops

(in Percent)

Crop	APY	2020-21	2021-22	2022-23	2023-24	2024-25
Wheat	Area	-0.7	-2.1	3.1	1.4	2.9
	Production	1.6	-1.7	2.6	2.5	3.7
	Yield	2.4	0.5	-0.5	1.1	0.8
Barley	Area	0.5	-23.5	38.6	-12.3	9.7
	Production	-3.8	-17.2	39.5	-11.2	8.5
	Yield	-4.2	8.2	0.6	1.2	-1.1
Gram	Area	3.1	7.4	-2.5	-8.4	0.2
	Production	7.5	13.7	-9.4	-10.0	2.7
	Yield	4.4	5.8	-7.1	-1.8	2.5
Lentil	Area	12.7	-3.8	15.9	6.5	-2.1
	Production	35.4	-15.1	22.8	14.9	-1.1
	Yield	20.1	-11.6	5.9	8.0	1.0
Rapeseed & Mustard	Area	-2.3	19.3	10.8	3.7	-6.0
	Production	11.9	17.2	5.7	4.9	-4.9
	Yield	14.5	-1.8	-4.6	1.1	1.2
Safflower	Area	8.2	40.7	38.1	-39.9	-12.5
	Production	-17.8	70.7	46.9	-44.2	-13.8
	Yield	-24.1	21.4	6.3	-7.0	-1.5
Total Cereals	Area	2.0	-1.6	3.9	1.3	4.8
	Production	3.9	1.1	5.3	1.5	6.7
	Yield	1.9	2.7	1.4	0.2	1.8
Total Pulses	Area	2.8	6.8	-6.0	-4.8	0.4
	Production	10.6	7.2	-4.6	-7.0	4.1
	Yield	7.5	0.3	1.6	-2.3	3.6



Total Foodgrains	Area	2.2	0.3	1.6	-0.1	3.9
	Production	4.4	1.6	4.5	0.8	6.5
	Yield	2.2	1.3	2.8	0.8	2.5
Total Oilseeds	Area	6.2	0.4	4.5	-0.2	0.2
	Production	8.2	5.6	8.9	-4.1	7.4
	Yield	1.9	5.2	4.3	-3.9	7.2

Source: Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare

Pulses

- 3.5 Gram, the major crop contributing about 45 percent to total pulses production in the country, showed high fluctuations in production during last five years. Gram production, which increased by 7.5 percent in 2020-21 and the growth further accelerated to 13.7 percent in 2021-22, declined by 9.4 percent in 2022-23 and 10 percent in 2023-24 due to negative growth both in area and yield. In 2024-25, production increased by 2.7 percent mainly driven by growth in yield (2.5%). Performance of lentil in terms of growth in area, production and yield has been impressive in three out of five years. However, large fluctuations in area and yield of both gram and lentil have led to instability in production.

Oilseeds

- 3.6 The production of rapeseed & mustard recorded significant growth up to 2023-24 and growth in production was largely attributed to area expansion except in 2020-21 when R&M yield recorded 14.5 percent increase. However, R&M production fell by 4.9 percent in 2024-25 due to 6 percent decline in area. Safflower contributing less than 0.5 percent to total area and production of oilseeds, recorded high variations in area, production and yield during last five years. Safflower production declined during the last two years due to decline in both yield and acreage. However, production of total oilseeds recorded significantly high growth of more than 5 percent during last five years, except in 2023-24 and contribution of yield was higher than area.
- 3.7 It is evident from the above analysis that production of *rabi* cereals, pulses and oilseeds showed an improvement during last five years, but growth has been highly volatile. Increase in area was the major source of production growth as the yield growth was low or stagnant in all crops except lentil. Therefore, there is a need to focus on improving crop yield and reducing volatility for ensuring food and nutritional security, particularly for pulses and edible oils. Recently, Government has taken various steps towards achieving self-sufficiency in pulses and oilseeds, details of which are given in Box 3.1. It is expected that these initiatives would help in improving yields and thus help in achieving the goal of self-sufficiency in pulses and oilseeds in the country.



Box 3.1: Recent Initiatives to Achieve Self-sufficiency in Oilseeds and Pulses

In view of the growing demand and increasing dependence on import for pulses and edible oils, Government has taken important steps with an objective to achieve self-reliance by enhancing production, ensuring remunerative prices to the farmers, and promoting crop diversification. These initiatives include the National Mission on Edible Oils-Oilseeds (NMEO-OS), National Mission on Edible Oils-Oil Palm (NMEO-OP), and Mission for Aatmanirbharta in Pulses.

NMEO-OS, being implemented for seven years from 2024-25 to 2030-31 with a financial outlay of ₹10,103 crore, aims to increase primary oilseeds production to 69.7 million tonnes by 2030-31. The Mission focuses on enhancing the production of key primary oilseed crops and increasing collection and extraction efficiency from secondary sources. NMEO-OP is targeted to bring additional area of 6.5 lakh hectares under oil palm (3.28 lakh ha in North-Eastern States and 3.22 lakh ha in rest of India) by 2025-26 and Crude Palm Oil production to 2.8 million tonnes by 2028-29.

The 'Mission on Aatmanirbharta in Pulses' is focused on enhancing productivity, improving post-harvest storage and management, and assured procurement of tur/arhar, urad and lentil. Further, the National Food Security & Nutrition Mission (NFSNM), erstwhile National Food Security Mission (NFSM) being implemented under Krishonnati Yojana, aims to increase production of pulses by increasing area, particularly in rice-fallow, improving yield and availability of quality seeds.

International Comparison of Yield of Major Rabi Crops

- 3.8 Though yield potential of crops differs across countries due to variations in climatic conditions, soil types and various other factors, yield comparison among countries still helps in identifying productivity efficiency gaps, setting productivity goals, and devising strategies to bridge the gaps. Yields of major rabi crops in India vis-à-vis world average and highest yielding country among the major producers are presented in Table 3.2.
- 3.9 As is evident from the Table, the all-India average yield is lower than the world average for wheat, barley, lentil, safflower, and R&M, while gram yield is marginally higher than world average. Yield gap is significant in the case of lentil and R&M, while all-India yield levels are close to the world average for wheat, safflower and barley. However, comparison of all-India average yields with the top-yielding countries shows substantial yield differences in all crops. The all-India average yield was less than half of the highest-yielding countries for all crops except gram. Similarly, the yields in highest-yielding State were also significantly lower than the world highest yields. For instance, State highest yields of barley and wheat in Punjab were only 52 percent and 57 percent of world highest yields in Germany and UK, respectively. Similarly, in case of gram, the average yield in Gujarat, highest yield in the country, was 82 percent of the yield in Australia, while for other crops, the highest yield recorded in the State was less than 55 percent of top-yielding country, indicating large efficiency gaps in yield.

Table 3.2: International Comparisons of Yield of Mandated *Rabi* Crops

(qtl/ha)

Crop	World Average	World Highest		All-India Average	State Highest	
Wheat	36.2	UK (1.8)	81.8	35.3	PB	46.7
Barley	31.5	Germany (7.3)	68.9	29.6	PB	35.6
Gram	11.7	Australia (5.7)	21.6	12.1	GJ	17.7
Lentil	11.7	China (2.6)	25.5	9.6	RJ	13.2
Rapeseed & Mustard	20.7	Germany (4.7)	36.8	14.8	GJ	19.5
Safflower	7.9	Mexico (8.3)	18.5	7.5	TG	9.7

Note: 1. TE2023 used for world average and all-India average yield and TE2023-24 for States average yield

2. Countries with production share $\geq 1\%$ in respective crops considered for the analysis

3. Figures in parenthesis indicates country's share in world total production

Sources: 1. FAOSTAT for world average, top yielding countries and all-India average yield

2. Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare for State average yield

Crop Yield Analysis in Major Producing States

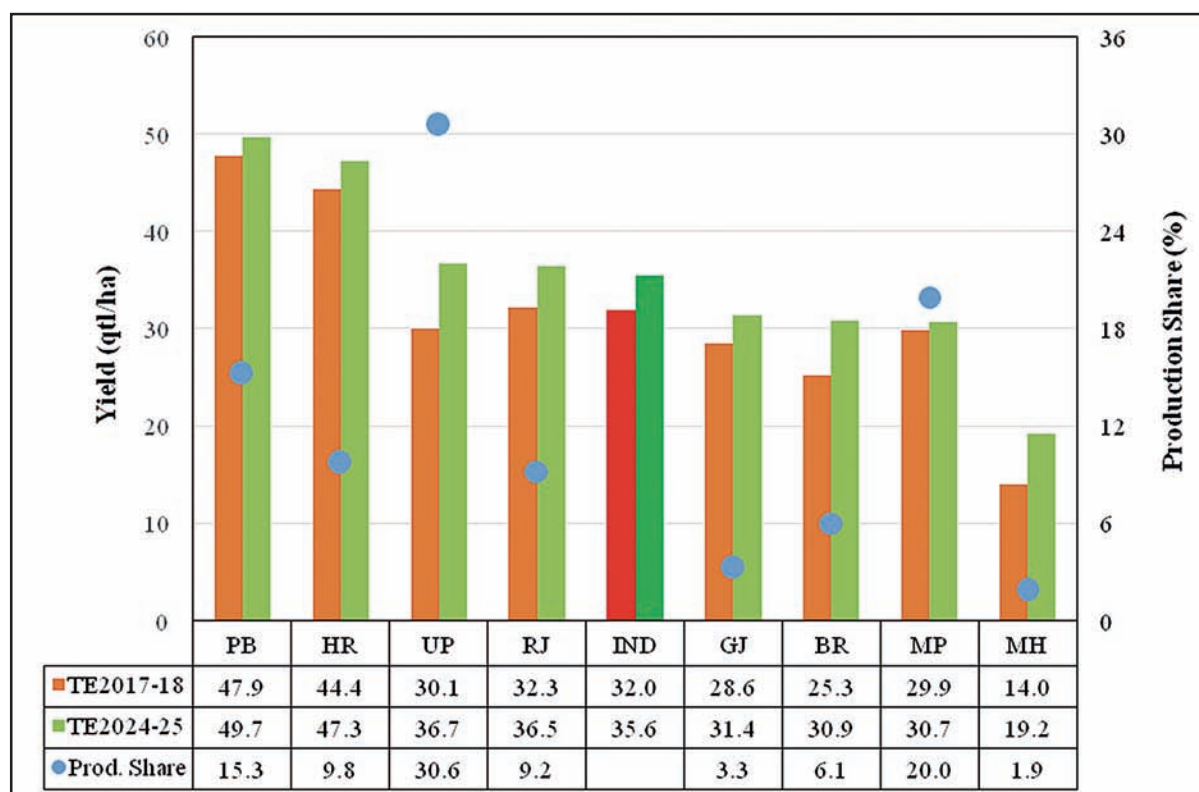
- 3.10 In this section, yields of mandated *rabi* crops in major States have been compared between recent two periods, i.e. TE2017-18 and TE2024-25 to see the performance of States in terms of yield.

Cereals

- 3.11 In Chart 3.1, yield trends of wheat in major producing States are compared between the two aforementioned periods. Punjab recorded the highest yield, followed by Haryana during both periods, but the increase during the last decade was low (3.8%) in Punjab compared to Haryana (6.5%). Among wheat producing States, yield increase was highest in Maharashtra (37.1%), followed by Bihar (22.1%), and Uttar Pradesh (21.9%). Madhya Pradesh, contributing 20 percent to the total production, recorded only a marginal improvement of 2.7 percent in the current period compared to the previous period. Though wheat yield has improved in Maharashtra, Madhya Pradesh, Bihar, and Gujarat, it is still significantly lower than all-India average.



Chart 3.1: Average Yield of Wheat in Major Producing States

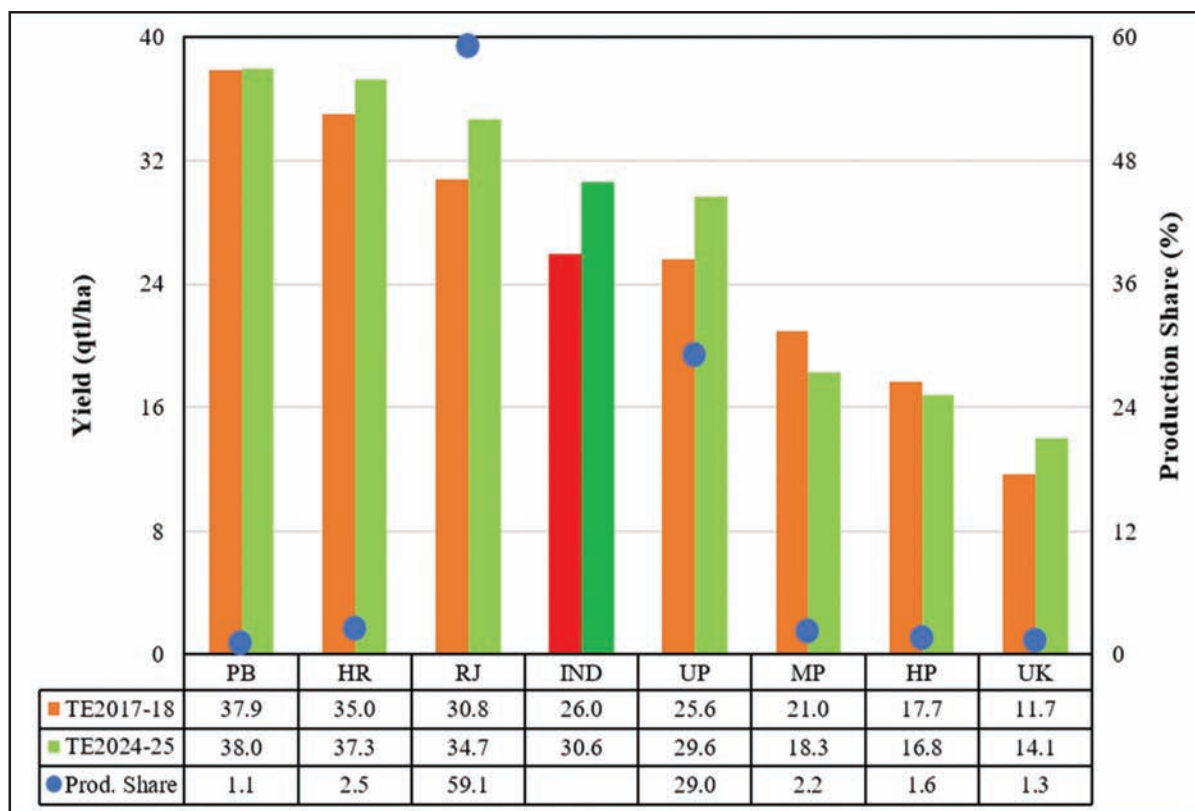


Source: Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare

- 3.12 Chart 3.2 compares the yield trends of barley in major producing States. Punjab, followed by Haryana recorded the highest yield but the increase was nominal (0.3%) in Punjab. It is encouraging to note that Rajasthan and Uttar Pradesh, the top two producers contributing 88 percent of barley production, recorded significant improvement in productivity during TE2024-25 compared to TE2017-18. As a result, an impressive growth of 17.7 percent in yield was recorded at the all-India level during the current period. Barley yield declined in Madhya Pradesh (-12.9%) and Himachal Pradesh (-5.1%).



Chart 3.2: Average Yield of Barley in Major Producing States



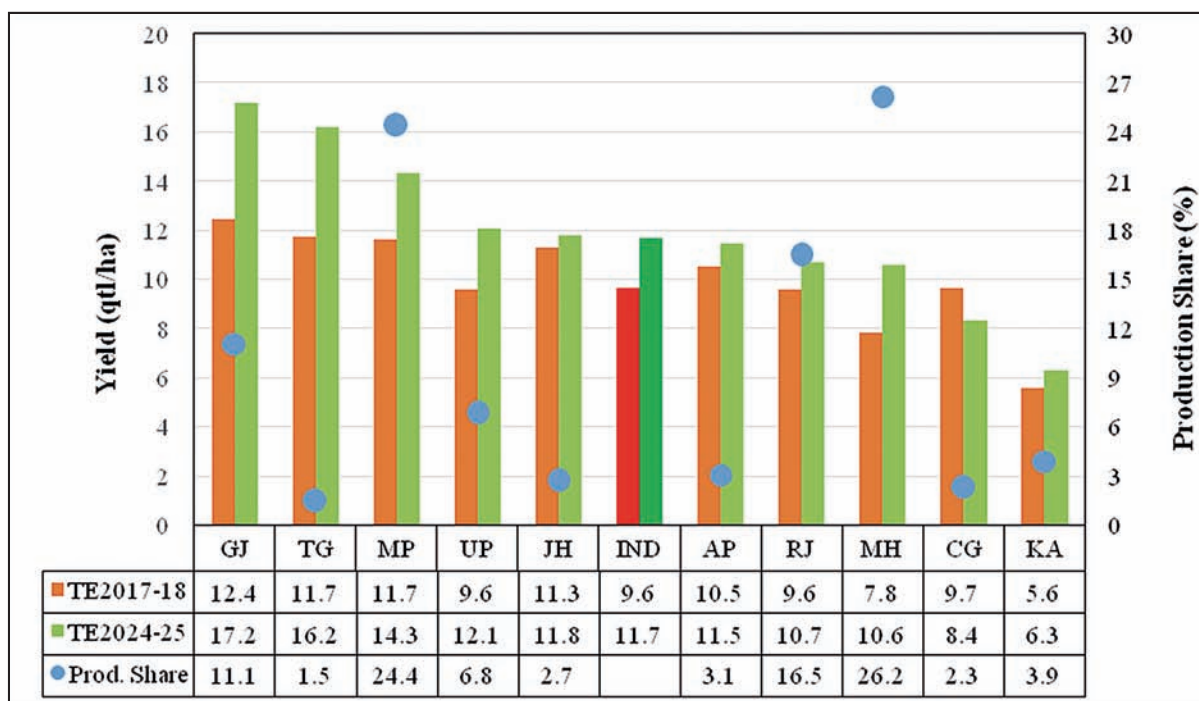
Source: Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare

Pulses

- 3.13 It is evident from Chart 3.3 that during TE2024-25, gram yield has improved in all the producing States, except Chhattisgarh. As a result, yield at all-India level registered a substantial growth of 21.9 percent. The absolute improvement in yield was more than 4 quintal per hectare in Gujarat and Telangana and more than 2.5 quintal per hectare in Maharashtra, Uttar Pradesh and Madhya Pradesh. Per hectare yield is significantly lower in Karnataka and Chhattisgarh than all-India average yield.
- 3.14 The average yield of lentil in the country increased by 14.8 percent from 8.8 quintal per hectare in TE2017-18 to 10.1 quintal per hectare in TE2024-25 (Chart 3.4). Madhya Pradesh and Uttar Pradesh, which together account for more than three-fourth of country's total lentil production, recorded yield improvement of more than 20 percent. Bihar and West Bengal, together contributing 15.8 percent of the production, witnessed a decline of 10 percent and 5 percent, respectively. Rajasthan showed a noteworthy improvement of 37.8 percent during the current period and secured the top position.

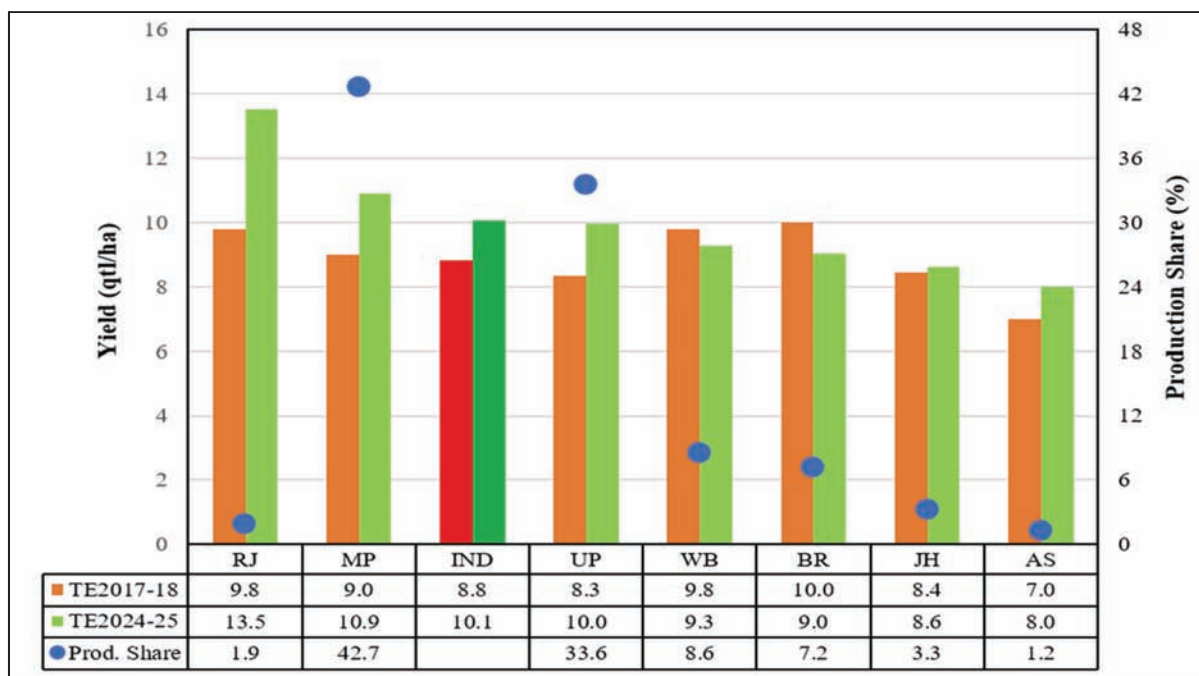


Chart 3.3: Average Yield of Gram in Major Producing States



Source: Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare

Chart 3.4: Average Yield of Lentil in Major Producing States



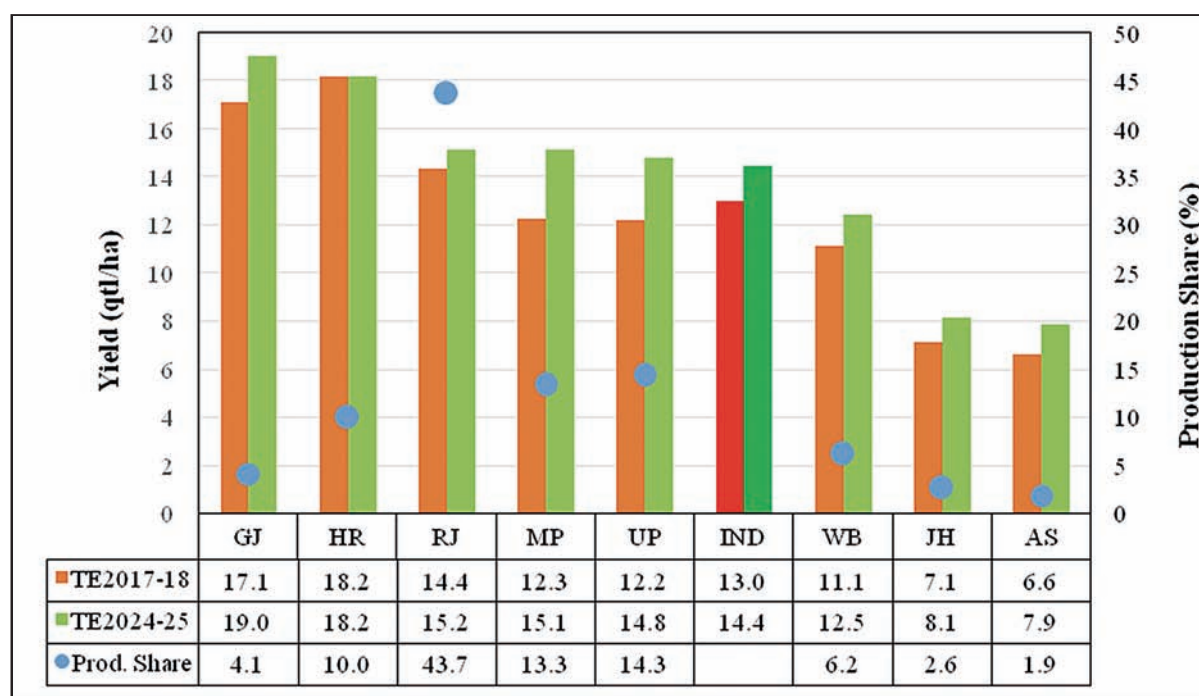
Source: Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare



Oilseeds

- 3.15 Yield of rapeseed & mustard in major producing States during TE2017-18 and TE2024-25 is presented in Chart 3.5. In TE2024-25, Gujarat had the highest yield, followed by Haryana, Rajasthan, and Madhya Pradesh, while Assam had the lowest. At the all-India level, R&M yield increased by 10.8 percent during last 10 years. The increase in yield was highest in Madhya Pradesh (22.8%), followed by Uttar Pradesh (21.3%), and Assam (19.7%). Rajasthan, the largest producer, also registered an improvement of 5.6 percent in yield.

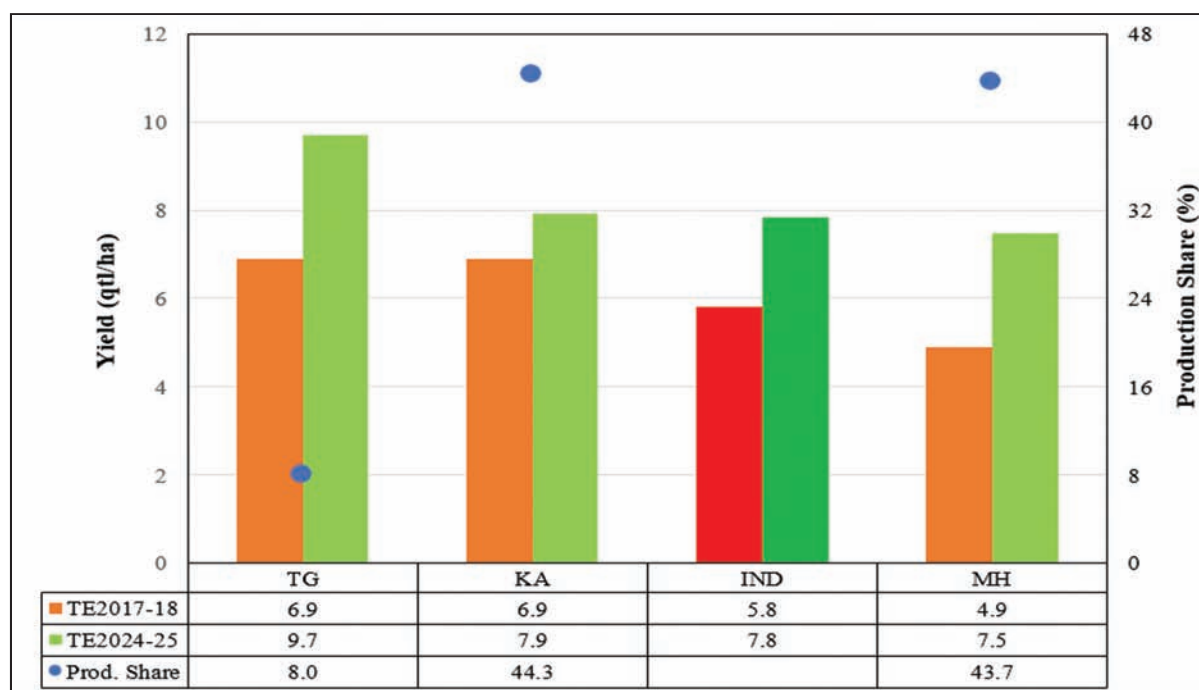
Chart 3.5: Average Yield of Rapeseed & Mustard in Major Producing States



Source: Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare



Chart 3.6: Average Yield of Safflower in Major Producing States



Note: Andhra Pradesh has been excluded as it has negligible production during TE2017-18

Source: Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare

- 3.16 Yield levels of safflower in major producing States are presented in Chart 3.6. Safflower is majorly concentrated in three States of the country. Karnataka and Maharashtra, with almost equal share, together contribute 88 percent of the country's total production. Telangana and Karnataka recorded yields higher than all-India average, while Maharashtra had lower than the national average yield. Safflower yield significantly improved in all these States with highest increase in Maharashtra.

Classification of States on the basis of Growth in Yield and Area

- 3.17 In this section, States have been classified into 'high' and 'low' categories based on the growth in area and yield to compare their performance between two time periods i.e. 2015-16 to 2019-20 and 2020-21 to 2024-25. States having production share ≥ 1 percent have been considered for the analysis. All-India average growth in area and yield from 2015-16 to 2024-25 (Y-o-Y) has been taken as the benchmark for classification. Based on this criterion, States have been classified into the following four categories. While classification of States is presented in Table 3.3 to Table 3.5, the State-wise average growth in area and yield for two time periods for major States is given in Annex Table 3.1.



		Yield Growth	
		≥ All-India	< All-India
Area Growth	≥ All-India	HH	HL
	< All-India	LH	LL

- 3.18 Table 3.3 compares the distribution of wheat and barley producing States according to growth in area and yield during two different periods. During last one decade, the all-India area under wheat increased at annual average growth of 0.5 percent and yield at 2.8 percent. Gujarat and Maharashtra, which were in HH category, slipped to HL category during the current period. Performance of Bihar improved from LH category to HH category, however, its yield growth decelerated during last five years compared to previous period (Annex Table 3.1). It is concerning to note that Haryana, Punjab and Uttar Pradesh, the leading wheat producing States, recorded decline in both area and yield growth and have shifted to LL category.
- 3.19 In case of barley, Haryana, Madhya Pradesh, and Rajasthan showed improvement in area growth but deceleration in yield growth, while Uttar Pradesh, though sustained area growth but recorded decline in yield growth. Uttarakhand shifted from HH to LL category due to lower growth in both area and yield during recent period. It is noteworthy that no State was in HH category during the current period. The position of Himachal Pradesh and Punjab remained unchanged during both periods.

Table 3.3: Categorization of States based on Area and Yield Growth in Wheat and Barley

Wheat		2015-16 to 2019-20		2020-21 to 2024-25	
		Yield Growth (2.8)			
		High	Low	High	Low
Area Growth (0.5)	High	GJ, MH	MP	BR	GJ, MP, MH, RJ
	Low	BR, HR, PB, RJ, UP	-	-	HR, PB, UP
Barley		Yield Growth (3.0)			
Area Growth (-0.2)	High	UP, UK	-	-	HR, MP, RJ, UP
	Low	HR, MP, RJ	HP, PB	-	HP, PB, UK

Note: States having production share $\geq 1\%$ in respective crop considered for the analysis

Source: Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare



- 3.20 Table 3.4 presents the status of gram and lentil producing States in terms of growth in area and yield during the two time periods. In case of gram, there were 7 States under HH category during initial five years but none of these could retain their position during the recent five years. Among these States, Andhra Pradesh, Karnataka, Rajasthan, Telangana and Uttar Pradesh performed low in terms of both area and yield growth, consequently shifted to least efficient (LL) category, while Gujarat and Maharashtra recorded decline in yield growth moving to HL category. Madhya Pradesh also lost its position due to lower growth in yield.
- 3.21 As far as lentil is concerned, among the major producers, Madhya Pradesh has improved its performance in terms of area growth, while Uttar Pradesh, which was under HH category, recorded deceleration in yield growth and shifted to HL category. Bihar showed improvement in yield growth, while Rajasthan performed better in terms of area growth but low in yield growth. Jharkhand and Assam remained in HL and LL category, respectively, while West Bengal shifted from HL to LL category due to significant decline in area and yield growth lower than all-India average.

Table 3.4: Categorization of States based on Area and Yield Growth in Gram and Lentil

Gram		2015-16 to 2019-20		2020-21 to 2024-25	
		Yield Growth (3.1)			
		High	Low	High	Low
Area Growth (1.8)	High	AP, GJ, KA, MH, RJ, TG, UP	CG, JH	CG	GJ, JH, MH
	Low	MP	-	-	AP, KA, MP, RJ, TG, UP
Lentil		Yield Growth (4.6)			
Area Growth (2.0)	High	UP	JH, WB	MP	JH, RJ, UP
	Low	MP, RJ	AS, BR	BR	AS, WB

Note: States having production share $\geq 1\%$ in respective crop considered for the analysis

Source: Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare

- 3.22 The classification of major R&M producing States based on area and yield growth is presented in Table 3.5. Haryana, Jharkhand, and Rajasthan were under the most preferred (HH) category during the previous period, while during the recent period, only Uttar Pradesh was under this category. Assam improved its position in the recent period due to higher yield growth, while Haryana, Rajasthan and West Bengal slipped to least efficient category (LL) on account of lower growth in area and yield than all-India average. Gujarat and Madhya Pradesh recorded higher area growth but lower yield growth and shifted from LH category to HL.



**Table 3.5: Categorization of States based on Area and Yield
Growth in Rapeseed & Mustard**

Rapeseed & Mustard		2015-16 to 2019-20		2020-21 to 2024-25	
		Yield Growth (3.3)			
		High	Low	High	Low
Area Growth (4.3)	High	HR, JH, RJ	WB	UP	GJ, JH, MP
	Low	GJ, MP, UP	AS	AS	HR, RJ, WB

Note: States having production share $\geq 1\%$ in respective crop considered for the analysis

Source: Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare

- 3.23 The growth in yield of mandated *rabi* crops in top three producing States have slowed down except lentil. Therefore, the yield needs to be increased by developing and disseminating high-yielding and multi-stress tolerant varieties and targeting areas based on suitability and relative profitability.

Area and Yield Performance at Disaggregated Level

- 3.24 Yield performance of *rabi* crops at district level for two time periods, i.e., TE2018-19 and TE2023-24 has been presented in this section. All-India average yield and median of area shares of districts in all-India area of the respective crops have been taken as benchmark for categorization of districts. Districts contributing 90 percent of all-India production were considered for the analysis and categorized into following four categories.

HH: Area share $\geq$ median area share, and average yield $\geq$ all-India yield

HL: Area share $\geq$ median area share, and average yield $<$ all-India yield

LH: Area share $<$ median area share, and average yield $\geq$ all-India yield

LL: Area share $<$ median area share, and average yield $<$ all-India yield

- 3.25 Table 3.6 demonstrates the categorization of wheat and barley producing districts based on the area share and yield during TE2018-19 and TE2023-24. In case of wheat, majority of the districts were under HH category during both periods, however, area share of the districts under this category reduced from 43.8 percent to 36.6 percent between the periods. On the other hand, number of districts (area share) under the HL category has increased from 24 (13.3%) in TE2018-19 to 36 (22.6%) in TE2023-24, indicating deceleration in yield during the TE2023-24. The area and yield quadrant-based potential districts for wheat cultivation are presented in Map 3.1. The LH category districts are mostly in Eastern Rajasthan and adjoining districts of Madhya Pradesh and Uttar Pradesh, part of Haryana and Western Uttar Pradesh, and a few districts of South-West Gujarat. The low potential (LL category) districts are mostly from Bihar and Eastern Uttar Pradesh.



- 3.26 In case of barley, the number of districts contributing 90 percent of the country's total production reduced from 83 in TE2018-19 to 65 in TE2023-24. The number of districts under HH category and their area share reduced in TE2023-24 over TE2018-19 from 26 to 12 and from 51.8 percent to 39.8 percent, respectively. Both number of districts and area share under HL and LL categories increased during current period as compared to previous period but increase was notable in HL category. These trends reflect a decline in yield in the current period compared to the previous period. Distribution of districts based on area share and productivity of barley in TE2023-24 is shown in Annex Map 3.1.

Table 3.6: Categorization of Districts based on Area Share and Yield of Rabi Cereals

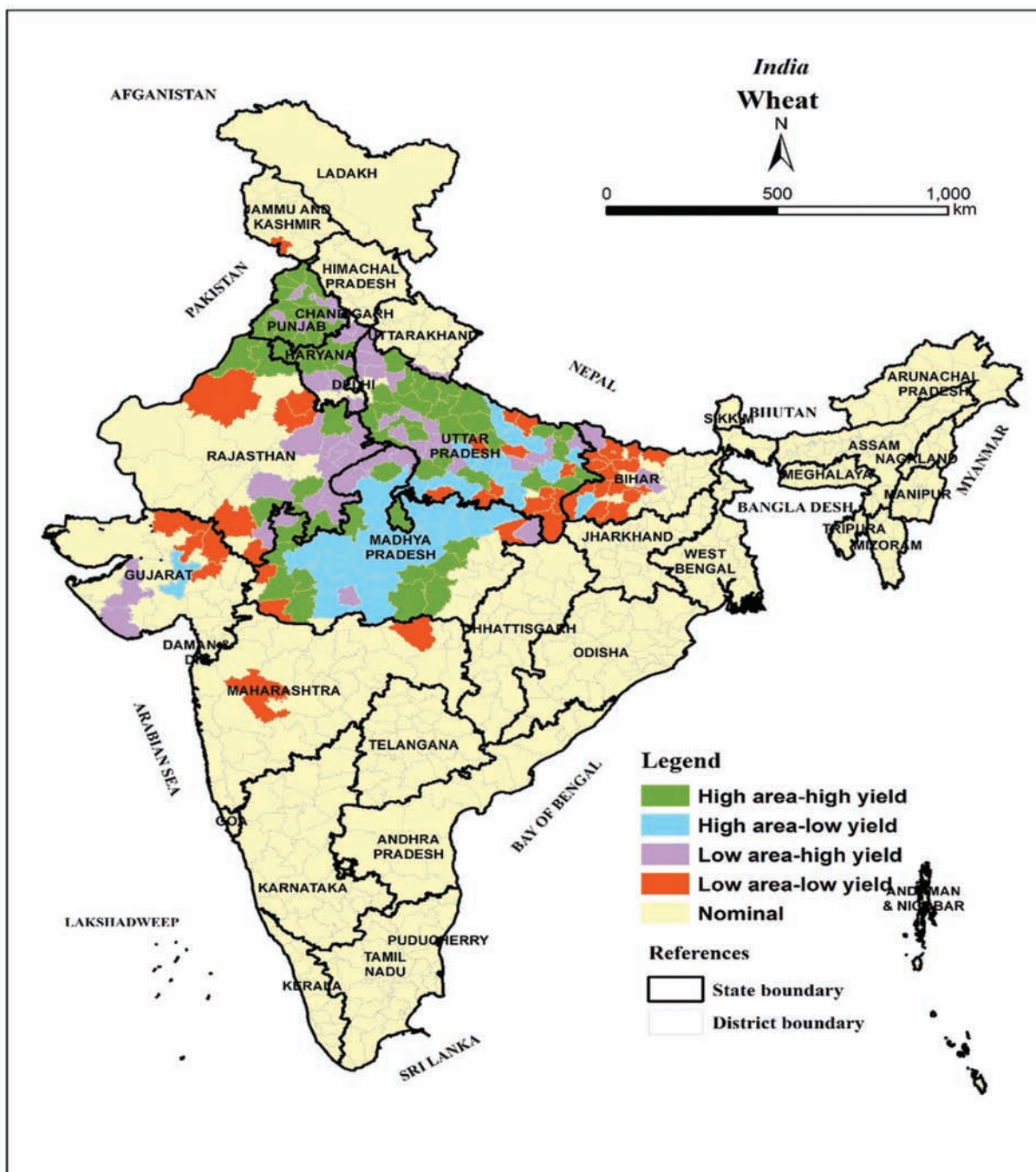
Wheat							
TE2018-19		Yield (33.6 qtl/ha)		TE2023-24		Yield (35.4 qtl/ha)	
		High	Low			High	Low
Area	High	74 (43.8)	24 (13.3)	Area	High	66 (36.6)	36 (22.6)
Share (0.4)	Low	63 (17.6)	35 (9.7)	Share (0.4)	Low	58 (15.1)	43 (11.5)
Barley							
TE2018-19		Yield (27.3 qtl/ha)		TE2023-24		Yield (30.5 qtl/ha)	
		High	Low			High	Low
Area	High	26 (51.8)	16 (17.3)	Area	High	12 (39.8)	21 (30.0)
Share (0.7)	Low	24 (8.3)	17 (8.8)	Share (0.9)	Low	13 (6.0)	19 (9.9)

Note: 1. Districts contributing 90% production of respective crops considered for the analysis
 2. Number in parenthesis shows the area share of the districts
 3. All-India Yield and Median of Area Shares of districts in all-India area used as benchmark for categorization

Source: Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare



Map 3.1: Classification of Major Wheat Producing Districts based on Area Share and Yield



Note:

1. Districts contributing 90% production of respective crops considered for the analysis
2. All-India Yield and Median of Area Shares of districts in all-India area used as benchmark for categorization
3. Count of districts under different categories may slightly deviate from Table 3.6 due to formation of some new districts in recent years

Source:

1. Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare
2. Map drawn by ICAR- National Bureau of Soil Survey and Land Use Planning, Nagpur



- 3.27 Classification of districts based on the area share and yield of gram and lentil during TE2018-19 and TE2023-24 has been exhibited in Table 3.7. In case of gram, maximum number of districts were under LH category and minimum were under LL category, while the highest area share was under HL category and lowest in LL category. No significant shift in number of districts under different categories was noticed between two periods. Map 3.2 shows the distribution of districts based on area share and productivity of gram in TE2023-24. Madhya Pradesh had the highest number of LH category districts, followed by Rajasthan (Southern and South-Eastern part of the State) indicating scope for increasing area under gram cultivation, while Maharashtra, followed by Rajasthan (Western and North-Western part of the State) had the highest number of HL category districts, where efforts are needed to improve gram productivity.
- 3.28 Almost a similar pattern was observed in lentil. However, geographical distribution was significantly different compared to gram (Annex Map 3.2). The 'high area-high yield' lentil producing districts were mainly concentrated in Madhya Pradesh and HL category districts were mostly in Uttar Pradesh. The LH category districts were spread across all lentil producing States.

Table 3.7: Categorization of Districts based on Area Share and Yield of Rabi Pulses

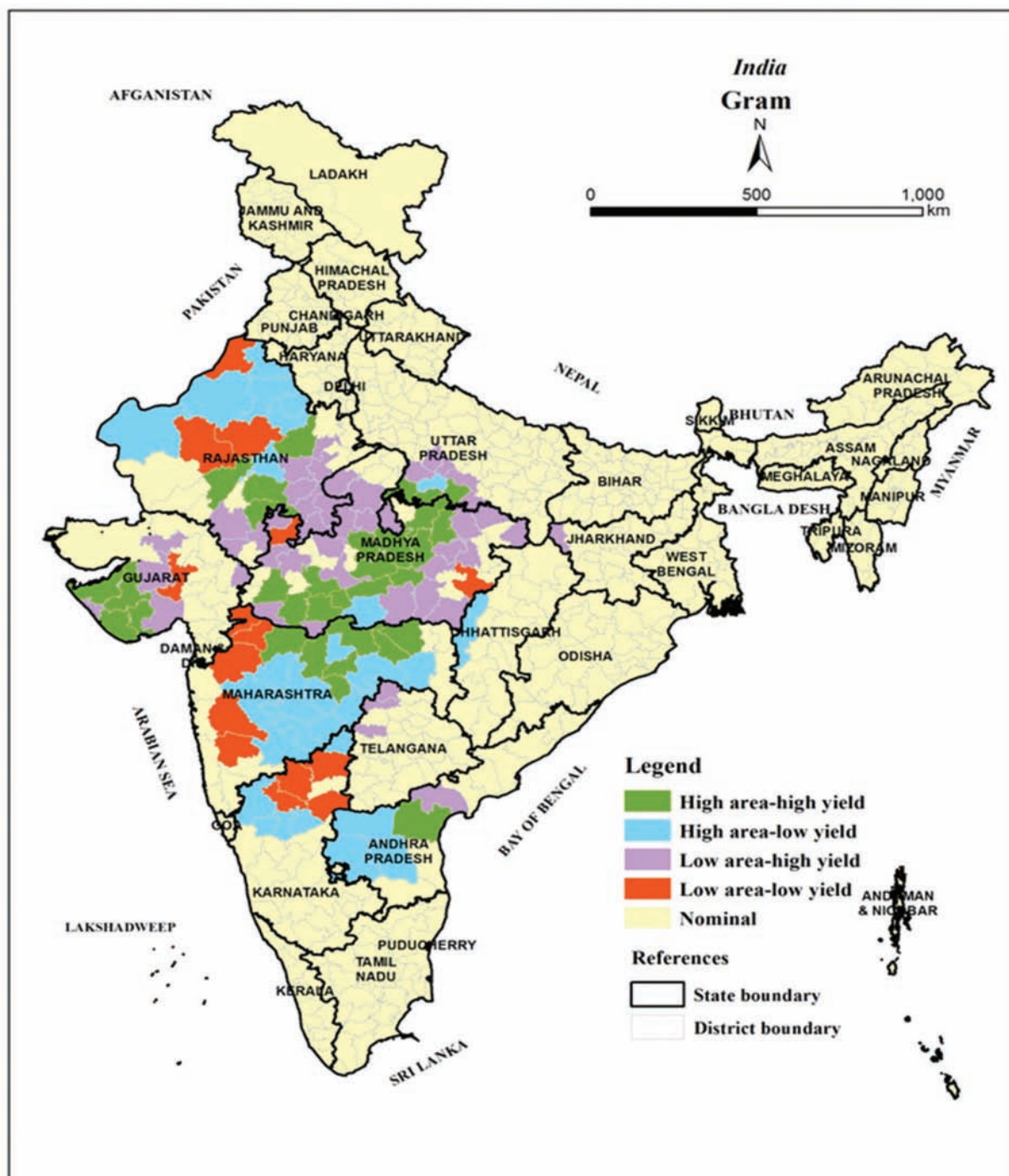
Gram							
TE2018-19		Yield (10.3 qtl/ha)		TE2023-24		Yield (12.0 qtl/ha)	
		High	Low			High	Low
Area	High	31 (33.8)	31 (35)	Area	High	33 (30.1)	33 (37.6)
Share (0.6)	Low	51 (16.5)	11 (4.9)	Share (0.5)	Low	50 (15.5)	16 (6.3)
Lentil							
TE2018-19		Yield (9.3 qtl/ha)		TE2023-24		Yield (9.6 qtl/ha)	
		High	Low			High	Low
Area	High	25 (43.4)	21 (27.3)	Area	High	24 (46.0)	18 (27.2)
Share (0.7)	Low	32 (11.4)	14 (5.4)	Share (0.6)	Low	29 (8.7)	13 (5.3)

Note: 1. Districts contributing 90% production of respective crops considered for the analysis
 2. Number in parenthesis shows the area share of the districts
 3. All-India Yield and Median of Area Shares of districts in all-India area used as benchmark for categorization

Source: Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare



**Map 3.2: Classification of Major Gram Producing Districts
based on Area Share and Yield**



- Note:**
1. Districts contributing 90% production of respective crops considered for the analysis
 2. All-India Yield and Median of Area Shares of districts in all-India area used as benchmark for categorization
 3. Count of districts under different categories may slightly deviate from Table 3.7 due to formation of some new districts in recent years

- Source:**
1. Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare
 2. Map drawn by ICAR- National Bureau of Soil Survey and Land Use Planning, Nagpur



- 3.29 Table 3.8 shows categorization of rapeseed & mustard and safflower producing districts in terms of area share and yield during TE2018-19 and TE2023-24. The number of districts contributing 90 percent of country's total R&M production has increased from 107 in TE2018-19 to 115 in TE2023-24. During TE2018-19, 31 districts were in HH category with the largest area share of 48.1 percent, while equal number of districts were under the least efficient category (LL), but their area share was only 8.3 percent. In TE2023-24, the number of districts under the HH category increased from 31 to 37, but their area share remained almost the same, while both number and area share of districts under the LL category increased. Mapping of districts based on area and yield level during TE2023-24 showed that Rajasthan had the highest number of districts under HH category, while Uttar Pradesh had the highest number in LL category (Map 3.3). More than half of HL category districts were from Rajasthan and close to half of the LH category districts were in Uttar Pradesh.
- 3.30 Safflower is grown in limited number of districts and 33 districts contributed 90 percent of country's total production during TE2018-19. The production further declined during TE2023-24 and only 19 districts contributed 90 percent of total production. Decrease in number of districts was sharpest in "high area-low yield" category, from 14 in TE2018-19 to four in TE2023-24. This indicates an urgent need for increasing yield to revive area under safflower. The spatial distribution of districts under various categories is depicted in Annex Map 3.3.

Table 3.8: Categorization of Districts based on Area Share and Yield of Rabi Oilseeds

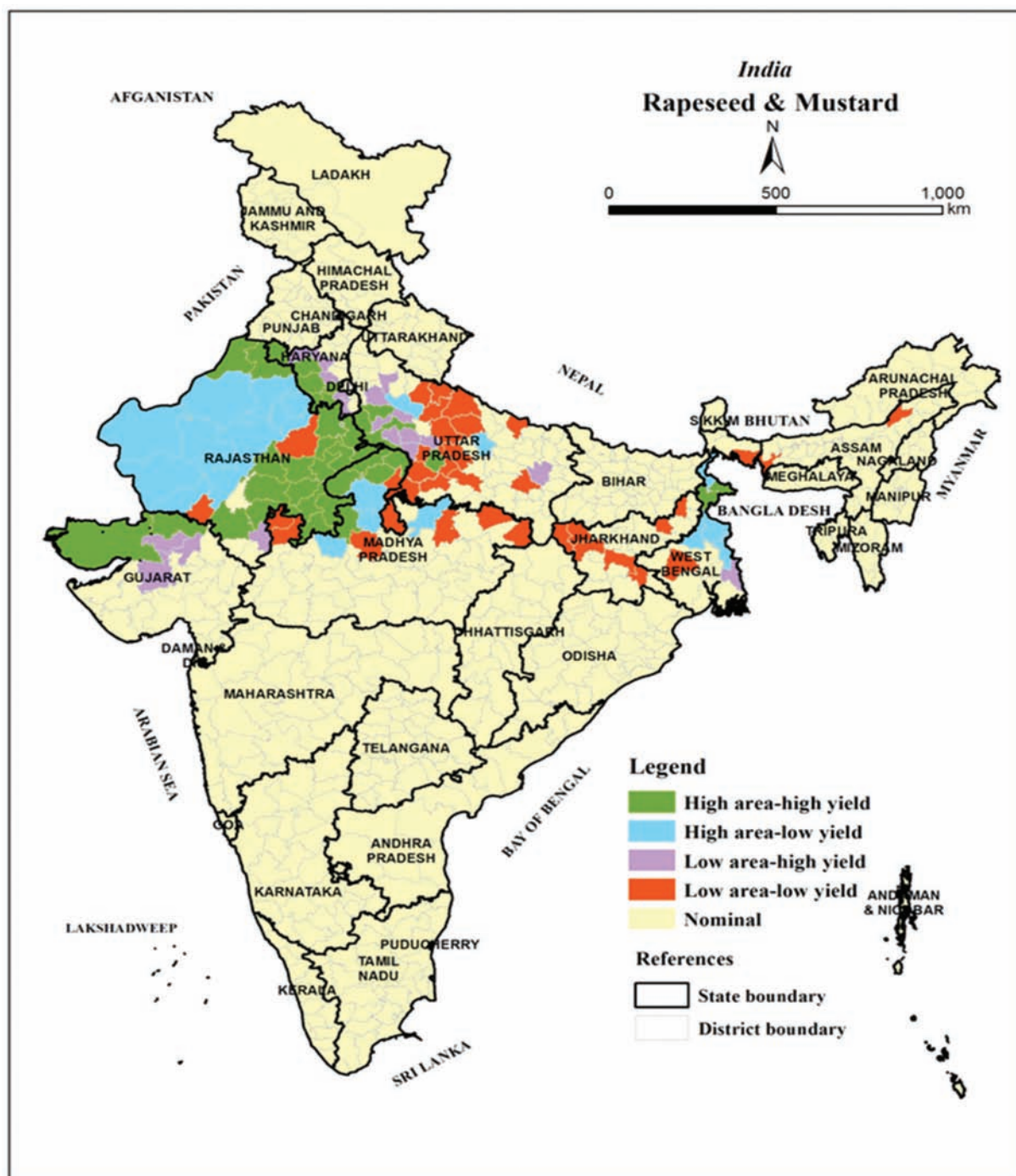
Rapeseed & Mustard							
TE2018-19		Yield (14.1 qtl/ha)		TE2023-24		Yield (14.5 qtl/ha)	
		High	Low			High	Low
Area	High	31 (48.1)	23 (21.6)	Area	High	37 (48.0)	21 (23.6)
Share (0.4)	Low	22 (4.8)	31 (8.3)	Share (0.4)	Low	19 (3.9)	38 (9.4)
Safflower							
TE2018-19		Yield (6.4 qtl/ha)		TE2023-24		Yield (8.0 qtl/ha)	
		High	Low			High	Low
Area	High	3 (15.6)	14 (56.3)	Area	High	4 (29.2)	6 (46.7)
Share (1.5)	Low	11 (8.3)	5 (5.1)	Share (2.3)	Low	6 (8.1)	3 (5.1)

Note: 1. Districts contributing 90% production of respective crops have been considered for the analysis
 2. The number in parenthesis shows the area share of the districts
 3. All-India Yield and Median of Area Shares of districts in all-India area have been used as benchmark for categorization

Source: Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare



Map 3.3: Classification of Major Rapeseed & Mustard Producing Districts based on Area Share and Yield



- Note:**
1. Districts contributing 90% production of respective crops considered for the analysis
 2. All-India Yield and Median of Area Shares of districts in all-India area used as benchmark for categorization
 3. Count of districts under different categories may slightly deviate from Table 3.8 due to formation of some new districts in recent years

- Source:**
1. Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare
 2. Map drawn by ICAR- National Bureau of Soil Survey and Land Use Planning, Nagpur



- 3.31 The classification of districts based on area and yield showed that there is ample scope for increasing production of *rabi* crops by strategically focusing on potential districts. In ‘high area-low yield’ districts, the focus should be on improving productivity by introducing improved varieties and promoting good agricultural practices (GAP). The LH category districts should be targeted for area expansion, while ‘low area-low yield (LL)’ category districts should be targeted for shifting cropping patterns towards alternative crops based on suitability, natural resource endowment, and relative profitability.
- 3.32 The ‘*Pradhan Mantri Dhan-Dhanya Krishi Yojana*’, announced in the recent budget, aims to enhance productivity and promote crop diversification and sustainable agricultural practices in targeted 100 districts with low productivity, moderate cropping intensity, and below-average credit penetration. While the scheme is expected to increase contribution of these 100 districts to national agricultural production, crop-specific identification of districts and targeted approach will further strengthen the efforts in increasing production of *rabi* crops.

Yield Gap Analysis

- 3.33 This section presents yield gap analysis in major producing States in the country. For this analysis, two types of yield gaps namely, Yield Gap (A) and Yield Gap (B) have been considered. Yield Gap (A) is computed by taking difference between potential farm yield (yield achieved under Front Line Demonstration (FLD) where best scientific and management practices are followed) and realized farm yield under farmers’ practices. While, Yield Gap (B) compares State average yield with potential yield achieved under FLD.
- 3.34 The potential, realized and State average yield of mandated *rabi* crops at all-India level have been displayed in Table 3.9. Significant differences can be seen between potential yield and all-India average yield for most of the mandated crops during TE2023-24, which created significant Yield Gap (B) ranging from 9.9 percent in R&M to 35.7 percent in barley. On the other hand, Yield Gap (A) was recorded in a narrow range from 10.9 percent in wheat to 19.9 percent in R&M. However, large inter-State variations in yield gap were observed (Annex Table 3.2). Among the major producing States, Yield Gap (B) was significantly high in Rajasthan for wheat, barley and gram, in Uttar Pradesh for lentil and R&M, and in Karnataka for safflower.



Table 3.9: Potential, Realized and Average Yield of Mandated *Rabi* Crops at All-India Level (TE2023-24)

Crop	Potential Yield (qtl/ha)	Realized Yield (qtl/ha)	All-India Average (qtl/ha)	Yield Gap A (%)	Yield Gap B (%)
Wheat	49.5	44.1	35.4	10.9	28.5
Barley	47.4	38.8	30.5	18.1	35.7
Gram	18.0	14.7	12.0	18.3	33.3
Lentil	13.1	10.6	9.6	19.1	26.7
R&M	16.1	12.9	14.5	19.9	9.9
Safflower	10.6	8.6	7.9	19.0	25.2

Source: 1. Respective ICAR Institutions

2. Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare

- 3.35 It can be inferred from the above analysis that there is ample scope for increasing production by reducing the yield gaps. Since, Yield Gap (B) arises due to technological and management constraints coupled with socio-economic constraints faced by farmers, effective extension services are required to educate farmers about the latest technologies, promote efficient farm management practices, and ensuring timely availability of quality inputs for enhancing production. In States like Punjab and Haryana, realized crop yields in wheat are close to potential yield and hence technological developments become crucial for sustained growth in wheat production.

Drivers of Yield Growth

- 3.36 Availability and quality of agricultural inputs like land, water, seeds and fertilizers, efficient use of inputs, best agricultural practices, timely availability of agricultural credit, etc. are the key factors influencing growth of agricultural productivity. Moreover, modern infrastructure facilities and supportive Government policies and programmes provide a favourable environment for agricultural development. This section highlights these important factors as well as policy initiatives and interventions by the Government in this regard.

Availability of Quality Seeds

- 3.37 The availability of quality seeds is one of the most crucial factors affecting crop yield. Studies have shown that quality seeds contribute 15-20 percent to total production, depending on crop management and complementary inputs. India has made significant strides in developing high-yielding as well as biotic and abiotic stress-tolerant varieties, and improving the quality traits of grains.
- 3.38 Seed Replacement Rates (SRR) of *rabi* crops in major States are shown in Table 3.10. For wheat, SRR was highest in Haryana (62.8%) but lower than the recommended rate of 33 percent in Madhya Pradesh (24.9%), the second largest producer of wheat. In case of gram, SRR varied from 19.8



percent in Rajasthan to 85 percent in Andhra Pradesh and Telangana, and was lower than recommended in Rajasthan, Madhya Pradesh, Tamil Nadu, and Bihar. SRR in lentil was lowest in Madhya Pradesh, the largest producer, and was highest in West Bengal. As far as R&M is concerned, except West Bengal and Madhya Pradesh, major producing States achieved SRR above 50 percent. States should ensure improvement in SRR in crops which have SRR below the recommended level.

Table 3.10: Seed Replacement Rate of Rabi Crops in Major Producing States (2024-25)

State	Wheat	Gram	Lentil	R&M
Andhra Pradesh	-	85	-	-
Bihar	39.9	29.7	27.8	76.6
Gujarat	51.3	51	-	100
Haryana	62.8	50.7	-	81.7
Karnataka	-	33.5	-	-
Madhya Pradesh	24.9	28.2	20	47.3
Maharashtra	36.8	33	-	-
Punjab	39.3	-	-	-
Rajasthan	35.3	19.8	-	70.2
Tamil Nadu	-	31.1	-	-
Telangana	-	85	-	-
Uttar Pradesh	47.6	34.2	37.6	69.9
West Bengal	-	38	38	49.2

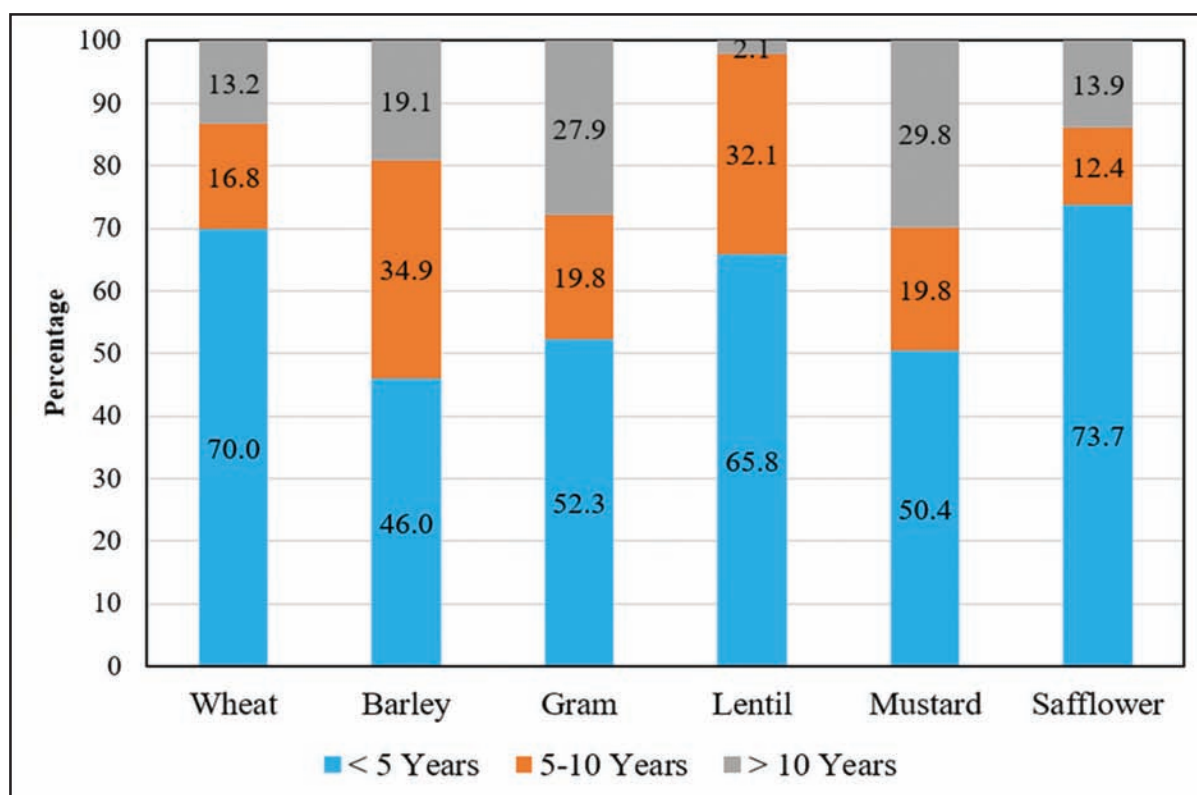
Note: SRR of 33% for wheat, gram and lentil and 50% for rapeseed & mustard are considered ideal

Source: Seed Division, Ministry of Agriculture and Farmers Welfare

- 3.39 During 2024-25, availability of seeds for all mandated crops was significantly higher than requirement (Annex Table 3.3). However, during the Commission's consultation with stakeholders, farmers raised the issue of non-availability of quality and new varieties seeds. The data on breeder seed production also indicates that though overall breeder seed production is more than seed indented by various seed multiplication agencies, but there are demand supply mismatches at variety level, particularly of new varieties.
- 3.40 Chart 3.7 presents the percentage distribution of breeder seed indents according to age of varieties in TE2024-25. The share of newer varieties (<5 years old) was encouraging in wheat (70%), lentil (65.8 %), and safflower (73.7%) but was low in barley (46%), mustard (50.4%), and gram (52.3%). Therefore, there is significant scope for improving agricultural extension, strengthening seed supply chains and effective seed production planning for timely delivery of quality seeds.



Chart 3.7: Share of Different Varieties (based on Age) in Total Breeder Seeds Indent



Source: ICAR-Indian Institute of Seed Research, Mau

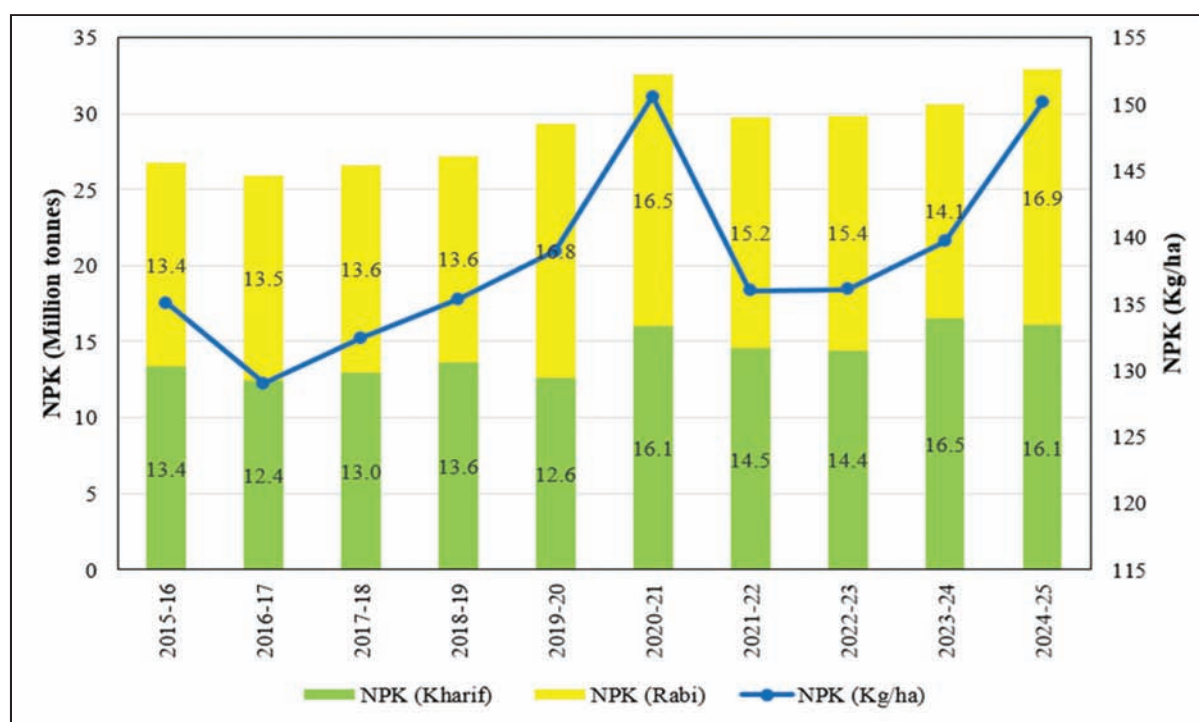
- 3.41 Investment in Research and Development (R&D) is crucial for developing high yielding and climate-resilient crop varieties. In Budget 2025-26, the Government has announced launching of National Mission on High Yielding Seeds aimed at strengthening the research ecosystem, targeted development and propagation of seeds with high yield, pest resistance and climate resilience. The mission also aims to ensure commercial availability of seeds of more than 100 varieties released since July 2024.
- 3.42 For ensuring the timely and quality seed availability for the farmers, the Government has launched SATHI (Seed Traceability, Authentication and Holistic Inventory) portal and Mobile Application, which provide a centralized online system for tracking seeds from production to distribution, ensuring authenticity, and preventing spurious seeds from entering the market. The movement of seed bags in the supply chain is recorded in real-time through QR-coded seed bags to provide end-to-end traceability and seed inventory from retailer to State and National level. As on 16th June 2025, more than 3 lakh seed growers, 15.6 thousand seed producing agencies, 6 thousand seed processing plants, and more than 57 thousand licensed dealers have been registered on this portal.



Fertilizer Use Efficiency and Soil Health

- 3.43 Fertilizers play a vital role in enhancing crop yields by providing essential nutrients for plant growth. Various initiatives have been taken by the Government to increase domestic production and ensure timely availability to farmers at affordable prices.
- 3.44 Total consumption and per hectare consumption of fertilizer nutrients during last 10 years have been exhibited in Chart 3.8. Total fertilizer consumption has increased from 26.8 million tonnes in 2015-16 to 33 million tonnes in 2024-25 at a CAGR of 2.4 percent. However, per hectare consumption grew at a much lower CAGR (1.1%) during the last 10 years. Growth rate in *kharif* consumption was higher (2.9%) than *rabi* season consumption (2%). During 2024-25, *rabi* consumption and per hectare consumption recorded a significant increase, while *kharif* consumption marginally declined.

Chart 3.8: Total and Per Hectare Consumption of Fertilizer Nutrient (NPK) in India

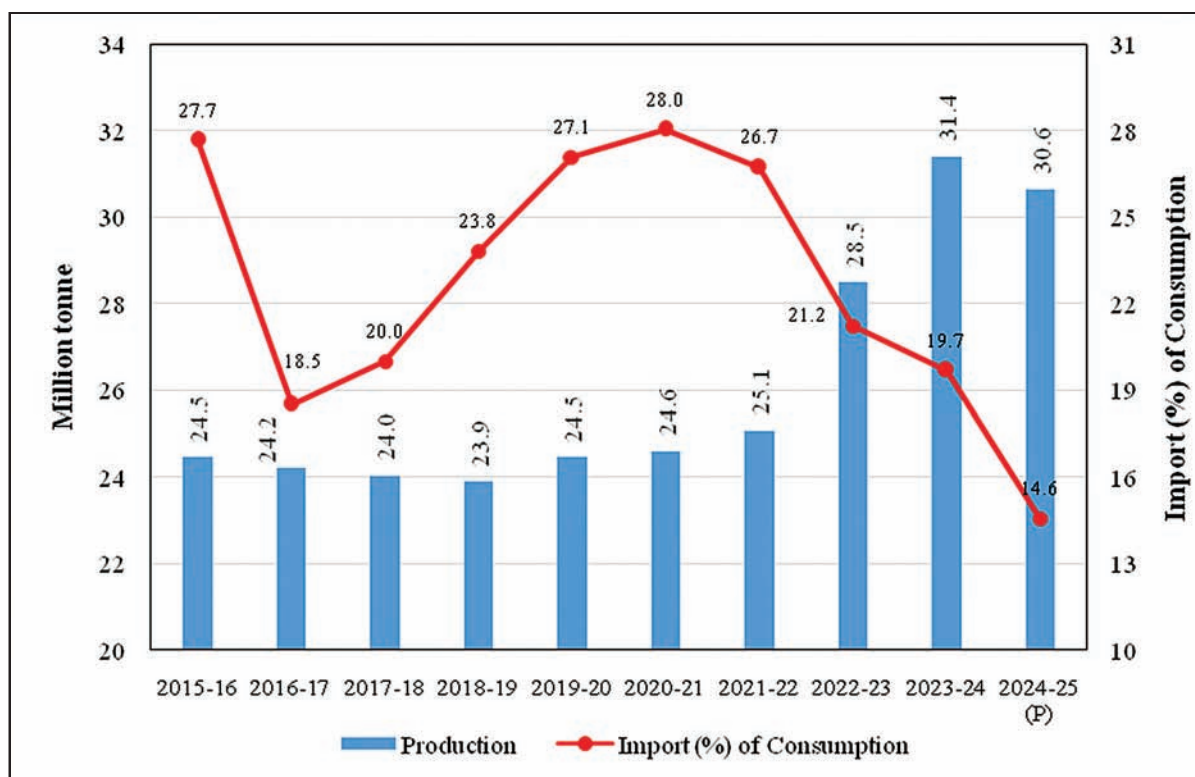


Source: Fertilizer Association of India

- 3.45 New Investment Policy (NIP) was announced in 2013 (amended in 2014) by Government to facilitate fresh investment in urea sector and make India self-reliant in the urea production. Six new Urea units, four by Joint Venture Companies (JVCs) of designated PSUs and two by private companies, have been set up under the policy. As is evident from Chart 3.9 that the dependence on urea imports has reduced significantly after 2020-21 as domestic production has recorded an increase of 6.8 million tonnes from 2020-21 to 2023-24. However, a marginal decline in urea production was witnessed in 2024-25.



Chart 3.9: Total Production and Share of Import in Total Consumption of Urea



Source: Fertilizer Association of India

- 3.46 For efficient and balanced use of fertilizers, improve soil health, protect environment, promote organic fertilizers, and prevent groundwater pollution, Government has launched various programmes and schemes such as Soil Health Card scheme, PM Programme for Revival, Awareness, Nutrition and Improvement of Mother Earth (PM-PRANAM), Galvanizing Organic Bio-Agro Resources Dhan (GOBARDhan), etc. Apart from this, Government is promoting liquid fertilizers, slow and controlled release fertilizers and organic fertilizers towards achieving this objective.
- 3.47 Despite these initiatives, fertilizer use is far from optimal level adversely affecting soil health and crop productivity. There are significant spatial variations and imbalances in nutrients use. In some States and districts, per hectare use is higher than required quantity, while in others, the use is very low. The nutrient consumption is highly tilted in favour of nitrogen, which is clearly visible from the nutrient use pattern shown in Table 3.11. Excessive nitrogen leads to groundwater pollution. Further, imbalanced and inefficient fertilizer use adversely affects the crop response to fertilizer application. The fertilizers response, measured in terms of foodgrains production per kilogram of NPK use, has reduced from 12 kilogram in 1960-69 to only 3 kilogram in 2020.

**Table 3.11: NPK Use Ratio during Last 10 Years**

Year	N : P : K
2015-16	7.2 : 2.9 : 1
2016-17	6.7 : 2.7 : 1
2017-18	6.1 : 2.5 : 1
2018-19	6.6 : 2.6 : 1
2019-20	7.3 : 2.9 : 1
2020-21	6.5 : 2.8 : 1
2021-22	7.7 : 3.1 : 1
2022-23	11.8 : 4.6 : 1
2023-24	10.9 : 4.4 : 1
2024-25*	9.3 : 3.5 : 1

Note: * Provisional

Source: Fertilizer Association of India

- 3.48 About 50 percent of total consumption is concentrated in 138 districts and per hectare average consumption of these districts is 27 percent higher than the all-India average. Table 3.12 presents the percentage of districts under different ranges of fertilizer nutrients (NPK) consumption during TE2023-24. Out of total 636 districts, consumption of fertilizers was more than 300 kilogram per hectare in 36 districts (5.7%) and in more than one-fourth of districts, the consumption was above 200 kilogram per hectare. Contrary to this, per hectare consumption was less than 50 kilogram in around one-fifth of the districts. A notable high percentage of districts with consumption above 300 kilogram per hectare is located in Andhra Pradesh (34.6%), Bihar (21.1%), Telangana (12.1%), and Tamil Nadu (10.8%). Per hectare consumption is alarmingly high in West Godavari (996 kg/ha), Vellore (662 kg/ha), East Godavari (598 kg/ha), and Nellore (561 kg/ha). Therefore, promoting judicious and balanced fertilizer use according to soil test-based recommendations will not only reduce the cost of cultivation but also result in efficient use and maintenance of soil fertility.



Table 3.12: Percentage of Districts under Different Ranges of Per Hectare NPK Consumption

States	>300 kg	200-300 kg	100-200 kg	50-100 kg	<50 kg
Andhra Pradesh	34.6	11.5	3.8	19.2	30.8
Assam	3.3	0.0	16.7	30.0	50.0
Bihar	21.1	39.5	39.5	0.0	0.0
Chhattisgarh	0.0	20.0	31.4	8.6	40.0
Gujarat	3.0	9.1	69.7	12.1	6.1
Haryana	9.1	54.5	27.3	4.5	4.5
Himachal Pradesh	0.0	0.0	33.3	16.7	50.0
Jammu & Kashmir	4.5	4.5	18.2	36.4	36.4
Jharkhand	0.0	12.5	29.2	20.8	37.5
Karnataka	0.0	22.6	48.4	22.6	6.5
Kerala	0.0	0.0	0.0	57.1	42.9
Madhya Pradesh	0.0	5.8	48.1	32.7	13.5
Maharashtra	0.0	0.0	52.9	38.2	8.8
Odisha	0.0	3.3	13.3	43.3	40.0
Punjab	9.1	77.3	13.6	0.0	0.0
Rajasthan	0.0	0.0	42.4	21.2	36.4
Tamil Nadu	10.8	18.9	48.6	8.1	13.5
Telangana	12.1	36.4	42.4	0.0	9.1
Uttar Pradesh	2.7	44.0	49.3	4.0	0.0
Uttarakhand	7.7	7.7	7.7	7.7	69.2
West Bengal	5.0	15.0	50.0	15.0	15.0
All-India	5.7	20.1	36.9	17.6	19.7

Source: Fertilizer Association of India

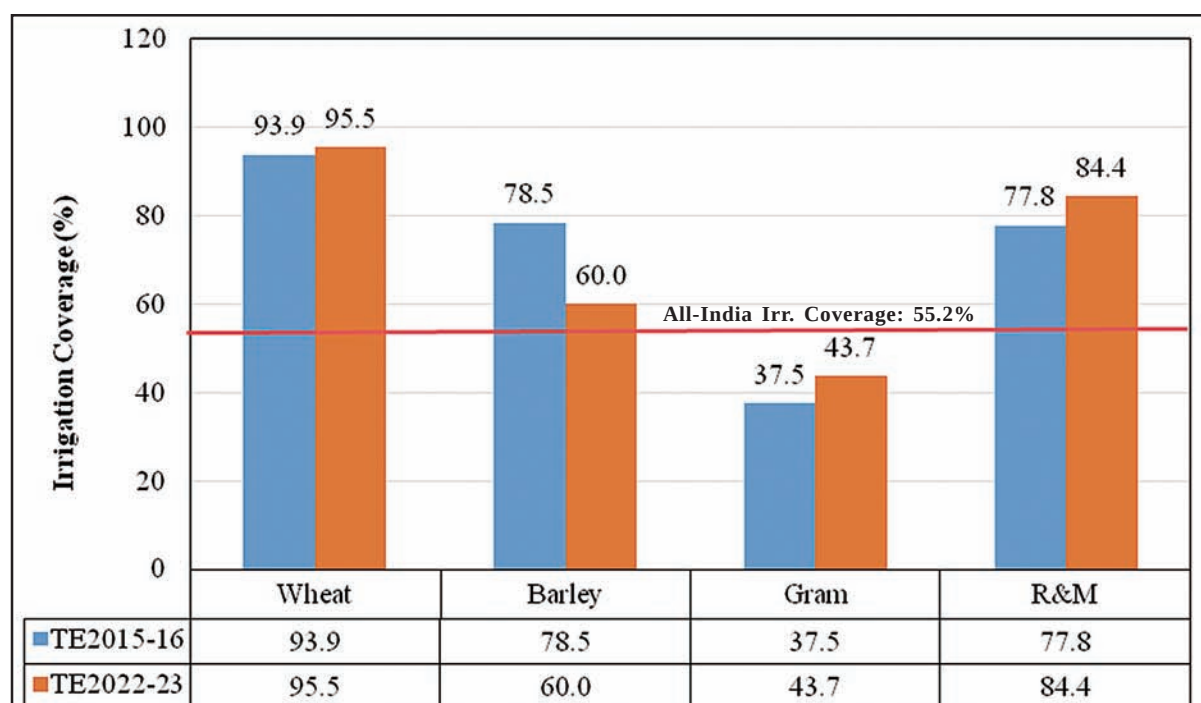
- 3.49 Keeping in view of deterioration in soil health, it is imperative to promote Integrated Nutrient Management (INM) approach through integration of chemical fertilizers, organic manure, bio-fertilizers, etc. to achieve optimum nutrient balance and thus ensure environmental sustainability.



Irrigation Development and Water Use Efficiency

- 3.50 Assured irrigation facilitates coping with challenges posed by climate change, and can ensure a more stable and secure food supply, particularly during adverse weather conditions. Significant progress has been made in irrigation coverage during the last few years. The irrigation intensity measured as gross irrigated area as a percentage of gross cropped area in the country has increased from 48.8 percent in TE2015-16 to 55.2 percent in TE2022-23. However, there are large disparities among States, with irrigation intensity above 95 percent in Haryana and Punjab, but less than 25 percent in the North-Eastern region (except Meghalaya), Jharkhand, Kerala, Maharashtra and Himachal Pradesh. During last ten years, there has been impressive increase in irrigation coverage in Telangana (22.2% points), Gujarat (16.4% points), Madhya Pradesh (13.4% points), Karnataka (8.5% points), and Bihar (8.2% points), but irrigation percentage has remained largely stagnant in North Eastern region, except Assam.
- 3.51 There were also considerable variations in irrigation coverage among crops. The crop-wise irrigation coverage depicted in Chart 3.10 shows that irrigation coverage under *rabi* crops, except barley, has increased during the last decade. However, more than half of the area under gram and 40 percent area under barley is still rainfed.

Chart 3.10: All-India Irrigation Coverage of Selected *Rabi* Crops during TE2015-16 and TE2022-23



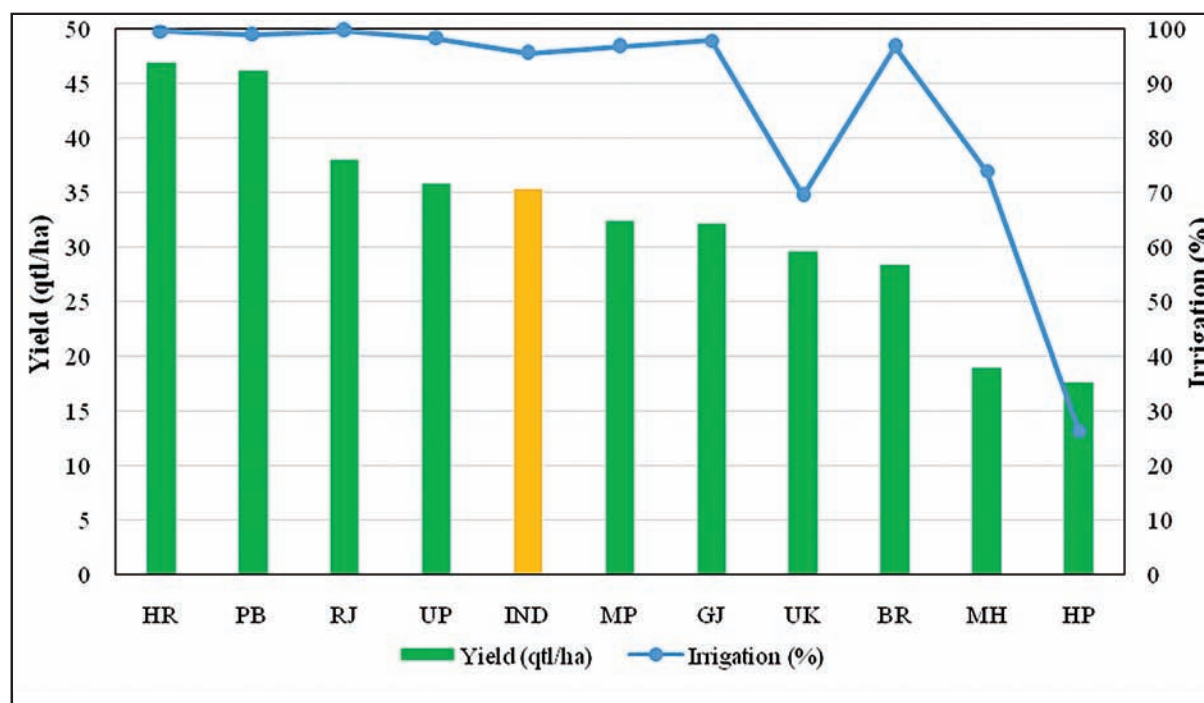
Source: Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare

- 3.52 Chart 3.11 shows irrigation coverage and yield level of wheat in selected States. Irrigation coverage in wheat was above 95 percent in Haryana, Punjab, Rajasthan, Uttar Pradesh, Madhya Pradesh, Gujarat and Bihar, relatively low in Maharashtra (73.9%), Uttarakhand (69.4%) and lowest in Himachal Pradesh (26.4%). Bihar, Madhya Pradesh and Gujarat, although have nearly 97-98 percent wheat area under



irrigation, recorded much lower yield than other major wheat producing States and all-India average level. This trend underscores the importance of other socio-economic, institutional and policy factors that contribute to yield improvement, in addition to irrigation.

Chart 3.11: Irrigation Coverage and Yield level of Wheat in Selected States



Note: 1. States having production share $\geq 1\%$ under respective crops considered
2. Irrigation and Yield (TE2022-23)

Source: Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare

- 3.53 Along with increasing irrigation coverage, efficiency in irrigation is equally important. Efficient irrigation not only helps increase crop yields and optimise water use, but also reduces pressure on depleting ground water resources, which is vital for agriculture. Efficient irrigation techniques such as drip and sprinkler irrigation reduce water wastage through precise application of water, and contribute to environmental sustainability.
- 3.54 With an objective to improve water use efficiency and expand irrigation coverage, 'Pradhan Mantri Krishi Sinchayee Yojana' (PMKSY) was launched in 2015. The scheme has four major components, Accelerated Irrigation Benefit Programme (AIBP) for faster completion of major and medium irrigation projects, 'Har Khet Ko Pani' (HKKP) for expanding irrigation coverage through micro irrigation projects, Watershed Development Component (WDC) for water harvesting and Per Drop More Crop (PDMC) for improving water use efficiency. From 2022-23, PDMC component of the scheme has been transferred to 'Rashtriya Krishi Vikas Yojana' (RKVY) to provide greater flexibility and autonomy to States. The Government of India, in collaboration with the National Bank for Agriculture and Rural Development (NABARD) has also created Micro Irrigation Fund (MIF) with the objective of facilitating States in mobilizing resources to expand coverage of micro irrigation. Due to these



initiatives, area under micro irrigation has almost doubled from 8.6 million hectares in December 2016 to 16.7 million hectares in March 2024, which has also contributed to increasing the overall irrigation coverage in the country. However, only five States, namely Andhra Pradesh, Gujarat, Karnataka, Maharashtra and Rajasthan, account for 70 percent of total area under micro-irrigation in the country, and coverage is still low in some of the groundwater stressed States like Punjab and Haryana.

- 3.55 It is pertinent to note that ‘*Pradhan Mantri Kisan Urja Suraksha Evam Utthaan Mahabhiyan*’ (PM KUSUM) Scheme was launched in 2019 with an aim to provide energy and water security to farmers, de-dieselisation of farm sector, increase farmers’ income, and promote use of renewable energy in agriculture. The Scheme has three components: (1) *Component A* for setting up 10,000 MW of decentralized grid connected renewable energy power plants by the farmers on their land, (2) *Component B* for installation of 14 lakh stand-alone solar agricultural pumps/replacement of diesel pumps by solar pumps in off-grid areas, and (3) *Component C* for solarisation of 35 lakh grid-connected agriculture pumps to provide reliable day-time solar power to farmers and incentivizing them for saving water by purchasing surplus solar power. The scheme is operational till 31st March 2026. However, the achievements so far under various components of the scheme are significantly lower than the targets. Therefore, there is a need to create awareness through various media to promote solarisation of irrigation pumps.

Access to Agricultural Credit

- 3.56 Access to institutional credit is extremely important for farmers as it meets their short-term agricultural needs as well as long-term investment requirements for improving crop productivity and profitability.
- 3.57 Government has taken various initiatives such as *Kisan Credit Card (KCC)*, *Modified Interest Subvention Scheme (MISS)*, collateral-free loans, loan against *Negotiable Warehouse Receipts (NWR)*/ *Electronic Negotiable Warehouse Receipts (e-NWR)*, etc. to improve access to farm credit among farmers and minimize distress sale. These initiatives have resulted in significant increase in agricultural credit during last few years. The total agricultural credit has increased from ₹15.8 lakh crore in 2020-21 to ₹28.7 lakh crore in 2024-25 (Chart 3.12) and the share of term loan has increased from 43.3 percent to 44.5 percent during the period. The share of small and marginal farmers in ground level credit to agriculture has also increased from 41 percent in 2014-15 to 56.5 percent in 2023-24, higher than their share in total area operated (47%) as per Agriculture Census 2015-16 but lower share of lending to small and marginal farmers by commercial banks remains an issue.



Chart 3.12: Institutional Credit Disbursement to Agriculture

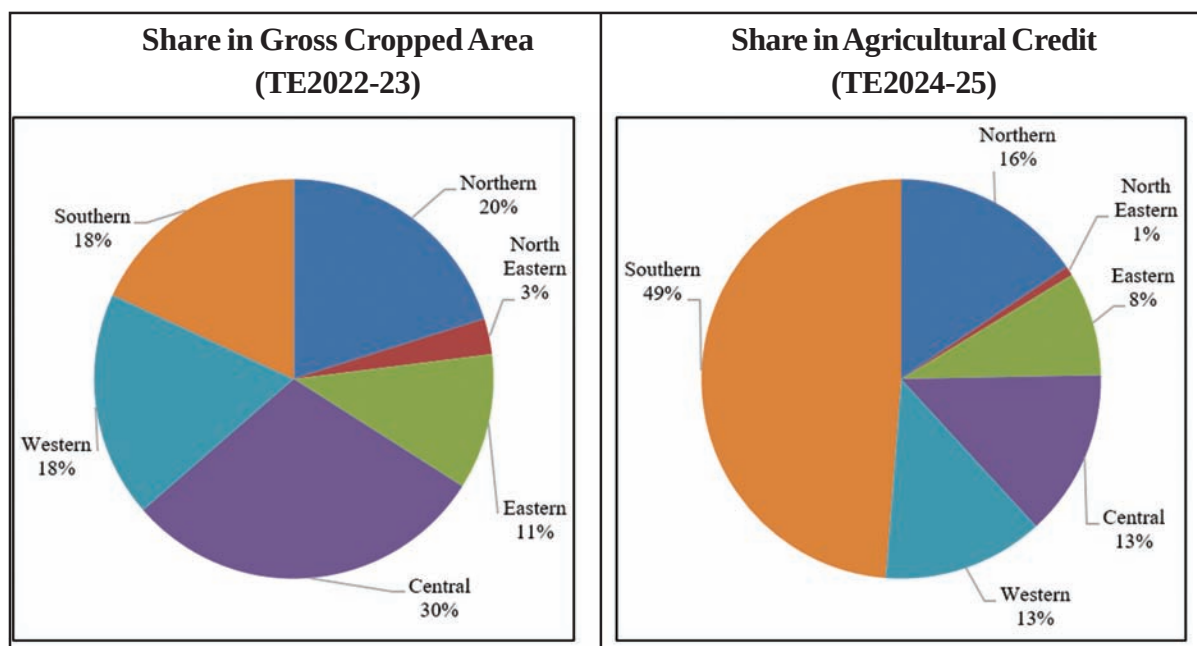


Source: National Bank for Agriculture and Rural Development

- 3.58 The 'Modified Interest Subvention Scheme' is an important initiative, under which Government provides concessional short-term loans to farmers at a 7 percent interest rate up to ₹3 lakh, with an additional 3 percent subvention for timely repayment, thus reducing the effective rate to 4 percent. In the Union Budget 2025-26, the loan limit under MISS has been increased from ₹3 lakh to ₹5 lakh. The interest subsidy under MISS has almost doubled from ₹11.5 thousand crore in 2018-19 to ₹22.6 thousand crore in 2024-25 (RE). In order to facilitate hassle-free access to credit for small and marginal farmer, loans up to ₹2 lakh have been made collateral free. On 16th December 2024, Government launched 'Credit Guarantee Scheme for e-NWR based Pledge Financing' (CGS-NPF) with a corpus of ₹1,000 crore for extending post-harvest financing to farmers and minimise distress sale.
- 3.59 Although institutional credit in agriculture has significantly increased during last 10 years, still wide disparities have been observed across States and regions. Share of different regions in total institutional credit disbursement vis-à-vis in total cropped area depicted in Chart 3.13 shows that almost half (49%) of the total agricultural credit goes to Southern region, which has 18 percent share in country's gross cropped area (GCA). On the contrary, share in credit is much lower than the share in GCA for other regions, particularly in Central region. The differences are significant within the regions. For example, the agricultural credit intensity per hectare of gross cropped area in Tamil Nadu is 4.5 times higher than Telangana (Annex Table 3.4).



Chart 3.13: Region-wise Distribution of Agricultural Credit



Source: 1. National Bank for Agriculture and Rural Development for agricultural credit
2. Department of Agriculture and Farmers Welfare for gross cropped area

Farm Mechanization

- 3.60 Human labour cost accounts for a significant share of the production cost due to constantly rising wage rates. Therefore, there is a need to increase level of mechanisation to improve the efficiency of agricultural operations and increase profitability. Various studies show that agricultural mechanisation can reduce costs by saving inputs, improving operational efficiency, and increasing cropping intensity.
- 3.61 Efforts are being made to increase level of mechanisation through implementation of 'Sub-Mission on Agricultural Mechanisation' (SMAM) as a component of RKVY. The Scheme aims at increasing reach of agricultural mechanization among small and marginal farmers by promoting custom hiring centres and setting up agricultural machinery banks to make equipments accessible and affordable. Under SMAM, financial assistance is provided to farmers and rural entrepreneurs for purchase of various types of agricultural implements and machineries. The Crop Residue Management (CRM) component of RKVY aims to support efforts of States of Punjab, Haryana, Uttar Pradesh, Madhya Pradesh, and NCT of Delhi for *in-situ* as well as *ex-situ* management of crop residue in the region. Progress of SMAM and CRM from 2014-15 to 2024-25 is presented in Annex Table 3.5.
- 3.62 For improving efficiency in use of inputs, safe use of chemical pesticides and timely operations, Government is promoting drones by providing subsidies to institutes for demonstration, and to Farmers Producers Organizations and individual farmers for application on the field. Further, 'Drone Didi' initiative aims to provide 15,000 drones to women-led self-help groups, thereby improving agricultural efficiency, empowering women and promoting the use of technology in agriculture.



- 3.63 The sustained efforts of the Government have increased the level of mechanisation in the country; however, it is still much lower than in other developed countries. There are wide inter-crop and inter-State disparities in the level of farm mechanization. Hence, there is a need to focus on developing suitable agricultural machinery and implements, especially for small farms and hilly region and post-harvest management. Machinery related to post-harvest and primary processing, such as for cleaning and sorting/grading of grains, seeds, fruits, and other produce, soil preparation, cultivation, harvesting, and threshing, attracts high GST. Therefore, given that labour costs contribute a major share to total operational costs, GST rates need to be rationalised to enhance affordability and adoption.

Crop Insurance

- 3.64 Crop insurance not only protects farmers from crop losses due to natural calamities, extreme weather and other non-preventable risks, but also encourages them to adopt modern agricultural practices, thereby contributing to sustainable production. As the climate change and weather-related uncertainties are increasing, resulting in increased risks in agriculture, crop insurance has become vital to protect farmers, especially small and marginal.
- 3.65 ‘*Pradhan Mantri Fasal Bima Yojana*’ (PMFBY), launched in 2016, is an affordable crop insurance product for farmers by providing comprehensive risk for crops against non-preventable natural risks. The Restructured Weather based Crop Insurance Scheme (RWBCIS), an integral part of PMFBY, is aimed to mitigate the hardship of the insured farmers against the likelihood of loss due to adverse weather conditions by linking pay-outs with deviation in weather indices. Recently, crop loss due to attack by wild animals has also been covered under the Scheme. The States can provide add-on coverage for crop loss due to the attack by wild animals as an option wherever the risk is perceived to be substantial and is identifiable. However, there is need to create awareness among the farmers about such provisions.
- 3.66 Table 3.13 presents progress of crop insurance under PMFBY during last five *rabi* seasons. The number of farmers covered under the scheme has increased from 94.9 lakh in 2020-21 to 118.6 lakh in 2023-24 and percentage of non-loanee farmers has increased from 10.7 percent to 18.2 percent during the period. Number of farmers covered and percentage of non-loanee farmers slightly declined during 2024-25.

**Table 3.13: Trend in Crop Insurance under PMFBY during Last Five Rabi Seasons**

Year	2020-21	2021-22	2022-23	2023-24	2024-25*
Farmers Covered (in lakh)	94.92	94.93	93.34	118.64	113.34
Non-Loanee Farmers (%)	10.73	9.50	6.03	18.16	14.67
Gross Premium (₹ '000 crore)	6.43	6.68	5.22	5.04	4.78
Farmers Premium (₹ '000 crore)	0.92	0.88	0.82	0.79	0.84
Sum Insured (₹ '000 crore)	61.40	58.94	56.22	68.25	68.16
Farmers Benefitted (in lakh)	24.30	57.41	67.48	50.10	-
Claims settlement (₹ '000 crore)	2.73	3.19	3.40	2.14	-

Note: 1. Information for mandated rabi crops only

2. *: Data is Provisional

Source: Credit Division, Department of Agriculture and Farmers Welfare, Ministry of Agriculture and Farmers Welfare

- 3.67 Hassel-free application process, coverage of risks and crops, transparency in yield estimation and loss assessment, and timely insurance pay-outs are important for instilling farmers' confidence and increasing uptake of crop insurance. The Government has taken several steps in this direction, including use of advanced technologies such as satellite imagery, drones, unmanned aerial vehicle (UAV), and remote sensing for crop-cutting experiments planning, yield estimation, loss assessment, digital claim calculation and settlement, etc. Apart from these, WINDS (Weather Information and Network Data System) and CROPIC (Collection of Real-Time Observations and Photographs of Crops) have been included in PMFBY to improve crop insurance and weather information for farmers. Whereas WINDS provides weather data, CROPIC allows for real-time monitoring of crop conditions.
- 3.68 For timely pay-outs, the Government has made mandatory for States to deposit their part of premium subsidy, equivalent to 50 percent of previous season's total liability on approved application in the 'Escrow Account' before the start of enrolment for the season. In addition to the existing channels for enrolment of farmers, a smart-phone app called AIDE (App for Intermediary Enrolment) has been developed for enrolment of farmers at their doorsteps through a large network of insurance intermediaries for the convenience of farmers.



Digital Agriculture Mission

- 3.69 In September 2024, Government approved ‘Digital Agriculture Mission’ aimed to establish a comprehensive farmer-centric digital and space-tech ecosystem to enhance transparency and efficiency in the execution of Government programmes, ensuring faster and more targeted delivery of services, and providing support for evidence-based policy making for farmers’ welfare. Two major components of the Mission are (1) Digital Public Infrastructure (DPI) which includes AgriStack, *Krishi* Decision Support System, Comprehensive Soil Fertility, and Profile Maps, and (2) Digital General Crop Estimation Survey (DGCES).
- 3.70 AgriStack follows a protocol-based approach, ensuring modularity, elasticity and adaptability to the evolving needs of the agricultural sector. By leveraging trustful data on farmers, farmlands and crops and the use of modern digital technologies, such as data analytics and artificial intelligence, AgriStack aims to make service delivery mechanisms more efficient and transparent in the interest of farmers. It is a collaborative initiative between agencies of Central and State Government, which consists of three foundational registries, i.e., the Farmers’ Registry, Geo-referenced village maps, and the Crop Sown Registry. The Government aims to create 11 crore digital identities like *Adhaar* (*Kisan Pehchan Patra*) of farmers by 2026-27 and collect plot-wise data of crops sown by farmers across all States/UTs from *Kharif* 2025. Further, DGCES is a nationwide initiative aimed at precisely assessing crop yield across a variety of crops.
- 3.71 As on 21st May 2025, a total of 6 crore farmer IDs have been generated, covering farmers from 14 States. Out of these 14 States, Madhya Pradesh, Rajasthan, Gujarat, Maharashtra, Andhra Pradesh, Karnataka, and Tamil Nadu have generated more than 50 percent of the farmer IDs of the target set for the respective States. To collect plot-wise data of crops sown by farmers, Digital Crop Survey (DCS) has been conducted in 492 districts of the country, covering more than 25.2 crore plots spread across 17 States. In Uttar Pradesh, Assam, Rajasthan, Madhya Pradesh, Karnataka, Bihar, Tamil Nadu, Odisha, and Chhattisgarh more than 85 percent of the targeted plots have been covered.

Extension Services

- 3.72 Dissemination of timely and quality information about latest varieties/breeds, technologies, improved management practices, schemes to the farmers are essential to increase crop productivity, quality and profitability. The ‘Sub-Mission on Agricultural Extension’ (SMAE) under *Krishonnati Yojana*, is playing crucial role in agricultural extension with an objective to strengthen, expand and upscale existing extension schemes. Under the Scheme, a decentralized farmer-driven and farmer-accountable extension system through an institutional arrangement, i.e. Agricultural Technology Management Agency (ATMA) at the district level, is promoted for technology dissemination. Recently, Government of India has launched a 15-days campaign, ‘*Viksit Krishi Sankalp Abhiyan*’, to reach farmers with the objective of transferring technologies, creating awareness about schemes, and getting feedback of farmers for demand driven research. Brief overview of the campaign along with achievements is given in Box 3.2.



Box 3.2: Viksit Krishi Sankalp Abhiyan

With the aim of bridging the gap between scientific research and its reach to farmers, ‘Viksit Krishi Sankalp Abhiyan’ was organized from 29th May 2025 to 12th June 2025. Under this nationwide campaign, scientists directly connected with farmers in several villages and educated them about the latest sustainable agricultural practices, understood the local challenges and developed solutions in collaboration with farmers. This initiative is a significant step towards strengthening agricultural extension. It is expected that intensive campaigns under this important initiative will empower and equip the farmers with the latest knowledge about modern technologies, improved and sustainable agricultural practices, which are essential to improve the crop productivity and profitability of farmers and thus make India a ‘Viksit Bharat’. Total 2,170 teams of scientists organized more than 60 thousand programmes benefitting more than 1.3 crore farmers from 1.4 lakh villages in 721 districts.

Source: Indian Council of Agricultural Research

- 3.73 India’s agricultural extension programmes have played a vital role in disseminating knowledge to farmers. However, agricultural extension is still considered as a weak link due to issues like resource constraints, lack of coordination among different agencies, insufficient regular capacity building of extension staff, etc. Notably, more than 40 percent of the sanctioned posts of extension workers are vacant in West Bengal, Assam, Himachal Pradesh, Haryana, Uttar Pradesh, Madhya Pradesh, Odisha and Punjab (Annex Table 3.6). Therefore, States need to strengthen agricultural extension by filling vacant posts, capacity building of extension workers and strengthening digital and ICT-based platforms.

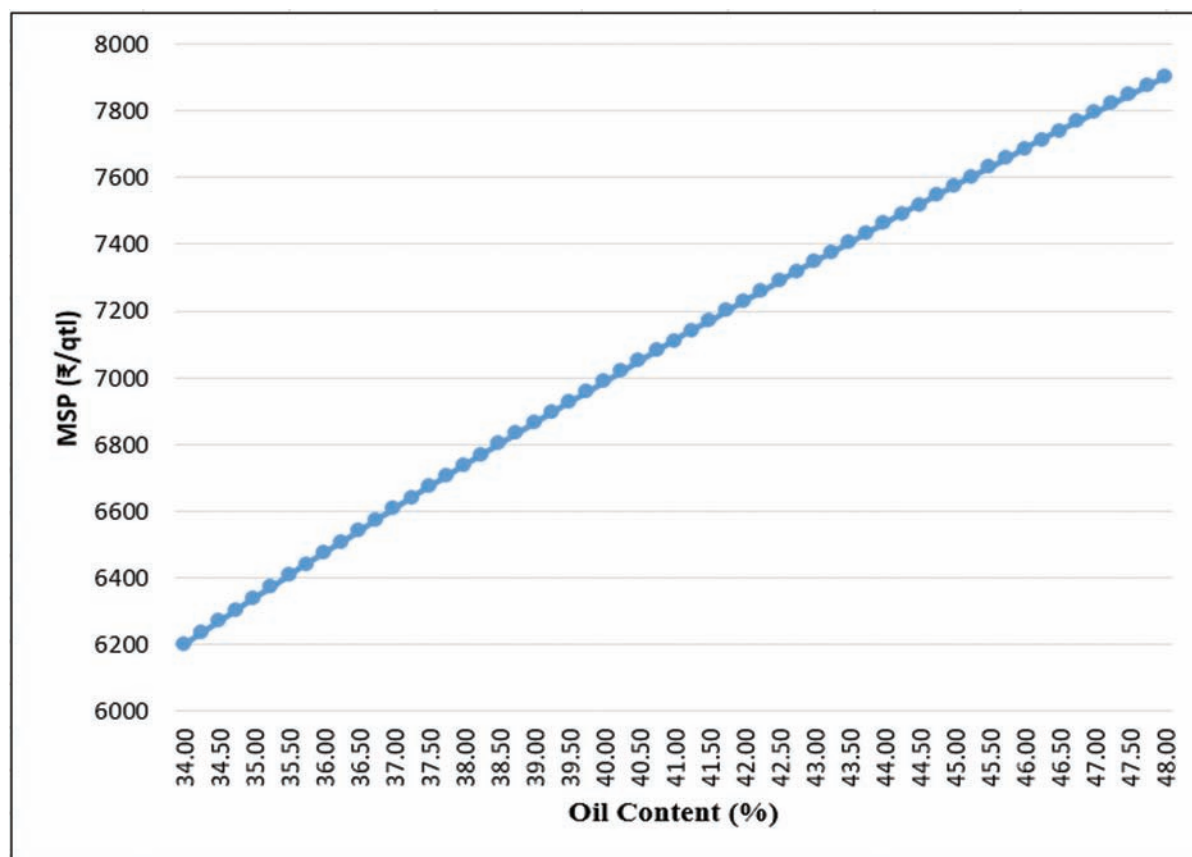
Linking MSP with Oil Content

- 3.74 Oilseed production in India between 2015-16 and 2024-25 has increased by more than 68 percent, but India’s dependence on edible oil imports has continued to grow. The Indian Council of Agricultural Research has developed a number of high oil content varieties and management practices to increase the oil yield in R&M and safflower. However, farmers currently have little incentive to adopt these varieties and management practices unless they receive premium for higher oil content. Therefore, it is essential to incentivise farmers for producing oilseeds with higher oil content. Therefore, based on extensive consultations with stakeholders, including farmers, processors, and scientists, the Commission recommends linking the MSP of R&M and safflower seeds with oil content with benchmark oil content of 34 percent in R&M and 28 percent in safflower. For every incremental increase of 0.25 percentage points in oil content beyond these levels, farmers should be provided with an additional incentive to promote the cultivation of high oil content varieties and increase area under these crops. The determination of incentives for higher oil content can be based on oil and oil cake content in the seeds and prices of oil cakes. Oil content-based MSP for R&M is illustrated in Para 3.75.
- 3.75 Considering that 100 kilograms of R&M seeds on an average contains 34 kilograms of oil and the remaining 65 kilograms de-oiled cake as around one percent is lost during the processing, the effective cost of oil extraction from one quintal of seeds, after adjusting for the value of the oil cake (assuming oil



cake prices ₹2,190/qrtl), comes out ₹4,777. Based on these calculations, a 0.25 percent point increase in oil content translates into an increase in MSP of ₹30.31 per quintal (illustrated in Chart 3.14). As the oil content rises, the unit cost of oil extraction declines marginally in proportion to the oil-to-cake ratio in R&M seeds (Annex Table 3.7).

Chart 3.14: Oil Content-Based MSP of Rapeseed & Mustard (Illustrative)



Recapitulation

- 3.76 Production of *rabi* crops has increased significantly during last five years, however, area expansion has been the primary source of production growth. The growth in yield of all the crops, except lentil, in top three producing States has decelerated during last five years compared to previous five years. The area under barley and safflower has declined to limited districts. Therefore, there is a need to develop varieties with high yield, multi-stress tolerant and quality attributes as per the market demand and target regions based on suitability and relative profitability.
- 3.77 The international comparisons showed that yield of *rabi* crops in the country is much lower than the world's highest yields, indicating considerable scope for improvement. Moreover, wide variations in yield have been observed across crops, States, and districts. The yield gaps can be minimised by ensuring quality seeds, increasing irrigation coverage, promoting good agricultural practices (GAP), etc. District-level comparison of area and yield revealed that there is considerable potential for increasing



production by adopting crop-specific strategies at disaggregated level. In ‘high area-low yield’ districts, the focus should be on improving productivity, while ‘low area-high yield’ category districts should be targeted for area expansion. The ‘low area- low yield’ (LL) category districts should be targeted for shifting cropping patterns towards alternative crops based on suitability, natural resource endowment, and relative profitability of crops.

- 3.78 Fertilizer consumption has increased in the country but there are significant imbalances in nutrient use as well as in nutrient distribution across States and districts. Therefore, promoting judicious and balanced fertilizer use as per soil test-based recommendations is imperative to reduce the cost of cultivation, improve use efficiency, and maintenance of soil fertility. Further, concerted efforts are needed to minimize regional disparities in institutional credit disbursement. Technological interventions in crop insurance are expected to improve transparency, accuracy in claims process and yield loss assessment. Further, AgriStack and DGCES are important initiatives towards driving digital transformation of Indian agriculture. The States need to strengthen and modernize the agricultural extension system for timely and quality advisory services and faster technology transfer.



CHAPTER 4



Performance and Outlook for India's Agricultural Trade

- 4.1 Amidst ongoing geopolitical tensions, trade conflicts and policy uncertainties, global trade is on a path toward gradual recovery, supported by easing of inflationary pressures and resurgence in demand. World merchandise trade volume not only increased by 2.9 percent in 2024, it was also the first year since 2017, excluding the post-pandemic rebound, when merchandise trade growth outpaced output, as per World Trade Organization (WTO). As regards India, while value of merchandise exports (in rupee term) increased by 2.3 percent, imports increased by 8.4 percent in 2024-25 over previous year. However, India's agri-exports increased by 8.6 percent and imports by 19.6 percent during 2024-25. Despite being one of the world's largest producer of pulses and oilseeds, India remains highly dependent on imports to meet the domestic demand for pulses and edible oils. In 2024-25, imports of pulses (in value) increased by over 50 percent and edible oils by 19 percent over previous year. Although, Government has taken various policy initiatives like National Mission on Edible Oils and Mission for *aatmanirbharta* in Pulses to increase domestic production but to make the mission successful, it is imperative to emphasize on effective implementation through participation of farmers and private sector, integrated value chains and ensuring remunerative and stable price to farmers. This chapter comprehensively analyses the recent trends in India's agricultural trade, domestic and international prices, trade policy changes and the global trade outlook of mandated crops.

Global Trade Performance

- 4.2 The world merchandise exports reached at US\$24.43 trillion in 2024, an increase of 2 percent over 2023, primarily attributed to increase in volume, as per World Trade Organisation's (WTO) Global Trade Outlook and Statistics 2025. Agricultural trade also increased by 3 percent in value term in 2024 over previous year due to decline in food prices, which led to rise in volume of exports. Amidst a surge in world tariffs and trade policy uncertainty, WTO anticipates a decline in volume of world merchandise trade by 0.2 percent in 2025 but a modest recovery of 2.5 percent in 2026.

India's Trade Performance

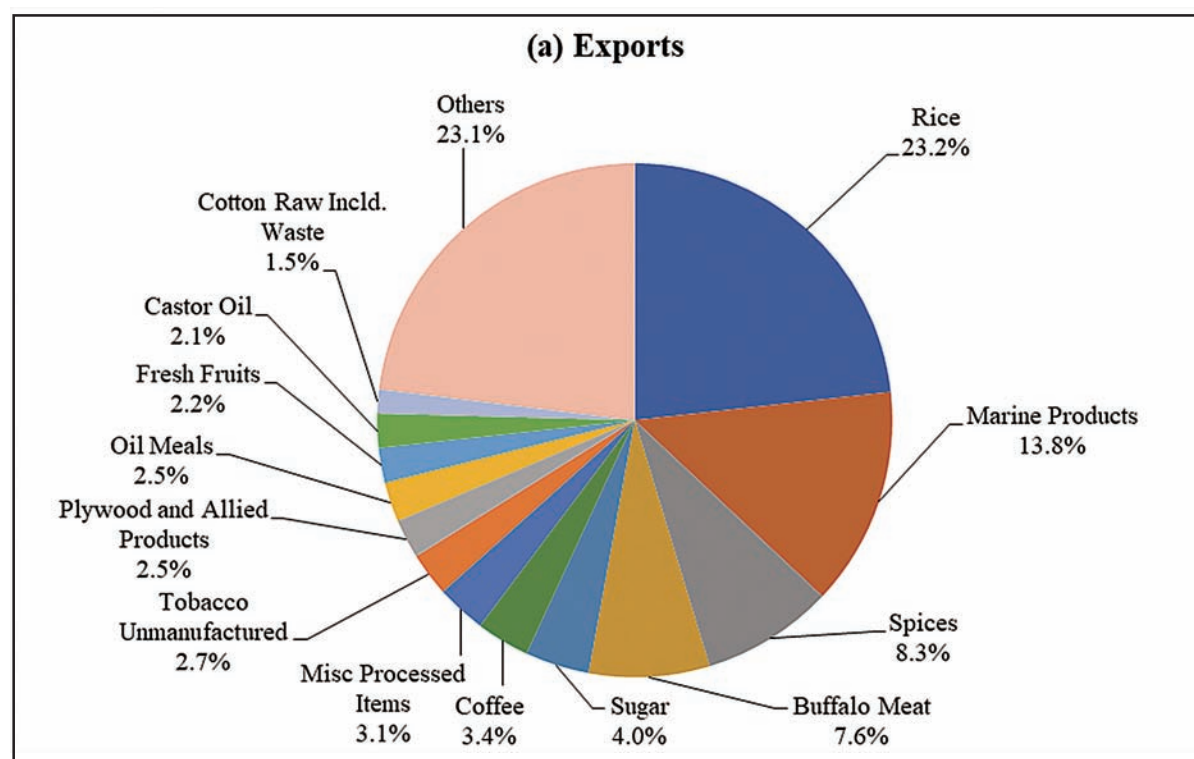
- 4.3 India has emerged as a significant player in global markets in the recent decade due to increase in agricultural production and supportive policies. As per Directorate General of Commercial Intelligence and Statistics (DGCIS), India's agri-exports were registered at ₹454.8 thousand crore in 2024-25, an increase of 8.6 percent over 2023-24, while imports of agricultural products were ₹347.9 thousand crore, a significant increase of 19.6 percent. As a result, the trade surplus in agriculture declined from

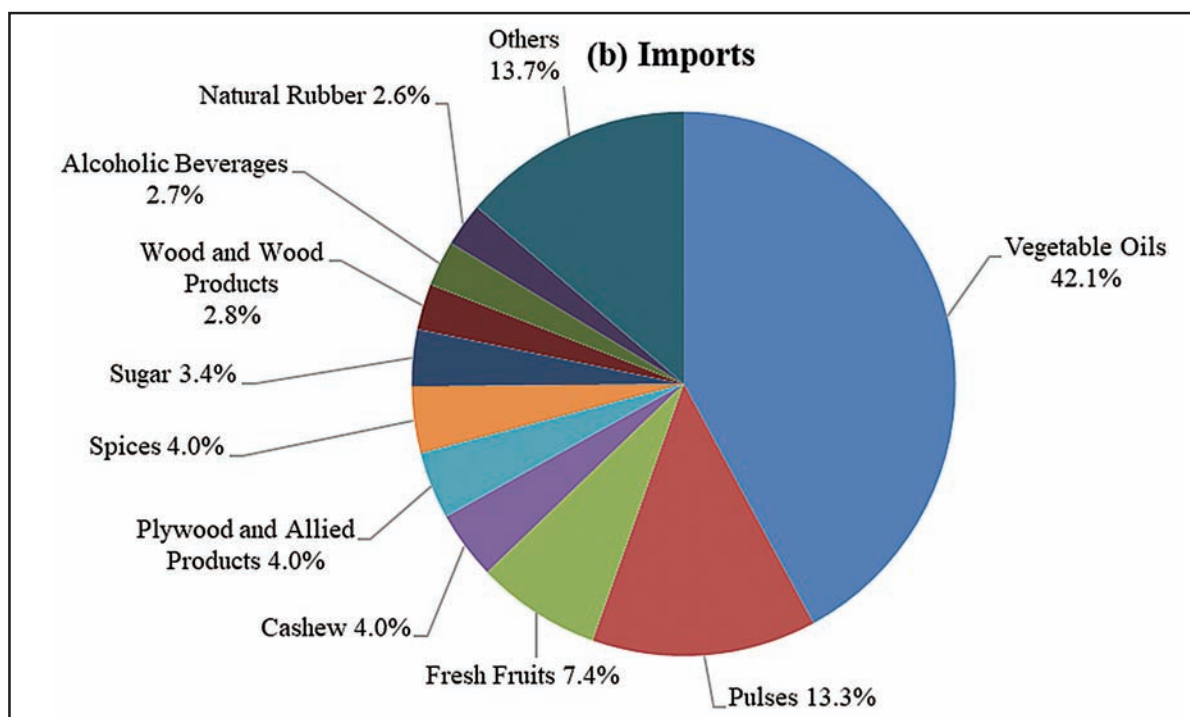


₹127.7 thousand crore in 2023-24 to ₹106.9 thousand crore in 2024-25. The share of agricultural exports in India's total merchandise exports increased from 11.6 percent in 2023-24 to 12.3 percent in 2024-25 and share of agricultural imports in total merchandise imports increased from 5.2 percent to 5.7 percent.

- 4.4 The composition of agricultural trade by major commodity groups for the year 2024-25 is shown in Chart 4.1(a) and 4.1(b). It can be seen that, rice (basmati and non-basmati) was the largest export item with a share of 23.2 percent in total agri-exports, followed by marine products (13.8%), spices (8.3%), buffalo meat (7.6%) and sugar (4%). Top five products accounted for nearly 57 percent of total agri-exports.
- 4.5 India is one of the major producers as well as consumers of wheat in the world. Due to high domestic demand, India occasionally exports wheat, mainly to neighboring countries. The share of leading importers of wheat from India in 2023-24 and 2024-25 are shown in Annex Table 4.1. After the Government imposed restrictions on exports of wheat in May 2022, wheat exports from India have declined significantly.
- 4.6 India's key agricultural imports included vegetable oils (42.1%), pulses (13.3%), fresh fruits (7.4%) and cashew (4.0%) in 2024-25 (Chart 4.1(b)). It is observed that the top five imported products accounted for over 70 percent of total agricultural imports. While the share of vegetable oils in total agricultural imports (in value term) almost remained same in 2023-24 and 2024-25, the share of pulses increased from 10.7 percent to 13.3 percent during the same period.

Chart 4.1: Share of India's Agricultural Exports and Imports, by Commodity/Group, in 2024-25 (in value)





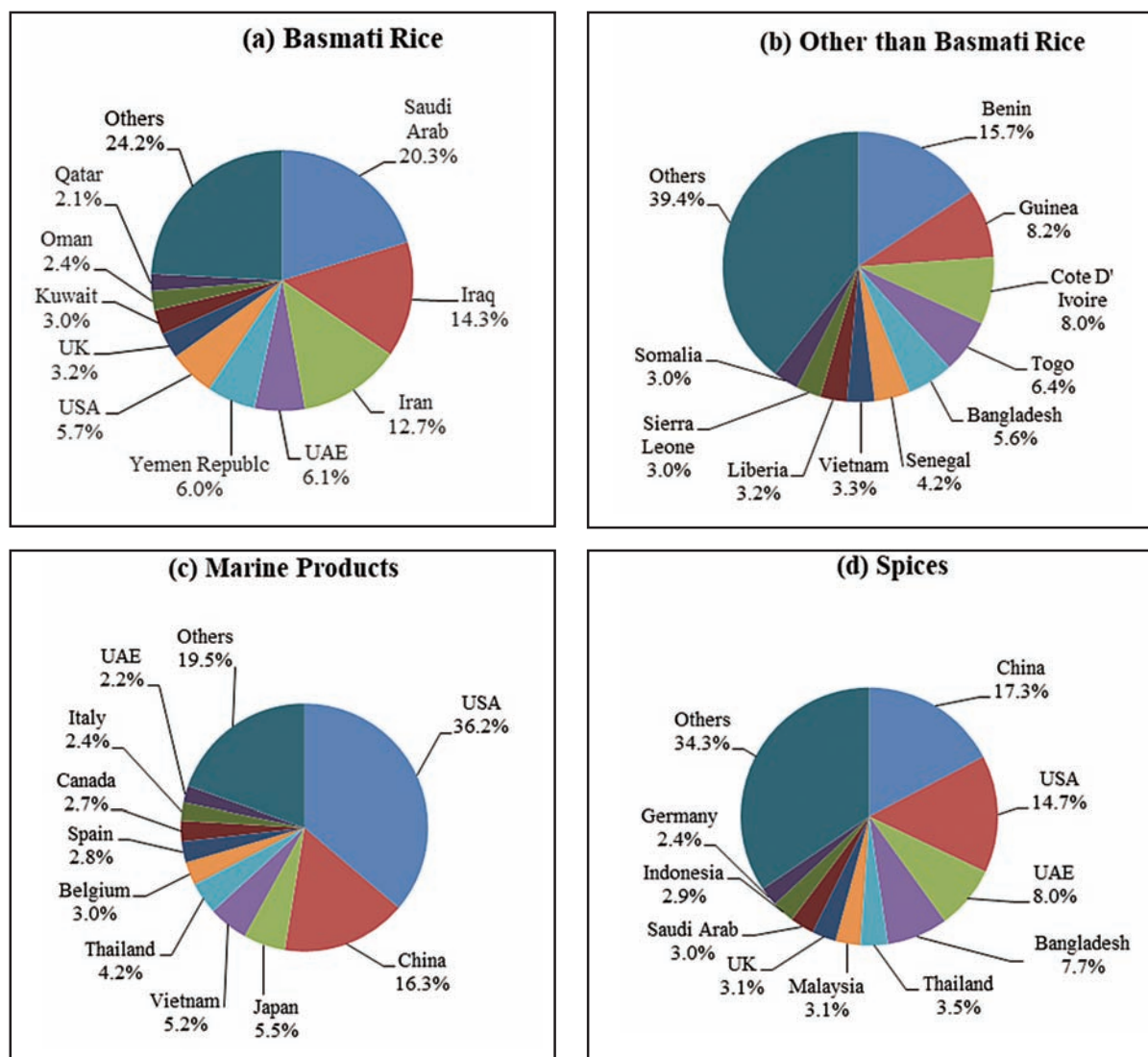
Source: Directorate General of Commercial Intelligence and Statistics, Ministry of Commerce and Industry

Agricultural Export Destinations

- 4.7 The top export destinations for major agricultural commodities from India in 2024-25 are shown in Chart 4.2. Saudi Arabia, Iraq and Iran were the top three destinations for India's basmati rice exports in 2024-25 and accounted for 47.2 percent of total exports (Chart 4.2 (a)). UAE, Yemen Republic and USA were the other important markets for basmati rice.
- 4.8 The top destinations for India's non-basmati rice were Benin, Guinea and Cote D' Ivore, which accounted for 31.8 percent of total exports (Chart 4.2(b)). It is worth noting that Benin was not only the top export destination of India's non-basmati rice but value of rice exports to Benin has also increased by 105 percent in 2024-25.
- 4.9 In 2024-25, India became the fourth largest marine products exporter and USA, China and Japan continued to be the major export destinations for Indian marine products (Chart 4.2 (c)). USA alone accounted for 36.2 percent of exports, followed by China (16.3%) and Japan (5.5%). India's marine exports have recorded a robust growth both in value and volume terms over the past decade.
- 4.10 India's diverse agro-climatic conditions allow cultivation of a wide variety of spices, which are very popular around the world. Major export destinations of Indian spices are China (17.3%), USA (14.7%), UAE (8%) and Bangladesh (7.7%) (Chart 4.2 (d)).



Chart 4.2: Top 10 Export Destinations of Major Export Commodities from India: 2024-25



Source: Directorate General of Commercial Intelligence and Statistics, Ministry of Commerce and Industry

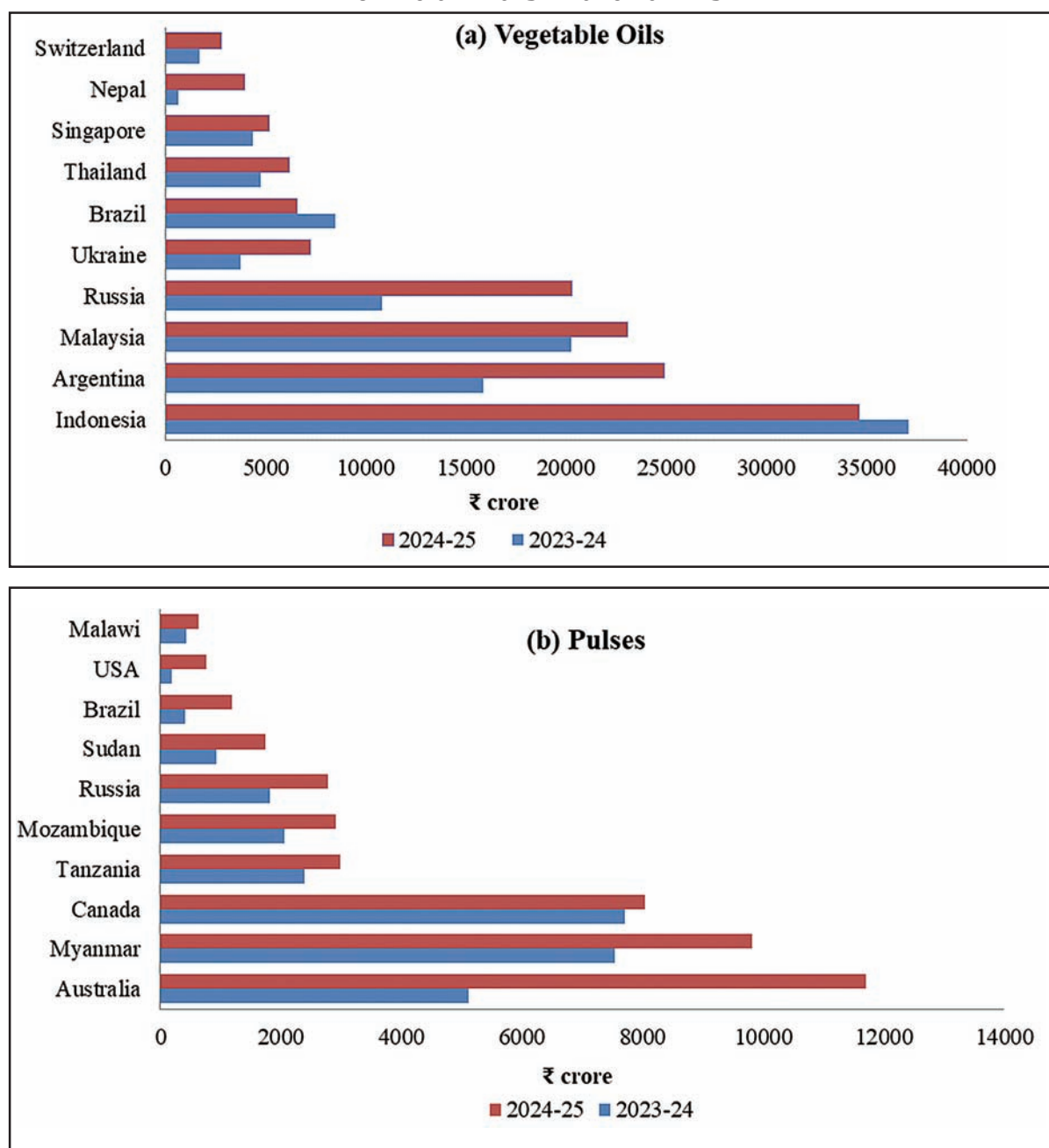
Origins of India's Agricultural Imports

- 4.11 In 2024-25, India continued to import two major products, edible oils and pulses, that represented over 55 percent of total India's agri-imports. India's vegetable oils and pulses imports from major countries in 2024-25 are shown in Chart 4.3. Indonesia remained the first source of origin, representing 23.6 percent of India's vegetable oil imports in 2024-25. Argentina was the second source, with import share rising from 12.9 percent in 2023-24 to 17 percent in 2024-25 and replaced Malaysia at second position in 2024-25. Malaysia was the third source, with a share of 15.8 percent of India's vegetable oil imports. Overall, the top three origins represented 56.4 percent of India's vegetable oil imports during 2024-25, which shows the level of concentration in vegetable oil imports.
- 4.12 Despite being the top producer of pulses, India has emerged as one of the largest importers of pulses in the world due to rising domestic demand. Although pulses production in the country has increased from 16-17 million tonnes to over 25 million tonnes over the last decade, but fast rise in demand for pulses has



led to increase in imports. In 2024-25, pulses imports increased by 50.1 percent in value and 53.8 percent in volume over 2023-24. Australia is a major exporter of lentil and gram, while Myanmar exports mainly *urad* and *tur*. Canada is the largest exporter of lentil to India. In order to achieve self-sufficiency in pulses production, in the Union Budget 2025-26, Government has announced six-year 'Mission for Aatmanirbharta in Pulses' with special focus on *tur/arhar*, *urad* and lentil, with an allocation of ₹1,000 crore for development and availability of climate resilient seeds, improving yields and post-harvest warehousing facilities and assuring remunerative prices to farmers through assured procurement.

Chart 4.3: Top Import Sources of Vegetable Oils and Pulses for India in 2023-24 and 2024-25



Note: Ranking of countries based on value of imports in 2024-25

Source: Directorate General of Commercial Intelligence and Statistics, Ministry of Commerce and Industry



Trade Performance and Price Trends

Cereals

Wheat

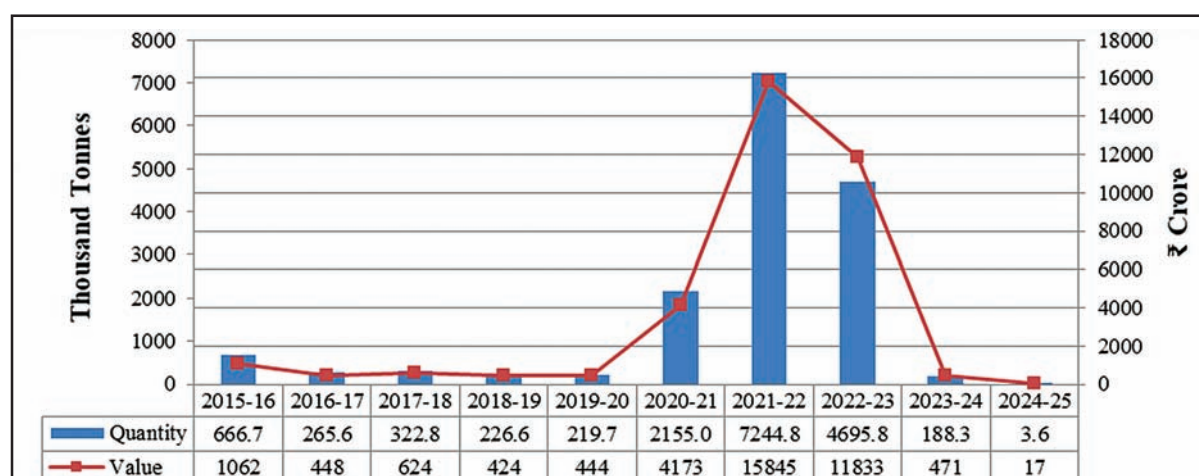
Global Production and Trade

- 4.13 Global production of wheat was 794.1 million tonnes and 27.1 percent was traded in TE2024-25, as per United States Department of Agriculture (USDA). China, with a share of 17.4 percent, was the largest producer of wheat in the world, closely followed by EU (16.5%), India (13.8%) and Russia (11.1%). Among the leading wheat exporters, Russia accounted for the largest share of 23 percent, followed by EU (15.4%), Canada (12.2%) and Australia (11.8%). With a share of 5.6 percent, Egypt was the largest importer of wheat in the world, followed by EU (5.5%), Indonesia (5.2%) and China (4.8%). The combined share of top five exporters (72%) compared to top five importers (25.3%) indicates that global wheat exports are heavily concentrated than imports and global wheat imports have become more diverse over time.

India's Trade

- 4.14 India's exports of wheat during the last 10 years are shown in Chart 4.4. India's wheat exports touched 7244.8 thousand tonnes in 2021-22, an all-time high but declined to 4695.8 thousand tonnes in 2022-23, and 188.3 thousand tonnes in 2023-24 due to export restriction imposed in May 2022 to check domestic prices and ensure sufficient domestic supplies. Despite record production of wheat during last three years, India's wheat exports were negligible (3.6 thousand tonnes) in 2024-25.

Chart 4.4: India's Exports of Wheat, 2015-16 to 2024-25

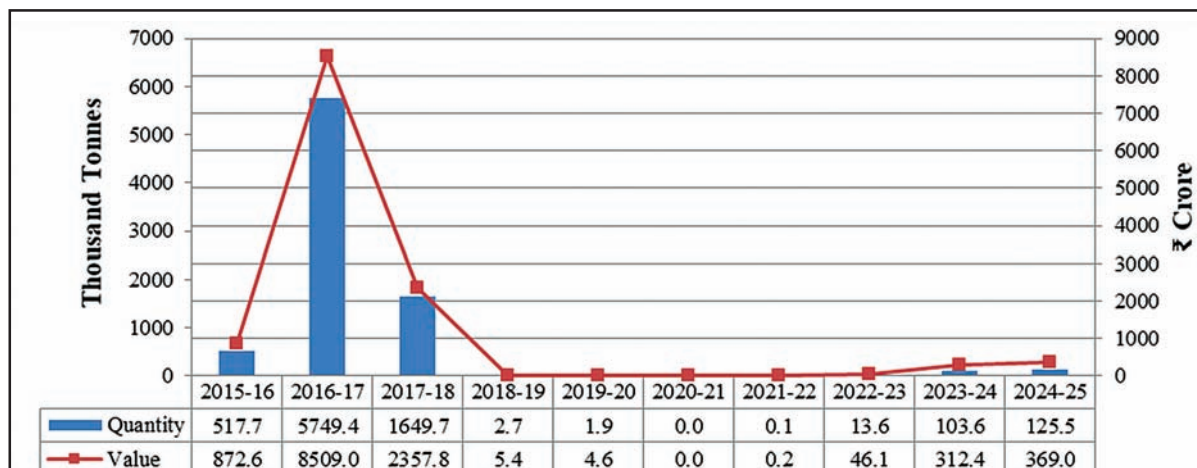


Source: Directorate General of Commercial Intelligence and Statistics, Ministry of Commerce and Industry

- 4.15 Chart 4.5 depicts India's imports of wheat during last 10 years, in both volume and value terms. It can be seen that India's imports have been negligible during the entire period, except 2016-17, when India imported 5749.4 thousand tonnes of wheat due to lower production in 2014-15 and 2015-16.



Chart 4.5: India's Imports of Wheat, 2015-16 to 2024-25



Source: Directorate General of Commercial Intelligence and Statistics, Ministry of Commerce and Industry

Trade Policy

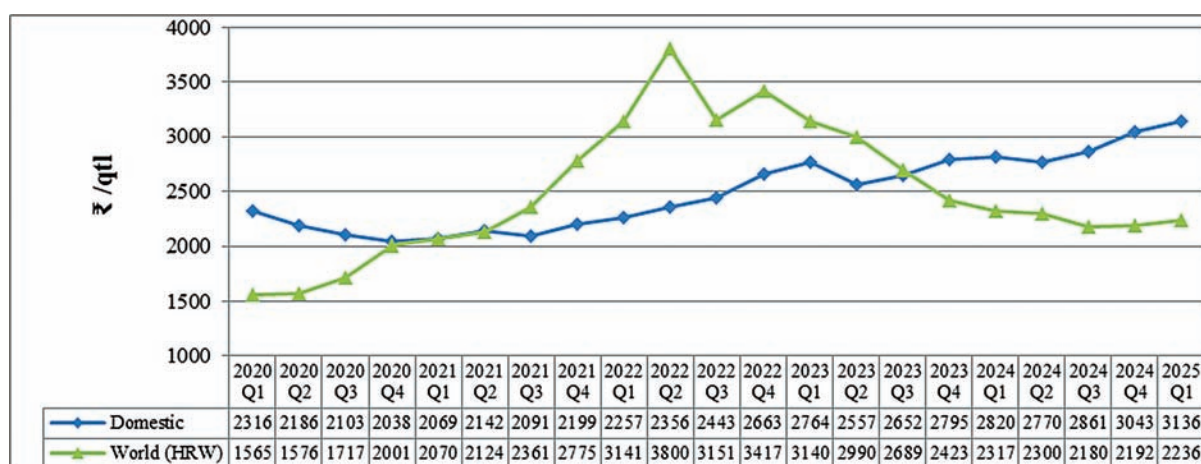
- 4.16 India's wheat trade policy is driven by various factors such as domestic food security concern, domestic production and prices as well as global market dynamics. Wheat exports from Central Pool were banned in August 2003 due to significant fall in production in 2002-03 and private wheat exports were also prohibited in February 2007 due to lower production in 2004-05 and 2005-06. However, ban was lifted in September 2011 and exports of 20 lakh tonnes of wheat were allowed under an Open General License (OGL) through Electronic Data Interchange (EDI)-enabled ports and in February 2012, unrestricted wheat exports were permitted under OGL, with no export authorization required from the Directorate General of Foreign Trade (DGFT).
- 4.17 Due to spike in global prices coupled with lower domestic production and rise in domestic prices in 2021-22, India amended wheat export policy from "free" to "prohibited" in May 2022 and export of *atta* (wheat) was restricted from 12th July 2022. Albeit, in special cases, wheat exports were permitted with Irrevocable Letters of Credit (LoC) issued beforehand and also for nations requesting wheat exports for food security concerns. In order to address the food security concerns in Nepal, Government approved exports of 2 lakh tonnes of wheat through the National Cooperative Exports Limited (NCEL) w.e.f. 4th January 2025.
- 4.18 Import duty on wheat was reduced from 25 percent to 10 percent in September 2016 and further to zero percent in December 2016 in order to control the rising prices caused by lower production in 2014-15 and 2015-16. However, Government imposed 10 percent import duty on wheat in March 2017, which was raised to 20 percent in November 2017 driven by significant increase in production in 2016-17 and record stocks in Central Pool. Subsequently, to protect farmers from cheaper imports of wheat due to fall in global prices, duty was increased to 30 percent in May 2018 and 40 percent in April 2019, which has remained at 40 percent since then. Moreover, imports of certain varieties of wheat are allowed only through Food Corporation of India (FCI).



Comparative Price Trends

- 4.19 Chart 4.6 depicts the comparative trends of domestic and international prices of wheat during 2020 to 2025(Q1). It is observed that between 2020(Q1) to 2020(Q4), domestic prices of wheat were higher than the international prices with significant divergence. However, both the prices moved in unison during the period 2020 (Q4) to 2021(Q2). Thereafter, both domestic and international prices followed upward trend with international prices remaining significantly higher than domestic prices due to global general commodity price inflation. International wheat prices hit the peak in 2022(Q2), caused by disruption in global wheat supplies consequent to Russia-Ukraine conflict. However, due to rise in global wheat supply, a steady and significant fall in international prices of wheat was noticed from 2022(Q2) to 2024(Q3), except 2022(Q4), followed by marginal rise in 2024(Q4). Although, steady growth in domestic prices of wheat was witnessed since 2021(Q2), except 2023(Q2) and 2024(Q2) but domestic prices remained below international prices till 2023(Q3). However, domestic prices of wheat moved above global prices in 2023(Q4) and have remained higher than world prices since then. In contrast to fall in world prices, domestic prices showed a rising trend since 2023(Q2).

Chart 4.6: Domestic and International Prices of Wheat, 2020 to 2025(Q1)



Note: Wheat (U.S.), no. 2 Hard Red Winter Gulf export price; June 2020 backwards, no. 1, Hard Red Winter, ordinary protein, export price delivered at the US Gulf port for prompt or 30 days shipment

Source: 1. Agriculture Marketing Division, Department of Agriculture and Farmers Welfare for domestic wholesale prices
2. World Bank for International prices

Barley

Global Production and Trade

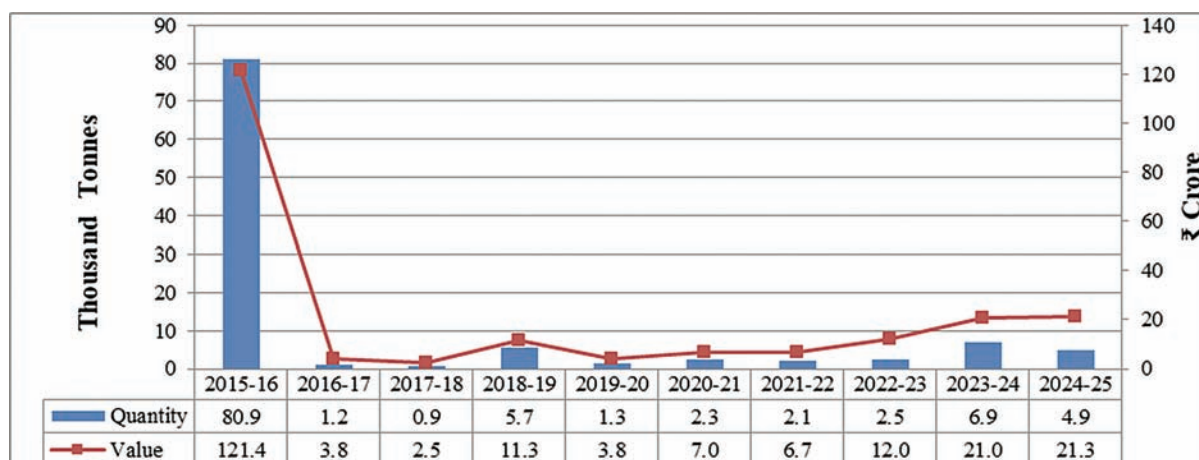
- 4.20 As per USDA, global production of barley was 146.1 million tonnes in TE2024-25, out of which 20.8 percent was traded. With a share of 34.2 percent, EU was the largest producer of barley, followed by Russia (13.3%) and Australia (8.7%). Australia was the largest exporter with a share of 24.3 percent, followed by EU (21.3%) and Russia (15.8%), while major importers of barley were China (36.7%), Saudi Arabia (10.1%) and Iran (5.5%). Global trade in barley is dominated by top two exporters and importers, which accounted for more than 45 percent of world trade in TE2024-25.



India's Trade

- 4.21 India's barley exports and imports from 2015-16 to 2024-25 are shown in Chart 4.7 and Chart 4.8. It is evident from the data, that India is a net importer of barley. During the last ten years, India's exports have been negligible, except in 2015-16, when India exported 80.9 thousand tonnes of barley worth ₹121.4 crore.

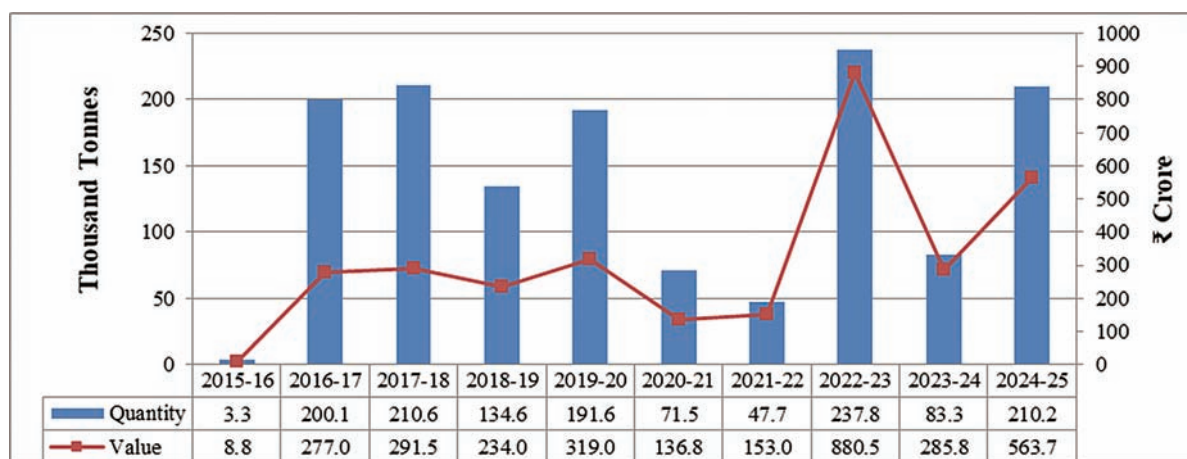
Chart 4.7: India's Exports of Barley, 2015-16 to 2024-25



Source: Directorate General of Commercial Intelligence and Statistics, Ministry of Commerce and Industry

- 4.22 Growing consumer preference for healthier and more diverse foods along with evolving market for distilled beverages including beer has led to increase in barley imports in India in the recent years as Indian barley is suitable for feed rather than malting purpose. India's barley imports have remained volatile during the last ten years, ranging from 3.3 thousand tonnes in 2015-16 to 237.8 thousand tonnes in 2022-23. Barley imports declined significantly during 2023-24 mainly due to reduced exportable surplus from key exporters. The trend reversed in 2024-25 and imports increased significantly both in volume (152%) and value (97%) over previous year. In order to reduce import dependency and increase domestic production of barley, there is a need to develop high yielding varieties suitable for malting purpose.

Chart 4.8: India's Imports of Barley, 2015-16 to 2024-25



Source: Directorate General of Commercial Intelligence and Statistics, Ministry of Commerce and Industry



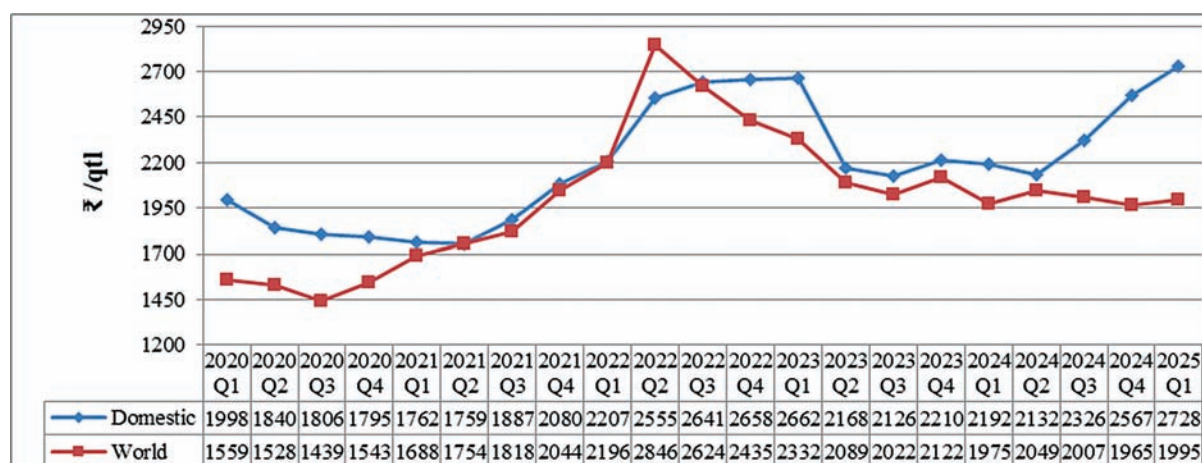
Trade Policy

- 4.23 At present, barley exports from India are under “free” category and do not require any export authorization. As regards imports, quantitative restrictions on imports were removed in 2002 and currently, imports of barley are “free” with zero import duty except seed quality of barley, which is “restricted”.

Comparative Price Trends

- 4.24 The comparative trends of domestic and international prices of barley during 2020 to 2025(Q1) are shown in Chart 4.9. It is observed that, international prices of barley remained below domestic prices during the entire period, except 2022(Q2). After witnessing a marginal fall between 2020(Q1) and 2020(Q3), international prices rose significantly and hit the peak in 2022(Q2). However, international prices declined steadily from 2022(Q2) onwards. On the other hand, domestic prices fell marginally between 2020(Q1) and 2021(Q2) but rebounded from 2021(Q3) and remained higher than international prices since 2022(Q3). The divergence between domestic and global prices increased sharply during the last three quarters. While India has been traditionally a net importer of barley as high domestic prices and lower quality make India less competitive in the international market.

Chart 4.9: Domestic and International Prices of Barley, 2020 to 2025(Q1)



Note: International Prices, Export, Australia (eastern states), Barley (feed), US\$/tonne

Source: 1. Agriculture Marketing Division, Department of Agriculture and Farmers Welfare for domestic wholesale prices
2. International Grain Council (FPMA) for International prices

Pulses

Global Production and Trade

- 4.25 India was the largest producer of pulses with about 28 percent share in global production during the last decade, as per the OECD-FAO Agricultural Outlook 2024-2033. Canada, China and the EU were the other major producers of pulses. Asian market, with a share of about 43 percent of global pulses production, accounts for 52 percent of global consumption and that makes it the most significant import destination of pulses in the world. About 20 percent of global production is traded and Canada

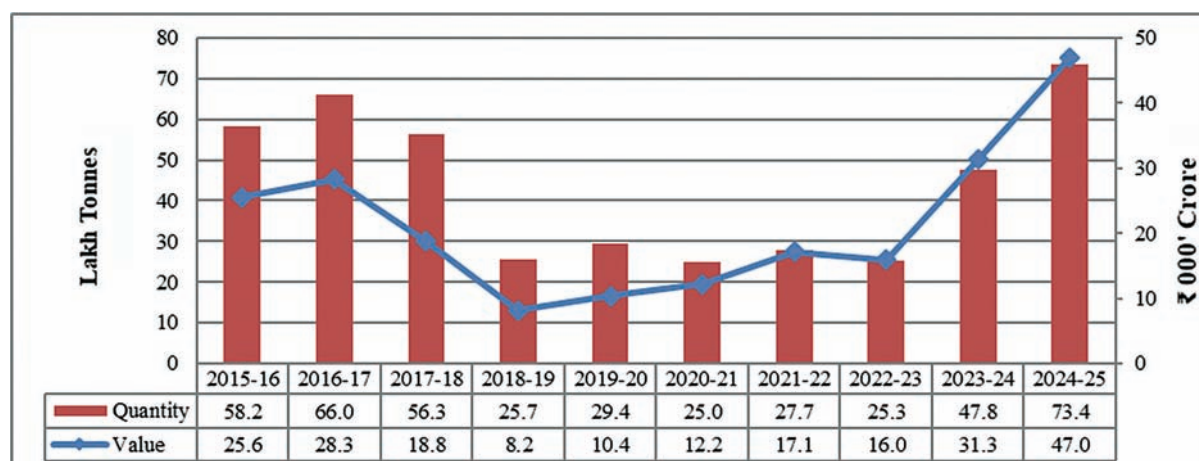


is the largest exporter with a share of 23 percent, while China is the largest importer in the world with a global share of 14 percent.

India's Trade

- 4.26 Despite being a major producer and consumer in the world, India imports large quantities of pulses to meet rising domestic demand. Chart 4.10 depicts India's imports of pulses during the period 2015-16 to 2024-25. India imported 58.2 lakh tonnes of pulses in 2015-16, which increased to 66 lakh tonnes in 2016-17 due to lower domestic production. In contrast, between 2016-17 and 2018-19, imports of pulses declined by 61 percent driven by significant increase in pulses production in the country, higher market and Minimum Support Prices (MSP) with large scale procurement and import restrictions. Between 2018-19 and 2022-23, an average of 26.6 lakh tonnes of pulses per year were imported by India. However, pulses imports reached 47.8 lakh tonnes in 2023-24, highest during last 6 years, caused by lower production due to *EL Nino* induced erratic monsoon and free imports of pulses at low/zero import duty. India's pulses imports surged to a historic high of 73.4 lakh tonnes (₹47 thousand crore) in 2024-25, 53.8 percent increase over 2023-24. The sharp jump in imports was driven largely by rising demand and liberal imports of pulses. *Tur/arhar*, lentil and *urad* accounted for 44.5 percent of total pulses imports in 2024-25.

Chart 4.10: India's Imports of Pulses, 2015-16 to 2024-25

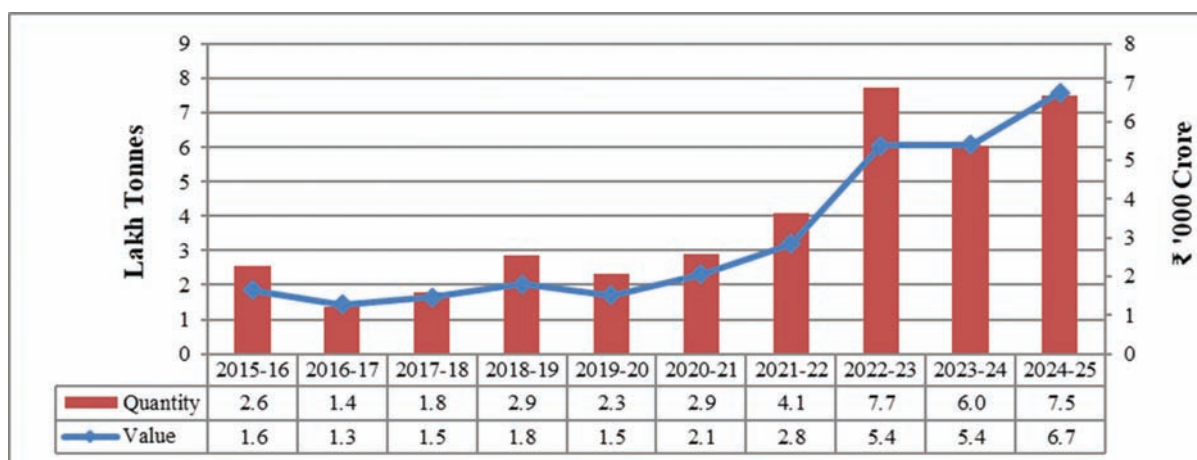


Source: Directorate General of Commercial Intelligence and Statistics, Ministry of Commerce and Industry

- 4.27 India exports small quantities of pulses mostly to neighboring countries due to high domestic demand. India's pulses exports during the period 2015-16 to 2024-25 are shown in Chart 4.11. India exported fairly small quantities of pulses ranging between 1.4 lakh tonnes and 2.9 lakh tonnes during 2015-16 and 2020-21. However, pulses exports witnessed steady growth in both volume and value from 2020-21 and reached the peak in 2022-23. India exported 7.5 lakh tonnes of pulses worth ₹6.7 thousand crore in 2024-25, about 25 percent higher than 2023-24.



Chart 4.11: India's Exports of Pulses, 2015-16 to 2024-25



Source: Directorate General of Commercial Intelligence and Statistics, Ministry of Commerce and Industry

- 4.28 The composition of India's imports of pulses (in volume) along with key sources of origin are shown in Table 4.1. It can be seen that with a share of 29.5 percent, yellow peas has the largest share in total pulses imports in 2024-25, followed by gram (22%), tur (16.7%), lentil (16.6%) and *urad* (11.2%). India primarily imports yellow peas from Canada, Russia and Turkey; gram from Australia, Tanzania and Russia; while Mozambique, Tanzania, and Myanmar are major suppliers of *tur* to India. Lentil is imported from Canada, Australia, and USA and *urad* imports are from Myanmar, Brazil, and Singapore. In 2024-25, the share of yellow peas imports has increased significantly with volume of imports doubled from 2023-24 level. The share of gram in pulses imports has also increased significantly but share of *tur* increased only marginally. In contrast, the share of lentil has declined drastically but share of *urad* declined only marginally. Rising share of gram imports in 2024-25 is driven by lower domestic production during last two years and duty-free imports.

Table 4.1: India's Imports of Pulses: Main Countries of Origin

Name	Import Quantity (Lakh Tonnes)		Share in total Pulses Imports (%)		Key Origin (Share in %) in 2024-25
	2023-24	2024-25	2023-24	2024-25	
Yellow Peas	11.7 (40.85)	21.6 (37.39)	24.5	29.5	Canada (50.2), Russia (31.0), Turkey (4.4)
Gram/Chickpea	2.8 (62.30)	16.1 (63.87)	6.0	22.0	Australia (87.1), Tanzania (4.7), Russia (2.2)
Tur/Arhar	7.7 (85.54)	12.2 (88.39)	16.1	16.7	Mozambique (28.6), Tanzania (26.1), Myanmar (21.4)
Lentil	16.8 (63.59)	12.2 (63.85)	35.1	16.6	Canada (51.3), Australia (38.5), USA (5.7)
Urad	6.2 (88.11)	8.2 (93.03)	13.1	11.2	Myanmar (87.3), Brazil (10.3), Singapore (1.5)
Moong	0.0 (47.19)	0.0 (60.06)	0.1	0.0	Tanzania (27.2), Egypt (19.1), UAE (18.4)
Others	2.5	3.0	5.1	4.1	-
Total	47.8	73.4	100.0	100.0	-

Note: () indicates average import value in (₹/kg)

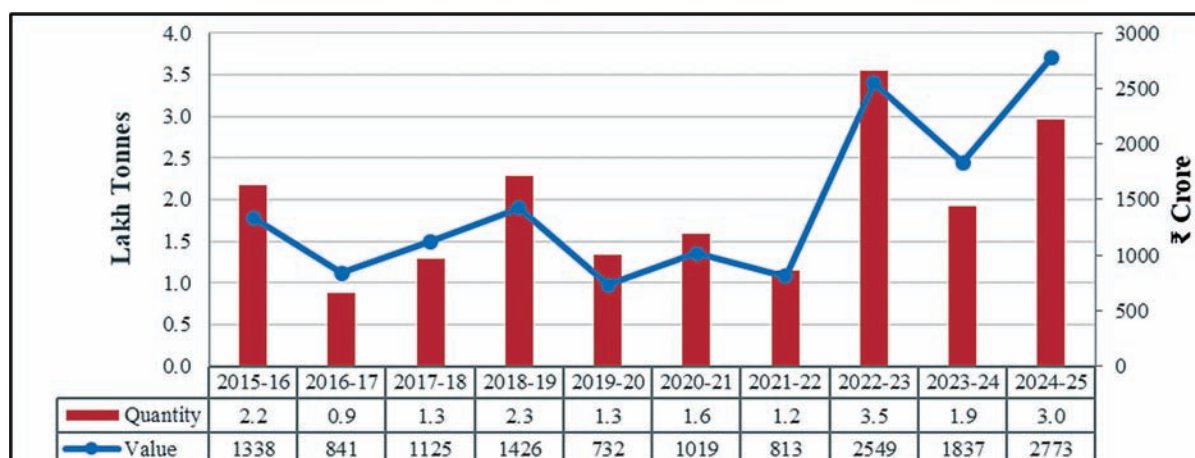
Source: Directorate General of Commercial Intelligence and Statistics, Ministry of Commerce and Industry



India's Trade in Gram

- 4.29 India's exports of gram during 2015-16 to 2024-25 are shown in Chart 4.12. It can be seen that India exports small quantities of gram with wide inter-year fluctuations. India exported an annual average of 1.9 lakh tonnes of gram during last ten years. It is worth noting that gram exports registered the highest growth, both in volume (213.5%) and value (207.8%), in 2022-23 over previous year. However, in 2023-24, exports of gram declined to 1.9 lakh tonnes owing to significant fall in domestic production and higher domestic prices. Exports of gram increased to 3 lakh tonnes in 2024-25.

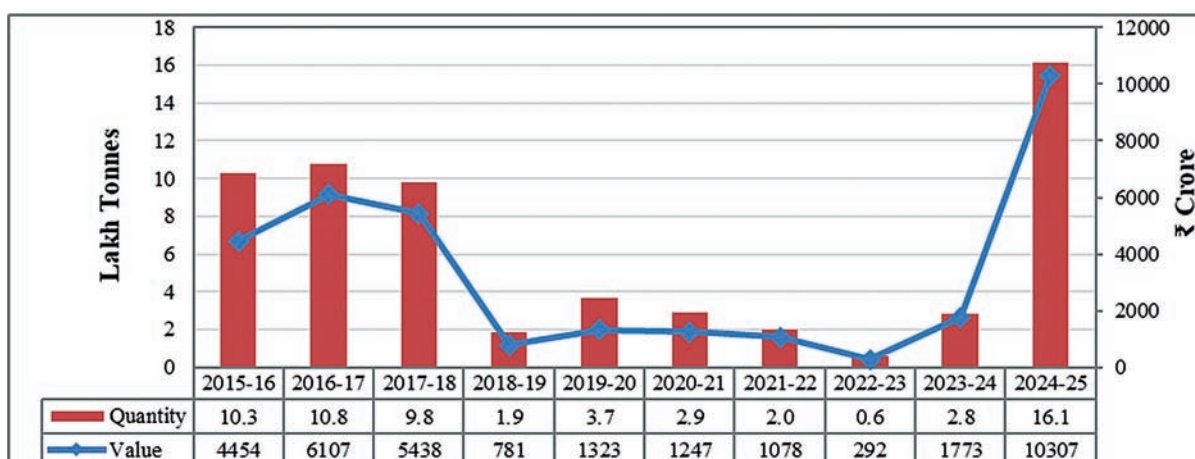
Chart 4.12: India's Exports of Gram, 2015-16 to 2024-25



Source: Directorate General of Commercial Intelligence and Statistics, Ministry of Commerce and Industry

- 4.30 India's imports of gram during 2015-16 to 2024-25 are shown in Chart 4.13. It is observed that between 2015-16 and 2017-18, India imported significant quantities of gram with an annual average 10.3 lakh tonnes. However, volume of imports steeply declined in 2018-19 and remained low during 2018-19 to 2023-24, importing about 2.3 lakh tonnes per year. In contrast, gram imports witnessed significant growth both in volume (467%) and value (481.2%) in 2024-25 over last year due to lower production and zero import duty.

Chart 4.13: India's Imports of Gram, 2015-16 to 2024-25



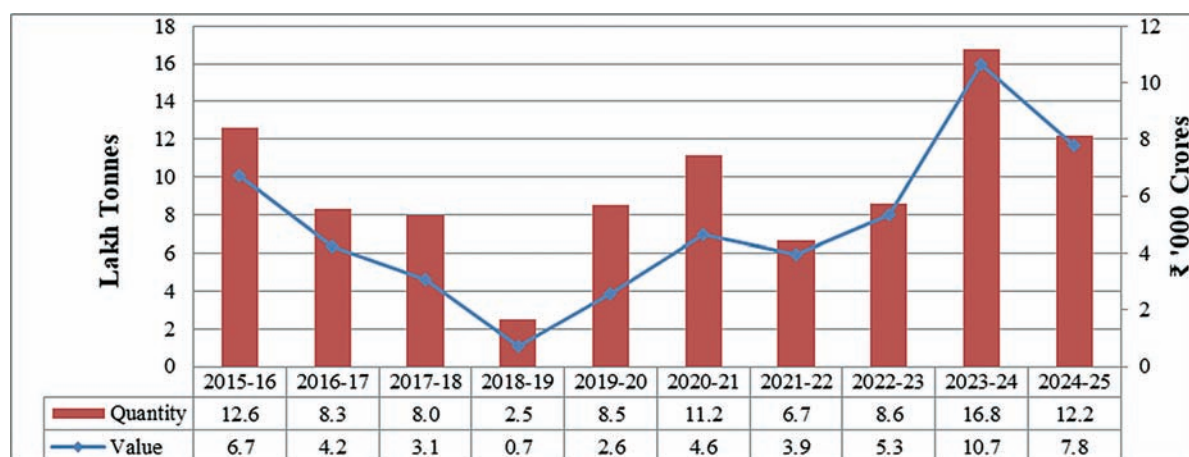
Source: Directorate General of Commercial Intelligence and Statistics, Ministry of Commerce and Industry



India's Trade in Lentil (Masur)

- 4.31 Despite a leading producer of lentil in the world, India relies heavily on imports due to rising domestic consumption. Chart 4.14 depicts India's imports of lentil from 2015-16 to 2024-25. Lentil imports declined steadily from 12.6 lakh tonnes in 2015-16 to 2.5 lakh tonnes in 2018-19, driven by high domestic production combined with hike in import duty in December 2017. However, lentil imports rebounded from 2018-19 and increased to 11.2 lakh tonnes in 2020-21, as a result of reduction in import duty from 30 percent to 10 percent between June and December 2020. Thereafter, lentil imports fell significantly in 2021-22, but strongly rebounded, reaching a record level of 16.8 lakh tonnes in 2023-24. Government's decision to reduce import duty to zero percent coupled with subdued international prices attributed to a steep rise in imports of lentil in 2023-24. However, imports declined to 12.2 lakh tonnes worth ₹7.8 thousand crore in 2024-25 due to improved domestic production during the last two years.

Chart 4.14: India's Imports of Lentil, 2015-16 to 2024-25



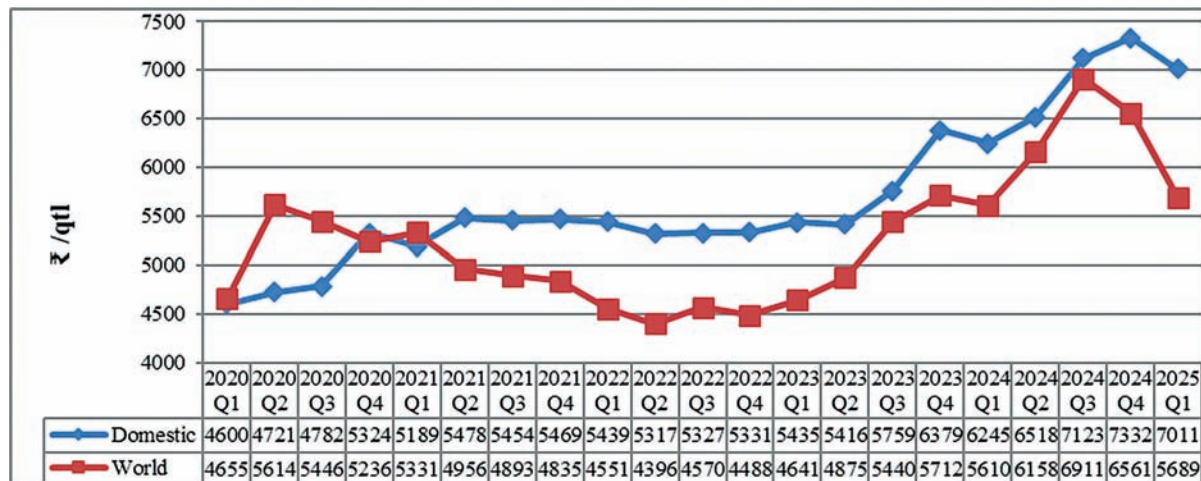
Source: Directorate General of Commercial Intelligence and Statistics, Ministry of Commerce and Industry

Comparative Price Trends in Gram and Lentil

- 4.32 Chart 4.15 shows comparative trends in domestic and international prices of gram for the period 2020 to 2025(Q1). It is observed that the domestic prices of gram have remained higher than the international prices during the whole period, except 2020(Q1 to Q3) and 2021(Q1). The domestic prices of gram followed an upward trend till 2021(Q2), barring a marginal fall in 2021(Q1). Thereafter, prices remained stagnant till 2023(Q2). However, between 2023(Q2) and 2024(Q4), significant increase was noticed in domestic prices, which was subsequently followed by marginal decline in 2025(Q1). However, the international prices of gram, which increased sharply from 2020(Q1) to 2020(Q2), followed a downward trajectory thereafter till 2022(Q2). From 2022(Q2) till 2024(Q3), international prices of gram continued to increase, except a few marginal falls in between. However, from 2024(Q3) to 2025(Q1), international prices recorded a steep decline compared to marginal decline in domestic prices.



Chart 4.15: Domestic and International Prices of Gram, 2020 to 2025(Q1)

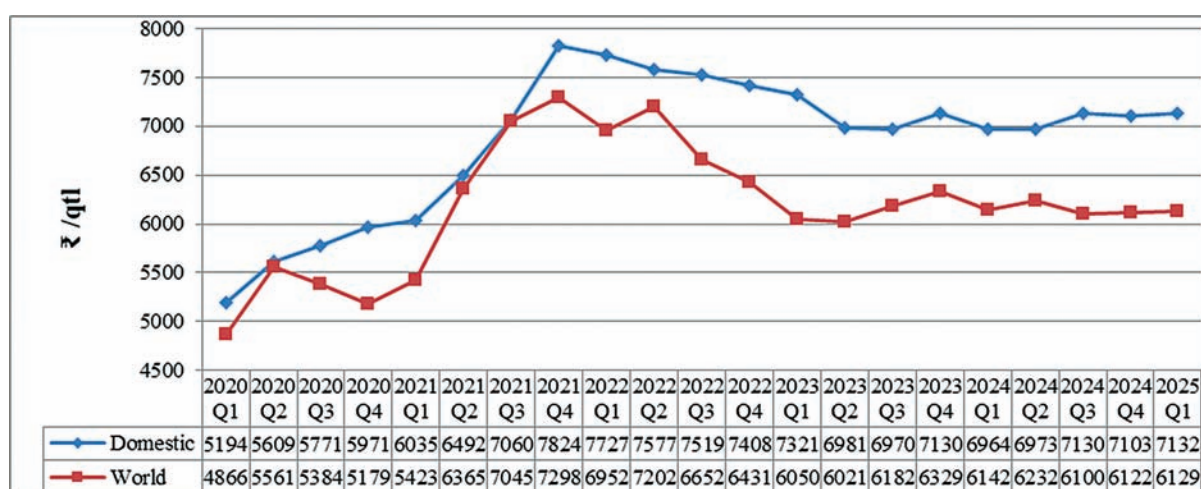


Note: International prices are for Australian gram till 2021(Q1) and Tanzanian gram from 2021 (Q2) to 2025 (Q1) delivered at Mumbai port

Source: 1. Agriculture Marketing Division, Department of Agriculture and Farmers Welfare for domestic wholesale prices
2. Agriwatch for International prices

4.33 The domestic and international prices of lentil from 2020 to 2025(Q1) are shown in Chart 4.16. It can be seen that the domestic prices remained higher than international prices during the entire period. Both the prices significantly increased from 2020(Q1) and hit the peak in 2021(Q4), except brief dip in international prices in 2020(Q3 & Q4). Thereafter, both the prices continued a declining trend from 2021(Q4) till 2023(Q2), except marginal increase in international prices in 2022(Q2). However, both the prices remained mostly stagnant from 2023(Q2) onwards. The correlation coefficient of 0.88 indicates a strong positive association between domestic and international prices during this period.

Chart 4.16: Domestic and International Prices of Lentil, 2020 to 2025(Q1)



Note: International prices are of Canadian lentil delivered at Mumbai port

Source: 1. Agriculture Marketing Division, Department of Agriculture and Farmers Welfare for domestic wholesale prices
2. Agriwatch for International prices



Trade Policy

- 4.34 Pulses trade policy in India is based on the twin objectives of achieving self-sufficiency and stabilizing domestic market prices. Major changes in export-import policy in pulses are given in Annex Table 4.3(a) and recent changes are summarized below:

Gram

- 4.35 In order to keep the domestic prices of pulses under control, Government decided to import pulses including gram at zero duty in June 2006, which continued at the same level till December 2017. Thereafter, to safeguard interest of domestic growers after bumper production, 30 percent Basic Custom Duty (BCD) was imposed on gram imports in December 2017, which was raised to 40 percent in February 2018 and to 60 percent in March 2018.
- 4.36 In February 2021, BCD on gram was reduced from 60 percent to 10 percent and an additional 50 percent Agriculture Infrastructure and Development Cess (AIDC) was imposed. Subsequently, in order to check rising domestic prices, Government reduced import duty on gram to zero percent w.e.f. 4th May 2024 till 31st March 2025. However, Government imposed 10 percent import duty on gram from 1st April 2025, which helped in improving market prices of gram.

Lentil

- 4.37 Due to more than 30 percent increase in domestic lentil production and to safeguard interest of farmers, import duty on lentil was raised to 30 percent in December 2017, which was further increased to 50 percent for lentil of USA origin in June 2019. However, due to continuous decline in production in 2018-19 and 2019-20, Government reduced the BCD on lentil to 10 percent (30% for lentil of US-origin) on 2nd June 2020, valid upto 31st August 2020, which was extended upto 31st December 2020. Subsequently, BCD on lentil was increased to 30 percent on 1st January 2021, which was later replaced with BCD of 10 percent and AIDC of 20 percent on 2nd February 2021. However, due to lower domestic production in 2021-22, followed by inflationary pressure, BCD on lentil was reduced to zero percent, while AIDC was reduced to 10 percent on 27th July 2021 and to zero percent on 13th February 2022 till 31st March 2023. The validity period of import duty exemption on lentil was extended till 31st March 2025. However, due to improved production in 2023-24 and 2024-25, Government increased import duty on lentil from zero percent to 10 percent (5% BCD plus 5% AIDC) from 8th March 2025.
- 4.38 At present, exports of both gram and lentil from India are under “free” category and do not require any export authorization from DGFT.

Edible Oils and Oilseeds

Global Production and Trade

- 4.39 As per USDA, global production of major oilseeds in TE2024-25 was 657.4 million tonnes, out of which about 31 percent was traded. Brazil was the largest producer accounting for 25.5 percent of global share, followed by USA (19.1%), China (10.2%), Argentina (7.2%) and India (6.4%) in TE2024-



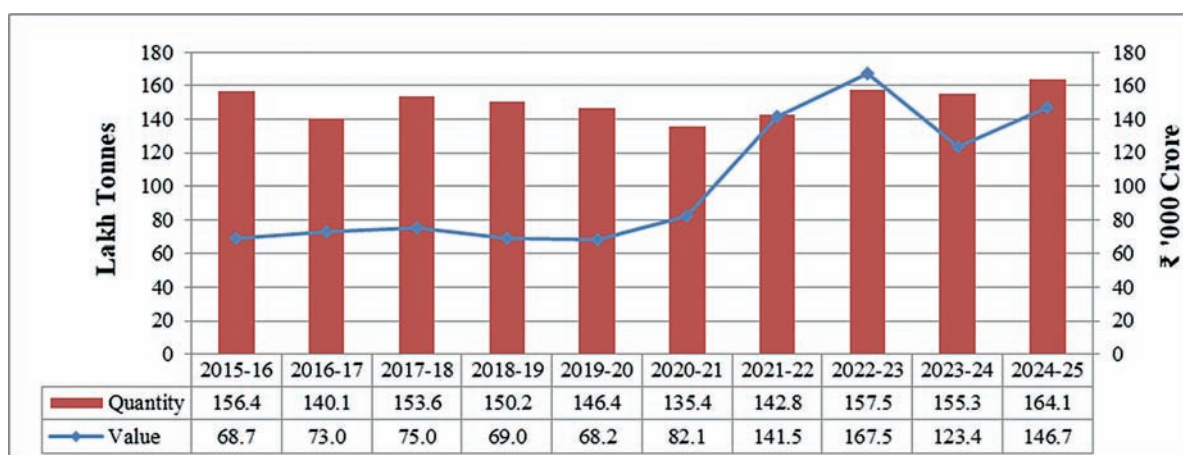
25. Brazil (49.7%) and USA (25%), the world's largest oilseeds exporters, together accounted for 74.7 percent of the global oilseeds exports. Among importers, China was the largest importer accounting for more than half of the global imports (56.8%), distantly followed by EU (10.8%) and Mexico (4%).

- 4.40 Global production of vegetable oils was 222.9 million tonnes in TE2024-25, out of which 37.7 percent was traded, as per USDA. Indonesia was the largest producer (22.8%) of the vegetable oils in the world in TE2024-25, followed by China (14%), Malaysia (9.6%) and EU (8.2%), while Indonesia (30.5%) and Malaysia (19.9%) accounted for more than half of the global exports of vegetable oils. It is pertinent to note that, in TE2024-25, India was the largest importer of vegetable oils, with a global share of 20 percent, followed by China (12%) and EU (11.2%).

India's Trade

- 4.41 Although India has witnessed impressive increase in domestic oilseeds production in the recent years, the country continues to depend on imports to meet the rising domestic demand for edible oils, which is mainly driven by higher incomes, changing dietary pattern, urbanization and liberal trade policies.
- 4.42 The trends of India's imports of edible oils from 2015-16 to 2024-25 are shown in Chart 4.17. It is observed that India's edible oils imports (in value term) are highly volatile due to large fluctuations in global prices and supply. Edible oil imports in India were recorded at 156.4 lakh tonnes in 2015-16, which significantly declined in 2016-17 but its import value continued to surge due to high global prices. Although, imports witnessed a noticeable growth in 2017-18, but from 2018-19, volume of imports followed a downward trajectory, reaching the decadal lowest at 135.4 lakh tonnes in 2020-21. Subsequently, the trend reversed and imports of edible oils reached 157.5 lakh tonnes in 2022-23. It is worth noting that import bill, which was ₹82.1 thousand crore in 2020-21 surged to ₹167.5 thousand crore in 2022-23, an increase of 104 percent primarily driven by very high world prices. On the contrary, in 2023-24, although volume of imports of edible oils declined marginally but significant fall was recorded in value of imports (26.4%) over previous year owing to steep fall in the international prices of edible oils and lower import duties. However, volume of imports increased by 5.7 percent but value of imports surged by 19 percent in 2024-25 over previous year due to higher global prices.

Chart 4.17: India's Imports of Edible Oils, 2015-16 to 2024-25



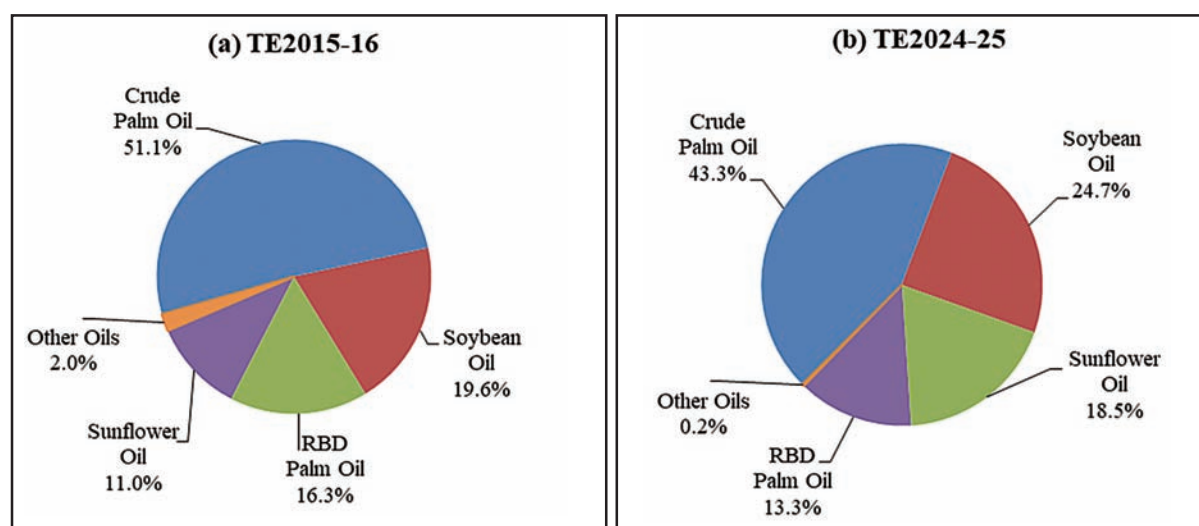
Source: Directorate General of Commercial Intelligence and Statistics, Ministry of Commerce and Industry



Changing Composition of Imports

- 4.43 Although, India is the world's largest importer of palm oil but composition of edible oil imports in the country has changed structurally over the years. The composition of imported edible oils in TE2015-16 and TE2024-25 is shown in Chart 4.18 (a) and (b), respectively. It is observed that the combined share of both RBD and crude palm oil has decreased from 67.4 percent in TE2015-16 to 56.6 percent in TE2024-25, while share of soft oil namely, soybean oil has increased from 19.6 percent to 24.7 percent and sunflower oil from 11 percent to 18.5 percent during the period. It indicates that the share of soft oils in total imports has increased significantly during the last 10 years.

**Chart 4.18: Share of Major Edible Oils in Total Edible Oils Imports (in volume):
TE2015-16 and TE2024-25**



Source: Directorate General of Commercial Intelligence and Statistics, Ministry of Commerce and Industry

- 4.44 With an objective to increase domestic oilseeds production and achieve self-reliance in edible oils, Government approved the National Mission on Edible Oils–Oilseeds (NMEO–Oilseeds) on 3rd October 2024 with a financial outlay of ₹10,103 crore and National Mission on Edible Oils–Oil Palm (NMEO–OP) in 2021, with an outlay of ₹11,040 crore. The NMEO–Oilseeds, being implemented over a seven-year period from 2024-25 to 2030-31, targets to increase oilseed production from 39 million tonnes in 2022-23 to 69.7 million tonnes by 2030-31. The major focus of the NMEO–Oilseeds is to enhance production of primary oilseed crops such as R&M, groundnut, soybean, sunflower, and sesamum, as well as increasing collection and extraction efficiency from secondary sources like cottonseed, rice bran, and tree borne oils and the main objective of NMEO–OP is to enhance area under oil palm cultivation and increase palm oil production in the country.

Rapeseed and Mustard (R&M) Seed

Global Production and Trade

- 4.45 As per USDA, global production of rapeseed was 88.3 million tonnes in TE2024-25, out of which 21.7 percent was traded. With a share of 21.5 percent each, EU and Canada were the largest producers

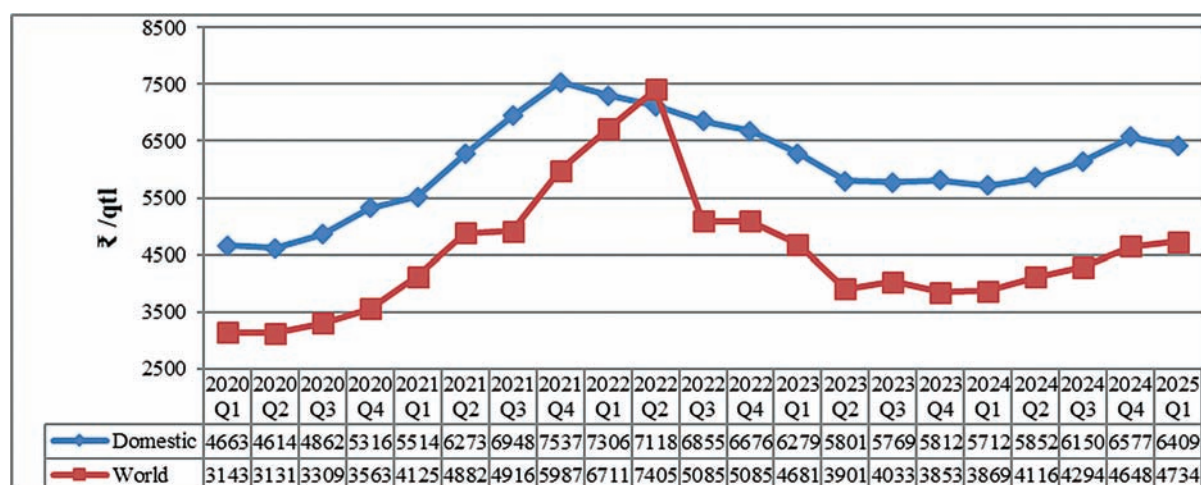


of rapeseed, followed by China (18.1%) and India (13%). Canada continued to be the largest exporter with a share of 41 percent, while EU was the largest importer (33.6%), followed by China (27.6%) and Japan (10.8%) in TE2024-25.

Comparative Trends in Prices

- 4.46 The comparative trends in domestic and international prices of R&M seed during the period 2020 to 2025(Q1) are shown in Chart 4.19. It is observed that domestic prices of R&M seed remained above the world prices during the entire period, except 2022(Q2). Barring a marginal dip in 2020(Q2), domestic prices of R&M seed not only exhibited an upward trend between 2020(Q1) to 2021(Q4) but also reached the historical high at ₹6,948/quintal in 2021(Q4). Similarly, international prices of R&M seed moved steeply upward from 2020(Q3) till 2022(Q2) and surpassed domestic prices by hitting the peak in 2022(Q2). However, from 2021(Q4), domestic prices followed a downward trend till 2023(Q3) and moved marginally upward thereafter, except 2024(Q1) and 2025(Q1). In case of international prices, a sharp decline of 47.3 percent was noticed between 2022(Q2) and 2023(Q2) but global prices recorded a rising trend since 2024(Q1).

Chart 4.19: Domestic and International Prices of R&M Seed, 2020 to 2025(Q1)



Note: International prices are R&M Oilseed, Hamburg CIF

Source: 1. Agriculture Marketing Division, Department of Agriculture and Farmers Welfare for domestic wholesale prices
2. FPMA for International prices (earlier taken from USDA Report where data on these prices are discontinued from September, 2023)

Rapeseed/Mustard Oil

Global Production and Trade

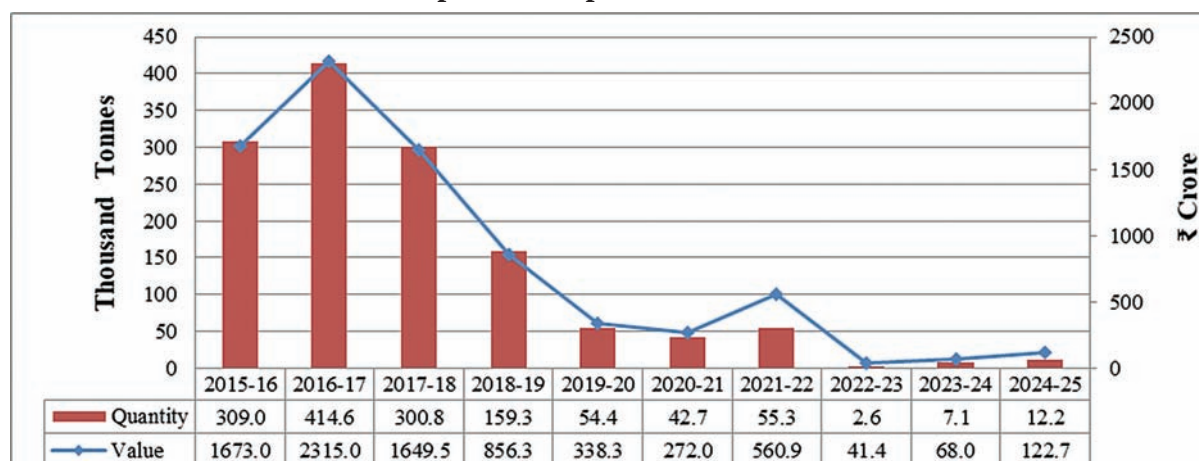
- 4.47 Global production of rapeseed oil was 33.8 million tonnes in TE2024-25, as per USDA, out of which 21 percent was traded. With a share of 29.8 percent, EU was the largest producer of rapeseed oil in the world, followed by China (22.8%), Canada (13.3%) and India (11.6%) in TE2024-25. While Canada was the largest exporter of rapeseed oil accounting for 46.2 percent of the global exports, China was the largest importer with a share of 28 percent in global imports in TE2024-25.



India's Trade

- 4.48 Chart 4.20 depicts the trends in India's imports of rapeseed oil during the period 2015-16 to 2024-25. It can be seen that imports of rapeseed oil increased significantly during 2016-17, but witnessed a declining trend thereafter due to increase in import duty on edible oils in November 2017 and increase in domestic production. In 2021-22, import volume grew by 29.5 percent but value of imports more than doubled over previous year due to steep rise in world prices. Imports of rapeseed oil remained low during 2022-23 but rose in the last two years.

Chart 4.20: India's Imports of Rapeseed Oil, 2015-16 to 2024-25



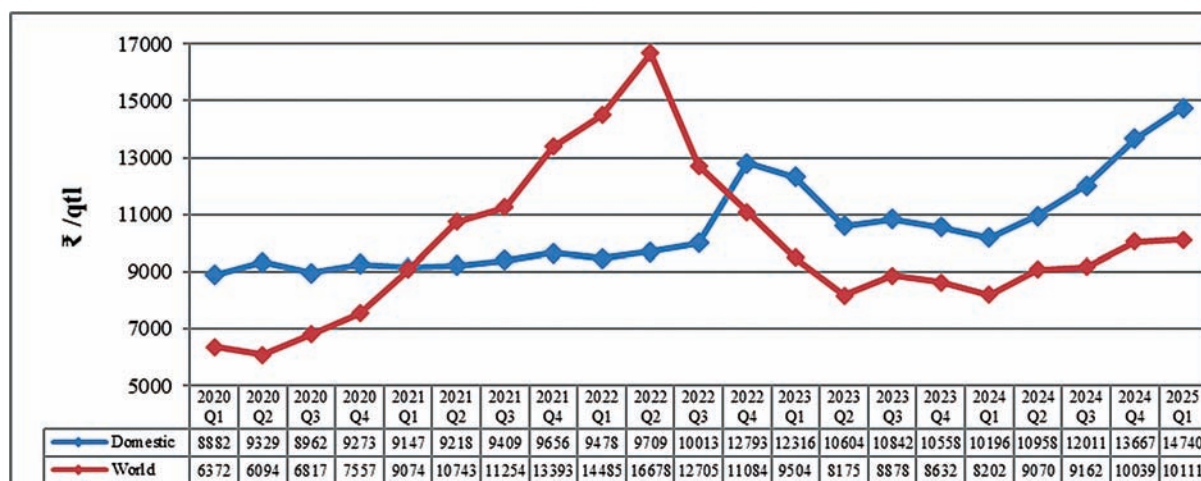
Source: Directorate General of Commercial Intelligence and Statistics, Ministry of Commerce and Industry

Comparative Trends in Prices

- 4.49 The comparative trends in domestic and international prices of rapeseed oil during the period 2020 to 2025(Q1) are shown in Chart 4.21. It is observed that domestic prices remained significantly higher than the international prices during 2020 and 2021(Q1) but global prices witnessed a steep rise, moved above domestic prices in 2021(Q2) and reached the record high in 2022(Q2). From 2022(Q2) to 2023(Q2), the trend reversed and international prices sharply declined but domestic prices continued to rise till 2022(Q4). Both domestic and world prices showed an increasing trend during the last five quarters and rate of increase in prices was much higher in domestic prices.



Chart 4.21: Domestic and International Prices of Rapeseed/Mustard Oil, 2020 to 2025(Q1)



Source: 1. Agriculture Marketing Division, Department of Agriculture and Farmers Welfare for domestic wholesale prices at Mumbai centre
2. World Bank for International prices

Trade Policy of Oilseeds and Edible Oils

- 4.50 Government reviews import duty structure and other trade policy measures for major edible oils from time to time in order to ensure remunerative prices to oilseeds growers, reduce import dependencies, achieve self-sufficiency in edible oils and protect interests of both consumers and producers. The following section discusses the recent changes in export-import policy of major edible oils, while changes in trade policy of edible oils since 2015 are given in Annex Table 4.3(b).

Import Policy

- The effective import duty on crude palm oil (CPO) was increased from 27.5 percent to 32.5 percent (15% BCD + 17.5% AIDC) in February 2021 but for crude soybean and sunflower oil, effective duty remained 35 percent (15% BCD + 20% AIDC), which narrowed the duty gap between palm oil and other edible oils. However, Government reduced the effective duty on crude palm oil from 32.5 percent to 27.5 percent (10% BCD + 17.5% AIDC) and on refined variant to 37.5 percent in June 2021.
- In August 2021, effective import duty on crude soybean and sunflower oil was reduced from 35 percent to 27.5 percent (7.5% BCD + 20% AIDC) and on refined variant of these oils from 45 percent to 37.5 percent.
- In September 2021, Government slashed the effective import duty on crude palm, soybean and sunflower oil to 22.5 percent (2.5% BCD + 20% AIDC) and on refined oils to 32.5 percent.
- In October 2021, due to high world prices, effective duty on refined variant of all these oils was further reduced to 17.5 percent and duty on crude palm oil was reduced to 7.5 percent (0% BCD + 7.5% AIDC) and in case of crude soybean and sunflower oil, the duty was reduced to 5 percent (0% BCD + 5% AIDC).



- Subsequently, Government reduced the effective import duty on refined palm oil to 12.5 percent in December 2021, and on crude palm oil to 5 percent (0% BCD + 5% AIDC) in February 2022. The effective import duty on refined soybean and sunflower oil was also reduced from 17.5 percent to 12.5 percent in June 2023.
- Government increased the duty on crude soybean and sunflower oil from 5 percent to 25 percent (20% BCD + 5% AIDC) and on refined variants from 12.5 percent to 32.5 percent in September 2024. Similarly, duty on crude palm oil was also increased from 5 percent to 25 percent (20% BCD + 5% AIDC) and on refined palm oil from 12.5 percent to 32.5 percent in September 2024.
- In May 2025, duty on crude palm, soybean and sunflower oil was reduced from 25 percent to 15 percent (10% BCD + 5% AIDC).
- In case of crude rapeseed and cottonseed oil, the effective import duty is 35 percent, while refined rapeseed and cottonseed oil attracts 45 percent import duty.
- Government has imposed 10 percent Social Welfare Cess, over and above the effective import duty on all edible oils from February 2018. The current import duty structure of edible oils is given in Table 4.2.

Table 4.2: Import Duty on Edible Oils w.e.f. 30th May 2025

Edible Oils	BCD (%)	AIDC (%)	Social Welfare Cess (%)	Effective Applicable Import Duty (%) *
Crude Palm Oil	10.0	5.0	10.0	16.50
RBD Palmolein	32.5	Nil	10.0	35.75
RBD Palm Oil	32.5	Nil	10.0	35.75
Crude Soybean Oil	10.0	5.0	10.0	16.50
Refined Soybean Oil	32.5	Nil	10.0	35.75
Crude Sunflower Oil	10.0	5.0	10.0	16.50
Refined Sunflower Oil	32.5	Nil	10.0	35.75
Crude Rapeseed Oil	35.0	Nil	10.0	38.50
Refined Rapeseed Oil	45.0	Nil	10.0	49.50
Crude Cottonseed Oil	35.0	Nil	10.0	38.50
Refined Cottonseed Oil	45.0	Nil	10.0	49.50

Note: * includes Social Welfare Cess at 10% of BCD plus AIDC

Source: Central Board of Indirect Taxes and Customs, Department of Revenue

Export Policy

- 4.51 Currently, oilseeds exports from India, except breeder/foundation/wild variety seeds, are under “free” category. However, edible oils export was banned from 17th March 2008 and export of rice bran oil in bulk was permitted from India w.e.f 6th February 2015. For groundnut oil, sesame oil, soybean oil



and maize (corn) oil, exports were permitted w.e.f 27th March 2017. Exports of all edible oils, except mustard oil were made free without quantitative restrictions, pack size etc. w.e.f. 6th April 2018. However, exports of mustard oil are permitted in pack of up to 5 kg with a Minimum Export Price (MEP) of US\$ 900 per tonne.

World Trade Outlook

Cereals

Wheat

- 4.52 Global wheat outlook projects record production, consumption, higher trade but contraction in ending stock in 2025-26 according to USDA. Wheat production is projected to increase due to larger crop in EU and India, offsetting decline in Kazakhstan, Australia, Pakistan and USA. Global trade is expected to expand due to higher imports from Turkey and China and higher exports from EU, Argentina, Australia and Russia. FAO Food Outlook also forecasts a higher production, consumption and trade but lower global wheat stock in 2025-26.
- 4.53 According to OECD-FAO Agricultural Outlook 2024-2033, global wheat production is estimated to increase by 83 million metric tonnes (Mt) to 872 Mt by 2033 of which 44 Mt will be from Asia. India is expected to provide the largest share of additional wheat, accounting for more than a quarter of the global production increase, driven by yield improvements and area expansion. Consumption of wheat is expected to be 11 percent higher in 2033 than in 2024 and more than half of this increase will originate from India, Pakistan, Egypt and China. Global wheat exports are projected to grow to 220 Mt by 2033, with Russia expected to maintain its position as the leading exporter with 26 percent share. The real prices of wheat are expected to decline over the medium-term.

Barley

- 4.54 Global outlook for barley in 2025-26 indicates higher production and consumption but lower trade, as per USDA estimates. The higher production will be due to larger crop in EU, Russia and China, while global trade will increase due to higher exports from Ukraine, EU and Russia. With global production slightly lower than global consumption, ending stocks are projected to decline modestly. As regards India, barley production is projected to decline but consumption is estimated to increase in 2025-26.

Pulses

- 4.55 Global supply of pulses is expected to increase by 25 Mt by 2033 and almost half of this increase will come from Asia, particularly from India, as per OECD-FAO Outlook. India's domestic production is projected to increase by an additional 8 Mt by 2033 due to sustained yield improvements. World trade in pulses, which grew from 15 Mt to 19 Mt over the past decade, is projected to reach 22 Mt by 2033. Canada is expected to remain the major exporter of pulses, followed by Australia and Russia.

Edible Oils and Oilseeds

- 4.56 According to FAO Food Outlook (June 2025), in 2024-25, global oilseed production is expected to increase due to increase in soybean, groundnut and cotton seed outputs, which will compensate for



lower rapeseed and sunflower seed production. However, global vegetable oils market will remain tight due to subdued growth in palm, rapeseed and sunflower oil output. As regards 2025-26, the Outlook although forecasts a continued rise in oilseed production but expects a slower pace in growth in global vegetable oils output.

- 4.57 According to USDA, global oilseed production is forecast to increase by 2 percent in 2025-26 compared to previous year, driven by higher soybean output and rebound in rapeseed and sunflower seed production. Global oilseed consumption is also forecast to increase in 2025-26 due to higher rapeseed, peanut and sunflower seed crush. Global oilseed trade is expected to increase with higher soybean exports, which will offset a decline in rapeseed exports. While record growth is expected in exports of soybean from Brazil, imports growth will be led by China, Argentina and Egypt.
- 4.58 Global vegetable oil production is forecast to increase by 2.8 percent in 2025-26 over previous year due to higher palm oil production in Indonesia and recovery in Malaysia, as per USDA. Soybean oil production is also forecast to reach another record high as China, USA, Argentina, and Brazil are likely to increase crushing of soybean. Global vegetable oil consumption is anticipated to grow, due to higher demand for food use than industrial use in 2025-26. Global vegetable oil trade is forecast to increase owing to recovery in palm and sunflower seed oil trade.
- 4.59 As per OECD-FAO Agricultural Outlook 2024-2033, vegetable oils use for food is projected to account for 55 percent of total consumption in 2033, driven by population growth and higher per capita use of vegetable oils in lower and middle-income countries. The two leading suppliers of palm oil, Indonesia and Malaysia, will continue to dominate the global vegetable oils trade. However, adoption of stringent environmental and sustainable agricultural norms (in line with the 2030 UN Agenda for Sustainable Development) by key importing nations are expected to adversely affect expansion of oil palm area in Indonesia and Malaysia and growth in production would primarily come from productivity improvements and speeding up of replanting. India is projected to maintain its high annual vegetable oils import growth due to growing domestic demand driven by population growth, urbanization and rising income. Oilseeds and its products prices are expected to increase slightly in nominal terms, but decline in real terms. However, world vegetable oils prices are projected to increase, driven by higher demand but the sector faces major uncertainty due to biofuel policies in the USA, EU and Indonesia.

Recapitulation

- 4.60 As per Global Trade Outlook and Statistics, merchandise trade volume, which increased by 2.9 percent in 2024, is expected to decline in 2025 before posting a modest recovery in 2026. As per USDA and FAO Food Outlook, global production, consumption and trade of wheat is forecast to increase in 2025-26. For barley, although production and consumption prospects are optimistic, but trade would remain lower. As regards major oilseeds, production of soybean, rapeseed and sunflower seed is projected to increase and world oilseeds trade is estimated to be higher due to rise in soybean exports in 2025-26.
- 4.61 Indian agricultural trade has exhibited strong resilience despite continued geopolitical tensions and



trade uncertainties. With higher increase in agri-imports (19.6%) than agri-exports (8.6%), trade surplus in agriculture has declined from ₹127.7 thousand crore in 2023-24 to ₹106.9 thousand crore in 2024-25. However, fast growing demand, lower domestic production and low/zero import duty have led to record high imports of pulses in 2024-25. In case of edible oils, there was significant rise in both volume and value of imports during 2024-25. In order to reduce India's heavy reliance on imports and ensure self-sufficiency in pulses and edible oils, sustained efforts are needed to increase acreage, improve yield and bridge the yield gap. Furthermore, a conducive trade policy to boost domestic production of pulses and oilseeds is the most crucial.



CHAPTER 5



Costs, Returns and Inter-Crop Parity

- 5.1 The Commission considers various factors such as cost of production, demand-supply situation, price trends, inter-crop parity, terms of trade, likely impact of price policy on the economy, rational utilization of land, water and other production resources, and a minimum 50 percent margin over cost of production, while recommending Minimum Support Price (MSP) for mandated crops. The Commission holds wide-ranging consultations with farmers and farmer associations/organizations, Central and State Governments, research organizations, industry associations, private food and agribusiness sector, etc.
- 5.2 The Commission uses State-wise cost estimates of crops compiled under 'Comprehensive Scheme (CS) for studying the Cost of Cultivation of Principal Crops in India', provided by the Economics, Statistics & Evaluation Division (ES&E Division) of Ministry of Agriculture and Farmers Welfare, Government of India. Since CS data are available up to the crop year 2023-24, cost projections are made for crop year 2025-26.
- 5.3 The projected Cost of Cultivation (CoC) for the crop year 2025-26 is calculated from the latest three-year actual cost estimates (2021-22 to 2023-24) for all States. The CoC estimates have not been projected for a crop in the State, where actual cost estimates are available for only one year. Additionally, projections have also not been made if the share of a State in all-India production or the crop share in the total production of the crop group within the State is negligible, or if the number of sample holdings under the CS for a crop is insufficient.
- 5.4 The projected CoC for crop year 2025-26 captures movement in overall input cost separately for each of the three years viz. 2021-22, 2022-23 and 2023-24 for each of the projected State. An assessment of likely changes in input cost for the crop year 2025-26, on the basis of 2021-22, 2022-23 and 2023-24, is made by constructing Composite Input Price Indices (CIPIs) (base 2011-12=100) for each State for every projected crop. The CIPIs are based on latest prices of different inputs like human labour, bullock labour, machine labour, fertilisers and manures, seeds, pesticides and irrigation as per data available from Labour Bureau, Ministry of Labour and Employment, State Governments and Office of the Economic Adviser, Ministry of Commerce and Industry. Based on these CIPIs, the Commission projected State-wise $CoC A_2$, $A_2 + FL$ and C_2 for each mandated *rabi* crop.



- 5.5 The State-wise Cost of Production (CoP) A_2 , A_2 +FL and C_2 projections for the crops are calculated by using respective projected CoC estimates, ratio of value of Main Product (MP) to Gross Value of Output (GVO) and estimated yield. Subsequently, all-India estimates of CoP A_2 , A_2 +FL and C_2 for a crop are derived as weighted average of State-wise CoPs using their production share in total production as weights. These projected all-India CoP estimates are then considered by the Commission for formulating price policy recommendations.
- 5.6 The Commission has undertaken cost projection exercise under certain implicit assumptions. First, it is assumed that fixed cost components will not undergo any significant change during the period for which actual cost estimates are available, and the crop year for which cost projections are being made. Second, given the inherent variability of yield from year to year, three-year average yield for TE2023-24 has been taken for smoothing fluctuations in yield and consequently in CoP. However, if there are wide fluctuations, Olympic average yield¹ has been used or yield has been estimated based on the time series yield data under Comprehensive Scheme. In a few cases, yield has been estimated based on yield data provided by ES&E Division, Ministry of Agriculture and Farmers Welfare, Government of India.

Costs and Returns during TE2023-24

- 5.7 The all-India average costs, Gross Value of Output (GVO) and gross returns during TE2023-24 in respect of mandated *rabi* crops have been analysed and are given in Table 5.1. The GVO is estimated at prevailing market prices of main product and by-products during harvest season in the village/cluster of villages where the crops are grown and harvested.

Cost of Cultivation

- 5.8 During TE2023-24, all-India per hectare average CoC A_2 and A_2 +FL was highest for barley at ₹37,225 and ₹55,507 and lowest for safflower at ₹16,150 and ₹20,772, respectively. In cereals, CoC A_2 for barley was marginally higher than wheat but A_2 +FL was significantly higher than that of wheat due to higher use of family labour in barley cultivation. In both the cereals, Himachal Pradesh had the lowest A_2 (Table 5.1(a)), due to low use of hired human labour and higher use of family labour. Rajasthan had the highest A_2 +FL in both the cereals, due to high usage of human, bullock and machine labour.
- 5.9 In pulses, CoC in lentil was lower than gram due to lower expenditure on farm inputs such as labour (human, bullock and machine), seed, fertilisers, manure, insecticides, irrigation etc. In case of gram, Haryana has the lowest A_2 and A_2 +FL (Table 5.1(a)), due to lower expenditure on labour (human, bullock and machine), while Jharkhand has the highest CoC A_2 +FL due to substantially high expenditure on labour (human, bullock and machine) compared to other States. In case of lentil, the CoC was highest in West Bengal due to substantially higher expenditure on human, bullock and machine labour compared to other States.
- 5.10 In oilseeds, the CoC was lower for safflower than rapeseed & mustard due to significantly lower cost incurred on farm inputs such as human and machine labour, seed, fertilizer & manure, irrigation. Gujarat had the highest CoC in case of rapeseed & mustard due to higher expenditure on labour (human,

¹Olympic average yield is calculated by dropping the highest and the lowest yield from latest five-year yields and calculating the average of the remaining three-year yields.



bullock and machine) and irrigation than other States. The details of actual costs and gross returns over actual CoC A_2 and A_2+FL of mandated *rabi* crops during TE2023-24 at all-India and State level are given in Annex Table 5.1 and Annex Table 5.2, respectively.

Gross Value of Output

- 5.11 All-India average GVO among mandated crops during TE2023-24 was highest for barley at ₹1,05,036 per hectare, due to higher value of by-products and lowest for safflower at ₹17,886 per hectare, primarily due to low yield. In case of wheat, four States had higher GVO than all-India average and highest GVO was realised for Rajasthan (₹1,16,105/ha) due to relatively higher price, while GVO was lowest for Chhattisgarh (₹55,105/ha) due to lower yield. In case of barley, highest GVO was recorded for Rajasthan (₹1,15,920/ha) due to higher yield and prices, and lowest for Himachal Pradesh (₹32,892/ha) due to lower yield and prices.
- 5.12 In case of pulses, GVO of gram (₹66,823/ha) was higher than lentil (₹56,386/ha), due to higher yield during TE2023-24. In case of gram, five States had higher GVO than all-India average with Andhra Pradesh (₹96,936/ha) at the top and Jharkhand (₹38,072/ha) at the bottom, primarily due to low yield. For lentil, highest GVO was recorded for West Bengal driven by higher yield, and lowest for Madhya Pradesh due to lower prices.
- 5.13 For oilseeds, GVO of rapeseed & mustard (₹87,877/ha) was significantly higher than safflower (₹17,886/ha) due to higher productivity level of rapeseed & mustard. For rapeseed & mustard, five out of 11 States had higher GVO than all-India average with Gujarat (₹1,22,369/ha) having the highest GVO due to higher yield and Himachal Pradesh with the lowest GVO (₹34,636/ha) due to lower yield.

Table 5.1: All-India Average Costs and Gross Returns over Cost of Cultivation (₹/ha) of Mandated Crops during TE2023-24

Crop	CoC A_2	CoC A_2+FL	GVO	Gross Returns over	
				A_2 (Col.4 - Col.2)	A_2+FL (Col.4 - Col.3)
(1)	(2)	(3)	(4)	(5)	(6)
Cereals					
Wheat	36290	45558	96014	59724	50456
Barley	37225	55507	105036	67812	49530
Pulses					
Gram	30802	39632	66823	36021	27192
Lentil	21687	28618	56386	34700	27769
Oilseeds					
Rapeseed & Mustard	28141	41025	87877	59736	46851
Safflower	16150	20772	17886	1736	-2885

Note: All-India CoC, GVO and gross returns of a crop are weighted average of respective CoCs, GVOs and gross returns of projected States

Source: CACP Calculations using CS data

**Table 5.1(a): Distribution of States according to Cost of Cultivation (CoC), TE2023-24**

Crop	CoC A ₂	
	$\geq$ All-India	$<$ All-India
Wheat	Maharashtra, Rajasthan, Uttar Pradesh, West Bengal	Punjab, Gujarat, Haryana, Bihar, Jharkhand, Madhya Pradesh, Chhattisgarh, Himachal Pradesh
Barley	Rajasthan	Uttar Pradesh, Himachal Pradesh
Gram	Maharashtra, Gujarat, Andhra Pradesh, Jharkhand	Chhattisgarh, Telangana, Madhya Pradesh, Karnataka, Rajasthan, Bihar, Uttar Pradesh, Haryana
Lentil	West Bengal, Bihar, Madhya Pradesh	Uttar Pradesh
Rapeseed & Mustard	Gujarat, West Bengal, Rajasthan, Punjab, Uttar Pradesh	Madhya Pradesh, Haryana, Bihar, Jharkhand, Assam, Himachal Pradesh
CoC A ₂ +FL		
Wheat	Rajasthan, Maharashtra, West Bengal, Uttar Pradesh	Gujarat, Jharkhand, Haryana, Bihar, Himachal Pradesh, Punjab, Madhya Pradesh, Chhattisgarh
Barley	Rajasthan	Uttar Pradesh, Himachal Pradesh
Gram	Jharkhand, Maharashtra, Gujarat, Andhra Pradesh, Rajasthan	Uttar Pradesh, Chhattisgarh, Madhya Pradesh, Karnataka, Bihar, Telangana, Haryana
Lentil	West Bengal	Uttar Pradesh, Bihar, Madhya Pradesh
Rapeseed & Mustard	Gujarat, West Bengal, Rajasthan	Uttar Pradesh, Punjab, Jharkhand, Haryana, Madhya Pradesh, Himachal Pradesh, Assam, Bihar

Note: States have been arranged in descending order of CoCs

Source: CACP Calculations using CS data

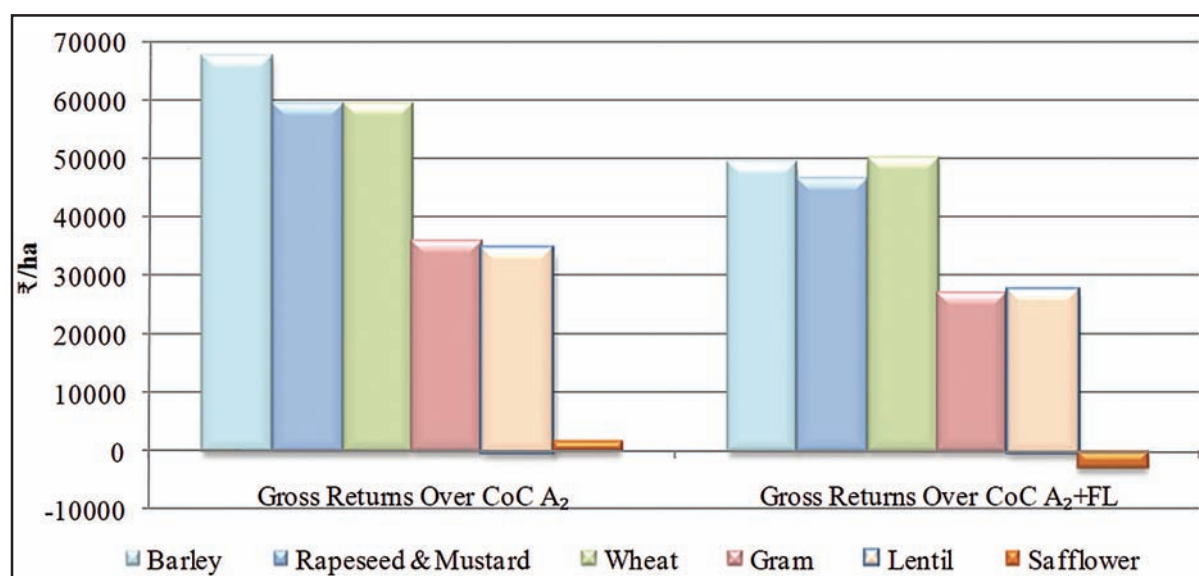
Gross Returns

- 5.14 Chart 5.1 represents all-India average gross returns over actual CoC A₂ and A₂+FL during TE2023-24 for mandated *rabi* crops. During TE2023-24, barley recorded the highest per hectare gross returns over CoC A₂ at ₹67,812, while wheat had the highest gross returns over CoC A₂+FL at ₹50,456. In contrast, safflower registered the lowest returns, with ₹1,736 over CoC A₂ and a negative return of (-) ₹2,885 over CoC A₂+FL. Among all crops, rapeseed & mustard showed the highest all-India average gross returns as a percentage of CoC A₂ (212.3%) and A₂+FL (114.2%), while safflower had the lowest in both cases.
- 5.15 Gross returns for wheat were highest for Haryana due to high yield and lower cost of production in the State and lowest for West Bengal due low yield and lower prices in the State (Table 5.1(b)). Per hectare returns over A₂ and A₂+FL for barley were highest for Rajasthan due to significantly lower cost whereas gross returns were lowest for Himachal Pradesh due to lower yield.



- 5.16 Per hectare returns of gram were highest for Andhra Pradesh due to significantly higher yield and higher output prices, while per hectare returns over A_2 and A_2+FL were lowest for Jharkhand, due to lower yield. For gram, seven out of 12 States have higher gross returns over CoC A_2 than all-India average, whereas six out of 12 States have higher gross returns over CoC A_2+FL than all-India average. Maharashtra, the second largest producer of gram, had lower returns over both CoC A_2 and A_2+FL compared with all-India average returns, due to significantly high cost in the State. In case of lentil, per hectare returns over A_2 were highest for Uttar Pradesh, the second largest producer, and lowest for West Bengal, while returns over A_2+FL were highest for Bihar and lowest for West Bengal. For rapeseed & mustard, per hectare returns over both A_2 and A_2+FL were highest for Haryana due to low cost of production and four out of 11 States have higher gross returns over CoC A_2 than all-India average, whereas five out of 11 States have higher gross returns over CoC A_2+FL than all-India average. The returns over A_2 were lowest for Himachal Pradesh and over A_2+FL , these were lowest for Jharkhand due to lower yield.

Chart 5.1: All-India Average Gross Returns of Crops, TE2023-24



Note: All-India CoC, GVO and gross returns of a crop are weighted average of respective CoCs, GVOs and gross returns of projected States

Source: CACP Calculations using CS data

Table 5.1(b): Distribution of States according to Gross Returns, TE2023-24

Crop	Gross Returns over CoC A_2	
	$\geq$ All-India	$<$ All-India
Wheat	Haryana, Rajasthan, Punjab	Uttar Pradesh, Madhya Pradesh, Gujarat, Bihar, Himachal Pradesh, Jharkhand, Maharashtra, Chhattisgarh, West Bengal
Barley	Rajasthan	Uttar Pradesh, Himachal Pradesh

(contd...)



Gram	Andhra Pradesh, Bihar, Gujarat, Chhattisgarh, Rajasthan, Madhya Pradesh, Uttar Pradesh	Telangana, Maharashtra, Haryana, Karnataka, Jharkhand
Lentil	Uttar Pradesh, Bihar	Madhya Pradesh, West Bengal
Rapeseed & Mustard	Haryana, Gujarat, Madhya Pradesh, Rajasthan	Punjab, Uttar Pradesh, West Bengal, Bihar, Assam, Jharkhand, Himachal Pradesh
Gross Returns over CoC A_2 + FL		
Wheat	Haryana, Punjab, Rajasthan	Madhya Pradesh, Gujarat, Uttar Pradesh, Bihar, Jharkhand, Himachal Pradesh, Chhattisgarh, Maharashtra, West Bengal
Barley	Rajasthan	Uttar Pradesh, Himachal Pradesh
Gram	Andhra Pradesh, Bihar, Chhattisgarh, Gujarat, Madhya Pradesh, Telangana	Maharashtra, Rajasthan, Haryana, Uttar Pradesh, Karnataka, Jharkhand
Lentil	Bihar, Madhya Pradesh, Uttar Pradesh	West Bengal
Rapeseed & Mustard	Haryana, Gujarat, Madhya Pradesh, Punjab, Uttar Pradesh	Rajasthan, Bihar, West Bengal, Assam, Himachal Pradesh, Jharkhand

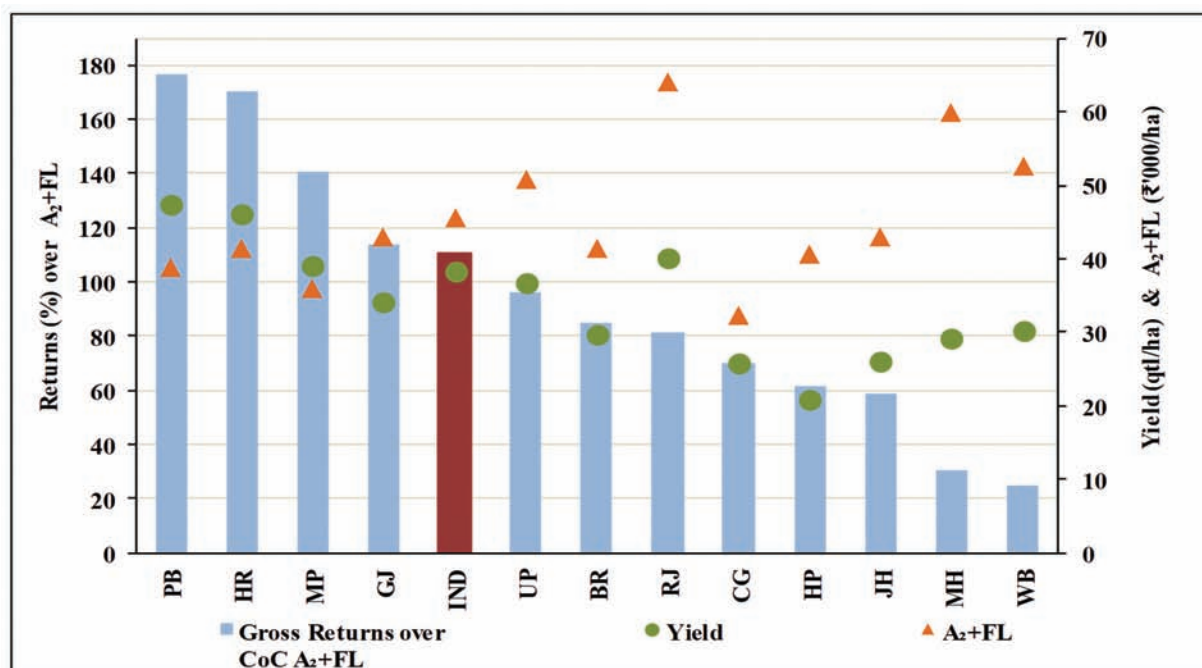
Note: States have been arranged in descending order of returns

Source: CACP Calculations using CS data

- 5.17 From the Chart 5.2(a), it is clear that the gross returns over CoC A_2 + FL from wheat are significantly higher in Punjab and Haryana due to substantially high yields, while States like West Bengal, Maharashtra, Jharkhand, Himachal Pradesh and Chhattisgarh have lower returns mainly due to lower yields and higher cost of cultivation. In case of gram (Chart 5.2(b)), gross returns over CoC A_2 + FL are higher in Bihar due to lower cost of cultivation and in Andhra Pradesh due to higher yield. In contrast Jharkhand records negative returns, while States like Uttar Pradesh, Karnataka, Maharashtra and Rajasthan have lower returns primarily due to lower yields and higher cost of cultivation. For rapeseed & mustard (Chart 5.2(c)), gross returns over CoC A_2 + FL are higher in Haryana and Madhya Pradesh due to high yields and low cost of cultivation, while States like Jharkhand, Himachal Pradesh, West Bengal and Assam have lower returns due to lower yields and higher cost of cultivation. To increase the farm profitability, efforts are required to enhance productivity and reduce cost in high cost States.



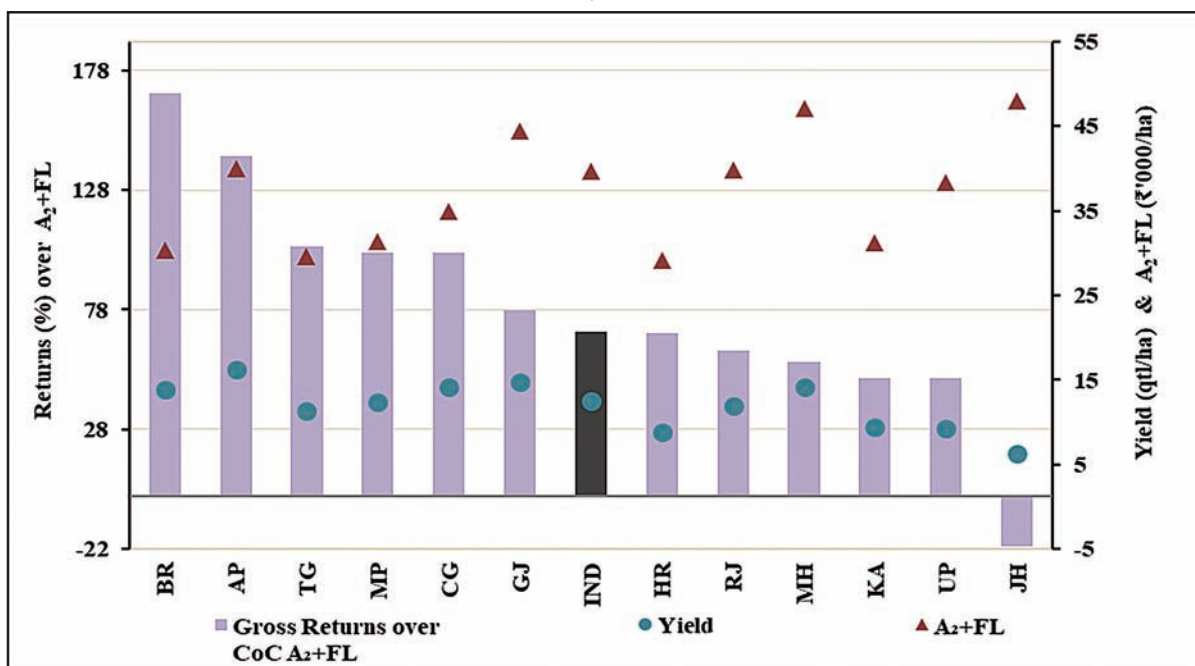
Chart 5.2(a): Inter-State Variations in Yield, Costs and Gross Returns for Wheat, TE2023-24



Note: All-India CoC, gross returns and yield of a crop are weighted average of respective CoCs, gross returns and yield of projected States

Source: CACP Calculations using CS data

Chart 5.2(b): Inter-State Variations in Yield, Costs and Gross Returns for Gram, TE2023-24

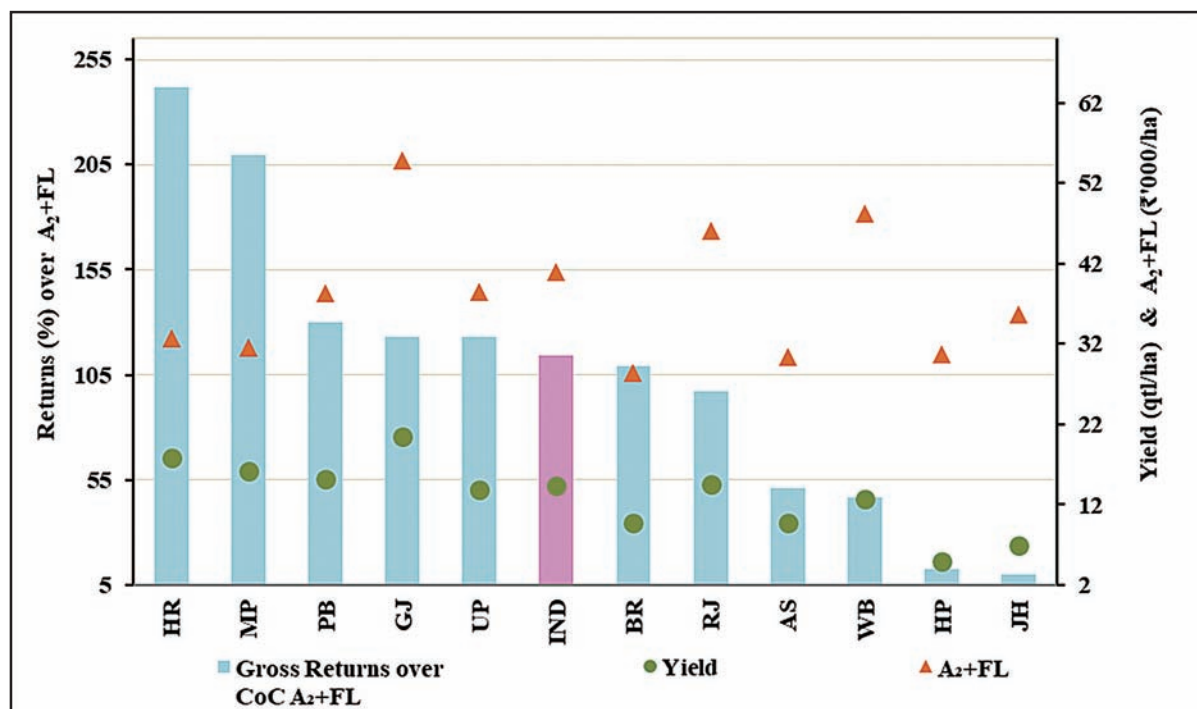


Note: All-India CoC, gross returns and yield of a crop are weighted average of respective CoCs, gross returns and yield of projected States

Source: CACP Calculations using CS data



Chart 5.2(c): Inter-State Variations in Yield, Costs and Gross Returns for Rapeseed & Mustard, TE2023-24



Note: All-India CoC, gross returns and yield of a crop are weighted average of respective CoCs, gross returns and yield of projected States

Source: CACP Calculations using CS data

Agricultural Wages and Farm Input Prices

Growth in Daily Wage Rates of Agriculture Labour

- 5.18 Average annual growth in daily wage rates at nominal terms of agriculture labour during *rabi* season in major States and at all-India level during 2022-23 to 2024-25 are given in Table 5.2. At all-India level, agriculture wage rate increased by 8.3 percent in 2022-23, 7.1 percent in 2023-24 and 6.6 percent in 2024-25, respectively, over the corresponding previous year. The highest increase in average daily wage rate in 2024-25 over 2023-24 was recorded in Himachal Pradesh at 10.4 percent, followed by Gujarat and Haryana (10.3%), and lowest in Punjab (1.6%). Assam, Bihar, Gujarat, Haryana, Himachal Pradesh, Maharashtra and Rajasthan recorded higher growth in average daily wages as compared to all-India average, while Andhra Pradesh, Karnataka, Madhya Pradesh, Punjab, Uttar Pradesh and West Bengal registered lower growth than all-India wage during 2024-25 over 2023-24.



Table 5.2: Growth (%) in Average Daily Wage Rates of Agricultural Labour during Rabi Season (2022-23 to 2024-25)

State	Growth (%)		
	2022-23	2023-24	2024-25
Andhra Pradesh	13.7	11.4	6.0
Assam	4.9	10.7	9.3
Bihar	8.5	8.0	7.0
Gujarat	9.0	5.3	10.3
Haryana	9.1	4.4	10.3
Himachal Pradesh	0.4	6.0	10.4
Karnataka	9.4	6.0	6.4
Madhya Pradesh	5.7	5.0	6.3
Maharashtra	5.2	6.2	7.1
Punjab	5.8	6.4	1.6
Rajasthan	4.1	10.0	9.1
Uttar Pradesh	7.2	5.0	6.1
West Bengal	5.3	3.3	5.7
All-India	8.3	7.1	6.6

Note 1: Average daily wage rates of agriculture labour are related to man only

2: Growth is for the period from October to April

3: All-India daily wage rate is weighted average of daily wage rates of States

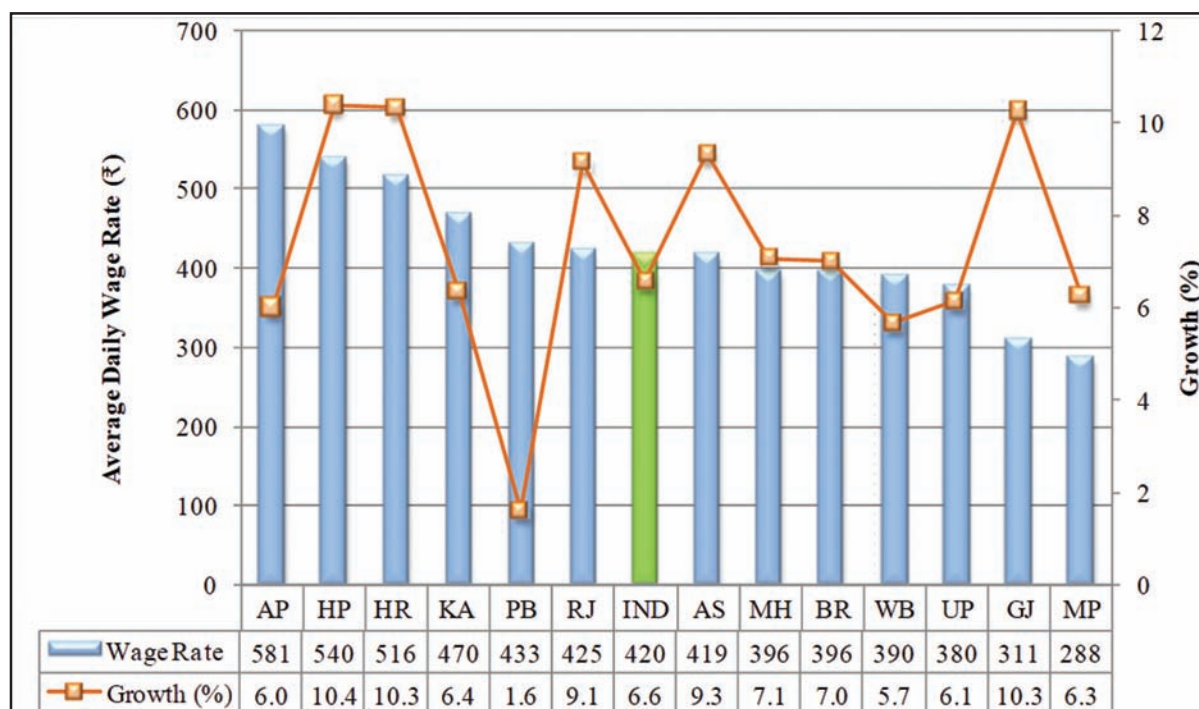
Source: Labour Bureau, Ministry of Labour and Employment and CACP Calculations

Daily Wages of Agriculture Labour

- 5.19 Chart 5.3 depicts State-wise average daily wages of agricultural labour in projected States during *rabi* season (October to April) in 2024-25 and growth in wages during *rabi* season in 2024-25 over 2023-24. All-India average daily wages of agricultural labour were ₹420 in 2024-25 and were highest in Andhra Pradesh (₹581) and lowest in Madhya Pradesh (₹288). The average daily wages in Andhra Pradesh, Himachal Pradesh, Haryana, Karnataka, Punjab and Rajasthan were higher in comparison to all-India average, while in Assam, Bihar, Gujarat, Madhya Pradesh, Maharashtra, Uttar Pradesh and West Bengal, wages were lower than all-India average.



Chart 5.3: Average Daily Wage Rates for Rabi Season 2024-25 and Growth over 2023-24 Season



Note 1: Average daily wage rates of agriculture labour are related to man only

2: Growth is for the period from October to April

3: All-India daily wage rate is weighted average of daily wage rates of States

Source: Labour Bureau, Ministry of Labour and Employment and CACP Calculations

- 5.20 Chart 5.3 clearly shows that the wage rates have increased significantly in most of the States, leading to increase in cost of cultivation/production. Strategies for promotion of mechanisation of farm operations are required to reduce dependence on human labour and thereby reducing cost of cultivation/production and enhancing farm profitability in the country. Annex Table 5.3 provides details on monthly average of daily wage rates for agriculture labour.

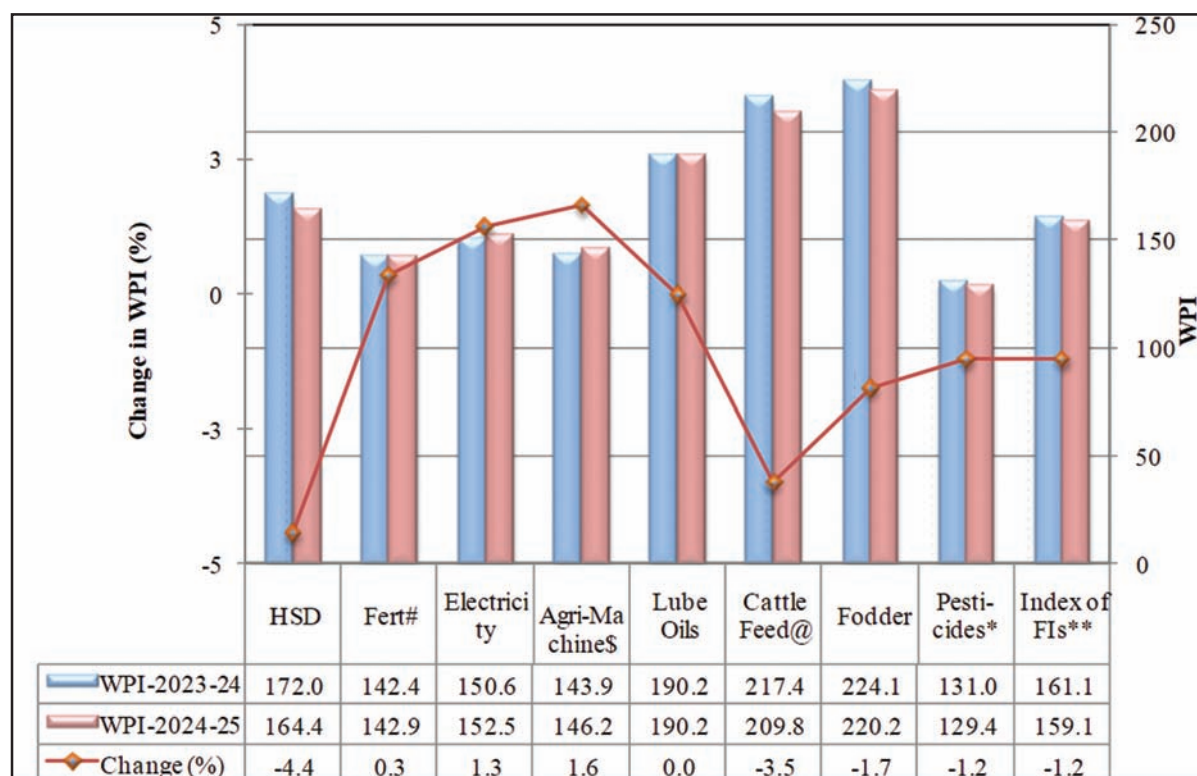
Movement in Farm Input Prices

- 5.21 The Commission analysed price trends of major farm inputs to assess the effect of changes in input prices on cost of cultivation/production. The changes in Wholesale Price Index (WPI) with base 2011-12 of major farm inputs during *rabi* season (October to April) in 2024-25 over 2023-24 are given in Chart 5.4. The indices of high speed diesel (-4.4%), cattle feed (-3.5%), fodder (-1.7%) and pesticides and other agrochemical products (-1.2%) showed a decline in 2024-25 over 2023-24, whereas indices of manufacture of agricultural and forestry machinery² (1.6%), electricity (1.3%), and manufacture of fertilizers & nitrogen compounds (0.3%), increased in 2024-25 over 2023-24. The weighted index of above-mentioned selected farm input prices decreased by (-)1.2 percent. The monthly wholesale price indices of various farm inputs from 2023 to 2025 are given in Annex Table 5.4.

²Manufacture of agricultural and forestry machinery consists of 'agricultural tractors', 'harvesters', 'agriculture implements', 'threshers' and 'soil preparation & cultivation machinery (other than tractors)'.



Chart 5.4: Movement in Wholesale Price Index (WPI) of Farm Inputs



Note 1: WPI of the Farm Input shown in the Chart is average of monthly WPIs from October to April

2: Index of Farm Inputs is weighted average of WPI of farm inputs mentioned in Chart

#: Fertilizers and Nitrogen Compounds

\$: Manufacture of Agricultural and Forestry Machinery

@: Gola & similar Cattle Feed

*: Pesticides and other Agrochemical Products

** : Index of Farm Inputs

Source: Office of the Economic Adviser, Ministry of Commerce and Industry and CACP Calculations

Cost Projections for Crop Year 2025-26

Projection of Farm Input Price Index

- 5.22 The Commission has computed farm input-wise all-India weighted 'input weights' for TE2023-24 and 'Composite Input Price Indices (CIPIs)' from 2022-23 to 2025-26 for mandated *rabi* crops (Base 2011-12=100) and are given in Table 5.3. The 'input price indices' and 'CIPIs' for crop year 2024-25 and 2025-26 for a State have been constructed using actual 'weights', 'input price indices' for TE2023-24, and latest farm input prices. Subsequently, based on these State-wise 'actual weights' and 'input price indices', crop-wise all-India weighted average 'input weights' and 'CIPIs' for all inputs, with weights being relative share of States in all-India area under the crop during TE2023-24, have been computed. Further, these crop-wise all-India weighted average 'input weights' and 'CIPIs' have been used to compute input-wise all-India weighted average 'input weights' and 'CIPIs', respectively, for all *rabi* crops, with weights being relative shares of crops in total production of mandated *rabi* crops at all-India level during TE2024-25. Eventually, these input-wise all-India weighted average 'input weights' and 'CIPIs' have been used to compute all-India weighted average 'Composite



Input Price Index (CIPi)' for all inputs of mandated *rabi* crops.

- 5.23 Table 5.3 shows that the all-India CIPi for *rabi* crops increased by 3.1 percent in 2023-24, 8.2 percent in 2024-25, and nine percent in 2025-26, while CIPi registered the highest increase in machine labour (12.7%), followed by irrigation charges (9.9%), seed (9.2%), manure (7.7%), bullock labour (7.5%), human labour (7.2%) and lowest increase in insecticides (1.3%) in 2025-26 over 2024-25.

Table 5.3: Trends in All-India *Rabi* Crops Farm Input Price Indices
(Base: 2011-12=100)

Inputs	Weights (TE2022-23)	Input Price Index				Percentage Change in Input Price Index 2025-26 over 2024-25
		2022-23	2023-24	2024-25	2025-26	
Human Labour	37.5	232.6	246.9	264.0	283.2	7.2
Bullock Labour	1.3	401.7	409.9	438.5	471.5	7.5
Machine Labour	27.8	243.6	248.6	275.6	310.5	12.7
Seeds	10.5	259.4	273.7	297.7	324.9	9.2
Fertilizers	10.1	138.6	140.6	142.9	145.2	1.6
Manures	0.3	266.8	263.1	282.8	304.7	7.7
Insecticides	1.8	164.5	151.9	153.5	155.5	1.3
Irrigation Charges	11.1	259.0	250.2	274.4	301.5	9.9
Composite Input Price Index (CIPi)		234.0	241.4	261.2	284.7	9.0
Percentage Change		-	3.1	8.2	9.0	-

Note 1: All-India Weights and CIPis are weighted average of respective weights and CIPis of projected States

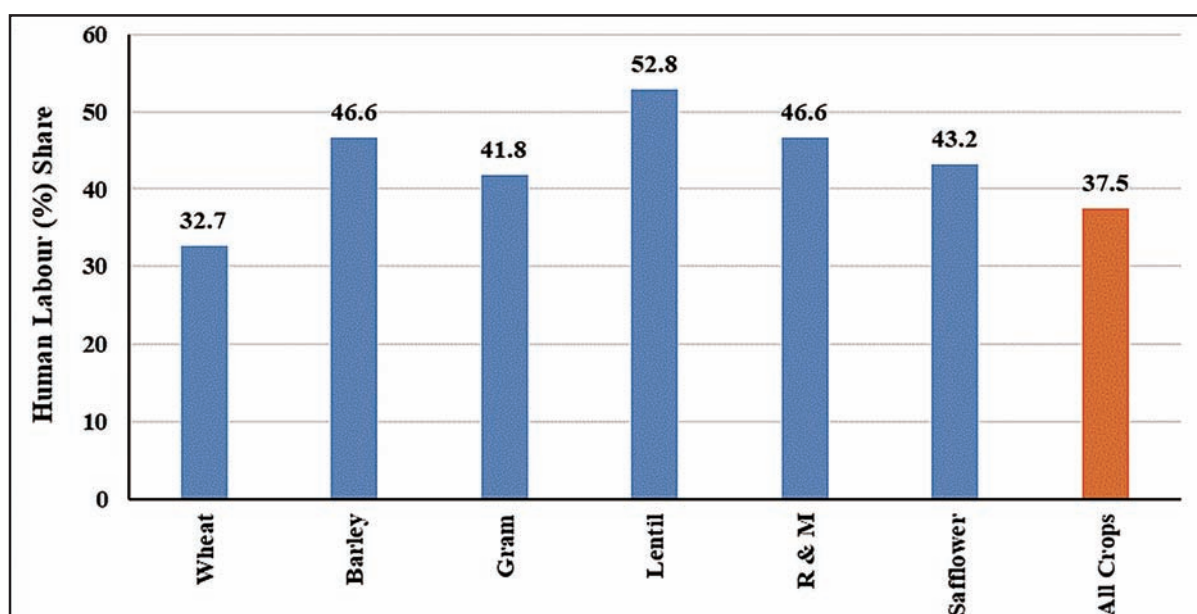
2: CIPis published in earlier reports may not be exactly comparable with the above Table due to change in weights, updation of data, exclusion/inclusion of some States and other factors

Source: CACP Calculations

- 5.24 As it is evident from the Chart 5.5, the share of human labour in total operational cost for all mandated crops except wheat is more than 40 percent. As the human labour shortage and rising wages have become a major constraint in improving profitability, it is imperative to formulate national and State-level strategies for promoting farm mechanisation, which would help in encouraging farmers to adopt mechanisation, thereby reducing costs and improve profitability.



Chart 5.5: Human Labour (%) Share in Total Operational Costs of Mandated Rabi Crops, TE2023-24



Note: Human labour share for all crops is weighted average of the human labour share of mandated crops with weights being respective crop area share

Cost of Production Projections

- 5.25 Based on State-wise actual cost estimates during TE2023-24 and projected CIPIs, State-wise estimates of CoC A_2 , A_2 +FL and C_2 for each of the mandated crop have been projected for the crop year 2025-26. Using these State-wise CoC estimates, average ratio of Main Product (MP) to Gross Value of Output (GVO) during TE2023-24, and estimated yield, State-wise CoP A_2 , A_2 +FL and C_2 for each crop for crop year 2025-26 were projected. Subsequently, all-India weighted average projected CoP A_2 , A_2 +FL and C_2 of mandated crops, with weights being the respective share of the States in all-India production during TE2024-25, have been worked out for crop year 2025-26 (*Rabi Marketing Season 2026-27*), and are given in Table 5.4.
- 5.26 During 2025-26, all-India projected CoP A_2 was lowest (₹912/ql) for barley, while A_2 +FL (₹1,239/ql) and C_2 (₹1,804/ql) were lowest for wheat. Safflower had the highest costs. In case of cereals, projected CoP A_2 +FL and C_2 for wheat were lower than barley but A_2 was higher for wheat than barley. In case of pulses, projected CoP A_2 was lower for lentil (₹2,825/ql) than gram (₹2,886/ql), while A_2 +FL and C_2 were lower for gram. In oilseeds, projected CoP A_2 (₹2,219/ql), A_2 +FL (₹3,210/ql), and C_2 (₹4,558/ql) for rapeseed & mustard were significantly lower than safflower, owing to higher yield.
- 5.27 The cost of production of some crops is high in some States due to low yields or high cost of cultivation or both low yield and high costs. Therefore, efforts are needed to improve productivity and optimise input costs to reduce production cost and improve profitability. State-wise and all-India projected costs and estimated yield for mandated *rabi* crops for crop year 2025-26, and production share



during TE2024-25 are given in Annex Table 5.5. State-wise break-up of actual CoC estimates for 2021-22, 2022-23, and 2023-24 for wheat, barley, gram, lentil, rapeseed & mustard, and safflower are provided in Annex Tables 5.6(a) to 5.6(f), respectively.

Table 5.4: Projected Cost of Production (CoP) for Mandated Crops for Crop Year 2025-26

Crop	Projected Cost of Production (₹/qtl)		
	A_2	$A_2 + FL$	C_2
Wheat	994	1239	1804
Barley	912	1361	1862
Gram	2886	3699	4875
Lentil	2825	3705	5243
Rapeseed & Mustard	2219	3210	4558
Safflower	3376	4360	5738

Note: All-India CoP of a crop is weighted average of CoPs of projected States

Source: CACP Calculations using CS data

Production vis-a-vis Cost Projection

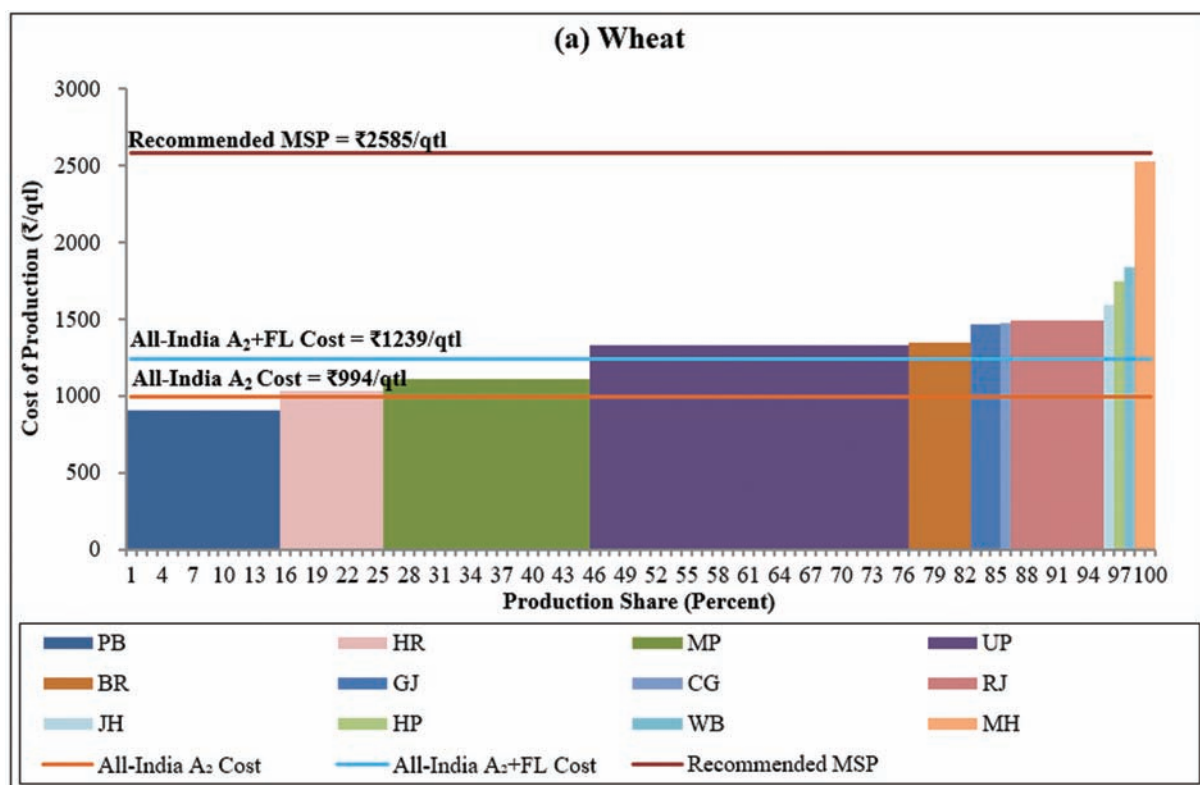
- 5.28 Charts 5.6(a) to (e) show crop-wise supply curves³ for $A_2 + FL$ CoP of projected States in ascending order with their corresponding shares in all-India production. The supply curve presented in Chart 5.6(a) shows that projected $A_2 + FL$ CoP for wheat is lowest in Punjab (₹905/qtl) mainly due to higher yield and highest in Maharashtra (₹2,529/qtl) primarily driven by low yield and significantly high cost of cultivation compared to other States. Among top-five producers of wheat in the country, which accounted for 86.8 percent production share of projected States during TE2024-25, Punjab has the lowest projected CoP, while Rajasthan has the highest CoP (₹1,488/qtl). The cost of production in Haryana, Madhya Pradesh and Punjab is lower than all-India projected costs, whereas, Bihar, Chhattisgarh, Gujarat, Himachal Pradesh, Jharkhand, Maharashtra, Rajasthan, Uttar Pradesh and West Bengal have higher costs than all-India cost of production. For barley, projected $A_2 + FL$ at ₹1,304 per quintal is lowest in Uttar Pradesh, and highest (₹2,296/qtl) in Himachal Pradesh mainly due to significantly low yield (Chart 5.6(b)).
- 5.29 In case of gram, as presented in Chart 5.6(c), Bihar has the lowest projected $A_2 + FL$ CoP at ₹2,613 per quintal, while Uttar Pradesh has the highest CoP (₹4,266/qtl). However, among the top-four producers of gram in the country, which account for 78.8 percent of production of projected States, Gujarat has the lowest projected CoP (₹3,324/qtl), while Maharashtra has the highest CoP (₹4,082/qtl) mainly due to higher cost of cultivation. Andhra Pradesh, Bihar, Chhattisgarh, Gujarat, Haryana, Madhya Pradesh, Rajasthan and Telangana have lower projected costs for gram, while Jharkhand, Karnataka, Maharashtra and Uttar Pradesh have higher costs than all-India. Supply curve for Lentil

³Supply curves for different crops are graphical representation of CoP, which represent the quantum of production of crop produced at different levels of CoP in various States.



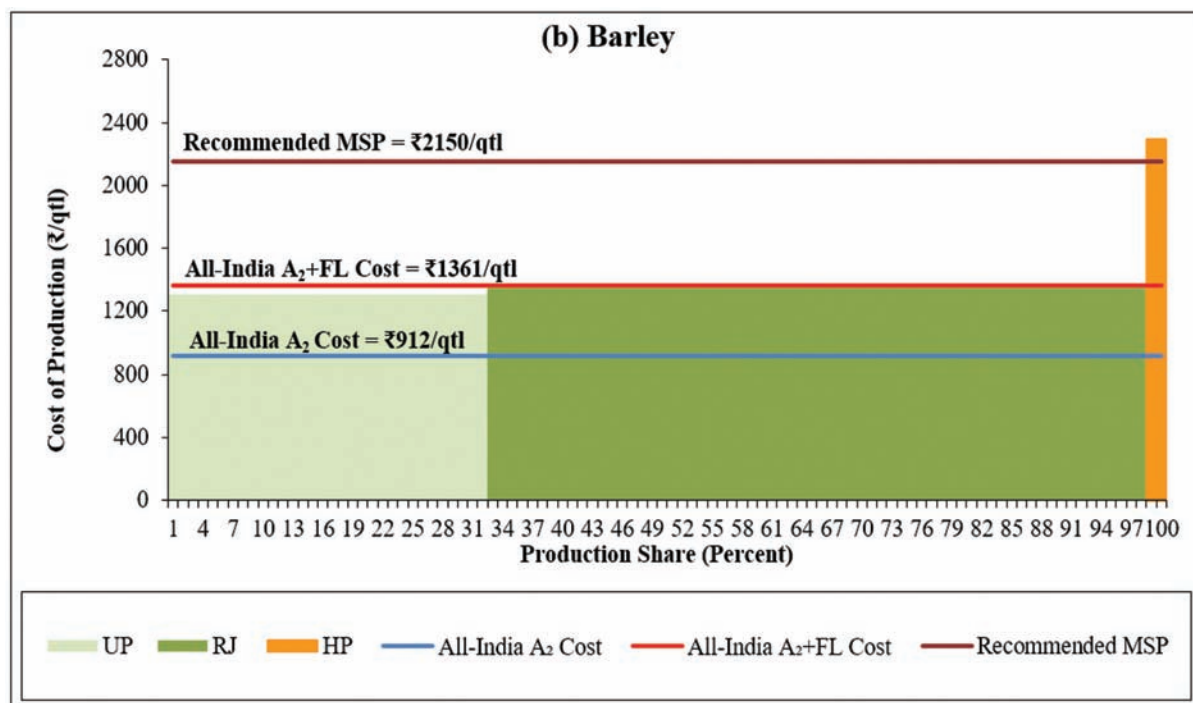
presented in Chart 5.6(d) shows that projected A_2 +FL cost was lowest in Bihar at ₹3,069 per quintal, and highest in West Bengal (₹4,396/ql).

- 5.30 Chart 5.6(e) shows that Madhya Pradesh has lowest projected CoP at ₹2,159 per quintal of rapeseed & mustard, while Jharkhand has the highest CoP (₹4,532/ql). Among top-four producers of rapeseed & mustard in the country, which account for 83.3 percent production of projected States, Madhya Pradesh has lowest CoP, and Rajasthan, the largest producer, has highest CoP (₹3,391/ql). Bihar, Gujarat, Haryana, Madhya Pradesh and Punjab have lower CoP than all-India average CoP, while Assam, Himachal Pradesh, Jharkhand, Rajasthan, Uttar Pradesh and West Bengal have higher CoPs than all-India projected cost.
- 5.31 All-India projected cost of production of mandated *rabi* crops for crop year 2024-25 and 2025-26 are given in Annex Table 5.7. The increase in projected A_2 +FL cost of production for crop year 2025-26 over 2024-25 was highest for safflower at 10.1 percent, and the lowest for lentil at 4.7 percent. In case of cereals, barley recorded higher increase (9.8%) in CoP than wheat (4.8%), while in pulses, lentil had marginally lower increase (4.7%) than gram (4.9%). In case of oilseeds, 10.1 percent increase in CoP was recorded in safflower compared to 6.6 percent in rapeseed & mustard.
- 5.32 The projected A_2 +FL cost of production was higher than all-India weighted CoP A_2 +FL in nine States, accounting for 54 percent of total production, out of 12 States for wheat, two out of three States for barley, four States with 40 percent production share out of 12 States for gram, two out of four States for lentil, and six States having 70 percent production share out of 11 States for rapeseed & mustard. Therefore, holistic and coordinated efforts are needed to reduce cost of cultivation and enhance productivity in high-cost States to improve farm profitability and income.
- 5.33 The share of production covered at the recommended MSP is 97.9 percent in wheat, barley (89.7%), gram (99.2%), lentil (92%), rapeseed & mustard (97.6%) and 44.3 percent for safflower. The MSP margins over all-India projected A_2 +FL cost of production was highest for wheat at 108.6 percent, followed by rapeseed & mustard (93.1 %), lentil (88.9%), gram (58.8%), barley (58%) and 50 percent for safflower. The recommended MSP covers C_2 cost of all producers of wheat except Maharashtra, barley except Himachal Pradesh, gram except Uttar Pradesh, rapeseed & mustard except Himachal Pradesh and all States for lentil and safflower included in the analysis. The highest MSP margin over projected CoP A_2 +FL for wheat was in Punjab (185.6%), barley in Uttar Pradesh (64.9%), gram (124.8%) and lentil (128.1%) in Bihar, and rapeseed & mustard in Madhya Pradesh (187.2%).

Chart 5.6: Supply Curve and Projected CoP for Mandated *Rabi* Crops, RMS2026-27

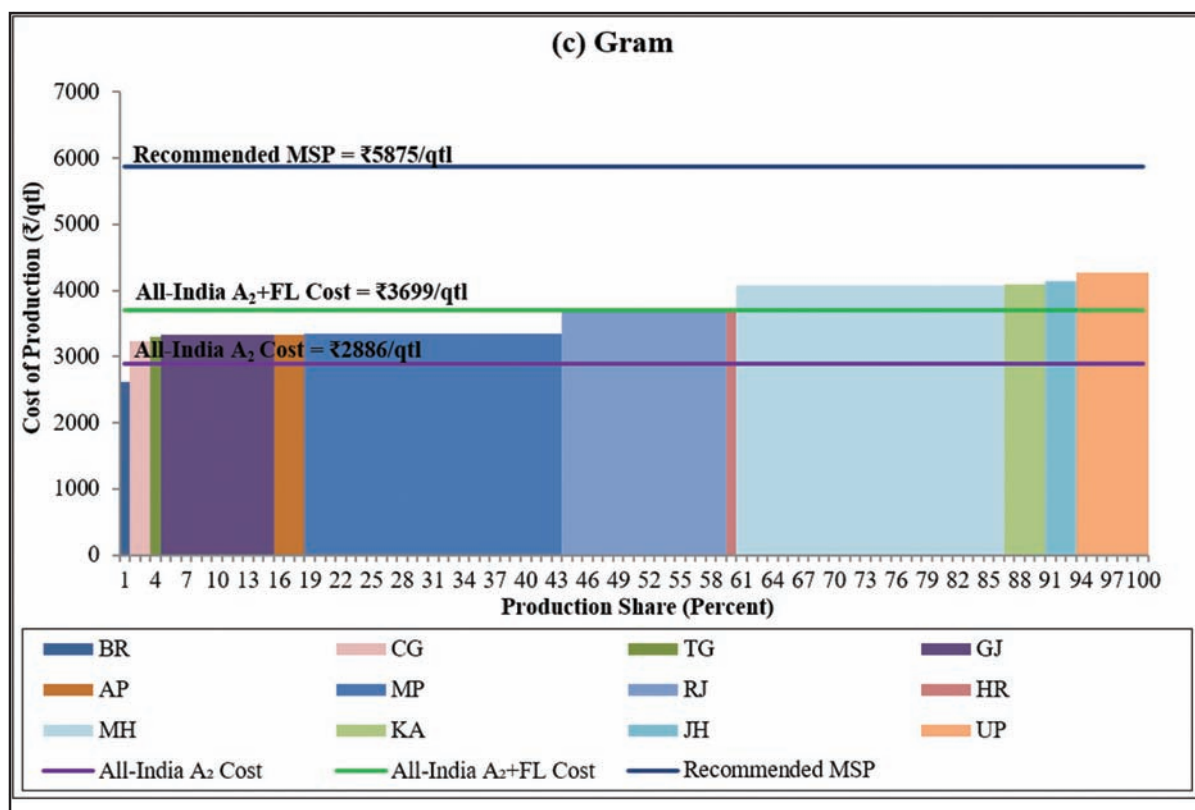
Note: All-India CoP is weighted average of CoPs of States mentioned in the Chart

Source: CACP Calculations



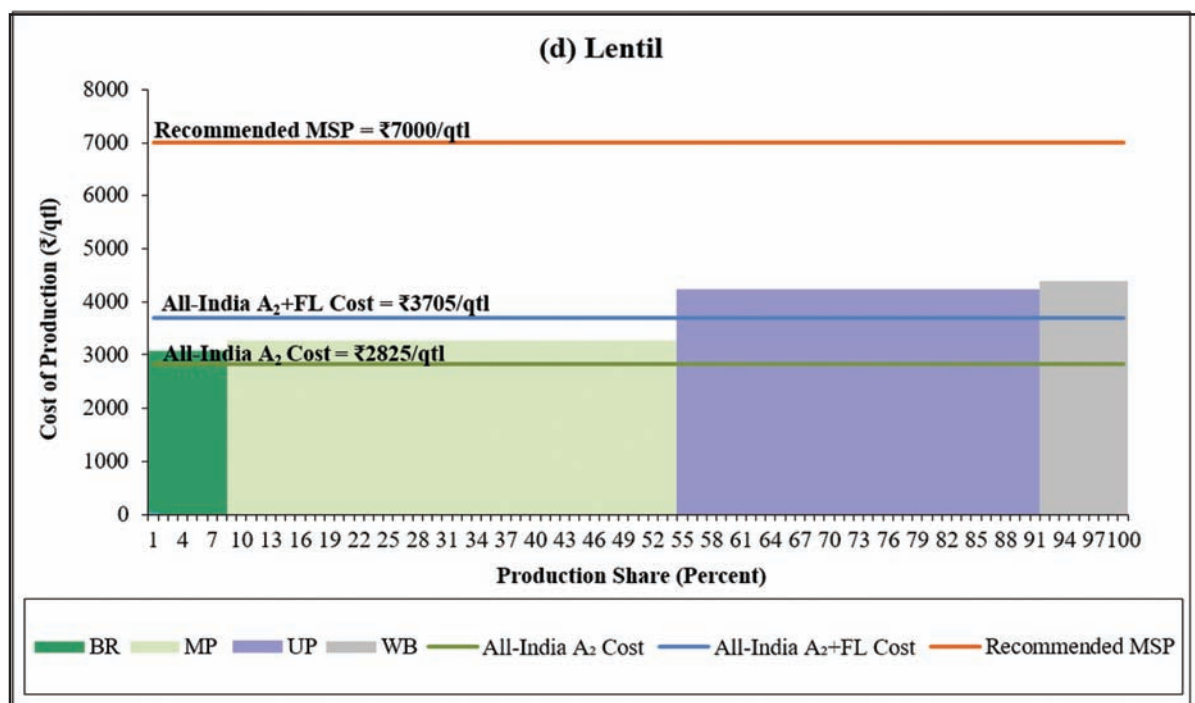
Note: All-India CoP is weighted average of CoPs of States mentioned in the Chart

Source: CACP Calculations



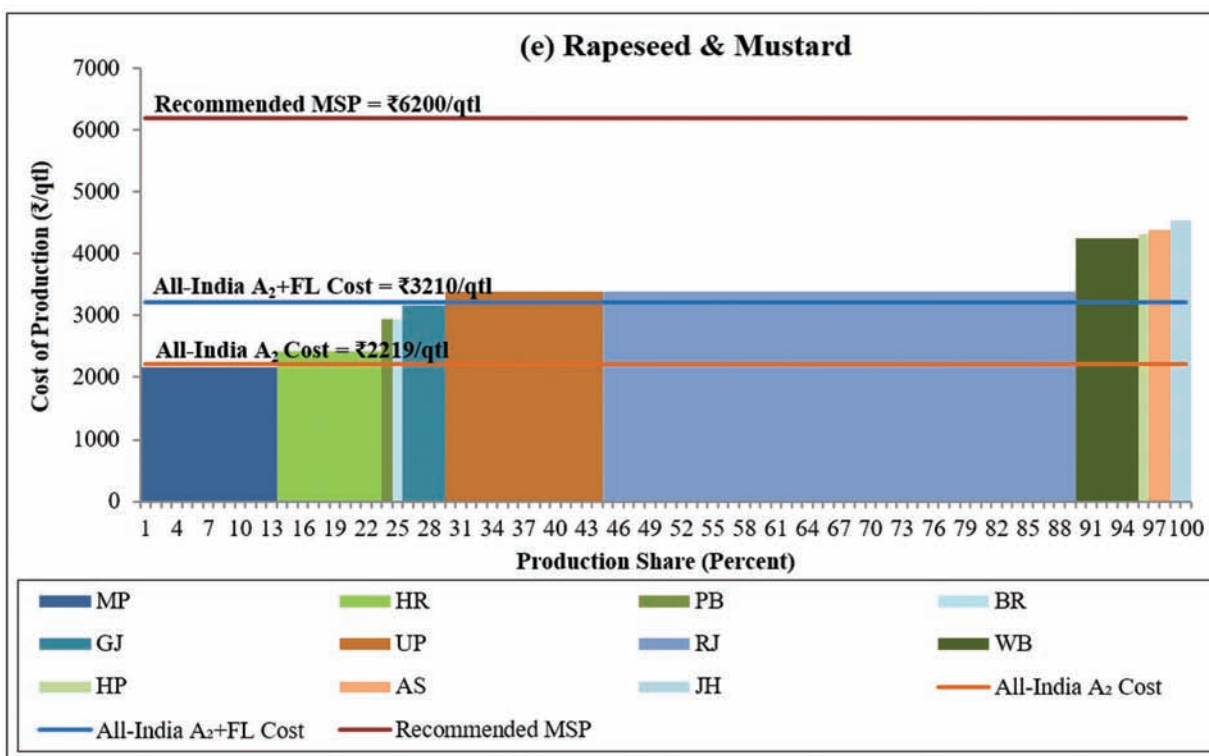
Note: All-India CoP is weighted average of CoPs of States mentioned in the Chart

Source: CACP Calculations



Note: All-India CoP is weighted average of CoPs of States mentioned in the Chart

Source: CACP Calculations



Note: All-India CoP is weighted average of CoPs of States mentioned in the Chart

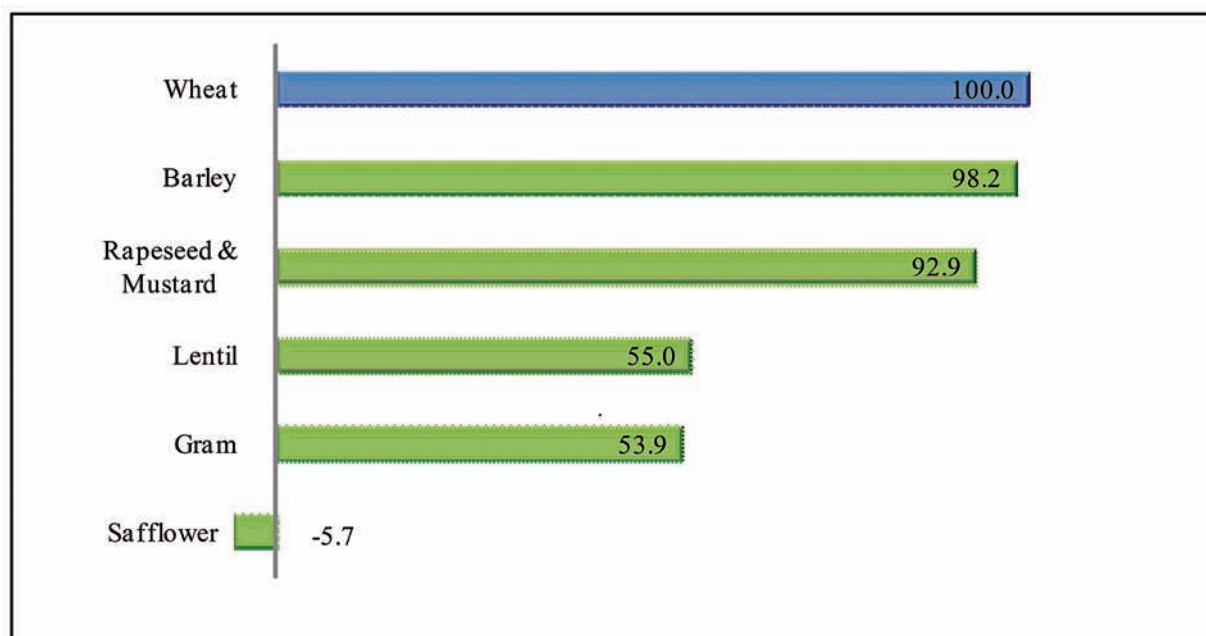
Source: CACP Calculations

Inter-Crop Parity in Returns

- 5.34 Inter-crop parity is an important factor for determination of MSP, therefore, the Commission analysed per hectare relative average gross returns of mandated crops in percentage terms with respect to wheat. Chart 5.7 shows per hectare relative average gross returns over CoC A₂+FL in percentage term for each of the mandated crop with reference to wheat during TE2023-24. It is evident from the Chart that wheat was the most profitable crop during TE2023-24 as compared to other *rabi* crops.
- 5.35 In case of cereals, relative returns over A₂+FL for barley were marginally lower than wheat but over A₂, barley was 13.5 percent more profitable than wheat. In case of pulses, the relative returns over A₂+FL for lentil were marginally higher than gram. Farmers incurred a net loss in cultivation of safflower mainly due to low yield. Wheat was the most profitable crop, followed by barley, rapeseed & mustard, lentil and gram.



Chart 5.7: Relative Average Gross Returns (%) over CoC A₂+FL with Respect to Wheat during TE2023-24



*Note: All-India gross returns of a crop are weighted average of respective gross returns of projected States
Source: CACP Calculations using CS data*

- 5.36 Relative average gross returns of mandated crops in projected States during TE2023-24 with respect to wheat over actual cost of cultivation are given in Annex Table 5.9. Among mandated crops, wheat was the most profitable crop in Himachal Pradesh, Jharkhand, Punjab and Uttar Pradesh during TE2023-24, while in Bihar, Chhattisgarh, and Maharashtra gram was more profitable crop. In Rajasthan, barley was the most profitable crop followed by wheat, rapeseed & mustard and gram. In Assam, Gujarat, Haryana, Madhya Pradesh and West Bengal, rapeseed & mustard recorded the highest profitability.

Comparison of CACP Cost Estimates with State Estimates

- 5.37 The estimates of cost of production/cultivation and yield provided by States and CACP projected costs, and Minimum Support Price (MSP) suggested by States for mandated *rabi* crops for RMS2026-27 are given in Annex Table 5.10. It is evident from the Table that there are significant variations in cost estimates provided by the States and CACP cost estimates primarily due to different methodologies and cost concepts used by the States and CACP.
- 5.38 The estimates of projected cost of production for wheat by Chhattisgarh, Gujarat, Haryana, Punjab and West Bengal were higher than CACP projections, while cost estimates by Madhya Pradesh and Uttar Pradesh were lower than CACP estimates. The cost estimates provided by Uttar Pradesh for barley is also less than CACP estimates. The cost of production estimates of gram by Andhra Pradesh, Gujarat and Haryana were higher than CACP estimates, but Chhattisgarh, Madhya Pradesh and Uttar Pradesh estimates were lower. For lentil, the cost estimates of Madhya Pradesh and Uttar Pradesh were lower than CACP estimates. In case of rapeseed & mustard, cost estimates by Gujarat, Haryana,



Madhya Pradesh and West Bengal were higher, while cost estimates of Assam and Uttar Pradesh were lower than CACP estimates.

Table 5.5: Classification of States based on Cost Estimates Provided by States and CACP Estimates

Crop	State Cost Estimates	
	$\geq$ CACP Estimates	$<$ CACP Estimates
Wheat	Chhattisgarh, Gujarat, Haryana, Punjab, West Bengal	Madhya Pradesh, Uttar Pradesh
Barley	-	Uttar Pradesh
Gram	Andhra Pradesh, Gujarat, Haryana	Chhattisgarh, Madhya Pradesh, Uttar Pradesh
Lentil	-	Madhya Pradesh, Uttar Pradesh
Rapeseed & Mustard	Gujarat, Haryana, Madhya Pradesh, West Bengal	Assam, Uttar Pradesh

Source: State Governments and CACP Calculations

- 5.39 The comparison of cost estimates provided by States for some crops could not be carried out as cost projections for these crops and States have not been undertaken by CACP due to either non-availability/inadequacy of data or thin sample size under the Comprehensive Scheme. Besides, some States have not provided cost estimates for few crops, therefore, cost comparison could not be undertaken for all crops/States.

Issues related to Sample Size under Comprehensive Scheme

- 5.40 The Commission uses cost estimates compiled under CS by ES&E Division of the Ministry of Agriculture and Farmers Welfare. It has been observed that there are some crops in the States whose share in the all-India crop area/production as well as in the area and production of crop group in the State is significant but are not covered under the CS. There are also instances, where sample size for certain crops in the State and/or all-India is inadequate, and may not be a representative sample for cost projections (Annex Table 5.11 and Annex Table 5.12).
- 5.41 Based on the above, the Commission recommends that sample size of gram in Haryana, Jharkhand and Telangana, rapeseed & mustard in Chhattisgarh, and safflower in Karnataka and Maharashtra should be increased. In case of safflower, although sample size has increased in the recent years, it is still small and can undermine the reliability and representativeness of cost of production estimates. Therefore, sample size for safflower in major producing States must be increased. Since the Commission uses three-year actual cost estimates in forecasting the cost of production, the CS data should be collected on continuous basis for gram in West Bengal, lentil in Jharkhand, rapeseed & mustard in Chhattisgarh and safflower in Maharashtra for reliable cost estimates.



- 5.42 The Commission analysed State-wise area and production of the mandated *rabi* crops during TE2024-25 and observed that there are certain crops in some States, which occupy reasonable share in area and production at all-India level or a crop group in the State, but CS data for these crops/States are not available (Annex Table 5.13). Inclusion of these States under Comprehensive Scheme would improve representation and reliability of cost projections at all-India level. The Commission also recommends that Comprehensive Scheme may be implemented in Jammu & Kashmir for wheat and rapeseed & mustard, and Uttarakhand for wheat, barley, lentil and rapeseed & mustard.

Recapitulation

- 5.43 Average CoC A_2 +FL per hectare during TE2023-24 was ₹45,558 for wheat; ₹55,507 for barley; ₹39,632 for gram; ₹28,618 for lentil; ₹41,025 for rapeseed & mustard; and ₹20,772 for safflower. Average GVO per hectare was ₹96,014 for wheat; ₹1,05,036 for barley; ₹66,823 for gram; ₹56,386 for lentil; ₹87,877 for rapeseed & mustard; and ₹17,886 for safflower. Average gross returns over CoC A_2 +FL per hectare during TE2023-24 were highest for wheat (₹50,456/ha), followed by barley (₹49,530/ha), rapeseed & mustard (₹46,851/ha), lentil (₹27,769/ha) and gram (₹27,192/ha), while safflower farmers incurred a net loss of ₹2,885 per hectare.
- 5.44 The all-India average daily wage rate of agricultural labour during *rabi* season of 2024-25 registered an increase of 6.6 percent, while all-India CIPI for mandated *rabi* crops registered a growth of nine percent. As labour availability and rising wages have become major constraints and share of human labour in operational cost is high (37.5%) in *rabi* crops, farmers should be encouraged to adopt farm mechanisation to reduce cost of cultivation/production and improve profitability.
- 5.45 The all-India per quintal A_2 +FL cost of production for crop year 2025-26 (RMS2026-27) has been projected at ₹1,239 for wheat; ₹1,361 for barley; ₹3,699 for gram; ₹3,705 for lentil; ₹3,210 for rapeseed & mustard and ₹4,360 for safflower. The increase in projected CoP A_2 +FL during crop year 2025-26 over 2024-25 ranged from 4.7 percent in lentil to 10.1 percent in safflower. The relative average gross returns were highest for wheat, followed by barley, rapeseed & mustard, lentil and gram.
- 5.46 Some States have significantly high cost of production due to low yields and high costs. Therefore, efforts should be made towards improving crop yield and promoting efficient use of inputs to reduce cost of cultivation. The CS data should be collected for States/crops that have significant share in all-India area and production, as well as in State's crop group. Ensuring adequate sample size is necessary to enhance representativeness and reliability of cost estimates.
- 5.47 The per quintal MSP recommended at ₹2,585 for wheat, ₹2,150 for barley, ₹5,875 for gram, ₹7,000 for lentil, ₹6,200 for rapeseed & mustard, and ₹6,540 for safflower, would cover projected A_2 +FL CoP for all States for Wheat, gram, lentil, rapeseed & mustard and safflower; and two out of three States for barley.



CHAPTER 6



Considerations and Recommendations for Price Policy

- 6.1 In making its recommendations on price policy and non-price measures, the Commission has considered various factors and issues set out in the Terms of Reference. These factors include cost of production, demand-supply situation, domestic and international prices, inter-crop parity, terms of trade between the agriculture and non-agriculture sector, likely impact of price policy on rest of the economy and competitiveness of agriculture, and a minimum of 50 percent margin over cost of production, among others. In addition, the Commission organized regional meetings with State Governments and farmers, as well as discussions with Central Ministries/Departments, farmers' associations, research institutions, procurement agencies, industry associations, etc. A brief discussion of the relevant factors is given below:

Demand-Supply Situation

Domestic Scenario

- 6.2 As per the 3rd Advance Estimates for 2024-25 by the Ministry of Agriculture & Farmers Welfare, wheat production in the country is estimated at a record 117.5 million tonnes, an increase of 3.7 percent over the previous year. Wheat stocks in the Central Pool were about 38 million tonnes on 1st June 2025 against the stocking norms of about 27.6 million tonnes (as on 1st July of each year) and 27 per cent higher compared to the corresponding period of the previous year.
- 6.3 Production of total pulses in 2024-25 is estimated at 25.2 million tonnes, an increase of 4.1 percent over 2023-24. Among major *rabi* pulses, gram production is estimated at 11.3 million tonnes, reflecting an increase of 2.7 percent over the previous year but 7.6 percent lower than 2022-23 record, while lentil production is estimated to fall by 1.1 percent in 2024-25 at 17.7 lakh tonnes. Stocks of pulses held by NAFED marginally increased to 1.5 million tonnes in 2024-25 but is lower than buffer stock norms of 2 million tonnes. Pulses consumption continues to increase faster than production, leading to sharp increase in imports from 2.7 million tonnes in 2022-23 to 7.3 million tonnes in 2024-25.
- 6.4 Despite significant increase in production of total oilseeds in the country, India's dependence on imports for edible oils remains substantially high due to fast rising domestic demand. Production of total oilseeds



is estimated at 42.6 million tonnes in 2024-25, an increase of 7.4 percent compared to 2023-24 but rapeseed and mustard production is estimated to decline to 12.6 million tonnes in 2024-25 and safflower production by 14 percent.

Global Scenario

- 6.5 According to latest forecasts by the USDA, FAO, and IGC, global wheat production and supply is projected to register a marginal increase (<1%) in 2025-26, while world trade (>5%) and utilization (>1.2%) are forecast to witness a higher growth, resulting in lower ending stocks and stocks-to-use ratio (SUR) in 2025-26. However, rising geopolitical conflicts and trade policy and economic uncertainties could adversely affect the trade outlook.
- 6.6 Canada, one of the world's leading exporters of pulses, is forecast to see 3.4 percent decline in total pulses production and 16.5 percent fall in exports, leading to 4.8 percent increase in domestic supply. In case of lentil, production is forecast to decrease by 4.4 percent, while exports, ending stocks and SUR are expected to remain unchanged in 2025-26. Dry peas production is forecast to increase by 4.3 percent but exports are expected to fall significantly, about 38 percent, leading to significant increase in both ending stocks (172%) and SUR (256.5%) in 2025-26.
- 6.7 In 2024-25, global oilseeds production and supply is forecast to increase due to higher production of rapeseed, soybean and sunflower seed. Global oilseeds utilization (3.4%) and trade (2.8%) are also forecast to grow. World ending stocks are forecast to increase by 1.4 percent but SUR is forecast marginally lower in 2025-26.
- 6.8 As per USDA's latest assessment, global production, supply, utilization, and trade of oilmeals and vegetable oils are projected to increase in 2025-26. Both USDA and ABARES estimates indicate higher ending stocks of protein meals and vegetable oils. However, SUR for vegetable oils is anticipated to be lower in 2025-26 as per USDA forecast but ABARES estimates forecast a higher SUR for vegetable oils.

Trade Performance

- 6.9 In 2024-25, India's agricultural imports grew much faster (19.6%) than exports (8.6%), resulting in decline in trade surplus from ₹127.7 thousand crore in 2023-24 to ₹106.9 thousand crore in 2024-25. The share of agricultural exports in India's total merchandise exports increased from 11.6 percent in 2023-24 to 12.3 percent in 2024-25, and the share of agri-imports in total merchandise imports marginally rose from 5.2 percent to 5.7 percent during the period. Rice was the largest agri-export item with a share of 23.2 percent in total agri-exports, followed by marine products (13.8%), spices (8.3%), buffalo meat (7.6%), and sugar (4.0%) in 2024-25. The largest category of agricultural imports was vegetable oils (42.1%), followed by pulses (13.3%), fresh fruits (7.4%), and cashew nuts (4.1%). Due to sharp increase in pulses imports during 2024-25, share of pulses in total agri-imports increased from 10.7 percent in 2023-24 to 13.3 percent in 2024-25.



Productivity Trends

- 6.10 Over the past three years, wheat production has shown an impressive growth primarily driven by area expansion as yield showed a marginal improvement. In 2024-25, wheat yield registered a modest increase of 0.8 percent, while barley yield declined by 1.1 percent. During the last five years, barley production showed wide fluctuations due to variations in crop acreage. Gram production rose by 2.7 percent in 2024-25, largely due to increase in yield (2.5%) but lentil production declined due to fall in area.
- 6.11 In case of *rabi* mandated oilseed crops, production of rapeseed & mustard declined by 4.9 percent in 2024-25 due to a significant decrease in the area under cultivation, while yield registered a positive growth of 1.2 percent. In case of safflower, both area and yield registered negative growth during the last two years, resulting in fall in safflower production in the country.
- 6.12 Over the last five years, yield of wheat, barley and gram registered an annual average growth rate of less than one percent, while safflower yield growth rate was negative. Lentil yield grew at an annual average growth rate of 4.7 percent and rapeseed & mustard yield by 2.1 percent over the last five years. Stagnation in yield of wheat, barley and gram is a matter of concern and needs urgent attention.

Food Inflation

Domestic Scenario

- 6.13 High food inflation has been a major concern over the last three years despite record production. The average annual rate of inflation of food articles (based on WPI; base year 2011-12) was 8.7 percent in 2022, declined to 5.9 percent in 2023 but rose to 8.2 percent in 2024, mainly driven by high inflation in cereals including wheat, pulses, vegetables and fruits. However, the rate of inflation based on WPI Food Articles showed a declining trend since December 2024 and fell from 8.53 percent to (-)1.56 percent in May 2025 but inflation for wheat remained elevated despite record production in 2024-25. The rate of inflation for oilseeds exhibited a continued deflationary trend, (-) 9.6 percent in 2023, and (-) 4.2 percent in 2024 but showed a rising trend during the last four months, from 0.11 percent in February 2025 to 2.79 percent in May 2025.
- 6.14 Consumer Food Price Index (CFPI) based inflation also witnessed a rising trend from about 6.8 percent in 2022 and 2023 to 8.4 percent in 2024 but significantly fell to 3 percent during January to May 2025. The high CFPI inflation was primarily driven by elevated inflation in cereals & products, pulses, fruits & vegetables and milk & products. During the last six months, inflation for pulses and products, vegetables, cereals and products and food and beverages showed a continuous decline but inflation for fruits, and oils and fats remained significantly high.

Global Scenario

- 6.15 The FAO Food Price Index (FFPI) marginally declined (0.8%) to 127.7 points in May 2025 from 128.7 points in April 2025. Cereal Price Index was 109 points in May 2025, registering a decline of 1.7 percent from April 2025. The FAO Vegetable Oil Price Index averaged 152.2 points in May



2025, a decline of 3.7 percent from April 2025, but 19.1 percent higher than May 2024 level. The decline in Vegetable Oil Price Index was driven by decline in world prices of palm, rapeseed, soy and sunflower oils.

- 6.16 International wheat prices remained generally steady in 2024-25 season with some fluctuations. In May 2025, world prices of wheat were 10.5 percent below their May 2024 level and 27.3 percent below their five-year average value but initial forecasts for 2025-26 season indicate higher prices due to higher global utilization, stronger import demand and reduction in stocks in major exporting countries as per FAO's latest assessment (June 2025).

Market Price Trends

- 6.17 The average market prices of wheat have consistently remained above the MSP since RMS2022-23, while market prices of barley generally showed an upward trend during 2024-25 and 2025-26 season and were higher than the MSP except in April 2024. Gram prices exhibited significant upward trend during RMS2023-24 and RMS2024-25 and moved above the MSP in March 2024 but fell below the MSP during RMS2025-26, despite some improvement in market prices after the imposition of 10 percent import duty on gram from April 1, 2025. Market prices of lentil and rapeseed & mustard also showed an upward trend during the last two seasons but remained below the MSP. Market prices of safflower displayed high volatility and were above the MSP during April-June 2025.

Procurement Operations

- 6.18 Wheat procurement during the RMS2025-26 stood at 30 million tonnes, higher than the previous year's level of 26.6 million tonnes, and number of beneficiary farmers increased by 11.5 percent, reaching 25.1 lakh. Procurement of *rabi* pulses, though increased during RMS2025-26 over RMS2024-25, but remained much below the sanctioned quantity under the PSS as market prices witnessed improvement. Procurement of rapeseed & mustard, which increased significantly to 11.1 lakh tonnes in RMS2023-24 and 12.1 lakh tonnes during RMS2024-25, declined to 2.9 lakh tonnes during RMS2025-26 (as on 5th June 2025) due to improvement in market prices of mustard.

Terms of Trade

- 6.19 The Farmers' Terms of Trade (FToT) index increased from 97.2 in 2022-23 to 99.5 in 2023-24(P) and Index of Agricultural Terms of Trade (AGRTToT), which includes both farmers and agricultural labourers, increased from 103.5 in 2022-23 to 105.5 in 2023-24 (P). The AGRTToT index has remained higher than the FToT over the years.

Cost of Production and Inter-Crop Parity

- 6.20 During TE2023-24, all-India per hectare average CoC A_2 (₹37,225) and A_2 +FL (₹55,507) were highest for barley, while these costs were lowest in safflower at ₹16,150 and ₹20,772, respectively. Average per hectare A_2 +FL was estimated at ₹45,558 for wheat, ₹55,507 for barley, ₹39,632 for gram, ₹28,618 for lentil, ₹41,025 for rapeseed & mustard, and ₹20,772 for safflower. The per hectare average GVO was highest for barley at ₹1,05,036, followed by wheat (₹96,014), rapeseed & mustard



(₹87,877), gram (₹66,823), lentil (₹56,386) and the lowest for safflower at ₹17,886. Average per hectare gross returns over A_2 +FL during TE 2023-24 were highest for wheat (₹50,456/ha), followed by barley (₹49,530/ha), rapeseed & mustard (₹46,851/ha), lentil (₹27,769/ha) and gram (₹27,192/ha), while safflower farmers incurred a net loss of ₹2,885 per hectare.

- 6.21 At all-India level, agricultural wage rate increased by 7.1 percent in 2023-24 and 6.6 percent in 2024-25 over the preceding years, with the highest increase in Himachal Pradesh (10.4%) and the lowest increase in Punjab (1.6%) in 2024-25 over the previous year. The all-India CIPI for *rabi* crops increased by 8.2 percent in 2024-25 and 9 percent in 2025-26. Among different components of CIPI, machine labour cost registered the highest increase of 12.7 percent and insecticide cost recorded the lowest increase (1.3%) percent in 2025-26 compared to 2024-25. The human labour accounts for more than 40 percent of the total operational costs in most mandated crops.
- 6.22 The all-India per quintal cost of production (CoP) A_2 +FL for crop season 2025-26 (RMS2026-27) is projected at ₹1,239 for wheat, ₹1,361 for barley, ₹3,699 for gram, ₹3,705 for lentil, ₹3,210 for rapeseed & mustard, and ₹4,360 for safflower. The rise in projected A_2 +FL cost during crop season 2025-26 over 2024-25 varied from 4.7 percent in lentil to 10 percent in safflower. Among the mandated *rabi* crops, wheat is the most profitable, followed by barley, rapeseed & mustard, lentil, and gram, while farmers cultivating safflower incur a net loss.

Non-Price Recommendations

- 6.23 Drawing upon the analysis presented in preceding chapters of this report and discussions with various stakeholders, the Commission makes the following non-price recommendations to encourage farmers to adopt improved technologies, sustainable use of resources, and facilitate crop diversification in tune with the emerging demand patterns in domestic and global markets.

Emphasis on Productivity Improvement

- 6.24 Productivity of wheat in Bihar, Madhya Pradesh and Maharashtra has increased during the last 10 years but yield levels are much lower than all-India average and other major producing States, while Punjab, Haryana and Uttar Pradesh experienced stagnation or slowdown in wheat yield. However, all-India average productivity growth in all mandated crops except lentil and rapeseed & mustard has experienced stagnation or slow down over the last five years. There are also considerable yield gaps across many States in most crops indicating large scope for increasing yields but wheat yield in Punjab and Haryana has almost reached the potential yield, indicating a need for major technological breakthrough for pushing productivity frontiers. Therefore, the Commission recommends strengthening technological interventions through Research and Development (R&D), availability of quality inputs and services, expansion of irrigation and better farm practices to improve productivity. In case of oilseeds, special emphasis should be given on improving seed yield and oil content to increase production and profitability.



Aatmanirbharta in Pulses: Role of Technology, Trade Policy and Incentives

- 6.25 India's spectacular growth in pulses production during 2015-16 and 2021-22 was driven by three pillars: (i) technology - ensuring timely availability of quality seed of improved varieties to farmers, (ii) remunerative prices to farmers through significant increase in MSP and effective procurement, and (iii) conducive trade policy - restricting cheap imports of pulses through high import duties, quantitative restrictions (QRs), import bans, etc. However, pulses production in the country has experienced stagnation during the last three years despite substantial increase in MSP. Unrestricted cheap imports of pulses, particularly, yellow peas, gram and lentil have adversely affected domestic market prices. In addition, technology fatigue has resulted in deceleration in pulses area and yield growth. Therefore, the Commission recommends two-pronged strategy – (i) special focus on developing new high-yielding climate resilient varieties and making quality seed available to farmers, and (ii) restrict cheap imports of gram and lentil through high tariffs and complete ban on yellow peas imports.

Strategies for Reducing Edible Oils Import Dependence

- 6.26 Despite significant increase in oilseeds production in the country during last 10 years, from 25.2 million tonnes in 2015-16 to 42.6 million tonnes in 2024-25, dependence on imports has remained high due to fast rising demand for edible oils. India imported 15.9 million tonnes of edible oils worth ₹1.46 lakh crore in TE2024-25. In mandated *rabi* oilseeds, performance of rapeseed & mustard has been remarkable in the recent period but safflower, which was an important and premium oil crop containing high oil content (28-30%) in Maharashtra and Karnataka, has witnessed significant decline in both area and production. The Commission recommends that a special programme for promoting safflower cultivation in Maharashtra, Karnataka and parts of Telangana and Andhra Pradesh should be launched under the NMEO-Oilseeds and concerted efforts should be made to develop high yielding varieties/hybrids with high oil content in mustard using conventional as well as biotechnological approaches. The Commission strongly believes that effective implementation of National Mission on Edible Oils - Oil Palm (NMEO-OP) will help in bridging the gap between demand and supply of edible oils by increasing area under oil palm and production of Crude Palm Oil (CPO).
- 6.27 World prices of vegetable oils in general have high volatility and sudden and large fall in global prices adversely affect domestic prices of oilseeds in the country. Therefore, the Commission recommends that a dynamic duty structure by linking import tariffs to international prices of vegetable oils should be adopted and in case of abrupt and sudden fall in global prices, trigger points can be identified for quick action to raise import duties.

Link Oilseeds MSP to Oil Content

- 6.28 There are significant variations in oil content of different varieties of oilseeds, therefore, a uniform MSP may not be optimal. To incentivise high oil content in oilseeds and ensure better returns to farmers, MSP of rapeseed & mustard and safflower should be linked to the oil content. The Commission recommends that for every 0.25 percentage point increase in oil content above the basic level, premium should be paid to farmers. Besides efforts should be made to develop seed varieties with high oil content, which are climate resilient and incentivise their use as a long term strategy.



Address Low Fertilizer Use Efficiency and Pricing Issues

- 6.29 It is worth mentioning that urea sector in India, which was highly dependent on imports, is moving in the direction of achieving self-sufficiency due to various initiatives taken by the Government. Production and consumption of fertilizers has increased in the country but imbalanced use of N, P and K resulting in low use efficiency, deficiency of micro and secondary nutrients and problem of excessive/low fertilizer use in some regions are major problems confronting Indian agriculture. Promoting balanced and judicious fertilizer application based on soil test recommendations is essential for reducing costs, enhancing nutrient use efficiency, and maintaining long-term soil fertility.
- 6.30 The Commission recommends strengthening soil testing laboratories, installing modern equipments and providing hands-on training to staff to provide real-time soil analysis and advisory services to farmers. As the current fertilizer subsidy structure has led to excessive use of urea, resulting in imbalanced nutrient use, the Commission recommends that urea price should be increased in a phased manner and subsidy savings on account of increase in urea prices should be used for providing higher subsidy on P & K fertilizers to address the problem of imbalance in use of nutrients. As problems of micronutrient deficiencies and low organic carbon in Indian soils are widespread, a nationwide intensive awareness campaign on balanced soil and crop nutrition with special focus on use of micronutrients and biofertilizers should be launched.

Strengthen PSS Procurement Operations at State Level

- 6.31 Procurement of pulses and oilseeds under PSS is implemented on the request of the State Governments/UTs and arrangements for setting up procurement centres, storage facilities, gunny bags, and other logistics are made by the States/UTs. However, the Scheme faces several challenges including delay in procurement operations, inadequate storage and warehousing infrastructure, availability of jute bags and other logistics constraints. The Commission recommends that States should make adequate arrangements for procurement including registration of farmers, details of purchase centers, quality specifications, scheduling of date and time for selling produce at purchase center, storage and warehousing, gunny bags, QR tags, timely payment, etc. well in advance and communicated to farmers.

Strategies for Reducing Costs

- 6.32 Rising cost in agriculture is a major concern, which adversely affects farm profitability and food security. Government has taken several initiatives including use of renewable energy sources, farm mechanization, nano-fertilizers, drones, etc. to reduce cost and promote efficient use of resources. However, adoption of nano-fertilizers and progress of PM-KUSUM is lower than anticipated. Therefore, Central/State Governments should organize awareness campaigns through various channels, including village-level meetings and demonstrations, and information dissemination through local media to promote solarizing of agricultural pumps and balanced use of fertilizers. The Commission recommends significant increase in allocation under SMAM and the issue of high GST rates on farm machinery and implements requires favourable consideration to encourage mechanization. In addition, farmers should be encouraged to shift from less profitable crops to more suitable and profitable crops.



Commodity Markets and Price Outlook

- 6.33 Commodity markets and price outlook reports providing comprehensive assessment and short-term forecasts of demand, supply, trade, stocks and prices for major commodities and policy developments in various countries can help in efficient functioning of markets and policy formulation. The Commission recommends that a comprehensive project on commodity markets and price outlook for providing reliable and timely information about markets and price forecasts should be launched to enable farmers to make informed decisions on production and marketing as well as help Government to take appropriate and timely policy decisions, which would lead to less volatility in markets and higher profitability. It will also help the Commission in price policy recommendations.

Improve Irrigation Water Use Efficiency

- 6.34 Various initiatives under *Pradhan Mantri Krishi Sinchayee Yojana* (PMKSY) have resulted in an impressive increase in irrigation potential and coverage under micro-irrigation. However, there are large disparities in irrigation intensity across States, ranging from as high as 98 percent in Punjab to less than 25 percent in States like Jharkhand, Assam, Kerala, and Maharashtra and area under micro-irrigation is concentrated in a few States. Therefore, future efforts should focus on reducing regional disparities in irrigation coverage and improve water use efficiency and water productivity. In order to encourage farmers to adopt micro-irrigation, awareness campaigns and training programmes should be organized and supply of good quality equipments along with post-installation maintenance services should be ensured to farmers. Recognising the importance of traditional water bodies like tanks and ponds, especially in regions with limited water availability, special emphasis should be placed on repair, renovation and restoration of such water structures with active involvement of district administration and gram *panchayats*.

Make Agricultural Credit More Inclusive

- 6.35 Government has taken various steps such as *Kisan Credit Card* (KCC), Interest Subvention Scheme, Credit Guarantee Scheme for e-NWR based pledge financing, priority sector lending, collateral-free loans etc. to enhance access to affordable institutional credit and avoid distress sale by farmers. These efforts have resulted in significant increase in flow of credit to agriculture as well as improved access to credit by small and marginal farmers, rising from 50.4 percent share in 2018-19 to 56.5 percent in 2023-24. However, large disparities across States persist and some regions are still credit starved due to low penetration and weak financial health of RFIs. Therefore, concerted efforts are needed to reduce inter-State and regional disparities and strengthen RFIs in Central, Eastern and North-eastern regions. In order to promote pledge financing against e-NWRs, publicity and awareness campaigns should be organised for various stakeholders including farmers, traders, bankers, APMC staff, etc.

Remove Market Distortions

- 6.36 Well-functioning agricultural markets are important as they help in price discovery and efficient allocation of resources but any distortions to competitive functioning of markets adversely affect these functions and reduce the efficiency of market outcomes. However, some States distort markets in the form of high market fees, taxes and other incidental charges on procurement resulting in high economic cost of



grains. Such interventions also restrict private sector participation and adversely affect competition. The Commission recommends that States should rationalise market fee and other charges to create efficient and competitive national market.

- 6.37 Wheat is currently stifled by various controls ranging from stock limits on various market participants to export restrictions. However, the Commission is of the view that such regulatory steps should be temporary in nature and efforts should be made to increase wheat production.

Measures to Address Crop Losses/Damages by Wild/Stray Animals

- 6.38 The incidences of crop loss/damages, particularly pulses, due to wild/stray animals have significantly increased in recent years. Under *Pradhan Mantri Fasal Bima Yojana* (PMFBY), crop losses due to wild animals, where the risk is substantial and identifiable, are covered but there is lack of awareness among farmers about such provision. Efforts should be made to spread awareness among farmers about such provisions under PMFBY, and feasibility of providing financial assistance to farmers for solar fencing of their fields could also be explored as an alternative method for minimizing crop damage from wild/stray animals.

Awareness about MSP and FAQ Specifications

- 6.39 Various initiatives have been taken by Central and State Governments to enhance awareness and participation of farmers and other stakeholders in procurement operations. The Department of Food and Public Distribution, Government of India, in collaboration with MyGov, hosted a series of activities to promote public awareness about MSP and procurement operations as well as community involvement in wheat procurement during RMS2025-26. Similarly, State Governments are also using different media platforms of publicity including print, electronic/digital, audio/video to spread awareness about MSP and procurement operations. During RMS2025-26, Government of Uttar Pradesh introduced mobile procurement centers to purchase wheat directly from villages. Such initiatives have resulted in increased awareness about MSP. However, farmers face main difficulty in meeting FAQ specifications primarily due to lack of knowledge or infrastructure facilities, particularly related to moisture content. Problem of high moisture content in mustard was raised by majority of farmers during the Commission's meetings and field visits. The Commission strongly recommends that basic infrastructure facilities for cleaning and drying of produce should be provided at farm/village level.

Sample Size and Coverage of States

- 6.40 A reasonable sample size is crucial for obtaining robust and reliable cost estimates but sample size in some cases under the Comprehensive Scheme is small. Therefore, the Commission recommends that sample size of gram in Haryana, Jharkhand and Telangana, rapeseed & mustard in Chhattisgarh, and safflower in Karnataka and Maharashtra should be increased. Since the Commission uses three-year actual cost estimates for forecasting costs, the CS data should be collected on continuous basis for gram in West Bengal, lentil in Jharkhand, rapeseed & mustard in Chhattisgarh and safflower in Maharashtra for reliable estimates. In addition, the Comprehensive Scheme should be implemented in Jammu & Kashmir for wheat and rapeseed & mustard, and Uttarakhand for wheat, barley, lentil and rapeseed & mustard.



Minimum Support Price (MSP) Recommendations

6.41 The Commission recommends the following MSPs for mandated *rabi* crops for RMS2026-27:

Table 6.1: Minimum Support Prices (MSPs) Recommended for RMS2026-27

Crop	Projected A ₂ +FL Cost for RMS2026-27	MSP for RMS2025-26	Recommended MSP for RMS2026-27	(₹/qtl) MSP as percent of A ₂ +FL
Wheat	1239	2425	2585 (6.6)	209
Barley	1361	1980	2150 (8.6)	158
Gram	3699	5650	5875 (4.0)	159
Lentil (<i>Masur</i>)	3705	6700	7000 (4.5)	189
Rapeseed & Mustard	3210	5950	6200 (4.2)	193
Safflower	4360	5940	6540 (10.1)	150

Note: Figures in parenthesis represent an increase in MSP (%) over the previous year

6.42 The Commission believes that these non-price and price policy recommendations will incentivise farmers to adopt modern technologies and practices, boost pulses and oilseeds production, re-orient production pattern to tap emerging market opportunities, promote efficient utilization of production resources and help in improving farmer's income and welfare.

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(Prem Chand)
Member (Official)

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Annex Table

Annex Table 1.1: All-India Estimates of Area under Principal Crops

(Lakh Ha)

Crop	Season	2012-13	2013-14	2014-15	2015-16	2016-17	2017-18	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25*
Rice	Kharif	389.1	394.5	398.3	396.6	398.5	393.5	399.6	390.1	403.6	410.4	404.0	407.3	434.1
	Rabi	38.4	46.9	42.8	38.4	41.5	44.3	41.9	46.5	54.1	52.4	43.3	40.8	44.7
	Summer											31.0	30.2	35.4
	Total	427.5	441.4	441.1	435.0	439.9	437.7	441.6	436.6	457.7	462.8	478.3	478.3	514.2
Wheat	Rabi	300.0	304.7	314.7	304.2	307.9	296.5	293.2	313.6	311.3	304.6	314.0	318.3	327.6
	Total	300.0	304.7	314.7	304.2	307.9	296.5	293.2	313.6	311.3	304.6	314.0	318.3	327.6
	Kharif	24.3	22.8	22.7	21.4	20.6	20.6	17.6	17.6	16.4	14.7	13.8	12.9	14.2
	Rabi	37.9	35.2	38.9	39.4	35.7	29.7	23.4	30.7	27.4	23.3	21.4	27.4	25.1
Bajra	Summer											0.2	0.5	0.5
	Total	62.1	57.9	61.6	60.8	56.2	50.2	40.9	48.2	43.8	38.0	35.4	40.8	39.8
	Kharif	73.0	78.1	73.2	71.3	74.6	74.8	71.1	75.4	76.5	68.4	71.7	69.7	68.4
	Summer											4.0	4.1	3.7
Maize	Total	73.0	78.1	73.2	71.3	74.6	74.8	71.1	75.4	76.5	68.4	75.7	73.8	72.1
	Kharif	72.1	73.1	75.6	71.8	78.4	74.3	73.3	75.5	77.6	77.9	80.5	83.3	84.3
	Rabi	14.6	17.6	16.2	16.3	17.9	19.5	17.0	20.2	21.4	21.7	21.8	21.8	27.2
	Summer											5.1	7.4	8.6
Ragi	Total	86.7	90.7	91.9	88.1	96.3	93.8	90.3	95.7	98.9	99.6	107.4	112.4	120.2
	Kharif	11.3	11.9	12.1	11.4	10.2	11.9	8.9	10.0	11.6	12.2	11.6	12.2	12.3
	Total	11.3	11.9	12.1	11.4	10.2	11.9	8.9	10.0	11.6	12.2	11.6	12.2	12.3
	Rabi	7.0	6.7	7.1	5.9	6.6	6.6	5.8	5.9	5.9	4.5	6.3	5.5	6.1
Barley	Total	7.0	6.7	7.1	5.9	6.6	6.6	5.8	5.9	5.9	4.5	6.3	5.5	6.1
	Kharif	116.1	119.6	113.8	110.6	111.5	112.8	102.0	107.6	109.0	99.6	101.5	99.5	99.3
	Rabi	37.9	35.2	38.9	39.4	35.7	29.7	23.4	30.7	27.4	23.3	21.4	27.4	25.1
	Summer											4.2	4.6	4.2
Shree Anna / Nutri Cereals	Total	154.0	154.8	152.8	149.9	147.2	142.5	125.4	138.3	136.3	122.9	127.0	131.5	128.6
	Kharif	188.2	192.7	189.5	182.4	190.0	187.2	175.4	183.1	186.5	177.4	182.0	182.8	183.6
	Rabi	59.4	59.5	62.2	61.6	60.1	55.7	46.1	56.7	54.7	49.6	49.5	54.6	58.4
	Summer											9.2	12.0	12.9
Nutri/Coarse Cereals	Total	247.7	252.2	251.7	243.9	250.1	242.9	221.5	239.9	241.2	227.0	240.7	249.4	254.9

(Contd.)



Annex Table

Annex Table 1.1: All-India Estimates of Area under Principal Crops

(Lakh Ha)

Crop	Season	2012-13	2013-14	2014-15	2015-16	2016-17	2017-18	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25*
Cereals	Kharif	577.4	587.2	587.8	578.9	588.4	580.6	575.0	573.3	590.1	587.8	586.0	590.1	617.7
	Rabi	397.9	411.1	419.7	404.2	409.5	396.5	381.2	416.8	420.0	406.6	406.8	413.7	430.8
	Summer											40.2	42.1	48.2
	Total	975.2	998.3	1007.5	983.1	997.9	977.1	956.2	990.1	1010.1	994.4	1033.0	1046.0	1096.7
Tur	Kharif	38.9	39.0	38.5	39.6	53.4	44.4	45.5	45.3	47.2	49.0	40.7	41.3	43.3
	Total	38.9	39.0	38.5	39.6	53.4	44.4	45.5	45.3	47.2	49.0	40.7	41.3	43.3
	Rabi	85.2	99.3	82.5	84.0	96.3	105.6	95.4	97.0	100.0	107.4	104.7	95.9	96.1
	Total	85.2	99.3	82.5	84.0	96.3	105.6	95.4	97.0	100.0	107.4	104.7	95.9	96.1
Urad	Kharif	24.6	23.5	24.9	27.2	34.8	43.5	47.3	37.0	32.1	36.3	31.0	26.8	21.0
	Rabi	6.9	7.2	7.6	9.0	10.0	9.3	8.8	8.3	9.3	10.1	6.4	5.9	6.2
	Summer											2.6	2.7	3.0
	Total	31.5	30.6	32.5	36.2	44.8	52.8	56.0	45.3	41.4	46.3	40.0	35.4	30.2
Moong	Kharif	19.8	23.4	20.3	27.6	33.7	32.6	38.3	35.2	38.2	38.4	34.9	31.7	33.9
	Rabi	7.4	10.5	9.9	10.7	9.6	9.8	9.2	10.6	13.1	17.1	1.5	1.4	1.3
	Summer											19.1	18.7	20.5
	Total	27.2	33.8	30.2	38.3	43.3	42.4	47.6	45.8	51.3	55.5	55.5	51.9	55.7
Lentil	Rabi	14.2	13.4	14.7	12.8	14.6	15.5	13.6	13.0	14.7	14.1	16.4	17.4	17.1
	Total	14.2	13.4	14.7	12.8	14.6	15.5	13.6	13.0	14.7	14.1	16.4	17.4	17.1
	Kharif	99.5	103.4	100.0	113.1	143.6	139.3	148.3	135.4	134.3	141.2	121.0	116.3	113.7
	Rabi	133.0	148.9	135.6	136.0	150.8	158.8	143.2	144.5	153.5	166.1	146.3	137.4	139.0
Total Pulses	Summer											21.7	21.4	23.5
	Total	232.6	252.2	235.6	249.1	294.5	298.1	291.5	279.9	287.8	307.3	289.0	275.1	276.2
	Kharif	676.9	690.6	687.8	692.1	732.0	720.0	723.3	708.6	724.4	729.0	707.0	706.4	731.4
	Rabi	530.9	559.9	555.3	540.1	560.3	555.3	524.5	561.3	573.5	572.7	553.1	551.1	569.8
Foodgrains	Summer											62.0	63.5	71.7
	Total	1207.8	1250.5	1243.0	1232.2	1292.3	1275.2	1247.7	1270.0	1298.0	1301.7	1322.0	1321.0	1372.9
	Kharif	39.3	46.5	40.1	38.4	45.8	41.4	41.3	41.6	51.7	49.1	42.6	40.4	50.0
	Rabi	7.9	8.6	7.6	7.6	7.6	7.5	6.0	6.7	8.4	7.9	4.2	3.4	3.4
Groundnut	Summer											2.8	3.2	4.2
	Total	47.2	55.1	47.7	46.0	53.4	48.9	47.3	48.3	60.2	57.1	49.6	47.1	57.5

(Contd.)

Annex Table 1.1: All-India Estimates of Area under Principal Crops

(Lakh Ha)

Crop	Season	2012-13	2013-14	2014-15	2015-16	2016-17	2017-18	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25*
Sesamum	Kharif	17.1	16.8	17.5	19.5	16.7	15.8	14.2	16.2	17.2	16.3	10.4	10.1	10.4
	Rabi											0.5	0.5	0.4
	Summer											4.3	4.7	4.5
Nigerseed	Total	17.1	16.8	17.5	19.5	16.7	15.8	14.2	16.2	17.2	16.3	15.2	15.3	15.3
	Kharif	3.1	3.0	2.3	2.5	2.6	2.2	1.6	1.4	1.3	1.0	0.9	0.8	0.5
	Total	3.1	3.0	2.3	2.5	2.6	2.2	1.6	1.4	1.3	1.0	0.9	0.8	0.5
Soybean	Kharif	108.4	117.2	109.1	116.1	111.8	103.3	111.3	121.9	129.2	121.5	130.8	132.6	129.6
	Total	108.4	117.2	109.1	116.1	111.8	103.3	111.3	121.9	129.2	121.5	130.8	132.6	129.6
	Kharif	3.0	2.6	2.2	1.6	1.7	1.4	1.2	1.3	1.0	1.4	2.2	0.7	0.8
Sunflower	Rabi	5.3	4.2	3.8	3.3	2.1	1.5	1.4	1.0	1.3	1.3	1.1	0.5	0.7
	Summer											0.4	0.3	0.2
	Total	8.3	6.8	5.9	4.9	3.8	2.8	2.6	2.3	2.3	2.7	3.6	1.5	1.7
Rapeseed &	Rabi	63.6	66.5	58.0	57.5	60.7	59.8	61.2	68.6	67.0	79.9	88.5	91.8	86.3
Mustard	Total	63.6	66.5	58.0	57.5	60.7	59.8	61.2	68.6	67.0	79.9	88.5	91.8	86.3
Safflower	Rabi	1.8	1.8	1.8	1.3	1.4	0.8	0.5	0.5	0.6	0.8	1.1	0.7	0.6
	Total	1.8	1.8	1.8	1.3	1.4	0.8	0.5	0.5	0.6	0.8	1.1	0.7	0.6
	Kharif	183.2	196.6	182.1	188.6	186.7	172.3	177.1	192.8	209.3	197.5	197.2	195.0	200.5
Oilseeds	Rabi	81.6	84.0	73.9	72.2	75.1	72.8	70.9	78.6	79.0	91.9	97.8	98.7	93.2
	Summer											7.4	8.3	8.9
	Total	264.8	280.5	256.0	260.9	261.8	245.1	247.9	271.4	288.3	289.5	302.4	301.9	302.7

Note: * Third Advance Estimates; from 2012-13 to 2021-22, summer included in rabi for rice, maize, jowar, rabi and summer included in kharif for sesame and kharif for bajra

Source: Economics, Statistics and Evaluation Division, Department of Agriculture & Farmers Welfare



Annex Table

Annex Table 1.2: All-India Estimates of Production of Principal Crops

(Lakh tonnes)

Crop	Season	2012-13	2013-14	2014-15	2015-16	2016-17	2017-18	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25*
Rice	Kharif	923.6	915.0	913.9	914.1	963.0	971.4	1020.4	1022.8	1052.1	1110.0	1105.1	1132.6	1218.5
	Rabi	128.7	151.5	140.9	130.0	134.0	156.2	144.4	165.9	191.6	184.7	150.0	146.0	156.7
	Summer											102.4	99.7	115.5
	Total	1052.3	1066.5	1054.8	1044.1	1097.0	1127.6	1164.8	1188.7	1243.7	1294.7	1357.6	1378.3	1490.7
Wheat	Rabi	935.1	958.5	865.3	922.9	985.1	998.7	1036.0	1078.6	1095.9	1077.4	1105.5	1132.9	1175.1
	Total	935.1	958.5	865.3	922.9	985.1	998.7	1036.0	1078.6	1095.9	1077.4	1105.5	1132.9	1175.1
	Kharif	28.4	23.9	23.0	18.2	19.6	22.7	17.4	17.0	19.9	16.0	14.8	15.1	17.5
	Rabi	24.4	31.5	31.5	24.2	26.0	25.3	17.4	30.8	28.3	25.5	23.2	31.9	30.9
Jowar	Summer											0.1	0.4	0.4
	Total	52.8	55.4	54.5	42.4	45.7	48.0	34.8	47.7	48.1	41.5	38.1	47.4	48.8
Bajra	Kharif	87.4	92.5	91.8	80.7	97.3	92.1	86.6	103.6	108.6	97.8	103.5	96.6	98.5
	Summer											10.8	10.5	10.2
	Total	87.4	92.5	91.8	80.7	97.3	92.1	86.6	103.6	108.6	97.8	114.3	107.2	108.6
Maize	Kharif	162.0	171.5	170.1	160.5	189.2	201.2	194.1	194.3	215.6	226.8	236.7	222.5	248.4
	Rabi	60.5	71.1	71.6	65.1	69.8	86.3	83.0	93.4	100.9	110.5	116.9	120.3	136.2
	Summer											27.2	33.9	38.2
	Total	222.6	242.6	241.7	225.7	259.0	287.5	277.2	287.7	316.5	337.3	380.9	376.7	422.8
Ragi	Kharif	15.7	19.8	20.6	18.2	13.9	19.9	12.4	17.6	20.0	17.0	16.9	16.7	18.3
	Total	15.7	19.8	20.6	18.2	13.9	19.9	12.4	17.6	20.0	17.0	16.9	16.7	18.3
Barley	Rabi	17.5	18.3	16.1	14.4	17.5	17.8	16.3	17.2	16.6	13.7	19.1	17.0	18.4
	Total	17.5	18.3	16.1	14.4	17.5	17.8	16.3	17.2	16.6	13.7	19.1	17.0	18.4
Shree Anna / Nutri Cereals	Kharif	135.9	140.6	139.3	121.0	135.2	139.1	119.7	141.9	152.0	134.5	139.0	132.9	138.7
	Rabi	24.4	31.5	31.5	24.2	26.0	25.3	17.4	30.8	28.3	25.5	23.2	31.9	30.9
	Summer											11.0	10.9	10.6
	Total	160.3	172.1	170.8	145.2	161.3	164.4	137.1	172.6	180.2	160.0	173.2	175.7	180.2
Nutri/ Coarse Cereals	Kharif	298.0	312.0	309.5	281.5	324.4	340.3	313.8	336.2	367.5	361.3	375.8	355.4	387.2
	Rabi	102.5	120.9	119.2	103.7	113.3	129.5	116.8	141.3	145.7	149.7	159.3	169.2	185.5
	Summer											38.2	44.9	48.8
	Total	400.4	433.0	428.6	385.2	437.7	469.7	430.6	477.5	513.2	511.0	573.2	569.4	621.4
Cereals	Kharif	1221.6	1227.0	1223.4	1195.6	1287.4	1311.6	1334.2	1358.9	1419.6	1471.3	1480.9	1487.9	1605.7
	Rabi	1166.3	1230.9	1125.3	1156.6	1232.4	1284.4	1297.1	1385.9	1433.2	1411.9	1414.8	1448.1	1517.3
	Summer											140.6	144.5	164.3
	Total	2387.8	2457.9	2348.7	2352.2	2519.8	2596.0	2631.3	2744.8	2852.8	2883.1	3036.3	3080.5	3287.2

(Contd.)

Annex Table 1.2: All-India Estimates of Production of Principal Crops

(Lakh tonnes)

Crop	Season	2012-13	2013-14	2014-15	2015-16	2016-17	2017-18	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25*
Tur	Kharif	30.2	31.7	28.1	25.6	48.7	42.9	33.2	38.9	43.2	42.2	33.1	34.2	35.6
	Total	30.2	31.7	28.1	25.6	48.7	42.9	33.2	38.9	43.2	42.2	33.1	34.2	35.6
Gram	Rabi	88.3	95.3	73.3	70.6	93.8	113.8	99.4	110.8	119.1	135.4	122.7	110.4	113.4
	Total	88.3	95.3	73.3	70.6	93.8	113.8	99.4	110.8	119.1	135.4	122.7	110.4	113.4
Urad	Kharif	15.0	11.5	12.8	12.5	21.8	27.5	23.6	13.3	15.1	18.7	17.7	16.0	13.0
	Rabi	4.7	5.5	6.8	7.0	6.6	7.4	7.0	7.5	7.2	9.1	6.3	4.9	5.3
	Summer											2.3	2.3	2.8
	Total	19.7	17.0	19.6	19.5	28.3	34.9	30.6	20.8	22.3	27.8	26.3	23.2	21.1
Moong	Kharif	7.9	9.6	8.7	10.0	16.4	14.3	17.8	18.3	20.0	14.8	17.2	11.5	17.5
	Rabi	4.0	6.5	6.4	5.9	5.2	5.9	6.7	6.8	10.9	16.9	1.1	1.0	1.0
	Summer											18.5	18.5	19.7
	Total	11.9	16.1	15.0	15.9	21.7	20.2	24.6	25.1	30.9	31.7	36.8	31.0	38.2
Lentil	Rabi	11.3	10.2	10.4	9.8	12.2	16.2	12.3	11.0	14.9	12.7	15.6	17.9	17.7
	Total	11.3	10.2	10.4	9.8	12.2	16.2	12.3	11.0	14.9	12.7	15.6	17.9	17.7
Total Pulses	Kharif	59.2	60.0	57.3	55.3	95.8	93.1	80.9	79.2	86.2	82.4	76.2	69.7	75.0
	Rabi	124.3	132.6	114.2	107.9	135.5	161.1	139.9	151.0	168.5	190.7	163.6	152.0	154.9
	Summer											20.8	20.8	22.5
	Total	183.4	192.6	171.5	163.2	231.3	254.2	220.8	230.3	254.6	273.0	260.6	242.5	252.4
Total Foodgrains	Kharif	1280.7	1287.0	1280.7	1250.9	1383.3	1404.7	1415.2	1438.1	1505.8	1553.6	1557.1	1557.7	1680.7
	Rabi	1290.5	1363.5	1239.6	1264.5	1367.8	1445.5	1436.9	1536.9	1601.7	1602.5	1578.4	1600.1	1672.2
	Summer											161.4	165.2	186.8
	Total	2571.3	2650.5	2520.3	2515.4	2751.1	2850.1	2852.1	2975.0	3107.4	3156.2	3296.9	3323.0	3539.6
Groundnut	Kharif	31.9	80.6	59.3	53.7	60.5	76.0	53.9	83.9	85.3	84.3	85.6	86.6	103.7
	Rabi	15.1	16.6	14.7	13.7	14.1	16.6	13.4	15.6	17.2	17.0	11.0	7.6	6.9
	Summer											6.3	7.7	8.4
	Total	47.0	97.1	74.0	67.3	74.6	92.5	67.3	99.5	102.4	101.4	103.0	101.8	119.0
Sesamum	Kharif	6.9	7.2	8.3	8.5	7.5	7.6	6.9	6.6	8.2	7.9	3.9	4.0	4.3
	Rabi											0.3	0.3	0.2
	Summer											3.8	4.2	3.8
	Total	6.9	7.2	8.3	8.5	7.5	7.6	6.9	6.6	8.2	7.9	8.0	8.5	8.3
Nigerseed	Kharif	1.0	1.0	0.8	0.7	0.9	0.7	0.5	0.4	0.4	0.3	0.3	0.3	0.2
	Total	1.0	1.0	0.8	0.7	0.9	0.7	0.5	0.4	0.4	0.3	0.3	0.3	0.2

(Contd.)



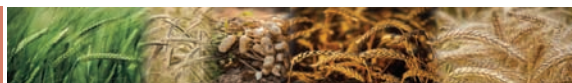
Annex Table

Annex Table 1.2: All-India Estimates of Production of Principal Crops

Crop	Season	2012-13	2013-14	2014-15	2015-16	2016-17	2017-18	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25*
		(Lakh tonnes)	(Lakh tonnes)	(Lakh tonnes)	(Lakh tonnes)	(Lakh tonnes)	(Lakh tonnes)	(Lakh tonnes)	(Lakh tonnes)	(Lakh tonnes)	(Lakh tonnes)	(Lakh tonnes)	(Lakh tonnes)	(Lakh tonnes)
Soybean	Kharif	146.7	118.6	103.7	85.7	131.6	109.3	132.7	112.3	126.1	129.9	149.9	130.6	151.8
	Total	146.7	118.6	103.7	85.7	131.6	109.3	132.7	112.3	126.1	129.9	149.9	130.6	151.8
Sunflower	Kharif	1.9	1.7	1.4	0.7	1.0	0.9	0.9	0.9	0.8	1.1	2.0	0.6	0.7
	Rabi	3.6	3.4	2.9	2.3	1.5	1.4	1.3	1.2	1.5	1.4	1.2	0.7	1.0
	Summer											0.4	0.4	0.3
Rapeseed & Mustard	Total	5.4	5.1	4.3	3.0	2.5	2.2	2.2	2.1	2.3	2.5	3.6	1.7	2.0
	Rabi	80.3	78.8	62.8	68.0	79.2	84.3	92.6	91.2	102.1	119.6	126.4	132.6	126.1
Safflower	Total	80.3	78.8	62.8	68.0	79.2	84.3	92.6	91.2	102.1	119.6	126.4	132.6	126.1
	Rabi	1.1	1.1	0.9	0.5	0.9	0.6	0.3	0.4	0.4	0.6	0.9	0.5	0.4
Oilseeds	Total	1.1	1.1	0.9	0.5	0.9	0.6	0.3	0.4	0.4	0.6	0.9	0.5	0.4
	Kharif	207.9	226.3	192.2	166.8	215.1	210.1	206.8	222.5	237.2	239.7	261.5	241.6	277.9
Oilseeds	Rabi	101.5	101.3	82.9	85.7	97.6	104.5	108.5	109.7	122.2	139.9	141.5	142.8	135.7
	Summer											10.5	12.3	12.4
Oilseeds	Total	309.4	327.5	275.1	252.5	312.8	314.6	315.2	332.2	359.5	379.6	413.6	396.7	426.1

Note: * Third Advance Estimates; from 2012-13 to 2021-22, summer included in rabi for rice, maize, jowar, rabi and summer included in kharif for sesame and kharif for bajra

Source: Economics, Statistics and Evaluation Division, Department of Agriculture & Farmers Welfare



Rabi Crops

Annex Table 1.3: All-India Estimates of Yield of Principal Crops

Crop	Season	2012-13	2013-14	2014-15	2015-16	2016-17	2017-18	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25*
Rice	Kharif	2373	2319	2295	2305	2417	2469	2553	2622	2607	2705	2735	2780	2807
	Rabi	3353	3232	3291	3382	3230	3531	3444	3569	3541	3525	3467	3582	3503
	Summer											3302	3302	3265
	Total	2461	2416	2391	2400	2494	2576	2638	2722	2717	2798	2838	2882	2899
Wheat	Rabi	3117	3145	2750	3034	3200	3368	3533	3440	3521	3537	3521	3559	3587
	Total	3117	3145	2750	3034	3200	3368	3533	3440	3521	3537	3521	3559	3587
	Kharif	1171	1050	1014	850	954	1104	989	967	1210	1090	1071	1170	1234
	Rabi	644	896	808	615	730	853	744	1002	1033	1093	1086	1165	1227
Jowar	Summer											811	789	830
	Total	850	957	884	697	812	956	849	989	1099	1092	1079	1162	1225
	Kharif	1198	1184	1255	1132	1305	1231	1219	1374	1420	1430	1443	1387	1441
	Summer											2705	2579	2717
Bajra	Total	1198	1184	1255	1132	1305	1231	1219	1374	1420	1430	1510	1453	1507
	Kharif	2246	2346	2249	2236	2413	2706	2648	2572	2779	2914	2940	2671	2947
	Rabi	4152	4050	4414	4006	3897	4436	4893	4631	4723	5084	5355	5529	4999
	Summer											5361	4603	4425
Maize	Total	2566	2676	2632	2563	2689	3065	3070	3006	3199	3387	3545	3351	3518
	Kharif	1392	1661	1706	1601	1363	1662	1390	1747	1724	1396	1454	1375	1492
	Total	1392	1661	1706	1601	1363	1662	1390	1747	1724	1396	1454	1375	1492
	Rabi	2521	2718	2280	2439	2663	2695	2837	2920	2796	3025	3044	3082	3049
Barley	Total	2521	2718	2280	2439	2663	2695	2837	2920	2796	3025	3044	3082	3049
	Kharif	1171	1175	1224	1094	1212	1233	1173	1318	1394	1351	1370	1336	1397
	Rabi	644	896	808	615	730	853	744	1002	1033	1093	1086	1165	1227
	Summer											2634	2381	2502
Shree Anna / Nutri Cereals	Total	1041	1111	1118	968	1096	1154	1093	1248	1322	1302	1364	1337	1400
	Kharif	1583	1619	1633	1544	1708	1818	1790	1836	1970	2037	2065	1944	2109
	Rabi	1725	2034	1915	1685	1885	2323	2532	2491	2667	3018	3218	3097	3174
	Summer											4133	3750	3794
Nutri/ Coarse Cereals	Total	1617	1717	1703	1579	1750	1934	1944	1991	2128	2251	2381	2283	2438
	Kharif	2116	2089	2081	2065	2188	2259	2320	2370	2406	2503	2527	2521	2599
	Rabi	2931	2995	2681	2862	3010	3239	3403	3325	3412	3472	3478	3500	3522
	Summer											3493	3429	3406
Cereals	Total	2448	2462	2331	2393	2525	2657	2752	2772	2824	2899	2939	2945	2997

(Contd.)

Annex Table



Annex Table

Annex Table 1.3: All-India Estimates of Yield of Principal Crops

Crop	Season	2012-13	2013-14	2014-15	2015-16	2016-17	2017-18	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25*
		(Kg/ha)	(Kg/ha)	(Kg/ha)	(Kg/ha)	(Kg/ha)	(Kg/ha)	(Kg/ha)	(Kg/ha)	(Kg/ha)	(Kg/ha)	(Kg/ha)	(Kg/ha)	(Kg/ha)
Tur	Kharif	776	813	729	646	913	967	729	859	914	861	814	827	823
	Total	776	813	729	646	913	967	729	859	914	861	814	827	823
Gram	Rabi	1036	960	889	840	974	1078	1041	1142	1192	1261	1172	1151	1180
	Total	1036	960	889	840	974	1078	1041	1142	1192	1261	1172	1151	1180
Urad	Kharif	610	490	516	459	626	632	500	359	469	515	571	598	620
	Rabi	679	768	892	773	656	798	796	904	778	903	988	828	849
	Summer											875	856	917
	Total	625	555	604	537	632	662	546	459	538	599	657	656	697
Moong	Kharif	398	410	428	363	488	440	466	519	522	385	493	364	515
	Rabi	539	620	640	554	546	600	727	645	833	987	730	733	733
	Summer											968	985	964
	Total	436	475	498	416	500	477	516	548	601	570	663	598	685
Lentil	Rabi	797	759	705	765	838	1047	901	847	1017	899	952	1028	1038
	Total	797	759	705	765	838	1047	901	847	1017	899	952	1028	1038
Total Pulses	Kharif	594	580	573	489	667	668	546	585	642	583	630	600	659
	Rabi	934	891	843	794	898	1015	976	1045	1097	1148	1118	1106	1114
	Summer											957	969	958
	Total	789	764	728	655	785	853	757	823	885	888	902	881	914
Total Foodgrains	Kharif	1892	1864	1862	1808	1890	1951	1957	2029	2079	2131	2203	2205	2298
	Rabi	2431	2435	2232	2341	2441	2603	2740	2738	2793	2798	2854	2903	2935
	Summer											2603	2601	2604
	Total	2129	2120	2028	2041	2129	2235	2286	2343	2394	2425	2494	2515	2578
Groundnut	Kharif	811	1735	1478	1399	1321	1834	1304	2016	1649	1717	2009	2142	2076
	Rabi	1908	1926	1948	1801	1861	2222	2238	2352	2034	2147	2619	2208	2005
	Summer											2279	2377	2019
	Total	995	1764	1552	1465	1398	1893	1422	2063	1703	1777	2075	2163	2067
Sesamum	Kharif	402	426	474	436	448	478	485	405	474	485	375	392	409
	Rabi											592	559	536
	Summer											888	896	845
	Total	402	426	474	436	448	478	485	405	474	485	527	553	541
Nigerseed	Kharif	325	328	328	295	332	321	290	303	317	337	325	323	292
	Total	325	328	328	295	332	321	290	303	317	337	325	323	292

(Contd.)

Annex Table 1.3: All-India Estimates of Yield of Principal Crops

(Kg/ha)

Crop	Season	2012-13	2013-14	2014-15	2015-16	2016-17	2017-18	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25*
Soybean	Kharif	1353	1012	951	738	1177	1058	1192	921	976	1069	1145	985	1172
	Total	1353	1012	951	738	1177	1058	1192	921	976	1069	1145	985	1172
Sunflower	Kharif	622	668	660	421	567	627	766	731	785	802	933	894	906
	Rabi	674	808	777	698	737	924	874	1174	1187	1048	1061	1397	1359
	Summer											1181	1247	1171
	Total	655	755	734	609	660	782	826	931	1011	923	996	1144	1129
Rapeseed & Mustard	Rabi	1262	1185	1083	1183	1304	1410	1511	1331	1524	1497	1428	1444	1461
	Total	1262	1185	1083	1183	1304	1410	1511	1331	1524	1497	1428	1444	1461
Safflower	Rabi	591	638	515	416	651	673	537	843	640	777	826	768	756
	Total	591	638	515	416	651	673	537	843	640	777	826	768	756
Total	Kharif	1135	1151	1056	884	1153	1219	1168	1154	1133	1214	1326	1239	1386
Oilseeds	Rabi	1244	1206	1122	1186	1300	1436	1531	1397	1547	1522	1446	1447	1456
	Summer											1423	1486	1402
	Total	1168	1168	1075	968	1195	1284	1271	1224	1247	1312	1368	1314	1408

Note: * Third Advance Estimates; from 2012-13 to 2021-22, summer included in rabi for rice, maize, jowar, rabi and summer included in kharif for sesame and kharif for bajra

Source: Economics, Statistics and Evaluation Division, Department of Agriculture & Farmers Welfare



Annex Table

Annex Table 1.4: Share of Major States in All-India Production of Mandated Rabi Crops, TE2024-25

Wheat	Barley		Gram		Lentil		Rapeseed & Mustard		Safflower	
	State	Share (%)	State	Share (%)	State	Share (%)	State	Share (%)	State	Share (%)
UP	RJ	59.1	MH	26.2	MP	42.7	RJ	43.7	KA	44.3
MP	UP	29.0	MP	24.4	UP	33.6	UP	14.3	MH	43.7
PB	HR	2.5	RJ	16.5	WB	8.6	MP	13.3	TG	8.0
HR	MP	2.2	GJ	11.1	BR	7.2	HR	10.0	AP	3.2
RJ	HP	1.6	UP	6.8	JH	3.3	WB	6.2	Oth*	0.9
BR	UK	1.3	KA	3.9	RJ	1.9	GJ	4.1		
GJ	PB	1.1	AP	3.1	AS	1.2	JH	2.6		
MH	Oth*	3.2	JH	2.7	Oth*	1.6	AS	1.9		
Oth*			CG	2.3			Oth*	3.9		
			TG	1.5						
			Oth*	1.6						

Note: * States having less than 1 percent share in total production have been clubbed as others

Source: Economics, Statistics and Evaluation Division, Department of Agriculture & Farmers Welfare

Annex Table 2.1: Annual Average Inflation Rate of Major Agricultural Commodities for Last Five Years based on Wholesale Price Index (Base 2011-12)

Commodity Name	2020-21	2021-22	2022-23	2023-24	2024-25
Wholesale Price Index	1.3	13.0	9.4	-0.7	2.3
Food Articles	3.1	4.1	7.3	6.6	7.3
Food Grains (Cereals+Pulses)	-0.2	2.6	9.2	8.5	8.4
Cereals	-2.6	1.6	11.3	7.2	7.9
Paddy	1.9	-0.7	5.2	9.3	8.4
Wheat	-4.0	4.7	15.5	4.5	7.7
<i>Jowar</i>	-4.2	-7.1	25.7	25.3	-7.7
<i>Bajra</i>	-16.8	4.2	26.9	9.9	6.2
Maize	-18.3	6.1	23.1	0.6	12.0
Barley	-10.6	9.3	38.2	-16.1	11.3
<i>Ragi</i>	3.3	1.5	-5.7	19.7	19.3
Pulses	11.6	6.9	0.6	14.4	10.7
Gram	5.6	9.4	-2.3	11.2	18.1
<i>Tur/Arhar</i>	10.7	7.1	6.1	32.8	11.8
<i>Moong</i>	19.0	-2.3	0.7	16.6	-0.4
<i>Masur</i>	14.7	20.1	4.1	-5.5	-0.8
<i>Urad</i>	25.0	-1.8	-0.9	12.6	7.1
Oilseeds	6.8	32.6	-4.3	-9.8	-1.9
Groundnut Seed	8.6	6.1	9.5	10.2	-9.7
Rape & Mustard Seed	12.8	28.6	-5.6	-16.7	9.2
Cotton Seed	6.1	12.5	2.5	-6.8	4.9
Gingelly Seed (Sesamum)	4.1	0.2	4.7	17.9	-1.4
Linseed	6.3	28.5	6.3	-23.2	7.2
Nigerseed	12.4	30.4	-2.2	16.6	6.0
Safflower (<i>Kardi</i> Seed)	-12.5	21.3	4.7	-14.3	6.1
Sunflower	7.0	34.6	0.6	-20.2	-4.7
Soyabean	6.5	60.8	-10.4	-14.8	-7.7

Source: Office of the Economic Adviser, Ministry of Commerce and Industry



Annex Table 2.2: All-India Item Wise Inflation Rates based on Consumer Price Index for Last Five Years (Financial Year wise) (Base 2012=100)

	2020-21	2021-22	2022-23	2023-24	2024-25
Consumer Price Index (General)	6.2	5.5	6.7	5.4	4.6
Consumer Food Price Index	7.7	3.8	6.6	7.5	7.3
Cereals and products	3.8	0.5	11.0	10.7	7.2
Rice_PDS	4.9	-2.7	-5.2	-17.4	15.4
Rice_other sources	4.1	0.5	7.9	12.2	8.6
Wheat/atta_PDS	-0.9	3.8	4.3	26.8	-2.9
Wheat/atta_other sources	0.9	0.4	17.1	7.8	7.5
<i>Jowar and its products</i>	11.0	-2.2	7.9	15.1	4.0
<i>Bajra and its products</i>	0.2	0.4	21.2	6.9	7.5
Maize and products	4.1	2.1	16.4	7.8	6.3
<i>Ragi and its products</i>	4.7	2.6	-0.2	10.8	13.1
Pulses and products	16.4	6.0	2.4	15.2	8.3
<i>Tur/Arhar</i>	18.4	5.3	4.5	33.4	10.3
Gram (split)	7.4	5.7	-1.5	7.6	17.6
Gram (whole)	8.7	7.0	1.0	9.5	15.4
Gram products	9.5	6.1	0.8	7.3	13.6
<i>Moong</i>	23.9	1.3	-0.4	10.3	3.8
<i>Masur</i>	18.9	11.1	5.9	-1.0	0.0
<i>Urad</i>	25.8	3.3	0.7	9.7	7.4
Mustard oil	21.4	35.4	-2.0	-19.1	7.6
Refined oil [sunflower; soyabean; saffola; etc.]	20.8	30.2	6.6	-23.1	5.4
Groundnut oil	21.4	16.5	8.2	-0.2	-1.4
Oilseeds (gm)	10.4	14.9	12.2	1.6	0.9
Groundnut	6.8	6.5	6.7	10.0	-0.7

Source: Ministry of Statistics and Programme Implementation

Annex Table 2.3: List of DCP States for Rice and Wheat

S. No.	States /UTs	DCP adopted for
1	Andhra Pradesh	Rice
2	Assam*	Rice
3	Bihar	Rice/Wheat
4	Chhattisgarh	Rice/Wheat
5	Gujarat	Rice/Wheat
6	Himachal Pradesh	Rice/Wheat
7	Jharkhand	Rice
8	Karnataka	Rice
9	Kerala	Rice
10	Madhya Pradesh	Rice/Wheat
11	Maharashtra	Rice/Wheat
12	Odisha	Rice
13	Punjab #	Wheat
14	Tamil Nadu	Rice
15	Telangana	Rice
16	Tripura	Rice
17	Uttarakhand	Rice/Wheat
18	West Bengal	Rice/Wheat

Note: 1. *: Assam is a DCP state only for two districts (Bongaigaon and Bishwanath).

2. #: Punjab was DCP State for Wheat from RMS 2014-15, but on request of State Govt, FCI is participating in procurement operations.

Source: Central Food Grains Procurement Portal, Department of Food and Public Distribution, Ministry of Consumer Affairs, Food and Public Distribution



Annex Table

Annex Table 2.4: Number of Farmers Benefitted from Procurement Operations in Major Wheat Producing States

(in '000)

States/ UTs	RMS2021-22	RMS2022-23	RMS2023-24	RMS2024-25	RMS2025-26*
Punjab	887.5	793.4	835.0	930.5	839.2
Madhya Pradesh	1725.1	591.2	796.7	616.3	900.5
Haryana	760.6	313.8	410.2	449.4	449.0
Uttar Pradesh	1152.3	81.1	47.9	152.3	143.9
Rajasthan	227.1	0.9	37.2	94.7	172.7
Bihar	96.7	0.6	0.2	3.4	5.2
Others	70.5	2.0	1.7	2.1	2.6
Total	4919.9	1783.2	2129.0	2248.7	2513.1

Note: *: As on 30th June 2025

Source: Food Corporation of India

Annex Table 2.5: Number of Pulses Farmers Benefitted during Last Three Rabi Marketing Seasons

Crop	State	Number of Farmers Benefitted		
		RMS2023-24	RMS2024-25	RMS2025-26*
Gram	Andhra Pradesh	37563	0	92
	Gujarat	168797	84	52446
	Haryana	0	0	3
	Karnataka	66043	0	0
	Madhya Pradesh	223818	14398	14753
	Maharashtra	392768	0	1077
	Rajasthan	97967	643	8518
	Telangana	35164	360	3666
	Uttar Pradesh	7270	0	0
	Total	1029390	15485	80555
Masur	Madhya Pradesh	39107	97796	59590
	Uttar Pradesh	5692	19111	12459
	Total	44799	116907	72049

Note: *:As on 5th June 2025

Source: National Agricultural Cooperative Marketing Federation of India (NAFED)



Annex Table 2.6: Number of Oilseeds Farmers Benefitted during Last Three Rabi Marketing Seasons

Crop	State	Number of Farmers Benefitted		
		RMS2023-24	RMS2024-25	RMS2025-26*
Mustard	Gujarat	24252	60068	32780
	Haryana	183758	172874	78854
	Madhya Pradesh	48964	98105	9926
	Rajasthan	183209	167005	21487
	Uttar Pradesh	12648	20354	113
	Assam	1066	3540	725
	Total	453897	521946	143885
Safflower	Karnataka	1292	288	0

Note: *: As on 5th June 2025

Source: National Agricultural Cooperative Marketing Federation of India (NAFED)

Annex Table 2.7: Time lags between the Central Governments' Approval for Procurement, Notification by the State and Commencement of Actual Procurement under PSS

Commodity	States	DA&FW I etter dated	State Govt. Notification/ letter dated	Procurement Period Approved by State Government	Date of Commencement of Procurement	Quantity procured (in MT)
Gram	Karnataka	24-12-2024	23.01.2025	23-01-2025 to 22-04-2025	07-04-2025	-
	Chhattisgarh	17-01-2025	10.01.2025	01-03-2025 to 30-05-2025	No procurement recorded	-
	Maharashtra	06-03-2025	26.03.2025	01-04-2025 to 29-06-2025	02-04-2025	1,358.2
	Madhya Pradesh	29-01-2025	20.02.2025	25-03-2025 to 31-05-2025	28-03-2025	46,875.0
	Telangana	21-02-2025	27.02.2025	01-03-2025 to 29-05-2025	06-03-2025	5,792.5
	Gujarat	07-03-2025	17.04.2025	21-04-2025 to 19-07-2025	21-04-2025	82,220.0
	Rajasthan	27-03-2025	09.04.2025	10-04-2025 to 08-07-2025	15-04-2025	20,240.0
	Uttar Pradesh	24-02-2025	17.03.2025	02-04-2025 to 30-06-2025	No procurement recorded	-
	Haryana	27-02-2025	21.02.2025	01-04-2025 to 10-05-2025	01-04-2025	5.3
	Andhra Pradesh	04-03-2025	18.03.2025	18-03-2025 to 15-06-2025	21-03-2025	183.1
Masur	Bihar	28-04-2025		To be decided by State		-
	Chhattisgarh	17-01-2025	10.01.2025	01-03-2025 to 30-05-2025	No procurement recorded	-
	Madhya Pradesh	29-01-2025	20.02.2025	25-03-2025 to 31-05-2025	28-03-2025	1,31,621.6
	Uttar Pradesh	24-02-2025	17.03.2025	02-04-2025 to 30-06-2025	15-04-2025	25,480.5
	Bihar	24-01-2025		To be decided by State		-
Mustard Seed	Chhattisgarh	31-01-2025	10.01.2025	15-02-2025 to 15-05-2025	No procurement recorded	-
	Madhya Pradesh	31-01-2025	20.02.2025	25-03-2025 to 31-05-2025	28-03-2025	28,319.1
	Gujarat	07-03-2025	17.04.2025	21-04-2025 to 19-07-2025	21-04-2025	42,544.3
	Rajasthan	27-03-2025	09.04.2025	10-04-2025 to 08-07-2025	15-04-2025	55,271.9
	Uttar Pradesh	24-02-2025	17.03.2025	02-04-2025 to 30-06-2025	20-04-2025	153.6
	Haryana	27-02-2025	21.02.2025	15-03-2025 to 01-05-2025	17-03-2025	1,61,185.0
	Assam	14-02-2025	19.02.2025	25-02-2025 to 24-06-2025	08-03-2025	2,793.0
	Bihar	28-04-2025		To be decided by State		-
	Odisha	30-04-2025	30.04.2025	18-05-2025 to 15-08-2025	No procurement recorded	-

Note: As on 2nd June 2025

Source: National Agricultural Cooperative Marketing Federation of India (NAFED)



Annex Table 3.1: Area and Yield Growth of Rabi Crops in Major Producing States

State	Area Growth		Yield Growth	
	2015-16 to 2019-20	2020-21 to 2024-25	2015-16 to 2019-20	2020-21 to 2024-25
Wheat				
Maharashtra	2.7	2.1	8.2	2.5
Madhya Pradesh	2.2	3.6	1.0	0.6
Gujarat	0.6	4.9	3.6	-0.8
Punjab	0.1	-0.1	3.2	0.8
Uttar Pradesh	0.0	-0.5	8.8	2.0
Bihar	0.0	0.9	7.8	4.1
Haryana	-0.5	-0.9	3.5	0.4
Rajasthan	-1.0	0.8	3.4	0.6
Barley				
Uttarakhand	3.1	-9.9	5.2	0.4
Uttar Pradesh	2.0	1.7	11.4	-0.4
Rajasthan	-0.5	6.2	3.3	0.8
Himachal Pradesh	-1.0	-2.9	-2.6	2.5
Madhya Pradesh	-10.3	0.6	12.7	1.0
Punjab	-10.7	-3.2	0.4	1.7
Haryana	-17.5	38.9	4.0	0.6
Gram				
Gujarat	37.2	24.8	5.6	2.6
Rajasthan	19.3	-7.3	8.8	2.9
Telangana	18.3	-9.8	10.3	1.4
Uttar Pradesh	13.9	-0.5	19.3	-3.1
Maharashtra	8.8	7.8	12.4	-0.7
Jharkhand	8.8	5.9	1.2	0.2
Andhra Pradesh	8.2	-8.2	17.2	-0.6
Chhattisgarh	6.5	3.9	-11.5	55.0
Karnataka	2.8	-3.8	6.5	-4.6
Madhya Pradesh	-5.5	-1.8	6.6	0.6
Lentil				
Jharkhand	49.8	6.0	-1.2	0.4
West Bengal	26.2	-1.6	-2.6	2.6
Uttar Pradesh	3.4	5.5	13.9	1.3
Assam	-3.0	1.6	0.0	0.1

(Contd...)

Annex Table 3.1: Area and Yield Growth of Rabi Crops in Major Producing States

State	Area Growth		Yield Growth	
	2015-16 to 2019-20	2020-21 to 2024-25	2015-16 to 2019-20	2020-21 to 2024-25
Bihar	-5.6	-1.0	-7.0	9.5
Rajasthan	-6.9	10.0	8.8	1.1
Madhya Pradesh	-8.2	13.0	5.3	10.3
Rapeseed & Mustard				
Jharkhand	9.9	10.6	5.1	0.2
West Bengal	7.1	0.9	1.9	1.4
Haryana	5.5	0.3	5.1	2.1
Rajasthan	5.4	3.6	4.0	3.2
Uttar Pradesh	4.2	13.9	7.0	3.5
Assam	0.5	1.6	-1.5	5.4
Madhya Pradesh	-0.6	14.2	9.1	0.7
Gujarat	-1.0	9.4	6.4	-0.9

Note: States having production share more than 1% in respective crop has been considered for the analysis

Source: Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare



Annex Table 3.2: Potential, Realized and Average Yield of Mandated Rabi Crops (TE2023-24)

Yield (qtl/ha)
Yield Gap (%)

State	Potential Yield	Realized Yield	State Average	Yield Gap A	Yield Gap B
Wheat					
Haryana	53.6	49.0	46.5	8.6	13.2
Jammu & Kashmir	27.7	18.8	20.6	32.1	25.6
Punjab	51.0	48.2	46.7	5.5	8.4
Rajasthan	51.1	41.9	37.5	18.0	26.6
Uttar Pradesh*	37.0	30.0	36.2	18.9	2.2
Barley					
Bihar	40.2	33.7	16.8	16.2	58.2
Himachal Pradesh	28.7	21.0	17.0	26.8	40.8
Haryana	43.2	39.9	35.1	7.6	18.8
Jammu & Kashmir	30.4	21.4	6.6	29.6	78.3
Madhya Pradesh	39.3	28.2	17.9	28.2	54.5
Punjab	44.6	39.0	35.6	12.6	20.2
Rajasthan	52.8	44.5	35.4	15.7	33.0
Uttar Pradesh	38.1	28.6	29.6	24.9	22.3
Gram					
Andhra Pradesh	18.1	15.8	11.5	12.7	36.5
Assam	11.6	8.9	7.5	23.3	35.3
Bihar	16.0	12.6	10.6	21.3	33.8
Chhattisgarh	7.5	5.0	8.2	33.3	-9.3
Gujarat	23.2	19.3	17.8	16.8	23.3
Haryana	18.5	15.6	10.7	15.7	42.2
Jammu & Kashmir	9.5	7.1	2.3	25.3	75.8
Karnataka	17.6	15.3	6.8	13.1	61.4
Maharashtra	16.6	13.7	10.9	17.5	34.3
Manipur	11.1	9.0	9.5	18.9	14.4
Madhya Pradesh	17.9	11.9	14.9	33.5	16.8
Punjab	18.4	12.9	11.7	29.9	36.4
Rajasthan	19.0	16.4	11.1	13.7	41.6
Tamil Nadu	12.1	10.0	9.3	17.4	23.1
Uttarakhand	16.8	16.0	7.9	4.8	53.0
Uttar Pradesh	16.4	13.3	12.7	18.9	22.6
West Bengal	14.3	10.6	12.5	25.9	12.6

(Contd...)

Annex Table 3.2: Potential, Realized and Average Yield of Mandated Rabi Crops (TE2023-24)

Yield (qtl/ha)
Yield Gap (%)

State	Potential Yield	Realized Yield	State Average	Yield Gap A	Yield Gap B
Lentil					
Assam	12.0	9.2	7.8	23.3	35.0
Bihar	16.7	14.0	9.2	16.2	44.9
Chhattisgarh	6.1	4.7	4.2	23.0	31.1
Jammu & Kashmir	11.1	8.6	6.9	22.5	37.8
Jharkhand	8.2	6.6	8.6	19.5	-4.9
Manipur	10.9	8.2	9.2	24.8	15.6
Madhya Pradesh	12.5	9.9	10.0	20.8	20.0
Rajasthan	16.9	14.0	13.1	17.2	22.5
Uttarakhand	14.8	12.1	8.5	18.2	42.6
Uttar Pradesh	14.1	11.7	9.5	17.0	32.6
West Bengal	9.7	7.9	9.0	18.6	7.2
Rapeseed & Mustard					
Assam	10.6	7.5	7.4	29.2	30.2
Gujarat	22.0	19.8	19.5	10.0	11.4
Himachal Pradesh	13.6	9.6	8.1	29.4	40.4
Haryana	24.4	21.2	18.1	13.1	25.8
Jammu & Kashmir	13.6	8.5	8.5	37.5	37.5
Jharkhand	12.2	8.3	8.2	32.0	32.8
Maharashtra	8.3	6.9	3.5	16.9	57.8
Madhya Pradesh	18.4	16.0	14.6	13.0	20.7
Punjab	20.5	19.3	16.0	5.9	22.0
Rajasthan	18.8	16.3	15.7	13.3	16.5
Uttarakhand	14.2	12.2	9.3	14.1	34.5
Uttar Pradesh	20.5	15.9	14.4	22.4	29.8
West Bengal	17.6	14.1	12.4	19.9	29.5
Safflower					
Andhra Pradesh	10.4	9.1	5.3	12.5	49.0
Chhattisgarh	9.3	7.8	3.4	16.1	63.4
Karnataka	14.4	11.1	7.7	22.9	46.5
Maharashtra	11.8	9.4	6.5	20.3	44.9
Telangana	10.5	8.2	8.9	21.9	15.2

Note : *FLDs conducted in Bundelkhand region only

Source: Various ICAR institutes



Annex Table 3.3: Crop-wise Requirement and Availability of Certified/Quality Seeds of Rabi Crops in India (2024-25)

(in '000 quintal)

Crop	Requirement	Availability
Wheat	14036.7	16066.5
Barley	242.5	329.9
Gram	2498.7	2602.7
Lentil	262.8	280.2
Rapeseed & Mustard	286.2	364.7
Safflower	5.4	8.9

Source: Seed Division, Ministry of Agriculture and Farmers Welfare

Annex Table 3.4: State-wise Institutional Agriculture Credit per Hectare of Gross Cropped Area (TE2024-25)

(₹ Lakh per Hectare)

State	Credit
Tamil Nadu	6.8
Kerala	6.3
Andhra Pradesh	4.3
Telangana	1.5
Karnataka	1.4
Haryana	1.4
All-India	1.2
Punjab	1.1
Odisha	1.1
Gujarat	1.1
Bihar	0.9
West Bengal	0.8
Maharashtra	0.7
Chhattisgarh	0.6
Uttar Pradesh	0.6
Rajasthan	0.6
Madhya Pradesh	0.4
North Eastern Region	0.3

Note : Credit Disbursement (TE2024-25), Cropped Area (2022-23)

Source: 1. National Bank for Agriculture and Rural Development for credit disbursement
2. Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare for gross cropped area



Annex Table 3.5: Custom Hiring Centres/Hi-Tech Hubs/FMBs Established under SMAM and CRM

Year	Custom Hiring Centres (CHCs)	Hi-Tech Hubs	Village Level Farm Machinery Banks	CRM (CHCs) (from April 2018)
2014-15	663	14	436	-
2015-16	268	13	218	-
2016-17	1458	13	1877	-
2017-18	1818	39	1942	-
2018-19	3871	106	1481	7382
2019-20	2155	60	1964	8475
2020-21	3979	55	5453	13805
2021-22	4612	103	3420	7528
2022-23	4194	72	3670	1889
2023-24	2533	119	3131	1505
2024-25	1653	52	1668	1699
Total	27204	646	25260	42283

Source: Mechanisation and Technology Division, Ministry of Agriculture and Farmers Welfare

Annex Table 3.6: State-wise Status of Extension Staff as on 31st March 2025

States	Sanctioned Strength	Actual Strength	Vacant (%)
Andhra Pradesh	9485	7990	15.8
Assam	4018	1419	64.7
Chhattisgarh	5164	3701	28.3
Gujarat	3332	2953	11.4
Haryana	2148	997	53.6
Himanchal Pradesh	1159	454	60.8
Jammu & Kashmir*	4559	2783	39.0
Madhya Pradesh	11950	6286	47.4
Odisha	8501	4913	42.2
Punjab	690	413	40.1
Rajasthan	9411	8707	7.5
Tamil Nadu	5330	4987	6.4
Uttarakhand	151	151	-
Uttar Pradesh	15321	7704	49.7
West Bengal	7150	2182	69.5

Note: *As on 31/12/2024

Source: State replies



Annex Table 3.7: Determination of Oil Content-Based MSP of Rapeseed & Mustard (Illustrative)

S. No.	Oil Content (%)	Oil cake (%) = {100 - Col.2} - 1 [^]	Realisation from oil cake [#] = {Col.3 * Price of oil cake} / 100	Cost of oil excluding oil cake ^{\$} = {MSP - Col.4}	Incentive for each 0.25 percent point of oil content (₹/qtl) = {Col.5 / Col. 2 * 0.25}	MSP at Oil content given at Col.2 (₹/qtl)
(1)	(2)	(3)	(4)	(5)	(6)	(7)
1	34.00	65.00	1423	4777	35.1	6200
2	34.25	64.75	1418	4782	34.9	6235
3	34.50	64.50	1412	4788	34.7	6270
4	34.75	64.25	1407	4793	34.5	6305
5	35.00	64.00	1401	4799	34.3	6339
6	35.25	63.75	1396	4804	34.1	6373
7	35.50	63.50	1391	4809	33.9	6408
8	35.75	63.25	1385	4815	33.7	6441
9	36.00	63.00	1380	4820	33.5	6475
10	36.25	62.75	1374	4826	33.3	6509
11	36.50	62.50	1369	4831	33.1	6542
12	36.75	62.25	1363	4837	32.9	6575
13	37.00	62.00	1358	4842	32.7	6608
14	37.25	61.75	1352	4848	32.5	6641
15	37.50	61.50	1347	4853	32.4	6673
16	37.75	61.25	1341	4859	32.2	6705
17	38.00	61.00	1336	4864	32.0	6738
18	38.25	60.75	1330	4870	31.8	6770
19	38.50	60.50	1325	4875	31.7	6801
20	38.75	60.25	1319	4881	31.5	6833
21	39.00	60.00	1314	4886	31.3	6865
22	39.25	59.75	1308	4892	31.2	6896
23	39.50	59.50	1303	4897	31.0	6927
24	39.75	59.25	1297	4903	30.8	6958
25	40.00	59.00	1292	4908	30.7	6989
26	40.25	58.75	1287	4913	30.5	7020
27	40.50	58.50	1281	4919	30.4	7050
28	40.75	58.25	1276	4924	30.2	7080
29	41.00	58.00	1270	4930	30.1	7111
30	41.25	57.75	1265	4935	29.9	7141
31	41.50	57.50	1259	4941	29.8	7171

(contd...)

Annex Table 3.7: Determination of Oil Content-Based MSP of Rapeseed & Mustard (Illustrative)

S. No.	Oil Content (%)	Oil cake (%) = {100- Col.2}-1 [^]	Realisation from oil cake [#] = {Col.3*Price of oil cake}/100	Cost of oil excluding oil cake ^s = {MSP- Col.4}	Incentive for each 0.25 percent point of oil content (₹/qtl) = {Col.5/ Col. 2*0.25}	MSP at Oil content given at Col.2 (₹/qtl)
(1)	(2)	(3)	(4)	(5)	(6)	(7)
32	41.75	57.25	1254	4946	29.6	7200
33	42.00	57.00	1248	4952	29.5	7230
34	42.25	56.75	1243	4957	29.3	7260
35	42.50	56.50	1237	4963	29.2	7289
36	42.75	56.25	1232	4968	29.1	7318
37	43.00	56.00	1226	4974	28.9	7347
38	43.25	55.75	1221	4979	28.8	7376
39	43.50	55.50	1215	4985	28.6	7405
40	43.75	55.25	1210	4990	28.5	7433
41	44.00	55.00	1204	4996	28.4	7462
42	44.25	54.75	1199	5001	28.3	7490
43	44.50	54.50	1193	5007	28.1	7519
44	44.75	54.25	1188	5012	28.0	7547
45	45.00	54.00	1183	5017	27.9	7575
46	45.25	53.75	1177	5023	27.8	7603
47	45.50	53.50	1172	5028	27.6	7630
48	45.75	53.25	1166	5034	27.5	7658
49	46.00	53.00	1161	5039	27.4	7685
50	46.25	52.75	1155	5045	27.3	7713
51	46.50	52.50	1150	5050	27.2	7740
52	46.75	52.25	1144	5056	27.0	7767
53	47.00	52.00	1139	5061	26.9	7794
54	47.25	51.75	1133	5067	26.8	7821
55	47.50	51.50	1128	5072	26.7	7848
56	47.75	51.25	1122	5078	26.6	7875
57	48.00	51.00	1117	5083	26.5	7901
Average increase in MSP with 0.25 percent increase in oil content					30.31	

Note: 1. #Burning loss of 1% during processing
 2. Assuming R&M DOC prices ₹2190/qtl
 3. \$ at MSP ₹ 6,200/qtl



Annex Table

Annex Table 4.1 : Major Importers of Wheat from India during 2023-24 and 2024-25

2023-2024			2024-2025		
Region/Country	Values (₹ lakh)	% share	Region/Country	Values (₹ lakh)	% share
Nepal	46,727	99.2	UAE	1,010	47.6
UAE	169	0.4	Iraq	593	43.8
Iraq	96	0.2	Nepal	94	7.0
Mongolia	66	0.1	Korea RP	23	1.7

Source : Directorate General of Commercial Intelligence and Statistics, Ministry of Commerce and Industry

Annex Table 4.2: Top Exporting Nations for Major Agricultural Products in India's Export Basket, 2024-25

Product	Export Markets (₹ crore)
Marine Products	USA (22672), China (10227), Japan (3452)
Rice (Non- Basmati)	Benin (8689), Guinea (4547), Cote D' Ivoire (4408)
Basmati Rice	Saudi Arabia (10192), Iraq (7201), Iran (6374)
Spices	China (6540), USA (5539), UAE (3009)
Sugar	Sudan (2251), Libya (2046), Somalia (1950)
Oil Meals	Bangladesh (1841), Korea (1290), Germany (1094)
Misc Processed Items	USA (2130), UAE (1513), Malaysia (1134)
Plywood and products	USA (4690), Netherlands (806), Germany (804)
Coffee	Italy (2758), Germany (1674), Belgium (1147)
Fresh Fruits	Iraq (1579), UAE (1509), Netherland (1379)
Cotton (raw including waste)	Bangladesh (5006), Vietnam (1085), Taiwan (183)
Tobacco (Unmanufactured)	Belgium (2878), UAE (2023), Indonesia (1265)
Castor Oil	China (4313), Netherlands (1145), USA (1010)

Source: Directorate General of Commercial Intelligence and Statistics, Ministry of Commerce and Industry



Rabi Crops



Annex Table

Annex Table 4.3(a) : Trade Policy of Pulses since 2015

Month	Trade Policy	Product	Description
20 th January 2016	Export of Roasted Gram	Gram	Export of roasted gram (whole/split) in consumer packs of 1 kg was permitted.
21 st August 2017	Revision of import policy from 'Free' to 'Restricted'	Moong, Urad	The import policy of moong & urad was revised from 'Free' to 'Restricted' subject to annual quota of 3 lakh metric tonnes (LMT).
22 nd November 2017	Removal of prohibition on export of all varieties of pulses	All Pulses	Export of all varieties of pulses including organic pulses made 'Free' for export without any quantitative ceilings till further order*.
21 st December 2017	Change in basic custom duty	Gram, Lentil	Basic custom duty (BCD) on chana (chickpeas) and lentil (masur) increased from 'zero' to 30 percent.
6 th February 2018	Change in basic custom duty	Gram	BCD on chana (chickpeas) increased from 30 to 40 percent.
1 st March 2018	Change in basic custom duty	Gram	BCD on gram was increased from 40 to 60 percent.
21 st March 2018	Benefit of Merchandise Exports from India Scheme (MEIS)	Gram	MEIS benefit of 7 percent was given for Bengal gram for exports made for the period 21 st March 2018 to 20 th June 2018.
16 th April 2018	Incorporation of new HS Codes	Gram	New EXIM Code for kabuli chana, Bengal gram and others were notified for incorporation in ITC(HS) 2017.
13 th July 2018	Extension of MEIS benefit	Gram	MEIS benefit of 7 percent for exports of Bengal gram, which was available upto to 20 th June 2018 was further extended upto 20 th September 2018.
15 th June 2019	Change in import duty	Lentil	BCD on lentil increased from 30 to 50 percent (on imports from U.S.A.).
18 th December 2019	Revision of import quota	Urad	Annual quota increased from 1.5 lakh tonnes to 4 lakh tonnes for import of urad for 2019-20 and was applicable till 31 st March 2020.
2 nd June 2020	Change in import duty	Lentil	BCD on lentils (country of origin other than U.S.A) was reduced from 30 to 10 percent, and (imports from U.S.A.) was reduced from 50 to 30 percent upto 31 st August 2020.
1 st September 2020	Change in import duty	Lentil	BCD on lentil increased from 10 to 30 percent (import from any country) and 30 to 50 percent (imports from U.S.A.).

(Contd...)

Month	Trade Policy	Product	Description
17 th September 2020	Change in import duty	Lentil	BCD on lentil reduced from 30 to 10 percent (import from any country) and 50 to 30 percent (imports from U.S.A.). The period of import of lentils extended at the reduced import duty @10 percent (import from any country) & 30 percent (import from U.S.A.), till 31 st October 2020 and till 31 st December 2020 respectively with cut-off date for Irrevocable Commercial Letter of Credit (ICLCs) till 31 st December 2020.
1 st October 2020	Allocation of quota for import	Urad	Procedure for allocation of quota for import of 1.5 lakh MT of urad notified upto 31 st March 2021, for the fiscal year 2020-21.
1 st January 2021	Change in import duty	Lentil	BCD on lentil increased from 10 to 30 percent (import from any country) and 30 to 50 percent (imports from U.S.A.).
2 nd February 2021	Change in import duty, AIDC imposed	Lentil	BCD on lentil decreased from 30 to 10 percent (import from any country) and 50 to 30 percent (imports from U.S.A.). Agriculture Infrastructure and Development Cess (AIDC) of 20 percent was imposed.
2 nd February 2021	Change in basic custom duty, AIDC imposed	Gram	BCD on Bengal gram decreased from 60 to 10 percent and AIDC of 50 percent was imposed. The prescribed import duty of 60 percent remained unchanged since then.
3 rd March 2021	Continuation of import quota	Urad	An annual quota of 4 lakh tonnes for import of urad was continued for fiscal year 2021-22.
15 th May 2021	Revision of import policy from 'Free' to 'Restricted'	Tur/Pigeon peas, Moong, Urad	The import policy for tur/pigeon peas, moong and urad was revised from 'Restricted' to 'Free' upto 31 st October 2021.
24 th June 2021	Memorandum of Understanding (MoU) between India and Myanmar	Tur, Urad	India provided an annual quota of 1 lakh tonnes of tur and 2.5 lakh tonnes of Urad, which was imported from Myanmar, through private trade over the next five financial years (2021-22 to 2025-26).
24 th June 2021	MoU between India and Malawi	Tur	India provided an annual quota of 50,000 tonnes of tur, which was imported from Malawi, through private trade over the next five financial years (2021-22 to 2025-26).

(Contd...)

Annex Table

Annex Table

Price Policy for



Rabi Crops

Month	Trade Policy	Product	Description
2 nd July 2021	Imposing stock limits on pulses applicable to wholesalers, retailers, millers and importers	All pulses	Different stock limit on all pulses (except moong) imposed up to 31 st October 2021.
26 th July 2021	Reduction in duty rates	Lentil	BCD on lentils (masur) was reduced from 10 to zero percent (import from any country) and from 30 to 10 percent (import from U.S.A.). AIDC was reduced from 20 to 10 percent. The effective import duty (BCD+AIDC) was reduced from 30 to 10 percent (import from any country) & from 50 to 20 percent (import from U.S.A.).
13 th September 2021	Extension of 'Free' import policy	Tur, Urad	The 'Free' import policy for tur and urad was extended for the period upto 31 st December 2021.
17 th September 2021	Change in duty rates	Lentil	BCD on lentils (masur) was increased from 10 to 20 percent (import from U.S.A.). The effective import duty increased from 20 to 30 percent for U.S. origin lentils.
7 th December 2021	Supply of essential commodities to Maldives	Dal	Export of 213.79 tonnes of dal from India to Maldives was permitted under bilateral trade agreement for each financial year 2021-22, 2022-23 and 2023-24.
20 th December 2021	Extension of 'Free' import policy	Tur, Urad	The 'Free' import policy for tur and urad was extended upto 31 st March 2022.
11 th February 2022	Revision of import policy from 'Free' to 'Restricted'	Moong	The import policy for moong was revised from 'Free' to 'Restricted' till further order.
12 th February 2022	Change in duty rates	Lentil	AIDC on lentil was declared as 'Zero'. The total import duty on lentil was notified as 'Zero' (import from any country) and reduced from 30 to 20 percent (import from U.S.A.). This became applicable till 30 th September 2022.
29 th March 2022	Extension of 'Free' import policy	Tur, Urad	'Free' import policy of tur and urad was extended up to 31 st March 2023.

(Contd...)

Month	Trade Policy	Product	Description
23 rd July 2022	Extension of 'Nil' AIDC	Lentil	AIDC on Lentil was 'Zero' and became applicable upto and inclusive of 31 st March 2023.
28 th December 2022	Extension of 'Free' import policy	Tur, Urad	The 'Free' import policy for tur and urad was extended upto 31 st March 2024.
29 th December 2022	Extension of 'Nil' AIDC	Lentil	AIDC on lentil was 'Zero' and became applicable upto and inclusive of 31 st March 2024.
3 rd March 2023	'Nil' Import Duty	Tur whole	Government reduced the import duty on 'tur whole' to 'Nil' w.e.f. 4 th March 2023.
5 th September 2023	Extension of Import Duty exemption	Lentil	The import duty on lentil imports from USA was removed w.e.f. 6 th September 2023.
8 th December 2023	'Restricted' to 'Free'	Yellow Peas	'Free' import of yellow peas upto 31 st March 2024.
21 st December 2023	Extension of Import Duty exemption	Lentil	Government extended validity period of import duty exemption on lentil till 31 st March 2025.
28 th December 2023	Extension of 'Free' import policy	Tur, Urad	'Free' import policy for tur and urad was extended upto 31 st March 2025.
23 rd February 2024	'Restricted' to 'Free'	Yellow Peas	'Free' import of yellow peas was allowed, where bill of lading (Shipped on Board) was issued on or before 30 th April 2024.
5 th April 2024	'Restricted' to 'Free'	Yellow Peas	'Free' import of yellow peas was allowed, where bill of lading (Shipped on Board) was issued on or before 30 th June 2024.
3 rd May 2024	Reduction in duty rates	Gram	Import duty on gram was reduced to zero w.e.f. 4 th May 2024 till 31 st March 2025.
8 th May 2024	'Restricted' to 'Free'	Yellow Peas	'Free' import of yellow peas was allowed, where bill of lading (Shipped on Board) was issued on or before 31 st October 2024.
13 th September 2024	Extension of 'Free' import policy	Yellow Peas	'Free' import policy for yellow peas was extended upto 31 st December 2024.

(Contd...)

Annex Table



Annex Table

Month	Trade Policy	Product	Description
24 th December 2024	Extension of 'Free' import policy	Yellow Peas	'Free' import policy for yellow peas was extended upto 28 th February 2025.
20 th January 2025	Extension of 'Free' import policy	Tur, Urad	'Free' import policy for tur and urad was extended upto 31 st March 2026.
7 th March 2025	Change in duty rates	Lentil	Import duty on lentil was increased to 10 percent (5% BCD plus 5% AIDC) from zero percent w.e.f. 8 th March 2025.
27 th March 2025	Change in duty rates	Gram	Import duty on gram was increased to 10 percent from zero percent w.e.f. 1 st April 2025.
30 th May 2025	Extension of 'Free' import policy	Yellow Peas	'Free' import policy for yellow peas was extended upto 31 st March 2026.

Note: 1. Social Welfare Cess (SWC) of 10 percent, over and above the effective import duty has been imposed on all types of pulses since February 2018

2. * exports of both gram and lentil are currently under 'Free' category and do not require any export authorization from DGFT

Source: 1. Directorate General of Foreign Trade, Ministry of Commerce and Industry

2. Central Board of Indirect Taxes and Customs, Department of Revenue

Annex Table 4.3(b): Trade Policy of Edible Oils since 2015

Month	Trade Policy	Product	Description
17 th September 2015	Increase in import duty	Crude and Refined R&M and Safflower oil	Basic customs duty (BCD) duty increased on crude edible oils from 7.5 to 12.5 percent and on refined edible oils from 15 to 20 percent.
23 rd September 2016	Change in import duty	Crude and Refined Palm oil	BCD on crude palm oil decreased from 12.5 to 7.5 percent and on refined palm oil from 20 to 15 percent.
11 th August 2017	Change in import duty	Crude and Refined Palm oil	BCD on crude palm oil increased from 7.5 to 15 percent and on refined palm oil from 15 to 25 percent.
17 th November 2017	Change in import duty	Crude and Refined edible oil	BCD increased on following edible oils: - Crude palm oil of edible grade from 15 to 30 percent; - Refined palm oil of edible grade from 25 to 40 percent; - Crude sunflower oil from 12.5 to 25 percent; - Refined sunflower oil from 20 to 35 percent; - Crude soya bean oil from 17.5 to 30 percent; - Refined soya bean oil from 20 to 35 percent; - Crude rapeseed oil including canola oil (Low erucic acid rapeseed oil), mustard oil and colza oil from 12.5 to 25 percent; - Refined rapeseed oil including canola oil (Low erucic acid rapeseed oil), mustard oil and colza oil from 20 to 35 percent.
2 nd February 2018	Increase in import duty	Crude and Refined Safflower oil	BCD on crude safflower oil increased from 12.5 to 30 percent and on refined safflower oil increased from 20 to 35 percent.
1 st March 2018	Increase in import duty	Crude and Refined Palm oil	BCD on crude palm oil increased from 30 to 44 percent and on refined palm oil increased from 40 to 54 percent.
6 th April 2018	'Free' export without any quantitative ceiling	Edible oils (except mustard oil)	Except mustard oil, exports of all edible oils were made 'Free' without any quantitative ceiling, pack size etc. w.e.f. 6 th April 2018. However, export of mustard oil permitted in packs of up to 5 kg with a Minimum Export Price (MEP) of US\$900/tonne.

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Annex Table

Month	Trade Policy	Product	Description
14 th June 2018	Increase in import duty	Crude and Refined Soybean, Sunflower, Safflower, Cottonseed, Rapeseed & Mustard oils	BCD increased on following edible oils: - Crude safflower, rapeseed and mustard oils increased from 30 to 35 percent. This is applicable till date. - Refined safflower, rapeseed and mustard oils increased from 35 to 45 percent. This is applicable till date. - Crude soybean oil increased from 30 to 35 percent. - Refined soybean oil increased from 35 to 45 percent. - Crude sunflower oil increased from 25 to 35 percent. - Refined sunflower oil increased from 35 to 45 percent. - Crude cottonseed oil increased from 30 to 35 percent. - Refined cottonseed oil increased from 35 to 45 percent.
1 st January 2019	Reduction in import duty	Crude Palm oil (CPO)	BCD on crude palm oil (CPO) of edible grade reduced from 44 to 40 percent. BCD on RBD palmolein reduced to 45 percent for imports from Malaysia and 50 percent for shipments from Indonesia.
4 th September 2019	Safeguard duty	RBD Palmolein	Safeguard duty of 5 percent was imposed on imports of RBD palmolein from Malaysia under Malaysia-India Comprehensive Economic Co-operation Agreement (MICECA), although imports under ASEAN agreement did not attract this safeguard duty.
1 st January 2020	Reduction in import duty	Crude and Refined Palm oil	BCD on crude and refined palm oil reduced to 37.5 percent and 45 percent respectively w.e.f. 1 st January, 2020 under ASEAN and MICECA agreement. Import policy of refined palm oil amended from 'Free' to 'Restricted' category.
8 th January 2020	Change in import policy	Refined Palm oil	Import policy of certain variants of refined palm oil revised from 'Free' to 'Restricted'.
26 th November 2020	Reduction in import duty	Crude Palm oil	BCD on crude palm oil reduced from 37.5 to 27.5 percent w.e.f. 27 th November 2020, in order to control the rise in price of palm oil and allow greater availability of oil in the market for consumption.

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Month	Trade Policy	Product	Description
Union Budget 2021-22	Import duty reduced & AIDC imposed	Crude Soybean and Sunflower oil	BCD on imports of crude soybean oil and crude sunflower oil was cut from 35 to 15 percent and additionally, 20 percent Agriculture Infrastructure and Development Cess (AIDC) was imposed on these two oils. Thus, the effective import duty on crude soybean oil and crude sunflower oil remained unchanged at 35 percent.
Union Budget 2021-22	Import duty reduced & AIDC imposed	Crude Palm oil	In the Union Budget 2021-22, BCD on crude palm oil was cut from 27.5 to 15 percent. In addition, 17.5 percent AIDC was imposed on imports of crude palm oil, which narrowed the duty gap between palm oil and other edible oils. The effective import duty on crude palm oil increased from 27.5 to 32.5 percent.
30 th June 2021	Reduction in import duty	Crude and Refined Palm oil	BCD on crude palm oil reduced from 15 to 10 percent. The effective import duty on crude and refined palm oil reduced to 27.5 percent (10% BCD + 17.5% AIDC) and 37.5 percent respectively.
30 th June 2021	Change in import policy	Refined Palm oil	Import Policy of different variants of refined palm oil revised from 'Restricted' to 'Free' category up to 31 st December 2021, subject to prohibition of imports through any port in Kerala.
19 th August 2021	Reduction in import duty	Crude and Refined Soybean and Sunflower oil	BCD on crude variants of soybean and sunflower oil reduced from 15 to 7.5 percent and on refined variants from 45 to 37.5 percent. The effective import duty on crude soybean and sunflower oil reduced to 27.5 percent (7.5% BCD + 20% AIDC).
11 th September 2021	Reduction in import duty and AIDC imposed	Crude and Refined Palm oil	BCD on crude palm oil reduced from 10 to 2.5 percent and refined palm oil from 37.5 to 32.5 percent. Simultaneously, AIDC on crude palm oil was increased to 20 percent. The effective import duty on crude and refined palm oil reduced to 22.5 percent (2.5% BCD + 20% AIDC) and 32.5 percent respectively.
		Crude Soybean and Sunflower oil	BCD on crude soybean oil and crude sunflower oil rationalized from 7.5 to 2.5 percent. The effective import duty on crude soybean and sunflower oil reduced to 22.5 percent (2.5% BCD + 20% AIDC).
		Refined Soybean and Sunflower oil	BCD on refined variants of soybean and sunflower oil reduced to 32.5 percent.

(Contd...)

Annex Table



Annex Table

Month	Trade Policy	Product	Description
13 th October 2021	AIDC reduced	Crude Soybean and Sunflower oil	The AIDC on crude soybean oil and crude sunflower oil was reduced from 20 to 5 percent along with elimination of BCD, earlier set at 2.5 percent w.e.f. 14 th October 2021. The effective duty reduced to 5 percent (0% BCD + 5% AIDC). The AIDC of 5 percent on crude soybean oil and crude sunflower oil was imposed till 31 st March 2022.
13 th October 2021	Reduction in import duty	Crude and Refined Palm oils	BCD slashed on crude palm oil from 2.5 percent to 'Nil' and refined palm oil from 32.5 to 17.5 percent. AIDC on crude palm oil reduced to 7.5 percent w.e.f. 14 th October 2021. The effective import duty on crude and refined palm oil reduced to 7.5 percent (0% BCD + 7.5% AIDC) and 17.5 percent respectively.
	Reduction in import duty	Refined Soybean and Sunflower oil	BCD on refined variants of soybean and sunflower oil was reduced to 17.5 percent.
20 th December 2021	Extension of 'Free' import policy	Refined Palm oil	'Free' import policy of different variants of refined palm oil extended up to 31 st December 2022. The import of refined palm oils was revised from 'Restricted' to 'Free' category till 31 st December 2021, with the exception on imports of these oils through any port in Kerala.
20 th December 2021	Reduction in import duty	Refined Palm oil	BCD on refined palm oil reduced to 12.5 percent. This was applicable till 31 st March 2024.
12 th February 2022	Reduction in AIDC	Crude Palm oil	AIDC on crude palm oil reduced from 7.5 to 5 percent w.e.f. 13 th February 2022. The effective import duty on crude palm oil reduced to 5 percent (0% BCD + 5% AIDC). This was applicable till 31 st March 2024.
24 th May 2022	Allocation of Tariff Rate Quota (TRQ)	Crude Soybean and Sunflower oil	Tariff Rate Quotas notified for allocation of 20 lakh tonnes of TRQ each for crude soybean oil and crude sunflower oil for the years 2022-23 and 2023-24. These TRQs provided exemption from whole of BCD and AIDC.
31 st August 2022	Import duty & AIDC	Crude and Refined Soybean, Sunflower oil	On crude soybean oil and sunflower oil, with zero percent BCD and 5 percent AIDC, the effective duty remained 5 percent. For refined soybean and sunflower oil, effective duty remained 17.5 percent. These concessional customs duties on edible oils imports extended till 31 st March 2023.

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Month	Trade Policy	Product	Description
31 st August 2022	Concessional import duty extended	Palm oil	The concessional import duty on palm oil import was extended for another six months from 30 th September 2022 to 31 st March 2023.
28 th December 2022	Extension of 'Free' import policy	Refined Palm oil	'Free' import policy of refined palm oil variants was extended beyond 31 st March 2022 until further orders.
11 th January 2023	Discontinuation of TRQ for Import	Crude Soybean oil	Last date of import of crude soybean oil under TRQ revised to 31 st March 2023. Further, no TRQs were allocated for import in financial year 2023-24.
1 st March 2023	Discontinuation of TRQ for Import	Crude Sunflower seed and Soybean oil	Last date of import of crude sunflower seed oil and crude soybean oil under TRQ revised to 31 st March 2023. Tariff Rate Quota for crude sunflower oil discontinued from 1 st April 2023.
14 th June 2023	Reduction in import duty	Refined Soybean and Sunflower oil	BCD on refined soybean and sunflower oil reduced from 17.5 percent to 12.5 percent w.e.f. 15 th June 2023. This duty was extended upto 31 st March, 2024.
21 st December 2023	Import duty and AIDC	Crude Palm oil	On crude palm oil, the effective import duty of 5 percent (0% BCD +5% AIDC) extended till 31 st March 2025.
21 st December 2023	Import Duty and AIDC	Crude and Refined Soybean and Sunflower seed oil	On crude soybean and sunflower oil, the effective import duty 5 percent (0% BCD +5% AIDC) and on refined soybean and sunflower oil, effective import duty 12.5 percent extended till 31 st March 2025.
14 th September 2024	Increase in import duty	Crude and Refined Palm oil	W.e.f. 14 th September 2024, BCD on crude palm oil increased from zero to 20 percent, making the effective duty on crude palm oil 25 percent (20% BCD +5% AIDC). Additionally, BCD on refined palm oil increased from 12.5 to 32.5 percent.
14 th September 2024	Increase in import duty	Crude soybean oil, crude sunflower oil and refined soybean oil and refined sunflower oil	W.e.f. 14 th September 2024, BCD on crude soybean oil and crude sunflower oil increased from zero to 20 percent, making the effective duty on these crude oils 25 percent (20% BCD +5% AIDC). Additionally, BCD on refined sunflower oil and refined soybean oil increased from 12.5 to 32.5 percent.



Annex Table

Month	Trade Policy	Product	Description
30 th May 2025	Reduction in import duty	Crude palm, soybean and sunflower oil	W.e.f. 30 th May 2025, BCD on crude palm, soybean and sunflower oil reduced from 20 to 10 percent, making the effective duty on these crude oils 15 percent (10% BCD +5% AIDC).

Note: Social Welfare Cess (SWC) of 10 percent, over and above the effective import duty has been imposed on all types of edible oils since February 2018

Source: 1. Directorate General of Foreign Trade, Ministry of Commerce and Industry

2. Central Board of Indirect Taxes and Customs, Department of Revenue

**Annex Table 5.1: Actual Costs and Gross Returns over Actual Cost of Cultivation
(₹/ha) of Rabi Crops for 2021-22, 2022-23 and 2023-24**

Year	CoC A ₂	CoC A ₂ +FL	GVO	Gross Returns over CoC	
				A ₂ (Col.4 - Col.2)	A ₂ +FL (Col.4 - Col.3)
(1)	(2)	(3)	(4)	(5)	(6)
Wheat					
2021-22	34737	43574	89598	54861	46023
2022-23	36841	46091	94891	58050	48800
2023-24	37291	47009	103553	66262	56544
TE2023-24	36290	45558	96014	59724	50456
Barley					
2021-22	33066	49554	105721	72655	56167
2022-23	38315	57190	105276	66961	48086
2023-24	40293	59776	104112	63819	44336
TE2023-24	37225	55507	105036	67812	49530
Gram					
2021-22	29663	37962	63798	34135	25837
2022-23	31405	40534	65018	33614	24485
2023-24	31338	40400	71653	40315	31254
TE2023-24	30802	39632	66823	36021	27192
Lentil					
2021-22	20834	27297	53632	32798	26335
2022-23	21513	28871	57100	35587	28230
2023-24	22714	29685	58427	35713	28741
TE2023-24	21687	28618	56386	34700	27769
Rapeseed & Mustard					
2021-22	26684	38573	98064	71380	59490
2022-23	28554	42119	82984	54430	40865
2023-24	29185	42384	82583	53398	40199
TE2023-24	28141	41025	87877	59736	46851
Safflower					
2021-22	13023	19305	16669	3646	-2636
2022-23	16725	20186	18270	1545	-1916
2023-24	18703	22824	18720	17	-4104
TE2023-24	16150	20772	17886	1736	-2885

Note: All-India CoC, GVO and gross returns of a crop are weighted average of respective CoCs, GVOs and gross returns of projected States

Source: CACP Calculations using CS data



**Annex Table 5.2: Average Gross Returns over Actual Cost of Cultivation
(₹/ha) of Rabi Crops in Selected States during TE2023-24**

State/Crop	CoC A ₂	CoC A ₂ +FL	GVO	Gross Returns over CoC	
				A ₂ (Col.4 - Col.2)	A ₂ +FL (Col.4 - Col.3)
(1)	(2)	(3)	(4)	(5)	(6)
Wheat					
Bihar	33107	41294	76204	43097	34910
Chhattisgarh	25990	32371	55105	29114	22734
Gujarat	35717	43063	91936	56218	48873
Haryana	34704	41362	111879	77175	70518
Himachal Pradesh	25077	40712	65796	40719	25084
Jharkhand	30689	43023	68272	37584	25249
Madhya Pradesh	28743	35860	86233	57490	50373
Maharashtra	48830	59787	78004	29175	18217
Punjab	35845	38925	107821	71976	68896
Rajasthan	41143	63971	116105	74962	52134
Uttar Pradesh	40930	50860	99591	58661	48731
West Bengal	40762	52643	65690	24928	13047
All-India	36290	45558	96014	59724	50456
Barley					
Himachal Pradesh	26084	48256	32892	6808	-15364
Rajasthan	38128	60594	115920	77791	55326
Uttar Pradesh	36912	48414	96936	60024	48521
All-India	37225	55507	105036	67812	49530
Gram					
Andhra Pradesh	36943	40060	96936	59993	56876
Bihar	21928	30392	81546	59618	51154
Chhattisgarh	30357	34916	70415	40058	35499
Gujarat	38190	44509	79066	40876	34557
Haryana	18415	29234	49152	30737	19918
Jharkhand	31621	48025	38072	6451	-9953
Karnataka	25790	31175	46645	20855	15470
Madhya Pradesh	25909	31322	63219	37310	31897
Maharashtra	40291	47188	73811	33520	26623
Rajasthan	24334	39918	64191	39857	24273
Telangana	26810	29581	60558	33749	30977
Uttar Pradesh	20905	38402	57368	36463	18966
All-India	30802	39632	66823	36021	27192

(Contd...)

**Annex Table 5.2: Average Gross Returns over Actual Cost of Cultivation
(₹/ha) of Rabi Crops in Selected States during TE2023-24**

State/Crop	CoC A ₂	CoC A ₂ +FL	GVO	Gross Returns over CoC	
				A ₂ (Col.4 - Col.2)	A ₂ +FL (Col.4 - Col.3)
(1)	(2)	(3)	(4)	(5)	(6)
Lentil					
Bihar	22213	27193	57073	34860	29880
Madhya Pradesh	21719	25787	54753	33035	28967
Uttar Pradesh	19153	27475	56253	37100	28778
West Bengal	29978	45250	62635	32657	17386
All-India	21687	28618	56386	34700	27769
Rapeseed & Mustard					
Assam	17063	30399	46005	28942	15606
Bihar	22892	28374	59320	36428	30947
Gujarat	42693	54826	122369	79676	67543
Haryana	24360	32635	111648	87288	79013
Himachal Pradesh	15888	30758	34636	18748	3878
Jharkhand	17429	35705	39344	21916	3640
Madhya Pradesh	25601	31573	97718	72116	66144
Punjab	29407	38360	88172	58764	49812
Rajasthan	29656	46040	90753	61097	44712
Uttar Pradesh	29346	38467	85793	56447	47326
West Bengal	33160	48254	70737	37577	22484
All-India	28141	41025	87877	59736	46851
Safflower					
Karnataka	16150	20772	17886	1736	-2885
All-India	16150	20772	17886	1736	-2885

Note: All-India CoC, GVO and gross returns of a crop are weighted average of respective CoCs, GVOs and returns of projected States

Source: CACP Calculations using CS data



**Annex Table 5.3: Monthly Average Daily Wage Rates (₹/day) for Agricultural Labour (Man)
during Rabi Season 2021-22 to 2024-25**

Year/Month	AP	AS	BR	GJ	HR	HP	KA	MP	MH	PB	RJ	UP	WB	IND
2021														
October	421	323	309	242	405	460	374	240	331	373	343	309	338	334
November	429	326	311	242	406	460	374	238	328	373	346	310	338	336
December	430	329	313	244	410	463	378	241	329	372	346	315	339	338
2022														
January	424	330	316	245	410	460	379	246	331	373	337	319	340	338
February	430	338	317	247	412	458	387	248	332	383	341	320	339	341
March	443	333	320	248	413	458	389	248	333	385	334	323	340	344
April	453	333	321	252	421	458	389	250	334	387	334	327	342	348
May	457	334	323	253	422	458	388	250	340	393	334	328	343	350
June	460	335	327	256	422	458	391	251	341	393	339	328	346	352
July	464	338	331	260	427	459	391	252	341	395	341	330	352	355
August	468	338	332	262	435	459	396	253	343	395	344	335	354	358
September	472	338	332	262	437	458	402	255	343	396	347	335	354	359
October	480	340	334	263	442	458	407	256	344	397	345	337	354	362
November	487	342	341	264	442	458	416	257	345	399	351	337	354	365
December	489	342	341	266	449	458	417	257	347	399	352	338	355	366
2023														
January	489	343	342	268	452	460	419	259	349	400	355	341	355	368
February	496	349	344	269	452	465	419	260	350	402	357	342	360	370
March	500	352	346	271	452	465	421	259	351	402	359	345	362	372
April	506	354	349	271	452	466	423	260	351	402	359	346	363	374
May	515	359	353	272	460	466	423	265	352	407	365	347	364	378
June	520	364	356	274	461	467	424	265	355	413	367	349	364	380
July	527	367	357	277	461	464	427	268	357	417	372	350	365	383
August	532	368	361	277	461	476	430	269	362	417	374	352	365	385
September	536	369	365	278	461	476	431	269	364	417	376	354	365	387
October	538	372	366	279	461	505	431	269	366	417	379	355	365	388
November	541	379	367	282	461	485	435	271	368	426	380	356	366	390
December	545	384	368	282	462	485	438	271	370	427	383	356	368	393

(Contd...)

Annex Table 5.3: Monthly Average Daily Wage Rates (₹/day) for Agricultural Labour (Man) during Rabi Season 2021-22 to 2024-25

Year/Month	AP	AS	BR	GJ	HR	HP	KA	MP	MH	PB	RJ	UP	WB	IND
2024														
January	550	386	368	283	466	485	442	272	371	427	390	357	371	395
February	552	386	372	283	472	485	447	272	371	428	396	358	371	396
March	555	387	373	282	474	485	451	272	372	428	398	359	371	398
April	558	389	374	283	481	492	451	273	372	428	400	363	371	399
May	561	391	377	285	487	492	453	273	374	428	402	365	372	401
June	567	391	380	291	491	513	457	274	376	428	404	365	374	404
July	571	394	382	297	496	517	458	276	377	428	406	368	377	407
August	575	400	386	299	502	522	458	279	386	430	409	369	381	410
September	575	406	386	302	507	526	463	280	389	430	412	372	383	412
October	576	410	386	302	512	526	465	281	391	432	415	374	384	414
November	578	415	388	308	512	533	465	284	392	432	419	376	386	416
December	580	418	388	312	512	541	466	285	395	432	424	378	387	417
2025														
January	581	420	395	313	512	544	467	287	396	432	425	381	389	420
February	582	423	402	313	517	545	471	293	397	432	427	382	390	422
March	585	423	405	313	520	545	477	293	399	432	430	382	393	424
April	587	426	406	313	528	545	482	295	402	438	436	384	402	428

Note 1: Daily Wage rate - Average of agricultural operations, such as, ploughing/tilling; sowing (including planting/transplanting/weeding); harvesting/winnowing/threshing; watering & irrigation and plant protection (applying pesticides, treating seeds, etc.)

2: All-India daily wage rate is weighted average of daily wage rates of States

Source: Labour Bureau, Ministry of Labour and Employment and CACP Calculations



Annex Table 5.4: Monthly Wholesale Price Index (Base: 2011-12=100) of Major Farm Inputs during Rabi Season 2023-24 and 2024-25

Year/ Month	High Speed Diesel	Manufacture of Fertilizers and Nitrogen Compounds	Electricity	Manufacture of Agricultural and Forestry Machinery	Lube Oils	Gola & similar Cattle Feed	Fodder	Manufacture of Pesticides and other Agrochemical Products	Index of Farm Inputs
2023									
October	175.1	142.2	142.5	142.5	190.2	218.6	242.5	133.0	160.5
November	174.3	143.0	153.3	143.2	190.2	218.9	230.9	132.8	163.1
December	172.0	142.5	158.4	143.9	190.2	219.1	225.6	132.3	163.7
2024									
January	171.8	141.4	157.2	144.1	190.2	218.3	222.7	131.2	162.9
February	172.1	142.0	157.0	144.1	190.2	215.9	220.5	130.0	162.8
March	170.3	142.6	144.6	143.9	190.2	216.3	219.3	130.5	158.4
April	168.5	143.3	141.1	145.3	190.2	214.7	207.2	127.1	156.2
May	167.7	143.3	135.7	145.2	190.2	214.5	204.4	127.8	154.1
June	166	143.3	132	145	190.2	214.2	204.9	128	152.5
July	167.4	143.4	132.9	143.5	190.2	214.1	205.6	128.9	153.2
August	166.7	143.2	134.8	144.3	190.2	214.8	208.8	129.8	153.8
September	165.1	142.7	136.7	145.2	190.2	214.9	210.2	128.8	153.9
October	164.2	142.9	147.4	145.3	190.2	214.0	213.2	129.3	157.1
November	164.4	143.5	149.4	145.6	190.2	212.1	218.9	129.3	158.2
December	164.6	143.0	157.5	145.8	190.2	211.1	219.6	128.7	160.7
2025									
January	165.6	142.9	155.8	146.6	190.2	209.0	226.4	129.2	160.8
February	166.6	142.8	154.4	147.0	190.2	207.7	227.9	129.2	160.8
March	165.4	142.4	152.5	146.5	190.2	206.8	222.4	129.9	159.5
April	160.0	142.9	150.4	146.8	190.2	207.9	212.9	130.2	156.8

Note: Index of Farm Inputs is weighted average of WPIs of farm inputs

Source: Office of the Economic Adviser, Ministry of Commerce and Industry and CACP Calculations

Annex Table 5.5: Projected Cost of Production (A_2 , A_2+FL & C_2) and Yield for Rabi Crops for RMS2026-27 and Production Shares during TE2024-25

State/Crop	Yield (qtl/ha)	Cost of Production (₹/qtl)			Share in Production (%)
		A_2	A_2+FL	C_2	
Wheat					
Bihar	29.7	1085	1350	1894	6.1
Chhattisgarh	25.9	1183	1470	1939	0.2
Gujarat	34.1	1219	1468	1936	3.3
Haryana	46.0	866	1031	1689	9.8
Himachal Pradesh	20.9	1080	1746	2451	0.6
Jharkhand	26.0	1140	1591	2096	0.4
Madhya Pradesh	39.1	892	1111	1550	20.0
Maharashtra	29.1	2070	2529	3091	1.9
Punjab	47.4	834	905	1565	15.3
Rajasthan	40.2	962	1488	1944	9.2
Uttar Pradesh	36.8	1070	1328	1944	30.6
West Bengal	33.1	1426	1837	2337	0.5
All India	38.4	994	1239	1804	97.9
Barley					
Himachal Pradesh	17.0	1249	2296	3027	1.6
Rajasthan	38.1	861	1364	1802	59.1
Uttar Pradesh	34.6	996	1304	1922	29.0
All India	36.2	912	1361	1862	89.7
Gram					
Andhra Pradesh	16.3	3075	3333	5106	3.1
Bihar	13.8	1893	2613	4199	0.6
Chhattisgarh	14.1	2808	3225	4358	2.3
Gujarat	14.8	2854	3324	4300	11.1
Haryana	8.8	2331	3686	5534	0.2
Jharkhand	11.8	2738	4143	5149	2.7
Karnataka	9.4	3390	4088	5355	3.9
Madhya Pradesh	12.5	2774	3349	4520	24.4
Maharashtra	14.2	3491	4082	5191	26.2
Rajasthan	12.0	2251	3670	4673	16.5

(Contd...)



Annex Table 5.5: Projected Cost of Production (A_2 , A_2+FL & C_2) and Yield for Rabi Crops for RMS2026-27 and Production Shares during TE2024-25

State	Yield (qtl/ha)	Cost of Production (₹/qtl)			Share in Production (%)
		A_2	A_2+FL	C_2	
Telangana	11.4	2993	3296	5091	1.5
Uttar Pradesh	9.2	2338	4266	6037	6.8
All India	12.8	2886	3699	4875	99.2
Lentil					
Bihar	10.1	2513	3069	4658	7.2
Madhya Pradesh	10.7	2757	3269	4417	42.7
Uttar Pradesh	9.0	2955	4220	6305	33.6
West Bengal	11.2	2912	4396	5688	8.6
All India	10.1	2825	3705	5243	92.0
Rapeseed & Mustard					
Assam	9.7	2483	4392	5543	1.9
Bihar	10.2	2378	2944	4756	0.8
Gujarat	20.5	2462	3154	4321	4.1
Haryana	17.8	1818	2430	4113	10.0
Himachal Pradesh	8.1	2241	4310	6671	0.1
Jharkhand	8.2	2224	4532	5867	2.6
Madhya Pradesh	16.1	1754	2159	3438	13.3
Punjab	15.3	2255	2936	4844	0.6
Rajasthan	15.3	2193	3391	4521	43.7
Uttar Pradesh	13.9	2586	3382	5264	14.3
West Bengal	13.2	2928	4243	5559	6.2
All India	14.8	2219	3210	4558	97.6
Safflower					
Karnataka	5.8	3376	4360	5738	44.3
All India	5.8	3376	4360	5738	44.3

Note: All-India CoP of a crop is weighted average of CoPs of projected States

Source: CACP Calculations using CS data

Annex Table 5.6(a): Wheat : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Bihar					Chhattisgarh				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	39361	40287	41533	2.4	3.1	31212	32476	32276	4.0	-0.6
Human Labour										
Casual	7066	7161	7362	1.3	2.8	1479	1549	1639	4.7	5.8
Attached	313	264	246	-15.7	-6.8	0	0	0	-	-
Family	8098	8151	8312	0.7	2.0	6113	6481	6547	6.0	1.0
Total	15476	15575	15920	0.6	2.2	7592	8030	8186	5.8	1.9
Bullock Labour										
Hired	0	0	0	-	-	0	0	0	-	-
Owned	0	0	39	-	-	0	0	0	-	-
Total	0	0	39	-	-	0	0	0	-	-
Machine Labour										
Hired	8459	8847	9251	4.6	4.6	10282	10589	10268	3.0	-3.0
Owned	99	322	317	224.8	-1.7	885	938	948	6.0	1.1
Total	8558	9169	9568	7.1	4.4	11167	11527	11216	3.2	-2.7
Seed	4318	4463	4734	3.4	6.1	3132	3286	3315	4.9	0.9
Fertilisers and Manure										
Fertilisers	4736	4817	5130	1.7	6.5	4067	4221	4194	3.8	-0.6
Manure	346	329	321	-4.8	-2.5	0	0	114	-	-
Total	5081	5146	5451	1.3	5.9	4067	4221	4309	3.8	2.1
Other Inputs										
Insecticides	100	101	100	0.7	-1.4	102	104	107	2.1	3.0
Irrigation charges	4861	4839	4696	-0.5	-3.0	4326	4451	4293	2.9	-3.6
Crop insurance	0	0	0	-	-	0	0	0	-	-
Interest on working capital	947	974	1007	2.8	3.4	761	788	780	3.6	-1.0
Miscellaneous	19	19	18	1.0	-2.1	65	69	71	6.1	2.9
Fixed Cost	21695	21001	20727	-3.2	-1.3	11955	12988	13738	8.6	5.8
Rental value of owned land	18273	17510	17087	-4.2	-2.4	9465	10322	11172	9.0	8.2
Rent paid for leased-in land	197	203	209	3.0	3.0	0	0	0	-	-
Land revenue, cesses & taxes	207	212	216	2.6	1.8	4	4	4	4.4	-1.3
Depreciation on implements & farm buildings	465	489	502	5.2	2.6	364	383	390	5.3	1.9
Interest on fixed capital	2552	2586	2713	1.3	4.9	2123	2279	2172	7.4	-4.7
Yield (Qt/ha)	29.3	29.5	30.2	0.9	2.4	25.3	27.2	25.1	7.5	-7.9

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare

(contd...)



Annex Table 5.6(a): Wheat : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Gujarat					Haryana				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	38418	42156	46200	9.7	9.6	39231	40657	41821	3.6	2.9
Human Labour										
<i>Casual</i>	5729	6636	8431	15.8	27.0	2324	2500	2538	7.6	1.5
<i>Attached</i>	129	129	0	-0.2	-	216	206	208	-4.6	0.9
<i>Family</i>	7029	7443	7565	5.9	1.6	6481	6716	6776	3.6	0.9
Total	12887	14208	15996	10.2	12.6	9021	9422	9522	4.4	1.1
Bullock Labour										
<i>Hired</i>	196	167	35	-14.9	-78.9	3	3	0	1.9	-
<i>Owned</i>	148	147	203	-0.5	38.5	2	2	0	3.8	-
Total	344	314	239	-8.8	-24.0	5	5	0	2.7	-
Machine Labour										
<i>Hired</i>	7886	8432	9276	6.9	10.0	12180	12491	12896	2.6	3.2
<i>Owned</i>	1979	2298	2244	16.1	-2.3	2789	2871	3065	3.0	6.7
Total	9866	10730	11520	8.8	7.4	14969	15362	15961	2.6	3.9
Seed	4646	5448	5865	17.3	7.6	3273	3462	3621	5.8	4.6
Fertilisers and Manure										
<i>Fertilisers</i>	4029	4324	4834	7.3	11.8	4914	5149	5353	4.8	4.0
<i>Manure</i>	389	459	450	18.0	-2.1	0	0	0	-	0.0
Total	4418	4783	5284	8.3	10.5	4914	5149	5353	4.8	4.0
Other Inputs										
Insecticides	590	673	738	14.0	9.8	1678	1737	1775	3.5	2.2
Irrigation charges	4530	4769	5208	5.3	9.2	4206	4313	4345	2.6	0.7
Crop insurance	0	0	0	-	-	57	55	57	-2.4	2.9
Interest on working capital	951	1052	1171	10.6	11.3	992	1029	1062	3.6	3.3
Miscellaneous	187	179	180	-4.2	0.4	117	123	125	4.7	1.4
Fixed Cost	16759	18724	20139	11.7	7.6	33244	35788	41271	7.7	15.3
Rental value of owned land	13085	14743	16288	12.7	10.5	27027	29737	35036	10.0	17.8
Rent paid for leased-in land	485	526	458	8.5	-12.9	131	136	142	4.0	4.3
Land revenue, cesses & taxes	9	9	5	1.0	-40.0	0	0	0	-	-
Depreciation on implements & Farm buildings	292	300	330	2.7	10.2	622	656	688	5.5	5.0
Interest on fixed capital	2888	3146	3057	8.9	-2.8	5464	5259	5404	-3.7	2.8
Yield (Qtl/ha)	32.5	34.3	35.5	5.3	3.6	44.2	46.7	47.1	5.5	0.9

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare

(contd...)

Annex Table 5.6(a): Wheat : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Himachal Pradesh					Jharkhand				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	39725	40322	39532	1.5	-2.0	41990	41977	42690	0.0	1.7
Human Labour										
<i>Casual</i>	810	799	1167	-1.3	46.0	6897	3740	3656	-45.8	-2.2
<i>Attached</i>	11	11	10	8.0	-10.7	0	0	0	-	-
<i>Family</i>	15453	15931	15523	3.1	-2.6	10843	13056	13105	20.4	0.4
Total	16273	16741	16701	2.9	-0.2	17741	16796	16761	-5.3	-0.2
Bullock Labour										
<i>Hired</i>	160	257	36	61.1	-86.0	0	0	0	-	-
<i>Owned</i>	3058	2186	1693	-28.5	-22.6	3870	4528	4706	17.0	3.9
Total	3218	2443	1729	-24.1	-29.2	3870	4528	4706	17.0	3.9
Machine Labour										
<i>Hired</i>	8060	8275	7671	2.7	-7.3	6976	6311	6513	-9.5	3.2
<i>Owned</i>	1135	1158	1992	2.0	72.0	181	148	143	-18.4	-3.1
Total	9195	9433	9663	2.6	2.4	7157	6459	6657	-9.8	3.1
Seed	2741	2962	3015	8.0	1.8	4018	4380	4439	9.0	1.4
Fertilisers and Manure										
<i>Fertilisers</i>	1689	2228	2159	31.9	-3.1	4519	4806	4994	6.4	3.9
<i>Manure</i>	4764	4564	4259	-4.2	-6.7	574	472	479	-17.8	1.4
Total	6453	6792	6419	5.3	-5.5	5093	5278	5472	3.6	3.7
Other Inputs										
Insecticides	326	328	354	0.7	8.0	0	0	2	-	-
Irrigation charges	694	793	827	14.3	4.4	3168	3660	3756	15.5	2.6
Crop insurance	0	0	0	-	-	0	0	0	-	-
Interest on working capital	736	739	728	0.5	-1.6	944	876	897	-7.1	2.3
Miscellaneous	91	91	97	0.1	6.7	0	0	0	-	-
Fixed Cost	21568	21785	23014	1.0	5.6	16587	20842	17241	25.6	-17.3
Rental value of owned land	16185	15827	15869	-2.2	0.3	11792	16384	12885	38.9	-21.4
Rent paid for leased-in land	51	39	17	-22.6	-55.8	0	0	0	-	-
Land revenue, cesses & taxes	10	11	11	3.1	4.0	59	60	59	2.1	-1.8
Depreciation on implements & farm buildings	679	738	1000	8.7	35.4	776	737	721	-5.1	-2.1
Interest on fixed capital	4643	5170	6116	11.3	18.3	3960	3660	3576	-7.6	-2.3
Yield (Qtl/ha)	21.5	20.4	20.8	-5.4	2.3	26.1	26.5	25.4	1.5	-4.2

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare

(contd...)



Annex Table 5.6(a): Wheat : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Madhya Pradesh					Maharashtra				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	33921	35475	36282	4.6	2.3	53560	62417	62012	16.5	-0.6
Human Labour										
<i>Casual</i>	3573	3803	3953	6.4	3.9	4826	6271	5131	29.9	-18.2
<i>Attached</i>	136	138	135	1.6	-2.3	900	783	676	-13.1	-13.6
<i>Family</i>	7047	7117	7189	1.0	1.0	9638	11497	11737	19.3	2.1
Total	10755	11057	11276	2.8	2.0	15365	18550	17545	20.7	-5.4
Bullock Labour										
<i>Hired</i>	28	30	29	5.4	-3.0	328	267	522	-18.6	95.9
<i>Owned</i>	509	511	494	0.3	-3.2	1801	1571	1404	-12.8	-10.6
Total	538	540	523	0.5	-3.2	2128	1837	1926	-13.7	4.8
Machine Labour										
<i>Hired</i>	9251	9777	10003	5.7	2.3	13095	15059	12313	15.0	-18.2
<i>Owned</i>	712	756	794	6.2	5.0	874	1567	3924	79.3	150.5
Total	9963	10533	10798	5.7	2.5	13968	16626	16238	19.0	-2.3
Seed	3579	3807	3866	6.4	1.5	5131	5804	6168	13.1	6.3
Fertilisers and Manure										
<i>Fertilisers</i>	3456	3719	3860	7.6	3.8	6516	7403	6817	13.6	-7.9
<i>Manure</i>	40	0	0	-	-	1000	1134	1215	13.4	7.2
Total	3496	3719	3860	6.4	3.8	7515	8537	8032	13.6	-5.9
Other Inputs										
Insecticides	182	190	196	4.4	3.1	849	988	1095	16.3	10.9
Irrigation charges	4214	4392	4494	4.2	2.3	7193	8459	9400	17.6	11.1
Crop insurance	262	255	263	-2.6	3.0	14	0	0	-	-
Interest on working capital	814	859	882	5.5	2.6	1331	1543	1523	15.9	-1.3
Miscellaneous	118	122	124	2.7	2.0	65	74	85	14.8	15.0
Fixed Cost	19571	19363	21257	-1.1	9.8	15361	16915	19384	10.1	14.6
Rental value of owned land	15633	15491	17236	-0.9	11.3	11223	13071	14715	16.5	12.6
Rent paid for leased-in land	0	0	0	-	-	0	0	0	-	-
Land revenue, cesses & taxes	4	4	4	-2.4	-2.2	15	14	16	-3.8	11.3
Depreciation on implements & Farm buildings	603	634	653	5.0	3.1	403	419	505	4.1	20.6
Interest on fixed capital	3330	3235	3364	-2.9	4.0	3720	3410	4147	-8.3	21.6
Yield (Qtl/ha)	40.8	39.2	37.3	-3.8	-4.8	26.3	31.0	30.0	17.8	-3.2

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare

(contd...)

Annex Table 5.6(a): Wheat : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Punjab					Rajasthan				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	31336	32525	33411	3.8	2.7	57191	63625	67926	11.3	6.8
Human Labour										
<i>Casual</i>	1305	1412	1481	8.2	4.9	5813	6436	5172	10.7	-19.6
<i>Attached</i>	655	670	680	2.3	1.6	177	276	15	56.0	-94.6
<i>Family</i>	2992	3061	3188	2.3	4.1	19926	22237	26321	11.6	18.4
Total	4952	5143	5349	3.9	4.0	25916	28948	31508	11.7	8.8
Bullock Labour										
<i>Hired</i>	0	0	0	-	-	0	0	0	-	-
<i>Owned</i>	8	8	7	3.2	-5.8	449	424	516	-5.5	21.7
Total	8	8	7	3.2	-5.8	449	424	516	-5.5	21.7
Machine Labour										
<i>Hired</i>	8934	9478	9737	6.1	2.7	10344	10261	9686	-0.8	-5.6
<i>Owned</i>	4370	4612	4678	5.5	1.4	1845	2542	3427	37.7	34.8
Total	13304	14090	14415	5.9	2.3	12189	12802	13113	5.0	2.4
Seed	3078	3144	3189	2.1	1.4	4734	5720	6117	20.8	7.0
Fertilisers and Manure										
<i>Fertilisers</i>	5649	5783	6019	2.4	4.1	3387	4023	3386	18.8	-15.8
<i>Manure</i>	19	20	20	7.1	0.0	469	372	311	-20.8	-16.4
Total	5667	5803	6039	2.4	4.1	3856	4395	3697	14.0	-15.9
Other Inputs										
Insecticides	2465	2414	2451	-2.1	1.5	318	337	270	6.1	-19.9
Irrigation charges	764	803	824	5.2	2.5	8515	9650	11373	13.3	17.9
Crop insurance	0	0	0	-	-	0	0	0	-	-
Interest on working capital	859	893	916	4.0	2.6	1129	1254	1261	11.1	0.5
Miscellaneous	239	227	222	-5.2	-2.1	84	94	71	12.0	-24.5
Fixed Cost	38117	42930	40612	12.6	-5.4	23757	23033	26982	-3.0	17.1
Rental value of owned land	26423	30710	28311	16.2	-7.8	18336	18046	20741	-1.6	14.9
Rent paid for leased-in land	5717	5908	6135	3.3	3.8	378	401	143	6.1	-64.4
Land revenue, cesses & taxes	0	0	0	-	-	1	0	0	-	-
Depreciation on implements & Farm buildings	548	586	610	6.9	4.0	680	717	851	5.3	18.7
Interest on fixed capital	5429	5726	5556	5.5	-3.0	4361	3870	5248	-11.3	35.6
Yield (Qt/ha)	42.4	48.0	51.8	13.2	7.8	41.6	38.8	40.2	-6.9	3.7

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare

(contd...)



Annex Table 5.6(a): Wheat : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Uttar Pradesh					West Bengal				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	47775	50033	50454	4.7	0.8	48671	55204	52676	13.4	-4.6
Human Labour										
<i>Casual</i>	6246	7228	6866	15.7	-5.0	4175	2025	3969	-51.5	96.0
<i>Attached</i>	7	6	52	-20.6	788.1	0	0	0	-	-
<i>Family</i>	9793	9789	10208	0.0	4.3	11311	12998	11335	14.9	-12.8
Total	16046	17023	17126	6.1	0.6	15486	15023	15304	-3.0	1.9
Bullock Labour										
<i>Hired</i>	6	1	0	-78.6	-	75	28	0	-62.3	-
<i>Owned</i>	56	75	85	33.3	13.8	99	684	0	590.9	-
Total	62	76	85	22.6	11.9	174	712	0	309.7	-
Machine Labour										
<i>Hired</i>	11927	12117	11783	1.6	-2.8	13545	14994	15517	10.7	3.5
<i>Owned</i>	855	936	1643	9.4	75.5	70	140	0	99.2	-
Total	12782	13053	13426	2.1	2.9	13616	15134	15517	11.2	2.5
Seed	4235	4765	5096	12.5	6.9	4839	5557	6253	14.8	12.5
Fertilisers and Manure										
<i>Fertilisers</i>	4868	5268	5619	8.2	6.7	7610	10565	7764	38.8	-26.5
<i>Manure</i>	35	40	44	13.0	11.2	1160	677	0	-41.7	-
Total	4903	5308	5664	8.2	6.7	8770	11242	7764	28.2	-30.9
Other Inputs										
Insecticides	197	209	219	6.2	4.6	200	616	755	208.2	22.7
Irrigation charges	8329	8292	7542	-0.4	-9.0	4449	5640	5827	26.8	3.3
Crop insurance	0	0	0	-	-	0	0	0	-	-
Interest on working capital	1151	1220	1220	6.0	0.0	1132	1279	1253	13.0	-2.1
Miscellaneous	70	88	77	25.9	-11.6	5	2	3	-58.6	58.3
Fixed Cost	28176	29304	34248	4.0	16.9	11727	18308	22506	56.1	22.9
Rental value of owned land	23385	24345	28584	4.1	17.4	10418	16984	21220	63.0	24.9
Rent paid for leased-in land	543	527	555	-3.0	5.4	456	211	126	-53.8	-40.3
Land revenue, cesses & taxes	2	3	5	15.8	75.2	0	0	0	-	-
Depreciation on implements & Farm buildings	805	885	992	10.0	12.1	222	178	187	-19.9	5.4
Interest on fixed capital	3441	3544	4112	3.0	16.0	632	936	974	48.1	4.1
Yield (Qtl/ha)	36.5	35.4	38.4	-3.0	8.5	21.0	34.2	35.8	62.8	4.7

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare

Annex Table 5.6(b): Barley: Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Himachal Pradesh					Rajasthan				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	47028	50663	45030	7.7	-11.1	53897	61563	63777	14.2	3.6
Human Labour										
<i>Casual</i>	158	68	69	-56.9	1.1	4066	6059	6578	49.0	8.6
<i>Attached</i>	127	57	0	-55.3	-	12	7	10	-38.6	45.3
<i>Family</i>	20680	22001	23833	6.4	8.3	20724	22760	23913	9.8	5.1
Total	20965	22126	23902	5.5	8.0	24802	28826	30501	16.2	5.8
Bullock Labour										
<i>Hired</i>	321	325	138	1.3	-57.7	0	0	0	-	-
<i>Owned</i>	14705	15957	9560	8.5	-40.1	348	295	186	-15.2	-36.8
Total	15025	16282	9697	8.4	-40.4	348	295	186	-15.2	-36.8
Machine Labour										
<i>Hired</i>	3600	3851	4218	7.0	9.5	8772	9110	9131	3.8	0.2
<i>Owned</i>	416	588	770	41.4	30.9	3097	3714	3985	19.9	7.3
Total	4016	4439	4988	10.5	12.4	11869	12824	13116	8.0	2.3
Seed	1877	2152	2276	14.6	5.7	4236	5135	6042	21.2	17.7
Fertilisers and Manure										
<i>Fertilisers</i>	93	53	94	-42.9	77.3	3005	2995	2775	-0.3	-7.4
<i>Manure</i>	4242	4730	3417	11.5	-27.8	760	914	0	20.2	-
Total	4335	4783	3511	10.3	-26.6	3765	3909	2775	3.8	-29.0
Other Inputs										
Insecticides	0	0	0	-	-	434	581	336	33.9	-42.2
Irrigation charges	0	0	2	-	-	7337	8703	9498	18.6	9.1
Crop insurance	0	0	0	-	-	0	0	0	-	-
Interest on working capital	798	869	642	8.8	-26.0	1005	1176	1208	17.0	2.7
Miscellaneous	11	11	11	6.3	1.5	102	114	115	11.7	0.9
Fixed Cost	10956	12167	13747	11.1	13.0	24774	21201	23058	-14.4	8.8
Rental value of owned land	7370	8167	9003	10.8	10.2	20781	16876	18397	-18.8	9.0
Rent paid for leased-in land	0	0	169	-	-	43	165	196	280.8	19.2
Land revenue, cesses & taxes	11	11	11	-4.2	2.1	0	0	0	-	-
Depreciation on implements & Farm buildings	589	584	672	-0.8	15.1	652	732	757	12.2	3.4
Interest on fixed capital	2987	3406	3893	14.0	14.3	3297	3428	3708	4.0	8.2
Yield (Qt/ha)	10.5	11.3	11.9	7.5	5.3	37.5	35.9	40.8	-4.3	13.5

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare

(contd...)



Annex Table 5.6(b): Barley: Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Uttar Pradesh				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	43279	48293	51242	11.6	6.1
Human Labour					
<i>Casual</i>	7945	9480	9844	19.3	3.8
<i>Attached</i>	0	0	0	-	-
<i>Family</i>	10881	11937	11689	9.7	-2.1
Total	18826	21417	21533	13.8	0.5
Bullock Labour					
<i>Hired</i>	0	0	0	-	-
<i>Owned</i>	0	0	0	-	-
Total	0	0	0	-	-
Machine Labour					
<i>Hired</i>	8069	9420	10515	16.7	11.6
<i>Owned</i>	2764	2507	4141	-9.3	65.2
Total	10833	11927	14656	10.1	22.9
Seed	3488	4405	4718	26.3	7.1
Fertilisers and Manure					
<i>Fertilisers</i>	4071	3917	4312	-3.8	10.1
<i>Manure</i>	0	0	6	-	-
Total	4071	3917	4318	-3.8	10.2
<i>Other Inputs</i>					
Insecticides	0	0	0	-	-
Irrigation charges	4991	5450	4819	9.2	-11.6
Crop insurance	0	0	0	-	-
Interest on working capital	982	1102	1199	12.2	8.8
Miscellaneous	89	76	0	-15.0	-
Fixed Cost	26204	33775	26478	28.9	-21.6
Rental value of owned land	22822	29792	21047	30.5	-29.4
Rent paid for leased-in land	35	0	1257	-	-
Land revenue, cesses & taxes	13	10	11	-18.6	2.6
Depreciation on implements & Farm buildings	316	387	399	22.3	3.2
Interest on fixed capital	3018	3585	3764	18.8	5.0
Yield (Qtl/ha)	30.9	35.9	35.4	16.2	-1.3

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare

Annex Table 5.6(c): Gram : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Andhra Pradesh					Bihar				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	37685	39600	41352	5.1	4.4	29869	29663	30099	-0.7	1.5
Human Labour										
<i>Casual</i>	6589	7400	7953	12.3	7.5	7690	6189	6122	-19.5	-1.1
<i>Attached</i>	156	104	0	-33.5	-	0	0	0	-	-
<i>Family</i>	3103	3225	3025	3.9	-6.2	7471	8752	9169	17.2	4.8
Total	9848	10728	10978	8.9	2.3	15161	14941	15291	-1.4	2.3
Bullock Labour										
<i>Hired</i>	547	979	557	78.9	-43.1	0	0	0	-	-
<i>Owned</i>	1671	884	494	-47.1	-44.2	0	0	0	-	-
Total	2218	1863	1050	-16.0	-43.6	0	0	0	-	-
Machine Labour										
<i>Hired</i>	8993	10419	11882	15.8	14.0	4611	4683	4866	1.6	3.9
<i>Owned</i>	1854	2292	2300	23.6	0.3	303	479	466	58.4	-2.8
Total	10848	12711	14182	17.2	11.6	4914	5163	5332	5.1	3.3
Seed	6378	6693	7231	4.9	8.0	5995	5658	5647	-5.6	-0.2
Fertilisers and Manure										
<i>Fertilisers</i>	4565	4527	4692	-0.8	3.6	2344	2540	2603	8.4	2.5
<i>Manure</i>	142	0	0	-	-	141	154	0	8.6	-
Total	4706	4527	4692	-3.8	3.6	2485	2694	2603	8.4	-3.4
Other Inputs										
Insecticides	2500	1754	1825	-29.9	4.1	284	226	237	-20.3	4.7
Irrigation charges	3	6	7	112.9	7.9	353	347	355	-1.7	2.5
Crop insurance	0	0	0	-	-	0	0	0	-	-
Interest on working capital	1048	1102	1161	5.2	5.4	679	634	634	-6.6	0.1
Miscellaneous	136	215	226	58.6	4.9	0	0	0	-	-
Fixed Cost	29202	28923	30561	-1.0	5.7	24716	23577	20385	-4.6	-13.5
Rental value of owned land	27911	27606	28166	-1.1	2.0	21354	20389	17046	-4.5	-16.4
Rent paid for leased-in land	0	0	977	-	-	0	0	0	-	-
Land revenue, cesses & taxes	0	0	0	-	-	103	118	124	14.5	5.0
Depreciation on implements & Farm buildings	187	187	194	0.1	4.0	408	388	406	-4.9	4.6
Interest on fixed capital	1105	1130	1224	2.2	8.4	2852	2682	2810	-6.0	4.8
Yield (Qtl/ha)	16.2	16.2	16.4	0.0	1.4	13.3	14.2	14.0	7.3	-1.8

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare

(contd...)



Annex Table 5.6(c): Gram : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Chhattisgarh					Gujarat				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	32893	34456	35163	4.8	2.1	39622	43175	47642	9.0	10.3
Human Labour										
<i>Casual</i>	4538	4753	4990	4.7	5.0	11020	14276	16424	29.5	15.1
<i>Attached</i>	0	0	0	-	-	0	0	0	-	-
<i>Family</i>	4380	4553	4745	4.0	4.2	6558	6178	6221	-5.8	0.7
Total	8917	9306	9735	4.4	4.6	17578	20453	22645	16.4	10.7
Bullock Labour										
<i>Hired</i>	0	0	0	-	-	395	151	451	-61.8	199.4
<i>Owned</i>	0	0	0	-	-	300	439	307	46.1	-30.1
Total	0	0	0	-	-	695	590	758	-15.2	28.5
Machine Labour										
<i>Hired</i>	7136	7472	7690	4.7	2.9	6893	6525	6907	-5.3	5.8
<i>Owned</i>	833	859	884	3.1	2.9	1551	2030	2204	30.9	8.6
Total	7970	8331	8573	4.5	2.9	8444	8555	9111	1.3	6.5
Seed	6679	6996	6890	4.7	-1.5	4576	5345	5630	16.8	5.3
Fertilisers and Manure										
<i>Fertilisers</i>	2760	2916	2959	5.6	1.5	2082	2127	2730	2.2	28.4
<i>Manure</i>	0	0	2	-	-	113	6	6	-95.0	5.1
Total	2760	2916	2961	5.6	1.5	2195	2133	2736	-2.8	28.3
Other Inputs										
Insecticides	2094	2209	2141	5.5	-3.1	2324	2164	2336	-6.9	7.9
Irrigation charges	3554	3735	3881	5.1	3.9	2719	2713	3061	-0.2	12.8
Crop insurance	0	0	0	-	-	0	0	0	-	-
Interest on working capital	864	906	922	4.9	1.7	1002	1121	1255	11.9	12.0
Miscellaneous	56	58	59	4.0	2.7	88	102	109	15.2	7.6
Fixed Cost	16429	17706	18183	7.8	2.7	14053	15910	18086	13.2	13.7
Rental value of owned land	11189	12119	12827	8.3	5.8	10395	11913	14674	14.6	23.2
Rent paid for leased-in land	0	0	0	-	-	906	1069	155	18.0	-85.5
Land revenue, cesses & taxes	1	1	1	1.4	-2.8	3	3	6	13.4	81.3
Depreciation on implements & Farm buildings	732	761	742	4.0	-2.5	300	309	337	2.8	9.4
Interest on fixed capital	4507	4824	4612	7.0	-4.4	2449	2615	2915	6.8	11.4
Yield (Qtl/ha)	13.6	14.7	14.1	8.4	-4.6	12.5	15.5	16.4	24.0	5.5

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare

(contd...)

Annex Table 5.6(c): Gram : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Haryana					Jharkhand				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	28117	29303	29857	4.2	1.9	47581	46941	47746	-1.3	1.7
Human Labour										
<i>Casual</i>	2523	2605	2677	3.2	2.8	5880	5370	5552	-8.7	3.4
<i>Attached</i>	0	0	0	-	-	0	0	0	-	-
<i>Family</i>	10351	10942	11166	5.7	2.0	16193	16485	16534	1.8	0.3
Total	12874	13546	13843	5.2	2.2	22073	21855	22086	-1.0	1.1
Bullock Labour										
<i>Hired</i>	61	0	0	-	-	0	0	64	-	-
<i>Owned</i>	0	0	0	-	-	10864	11488	11613	5.7	1.1
Total	61	0	0	-	-	10864	11488	11678	5.7	1.6
Machine Labour										
<i>Hired</i>	7063	7543	7836	6.8	3.9	3000	3038	3096	1.3	1.9
<i>Owned</i>	110	107	109	-3.2	2.4	0	0	25	-	-
Total	7173	7650	7945	6.6	3.9	3000	3038	3120	1.3	2.7
Seed	2389	2442	2498	2.2	2.3	6609	6207	6374	-6.1	2.7
Fertilisers and Manure										
<i>Fertilisers</i>	2016	2070	2009	2.7	-3.0	2880	3027	3048	5.1	0.7
<i>Manure</i>	0	0	0	-	-	0	0	85	-	-
Total	2016	2070	2009	2.7	-3.0	2880	3027	3133	5.1	3.5
Other Inputs										
Insecticides	117	123	126	4.4	2.7	1204	259	261	-78.5	1.0
Irrigation charges	2920	2888	2842	-1.1	-1.6	0	145	148	-	1.8
Crop insurance	0	0	0	-	-	0	0	0	-	-
Interest on working capital	538	556	566	3.4	1.8	951	923	946	-3.0	2.5
Miscellaneous	28	28	28	-1.5	0.6	0	0	0	-	-
Fixed Cost	18112	17582	17489	-2.9	-0.5	15605	10402	13046	-33.3	25.4
Rental value of owned land	16314	15673	15635	-3.9	-0.2	13582	8229	10970	-39.4	33.3
Rent paid for leased-in land	0	0	0	-	-	0	0	0	-	-
Land revenue, cesses & taxes	0	0	0	-	-	74	69	70	-6.5	1.0
Depreciation on implements & Farm buildings	138	145	143	5.5	-1.7	505	550	537	8.8	-2.4
Interest on fixed capital	1660	1764	1711	6.2	-3.0	1443	1554	1469	7.7	-5.5
Yield (Qtl/ha)	8.9	8.5	9.1	-4.0	6.7	7.6	5.3	6.0	-29.7	13.0

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare

(contd...)



Annex Table 5.6(c): Gram : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Karnataka					Madhya Pradesh				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	30486	30657	31540	0.6	2.9	29990	30804	31515	2.7	2.3
Human Labour										
<i>Casual</i>	6771	6497	6887	-4.0	6.0	4428	4665	4839	5.4	3.7
<i>Attached</i>	166	178	9	7.2	-94.9	113	118	118	4.6	0.3
<i>Family</i>	4950	5733	5473	15.8	-4.5	5438	5406	5396	-0.6	-0.2
Total	11887	12408	12369	4.4	-0.3	9978	10189	10353	2.1	1.6
Bullock Labour										
<i>Hired</i>	616	1093	918	77.5	-16.0	27	28	29	1.8	3.6
<i>Owned</i>	730	440	618	-39.7	40.4	711	737	716	3.7	-2.9
Total	1346	1534	1536	13.9	0.2	739	765	745	3.6	-2.6
Machine Labour										
<i>Hired</i>	5104	5052	5625	-1.0	11.3	6422	6670	6922	3.9	3.8
<i>Owned</i>	1763	1283	1458	-27.2	13.6	369	386	393	4.6	1.9
Total	6867	6335	7083	-7.7	11.8	6791	7055	7315	3.9	3.7
Seed	4351	4256	4304	-2.2	1.1	5463	5705	5962	4.4	4.5
Fertilisers and Manure										
<i>Fertilisers</i>	3639	3496	3067	-3.9	-12.3	2026	2076	2132	2.5	2.7
<i>Manure</i>	0	0	191	-	-	0	0	7	-	-
Total	3639	3496	3259	-3.9	-6.8	2026	2076	2139	2.5	3.1
<i>Other Inputs</i>										
Insecticides	1373	1575	1782	14.7	13.1	1487	1460	1507	-1.8	3.2
Irrigation charges	147	189	197	28.9	3.9	2527	2542	2461	0.6	-3.2
Crop insurance	0	0	99	-	-	153	159	156	3.6	-1.5
Interest on working capital	774	755	790	-2.4	4.6	744	770	791	3.4	2.8
Miscellaneous	102	108	121	6.2	11.8	81	83	85	2.3	2.2
Fixed Cost	11982	11895	13623	-0.7	14.5	15151	15073	17289	-0.5	14.7
Rental value of owned land	10447	10393	11680	-0.5	12.4	12727	12600	14820	-1.0	17.6
Rent paid for leased-in land	0	0	234	-	-	0	0	0	-	-
Land revenue, cesses & taxes	4	5	5	9.8	0.6	4	4	4	1.4	3.0
Depreciation on implements & Farm buildings	185	190	220	2.6	15.8	537	550	558	2.5	1.5
Interest on fixed capital	1345	1307	1484	-2.9	13.5	1883	1919	1906	1.9	-0.6
Yield (Qtl/ha)	9.2	9.7	9.3	4.8	-3.6	11.9	12.9	12.6	8.5	-2.1

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare

(contd...)

Annex Table 5.6(c): Gram : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Maharashtra					Rajasthan				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	43815	49414	46433	12.8	-6.0	37553	39120	40484	4.2	3.5
Human Labour										
<i>Casual</i>	10188	11355	9934	11.4	-12.5	4342	3609	3970	-16.9	10.0
<i>Attached</i>	454	654	741	44.2	13.3	91	173	0	91.0	-
<i>Family</i>	6303	7398	6990	17.4	-5.5	13910	16138	16706	16.0	3.5
Total	16945	19408	17665	14.5	-9.0	18342	19920	20675	8.6	3.8
Bullock Labour										
<i>Hired</i>	514	235	449	-54.3	91.3	0	0	0	-	-
<i>Owned</i>	1493	1641	1743	9.9	6.2	101	80	0	-20.5	-
Total	2007	1876	2192	-6.5	16.8	101	80	0	-20.5	-
Machine Labour										
<i>Hired</i>	7900	11130	9324	40.9	-16.2	7755	6224	6706	-19.7	7.8
<i>Owned</i>	3205	1643	1647	-48.7	0.2	1078	2627	2106	143.7	-19.8
Total	11105	12773	10971	15.0	-14.1	8833	8850	8813	0.2	-0.4
Seed	5154	5475	6112	6.2	11.6	4233	4102	4629	-3.1	12.9
Fertilisers and Manure										
<i>Fertilisers</i>	3294	4052	3488	23.0	-13.9	878	1102	1194	25.5	8.3
<i>Manure</i>	113	136	145	20.3	6.9	0	0	0	-	-
Total	3407	4187	3633	22.9	-13.2	878	1102	1194	25.5	8.3
<i>Other Inputs</i>										
Insecticides	1359	1376	1496	1.3	8.7	127	98	108	-23.1	10.0
Irrigation charges	2592	2957	3073	14.1	3.9	4289	4227	4299	-1.4	1.7
Crop insurance	29	0	0	-	-	0	0	0	-	-
Interest on working capital	1137	1273	1195	12.0	-6.1	716	696	721	-2.8	3.5
Miscellaneous	82	88	97	7.8	9.7	33	45	45	35.4	1.5
Fixed Cost	15844	16265	17840	2.7	9.7	14377	12855	14821	-10.6	15.3
Rental value of owned land	10950	11913	14047	8.8	17.9	10382	8929	11022	-14.0	23.4
Rent paid for leased-in land	0	0	0	-	-	445	584	95	31.1	-83.7
Land revenue, cesses & taxes	19	21	19	11.7	-11.4	1	0	0	-	-
Depreciation on implements & Farm buildings	636	590	618	-7.2	4.8	467	462	543	-1.1	17.7
Interest on fixed capital	4240	3741	3157	-11.8	-15.6	3081	2880	3161	-6.5	9.7
Yield (Qt/ha)	13.8	15.1	13.6	9.4	-9.9	13.2	11.3	11.5	-14.9	2.1

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare



Annex Table 5.6(c): Gram : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Telangana					Uttar Pradesh				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	27035	29531	31048	9.2	5.1	35527	39109	38665	10.1	-1.1
Human Labour										
<i>Casual</i>	7113	6787	7510	-4.6	10.6	1971	1706	2461	-13.4	44.2
<i>Attached</i>	918	826	1319	-10.0	59.6	8	0	2	-	-
<i>Family</i>	1872	3276	3166	75.0	-3.4	16321	19093	17077	17.0	-10.6
Total	9902	10890	11995	10.0	10.1	18300	20799	19540	13.7	-6.1
Bullock Labour										
<i>Hired</i>	24	0	448	-	-	0	0	0	-	-
<i>Owned</i>	1026	1184	695	15.4	-41.3	0	0	0	-	-
Total	1051	1184	1143	12.7	-3.4	0	0	0	-	-
Machine Labour										
<i>Hired</i>	4396	4210	6443	-4.2	53.0	6517	7104	6648	9.0	-6.4
<i>Owned</i>	1392	1904	144	36.8	-92.5	491	634	1658	29.1	161.6
Total	5788	6115	6587	5.6	7.7	7008	7738	8306	10.4	7.3
Seed	4745	5180	5410	9.2	4.4	5980	6559	6275	9.7	-4.3
Fertilisers and Manure										
<i>Fertilisers</i>	2595	3219	2802	24.0	-13.0	1442	1543	1733	7.0	12.3
<i>Manure</i>	0	0	0	-	-	0	0	0	-	-
Total	2595	3219	2802	24.0	-13.0	1442	1543	1733	7.0	12.3
<i>Other Inputs</i>										
Insecticides	1802	1742	1837	-3.3	5.4	0	0	118	-	-
Irrigation charges	309	322	340	4.3	5.6	2195	1829	2002	-16.7	9.4
Crop insurance	0	0	0	-	-	0	0	0	-	-
Interest on working capital	763	796	845	4.3	6.2	582	607	654	4.2	7.9
Miscellaneous	79	83	89	4.2	8.0	21	34	36	66.8	5.4
Fixed Cost	20386	20336	21649	-0.2	6.5	17191	16383	20787	-4.7	26.9
Rental value of owned land	17714	17725	18781	0.1	6.0	14186	13158	17342	-7.3	31.8
Rent paid for leased-in land	0	0	313	-	-	5	32	0	563.9	-
Land revenue, cesses & taxes	0	0	0	-	-	4	4	8	13.9	78.7
Depreciation on implements & Farm buildings	256	279	282	9.0	0.9	631	600	623	-4.9	3.9
Interest on fixed capital	2417	2332	2274	-3.5	-2.5	2366	2589	2814	9.4	8.7
Yield (Qtl/ha)	11.5	11.0	11.6	-4.4	5.9	9.2	9.3	9.2	1.1	-1.1

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare

Annex Table 5.6 (d) : Lentil (Masur) : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Bihar					Madhya Pradesh				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	25975	26903	27213	3.6	1.2	24526	25325	26133	3.3	3.2
Human Labour										
<i>Casual</i>	7577	7784	7727	2.7	-0.7	5182	5429	5687	4.8	4.7
<i>Attached</i>	49	34	0	-31.4	-	101	0	237	-	-
<i>Family</i>	4374	5146	5419	17.7	5.3	3981	4076	4147	2.4	1.8
Total	12000	12964	13146	8.0	1.4	9264	9505	10071	2.6	6.0
Bullock Labour										
<i>Hired</i>	0	0	0	-	-	139	143	148	3.2	3.1
<i>Owned</i>	0	0	0	-	-	269	275	282	2.2	2.8
Total	0	0	0	-	-	407	418	430	2.6	2.9
Machine Labour										
<i>Hired</i>	5757	5897	6062	2.4	2.8	6296	6451	6622	2.5	2.7
<i>Owned</i>	85	188	195	121.4	4.1	124	130	136	5.1	4.1
Total	5841	6084	6258	4.2	2.8	6420	6581	6758	2.5	2.7
Seed	4092	3972	3994	-2.9	0.5	2791	2945	2979	5.5	1.2
Fertilisers and Manure										
<i>Fertilisers</i>	2470	2656	2609	7.6	-1.8	1287	1364	1416	6.0	3.8
<i>Manure</i>	627	249	237	-60.4	-4.7	0	0	0	-	-
Total	3097	2905	2847	-6.2	-2.0	1287	1364	1416	6.0	3.8
Other Inputs										
Insecticides	204	221	210	8.5	-4.9	962	1007	1038	4.7	3.1
Irrigation charges	77	96	98	25.0	1.8	2597	2680	2593	3.2	-3.2
Crop insurance	0	0	0	-	-	123	126	128	2.2	1.8
Interest on working capital	655	659	660	0.7	0.2	623	644	666	3.4	3.5
Miscellaneous	8	0	0	-	-	52	54	52	3.7	-2.3
Fixed Cost	18275	17173	16333	-6.0	-4.9	13006	13068	14020	0.5	7.3
Rental value of owned land	16199	14969	14031	-7.6	-6.3	11448	11419	12321	-0.3	7.9
Rent paid for leased-in land	0	0	0	-	-	0	0	0	-	-
Land revenue, cesses & taxes	176	180	187	2.5	3.5	3	3	4	2.7	2.9
Depreciation on implements & Farm buildings	302	313	328	3.6	4.7	431	462	472	7.3	2.1
Interest on fixed capital	1598	1710	1788	7.0	4.6	1124	1183	1223	5.3	3.4
Yield (Qt/ha)	9.5	10.3	10.61	7.7	3.4	10.1	10.8	11.2	6.5	4.5

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare

(contd...)



Annex Table 5.6 (d) : Lentil (Masur) : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Uttar Pradesh					West Bengal				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	25185	27312	28734	8.4	5.2	40866	46436	45447	13.6	-2.1
Human Labour										
<i>Casual</i>	5945	5715	5641	-3.9	-1.3	6899	7546	7731	9.4	2.4
<i>Attached</i>	0	0	19	-	-	0	0	0	-	-
<i>Family</i>	6798	9451	8718	39.0	-7.8	15370	15680	14764	2.0	-5.8
Total	12743	15166	14378	19.0	-5.2	22269	23226	22495	4.3	-3.1
Bullock Labour										
<i>Hired</i>	0	0	0	-	-	796	414	0	-48.0	-
<i>Owned</i>	142	0	0	-	-	398	2286	0	474.0	-
Total	142	0	0	-	-	1195	2700	0	126.0	-
Machine Labour										
<i>Hired</i>	2916	2662	4985	-8.7	87.3	7389	6749	7230	-8.7	7.1
<i>Owned</i>	3672	3684	2274	0.3	-38.3	2	57	0	3159.4	-
Total	6588	6346	7259	-3.7	14.4	7390	6806	7230	-7.9	6.2
Seed	3265	3250	3965	-0.5	22.0	4727	8301	8872	75.6	6.9
Fertilisers and Manure										
<i>Fertilisers</i>	987	1068	1434	8.2	34.2	3705	3139	3723	-15.3	18.6
<i>Manure</i>	0	0	0	-	-	109	0	941	-	-
Total	987	1068	1434	8.2	34.2	3814	3139	4664	-17.7	48.6
Other Inputs										
Insecticides	0	0	6	-	-	364	308	319	-15.3	3.6
Irrigation charges	884	918	1065	3.8	16.0	322	1004	938	212.1	-6.6
Crop insurance	0	0	0	-	-	0	0	0	-	-
Interest on working capital	557	541	607	-2.9	12.1	773	932	930	20.6	-0.2
Miscellaneous	20	23	22	15.3	-2.9	13	20	0	60.7	-
Fixed Cost	19741	18355	20909	-7.0	13.9	15535	17335	14446	11.6	-16.7
Rental value of owned land	17705	15750	18082	-11.0	14.8	14548	16384	12148	12.6	-25.9
Rent paid for leased-in land	0	0	0	-	-	101	143	1462	41.4	923.4
Land revenue, cesses & taxes	14	11	11	-20.2	1.7	0	0	0	-	-
Depreciation on implements & Farm buildings	360	394	405	9.6	2.7	488	396	409	-18.8	3.4
Interest on fixed capital	1663	2199	2410	32.3	9.6	398	412	426	3.6	3.4
Yield (Qtl/ha)	9.3	9.0	8.7	-2.9	-3.5	9.2	10.5	11.4	14.4	8.3

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare

Annex Table 5.6(e): Rapeseed & Mustard : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Assam					Bihar				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	29607	30100	29162	1.7	-3.1	27247	27732	28404	1.8	2.4
Human Labour										
<i>Casual</i>	3140	2455	2388	-21.8	-2.7	8249	8945	9138	8.4	2.2
<i>Attached</i>	12	7	0	-39.8	-	46	146	0	216.2	-
<i>Family</i>	12562	13517	13928	7.6	3.0	5905	5155	5385	-12.7	4.5
Total	15713	15979	16316	1.7	2.1	14200	14246	14524	0.3	2.0
Bullock Labour										
<i>Hired</i>	44	43	23	-3.0	-46.3	0	0	0	-	-
<i>Owned</i>	2681	2730	1719	1.8	-37.0	0	0	0	-	-
Total	2725	2773	1742	1.8	-37.2	0	0	0	-	-
Machine Labour										
<i>Hired</i>	7127	6902	7042	-3.2	2.0	4698	4708	4939	0.2	4.9
<i>Owned</i>	281	478	547	69.7	14.5	122	316	322	158.2	2.0
Total	7408	7379	7589	-0.4	2.8	4821	5024	5261	4.2	4.7
Seed	691	723	743	4.7	2.7	1133	1243	1291	9.7	3.8
Fertilisers and Manure										
<i>Fertilisers</i>	1988	1931	1662	-2.9	-13.9	4232	4201	4325	-0.7	2.9
<i>Manure</i>	420	668	523	58.9	-21.8	151	150	155	-0.8	3.1
Total	2409	2599	2185	7.9	-15.9	4383	4351	4480	-0.7	3.0
Other Inputs										
Insecticides	90	86	59	-3.7	-32.0	28	26	27	-6.0	2.5
Irrigation charges	2	2	1	-4.8	-22.2	2032	2158	2124	6.2	-1.6
Crop insurance	0	0	0	-	-	0	0	0	-	-
Interest on working capital	517	503	462	-2.7	-8.1	647	684	698	5.8	2.0
Miscellaneous	53	56	65	5.9	16.4	4	0	0	-	-
Fixed Cost	11711	11200	12994	-4.4	16.0	21002	19069	18819	-9.2	-1.3
Rental value of owned land	10157	9574	10578	-5.7	10.5	18278	16229	15888	-11.2	-2.1
Rent paid for leased-in land	0	0	649	-	-	0	0	36	-	-
Land revenue, cesses & taxes	132	144	145	8.7	0.7	180	189	192	5.0	1.6
Depreciation on implements & Farm buildings	396	391	471	-1.2	20.7	365	379	396	3.8	4.4
Interest on fixed capital	1027	1092	1151	6.3	5.4	2178	2272	2307	4.3	1.5
Yield (Qt/ha)	9.8	9.2	10.1	-5.7	9.0	10.6	9.7	10.3	-8.4	6.0

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare

(contd...)



Annex Table 5.6(e): Rapeseed & Mustard : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Gujarat					Haryana				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	50785	56267	55795	10.8	-0.8	30865	31836	32318	3.1	1.5
Human Labour										
<i>Casual</i>	13987	15970	12863	14.2	-19.5	3700	3762	3840	1.6	2.1
<i>Attached</i>	0	0	0	-	-	3	3	3	0.8	-4.5
<i>Family</i>	10824	11905	13669	10.0	14.8	8135	8414	8277	3.4	-1.6
Total	24811	27875	26532	12.3	-4.8	11838	12179	12120	2.9	-0.5
Bullock Labour										
<i>Hired</i>	30	0	0	-	-	4	0	0	-	-
<i>Owned</i>	4	0	0	-	-	22	25	0	12.9	-
Total	34	0	0	-	-	25	25	0	-3.1	-
Machine Labour										
<i>Hired</i>	4705	6666	7736	41.7	16.1	7470	7679	7886	2.8	2.7
<i>Owned</i>	5169	4144	3740	-19.8	-9.8	2210	2285	2322	3.4	1.6
Total	9873	10810	11476	9.5	6.2	9681	9964	10208	2.9	2.4
Seed	2609	2792	3079	7.0	10.3	1562	1607	1652	2.9	2.8
Fertilisers and Manure										
<i>Fertilisers</i>	3892	4047	4383	4.0	8.3	3573	3763	3940	5.3	4.7
<i>Manure</i>	1533	2021	2000	31.9	-1.1	0	0	0	-	-
Total	5424	6069	6383	11.9	5.2	3573	3763	3940	5.3	4.7
Other Inputs										
Insecticides	484	424	592	-12.4	39.5	305	313	322	2.7	2.7
Irrigation charges	6127	6709	6239	9.5	-7.0	3108	3190	3260	2.6	2.2
Crop insurance	0	0	0	-	-	0	0	0	-	-
Interest on working capital	1211	1344	1277	11.0	-5.0	689	710	729	3.0	2.6
Miscellaneous	211	243	219	15.1	-10.0	84	86	88	1.6	3.2
Fixed Cost	27885	23828	24233	-14.5	1.7	27798	31155	36846	12.1	18.3
Rental value of owned land	22586	18508	19967	-18.1	7.9	22377	25492	30939	13.9	21.4
Rent paid for leased-in land	83	82	0	-1.9	-	48	50	52	2.6	3.7
Land revenue, cesses & taxes	3	4	4	7.0	7.1	0	0	0	-	-
Depreciation on implements & Farm buildings	512	523	421	2.1	-19.4	872	920	945	5.5	2.7
Interest on fixed capital	4701	4712	3841	0.2	-18.5	4501	4693	4911	4.3	4.6
Yield (Qtl/ha)	21.2	20.3	19.9	-4.5	-1.8	18.0	18.1	17.2	0.3	-4.8

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare

(contd...)

Annex Table 5.6(e): Rapeseed & Mustard : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Himachal Pradesh					Jharkhand				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	30784	30416	29157	-1.2	-4.1	34945	35672	35514	2.1	-0.4
Human Labour										
<i>Casual</i>	1173	1038	593	-11.5	-42.9	976	1268	1339	29.9	5.6
<i>Attached</i>	0	0	1250	-	-	0	0	0	-	-
<i>Family</i>	14455	14924	15231	3.2	2.1	18283	18501	18044	1.2	-2.5
Total	15628	15962	17074	2.1	7.0	19259	19769	19384	2.6	-1.9
Bullock Labour										
<i>Hired</i>	141	290	0	105.9	-	0	0	1	-	-
<i>Owned</i>	5796	5286	2830	-8.8	-46.5	4827	5400	5477	11.9	1.4
Total	5936	5576	2830	-6.1	-49.2	4827	5400	5478	11.9	1.4
Machine Labour										
<i>Hired</i>	2629	2194	2804	-16.5	27.8	3316	2853	2937	-14.0	3.0
<i>Owned</i>	1012	1576	1420	55.8	-9.9	10	14	14	44.4	-1.3
Total	3641	3770	4224	3.6	12.0	3326	2867	2951	-13.8	2.9
Seed	898	985	1025	9.7	4.0	1499	1447	1484	-3.5	2.6
Fertilisers and Manure										
<i>Fertilisers</i>	355	224	255	-36.9	13.9	3620	3776	3742	4.3	-0.9
<i>Manure</i>	3019	2774	2652	-8.1	-4.4	188	169	177	-10.2	4.6
Total	3374	2999	2907	-11.1	-3.0	3808	3945	3919	3.6	-0.7
Other Inputs										
Insecticides	0	0	0	-	-	271	174	166	-35.9	-4.7
Irrigation charges	801	641	660	-20.0	2.9	1450	1549	1604	6.8	3.5
Crop insurance	0	0	0	-	-	0	0	0	-	-
Interest on working capital	495	469	422	-5.1	-10.1	505	520	529	3.0	1.7
Miscellaneous	12	14	15	15.3	10.7	0	0	0	-	-
Fixed Cost	12909	12306	12787	-4.7	3.9	13063	11894	10268	-9.0	-13.7
Rental value of owned land	9149	8222	8380	-10.1	1.9	11616	10396	8822	-10.5	-15.1
Rent paid for leased-in land	29	28	0	-2.2	-	0	0	0	-	-
Land revenue, cesses & taxes	10	11	12	14.0	4.7	57	61	58	6.2	-4.1
Depreciation on implements & Farm buildings	569	585	672	2.9	14.9	259	275	272	6.2	-1.3
Interest on fixed capital	3153	3459	3722	9.7	7.6	1131	1162	1115	2.7	-4.0
Yield (Qt/ha)	5.0	4.8	4.9	-4.2	1.0	6.5	6.9	7.3	5.3	6.3

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare

(contd...)



Annex Table 5.6(e): Rapeseed & Mustard : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Madhya Pradesh					Punjab				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	29869	31263	31564	4.7	1.0	32654	33234	34313	1.8	3.2
Human Labour										
<i>Casual</i>	5955	6275	6302	5.4	0.4	3860	4244	4194	9.9	-1.2
<i>Attached</i>	27	29	0	5.8	-	3382	3253	3377	-3.8	3.8
<i>Family</i>	5752	6008	6156	4.5	2.5	8906	8790	9160	-1.3	4.2
Total	11734	12312	12458	4.9	1.2	16148	16287	16732	0.9	2.7
Bullock Labour										
<i>Hired</i>	0	0	0	-	-	8	8	0	2.4	-
<i>Owned</i>	14	15	15	2.6	3.1	18	18	24	0.6	35.2
Total	14	15	15	2.6	3.1	26	26	24	1.1	-5.5
Machine Labour										
<i>Hired</i>	8915	9374	9215	5.1	-1.7	1611	1715	1783	6.5	3.9
<i>Owned</i>	82	93	97	13.8	3.7	7991	8186	8585	2.4	4.9
Total	8997	9467	9312	5.2	-1.6	9602	9902	10367	3.1	4.7
Seed	3039	3141	3223	3.4	2.6	1506	1480	1521	-1.7	2.8
Fertilisers and Manure										
<i>Fertilisers</i>	2855	3058	3183	7.1	4.1	2884	3075	3226	6.6	4.9
<i>Manure</i>	17	0	0	-	-	0	0	0	-	-
Total	2872	3058	3183	6.5	4.1	2884	3075	3226	6.6	4.9
Other Inputs										
Insecticides	221	232	240	4.6	3.5	943	905	887	-4.1	-2.0
Irrigation charges	2059	2068	2153	0.5	4.1	592	590	564	-0.4	-4.4
Crop insurance	103	106	111	3.1	4.4	0	0	0	-	-
Interest on working capital	731	765	770	4.7	0.6	720	741	762	2.9	2.9
Miscellaneous	101	99	101	-1.8	2.0	234	230	230	-1.8	0.2
Fixed Cost	22057	21301	24424	-3.4	14.7	34776	35562	32762	2.3	-7.9
Rental value of owned land	18145	17189	20233	-5.3	17.7	26039	26501	23422	1.8	-11.6
Rent paid for leased-in land	0	0	0	-	-	4026	4099	4274	1.8	4.3
Land revenue, cesses & taxes	8	8	8	3.8	-2.8	0	0	0	-	-
Depreciation on implements & Farm buildings	637	668	696	4.9	4.1	786	839	854	6.8	1.8
Interest on fixed capital	3267	3436	3488	5.2	1.5	3925	4123	4212	5.0	2.2
Yield (Qtl/ha)	15.2	17.0	16.3	11.7	-4.1	15.0	15.2	15.6	1.3	2.5

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare

(contd...)

Annex Table 5.6(e): Rapeseed & Mustard : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Rajasthan					Uttar Pradesh				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	41660	45969	47184	10.3	2.6	36238	36626	36707	1.1	0.2
Human Labour										
<i>Casual</i>	4859	4747	5174	-2.3	9.0	6037	5407	5779	-10.4	6.9
<i>Attached</i>	102	158	0	54.2	-	29	1	14	-95.9	1124.1
<i>Family</i>	14817	16991	17346	14.7	2.1	9189	9483	8692	3.2	-8.3
Total	19778	21896	22520	10.7	2.8	15255	14891	14485	-2.4	-2.7
Bullock Labour										
<i>Hired</i>	0	0	0	-	-	0	0	0	-	-
<i>Owned</i>	63	72	66	14.4	-8.5	118	106	0	-10.0	-
Total	63	72	66	14.4	-8.5	118	106	0	-10.0	-
Machine Labour										
<i>Hired</i>	7968	7601	8470	-4.6	11.4	8950	8175	8694	-8.7	6.4
<i>Owned</i>	2324	2877	2715	23.8	-5.6	1023	2337	1554	128.6	-33.5
Total	10291	10477	11185	1.8	6.8	9973	10512	10248	5.4	-2.5
Seed	2485	2964	3037	19.3	2.5	2092	2996	2834	43.2	-5.4
Fertilisers and Manure										
<i>Fertilisers</i>	2672	2815	2589	5.4	-8.0	4533	4580	4927	1.0	7.6
<i>Manure</i>	22	31	33	39.6	7.9	0	0	57	-	-
Total	2694	2846	2622	5.6	-7.9	4533	4580	4984	1.0	8.8
Other Inputs										
Insecticides	115	158	170	37.5	7.2	2	2	3	18.3	25.1
Irrigation charges	5372	6633	6634	23.5	0.0	3402	2668	3254	-21.6	22.0
Crop insurance	0	0	0	-	-	0	0	0	-	-
Interest on working capital	813	878	904	8.0	3.0	820	822	849	0.3	3.2
Miscellaneous	48	44	45	-8.5	2.8	44	47	49	6.6	4.0
Fixed Cost	22385	17099	17935	-23.6	4.9	29715	30133	29016	1.4	-3.7
Rental value of owned land	18137	12700	12730	-30.0	0.2	24371	24459	23037	0.4	-5.8
Rent paid for leased-in land	391	474	852	21.0	79.8	1576	940	997	-40.4	6.1
Land revenue, cesses & taxes	0	0	0	-	-	9	10	7	20.3	-33.9
Depreciation on implements & Farm buildings	484	540	567	11.4	5.2	703	755	834	7.4	10.4
Interest on fixed capital	3372	3386	3787	0.4	11.8	3057	3969	4141	29.8	4.3
Yield (Qt/ha)	16.5	12.7	14.6	-23.0	14.7	14.2	14.3	13.3	0.9	-6.8

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare

(contd...)



Annex Table 5.6(e): Rapeseed & Mustard : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	West Bengal				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	43051	49082	50280	14.0	2.4
Human Labour					
<i>Casual</i>	10163	7413	7554	-27.1	1.9
<i>Attached</i>	0	0	0	-	-
<i>Family</i>	12617	16488	16176	30.7	-1.9
Total	22780	23902	23730	4.9	-0.7
Bullock Labour					
<i>Hired</i>	160	69	31	-56.9	-54.8
<i>Owned</i>	140	64	111	-54.2	73.1
Total	300	133	142	-55.6	7.0
Machine Labour					
<i>Hired</i>	8001	9252	10233	15.6	10.6
<i>Owned</i>	47	43	138	-8.2	217.6
Total	8048	9295	10370	15.5	11.6
Seed	1120	1287	1389	14.9	8.0
Fertilisers and Manure					
<i>Fertilisers</i>	5584	8023	8868	43.7	10.5
<i>Manure</i>	603	1124	793	86.3	-29.4
Total	6187	9147	9661	47.8	5.6
<i>Other Inputs</i>					
Insecticides	1326	772	832	-41.8	7.7
Irrigation charges	2335	3511	3071	50.4	-12.5
Crop insurance	0	0	0	-	-
Interest on working capital	922	988	1033	7.1	4.6
Miscellaneous	32	48	51	50.8	5.5
Fixed Cost	18346	18622	18738	1.5	0.6
Rental value of owned land	17560	17883	16682	1.8	-6.7
Rent paid for leased-in land	12	0	1290	-	-
Land revenue, cesses & taxes	0	0	0	-	-
Depreciation on implements & Farm buildings	333	346	368	3.8	6.3
Interest on fixed capital	441	393	397	-10.9	1.1
Yield (Qtl/ha)	11.1	12.8	14.1	16.0	9.9

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare

Annex Table 5.6 (f) : Safflower : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Karnataka				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	19127	20014	20062	4.6	0.2
Human Labour					
<i>Casual</i>	1446	4960	4507	243.1	-9.2
<i>Attached</i>	0	0	67	-	-
<i>Family</i>	6281	3460	4121	-44.9	19.1
Total	7727	8421	8695	9.0	3.3
Bullock Labour					
<i>Hired</i>	127	448	521	252.2	16.1
<i>Owned</i>	1697	1232	965	-27.4	-21.6
Total	1824	1680	1486	-7.9	-11.5
Machine Labour					
<i>Hired</i>	2747	3345	3624	21.8	8.4
<i>Owned</i>	1228	1334	578	8.6	-56.7
Total	3975	4679	4202	17.7	-10.2
Seed	505	579	644	14.7	11.2
Fertilisers and Manure					
<i>Fertilisers</i>	2974	2296	2593	-22.8	12.9
<i>Manure</i>	0	0	0	-	-
Total	2974	2296	2593	-22.8	12.9
<i>Other Inputs</i>					
Insecticides	1558	1647	1697	5.7	3.0
Irrigation charges	76	97	141	27.4	45.4
Crop insurance	0	0	0	-	-
Interest on working capital	389	502	483	28.9	-3.7
Miscellaneous	98	114	123	15.8	7.6
Fixed Cost	6036	6106	7131	1.2	16.8
Rental value of owned land	4167	4314	2870	3.5	-33.5
Rent paid for leased-in land	0	0	2574	-	-
Land revenue, cesses & taxes	5	5	6	2.3	10.2
Depreciation on implements & Farm buildings	173	166	182	-3.7	9.5
Interest on fixed capital	1691	1621	1499	-4.1	-7.5
Yield (Qtl/ha)	3.6	3.8	4.4	5.3	18.1

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare



Annex Table 5.7: All-India Projected Cost of Production(CoP) of *Rabi* Crops for Crop Season 2024-25 and 2025-26

Crop	Cost of Production (₹/qtl)						Percentage Change (2025-26 over 2024-25)		
	2024-25			2025-26					
	A_2	A_2+FL	C_2	A_2	A_2+FL	C_2	A_2	A_2+FL	C_2
Wheat	947	1182	1720	994	1239	1804	5.0	4.8	4.9
Barley	816	1239	1737	912	1361	1862	11.8	9.8	7.2
Gram	2747	3527	4662	2886	3699	4875	5.1	4.9	4.6
Lentil	2649	3537	5061	2825	3705	5243	6.6	4.7	3.6
Rapeseed & Mustard	2066	3011	4294	2219	3210	4558	7.4	6.6	6.1
Safflower	3052	3960	5550	3376	4360	5738	10.6	10.1	3.4

Source: CACP Calculations using CS data

Annex Table 5.8: Relative Average Gross Returns of *Rabi* Crops with respect to Wheat at All-India Level for 2021-22, 2022-23 and 2023-24

Year/ Crop	Actual Average Gross Returns Over Actual CoC (₹/ha)		Relative Average Gross Returns Over Actual CoC (%)	
	A_2	A_2+FL	A_2	A_2+FL
Wheat				
2021-22	54861	46023	100.0	100.0
2022-23	58050	48800	100.0	100.0
2023-24	66262	56544	100.0	100.0
TE2023-24	59724	50456	100.0	100.0
Barley				
2021-22	72655	56167	132.4	122.0
2022-23	66961	48086	115.3	98.5
2023-24	63819	44336	96.3	78.4
TE2023-24	67812	49530	113.5	98.2
Gram				
2021-22	34135	25837	62.2	56.1
2022-23	33614	24485	57.9	50.2
2023-24	40315	31254	60.8	55.3
TE2023-24	36021	27192	60.3	53.9
Lentil				
2021-22	32798	26335	59.8	57.2
2022-23	35587	28230	61.3	57.8
2023-24	35713	28741	53.9	50.8
TE2023-24	34700	27769	58.1	55.0
Rapeseed & Mustard				
2021-22	71380	59490	130.1	129.3
2022-23	54430	40865	93.8	83.7
2023-24	53398	40199	80.6	71.1
TE2023-24	59736	46851	100.0	92.9
Safflower				
2021-22	3646	-2636	6.6	-5.7
2022-23	1545	-1916	2.7	-3.9
2023-24	17	-4104	0.0	-7.3
TE2023-24	1736	-2885	2.9	-5.7

Note: All-India gross returns of a crop are weighted average of respective gross returns of projected States

Source: CACP Calculations using CS data



Annex Table 5.9: Relative Average Gross Returns of Rabi Crops with respect to Wheat in Selected States during TE2023-24

Crop/ State	Gross Returns Over (₹/ha)		Relative Gross Returns Over (%)	
	A_2	A_2+FL	A_2	A_2+FL
Assam				
Wheat*	24928	13047	100.0	100.0
Rapeseed & Mustard	28942	15606	116.1	119.6
Bihar				
Wheat	43097	34910	100.0	100.0
Gram	59618	51154	138.3	146.5
Lentil	34860	29880	80.9	85.6
Rapeseed & Mustard	36428	30947	84.5	88.6
Chhattisgarh				
Wheat	29114	22734	100.0	100.0
Gram	40058	35499	137.6	156.1
Gujarat				
Wheat	56218	48873	100.0	100.0
Gram	40876	34557	72.7	70.7
Rapeseed & Mustard	79676	67543	141.7	138.2
Haryana				
Wheat	77175	70518	100.0	100.0
Gram	30737	19918	39.8	28.2
Rapeseed & Mustard	87288	79013	113.1	112.0
Himachal Pradesh				
Wheat	40719	25084	100.0	100.0
Barley	6808	-15364	16.7	-61.2
Rapeseed & Mustard	18748	3878	46.0	15.5
Jharkhand				
Wheat	37584	25249	100.0	100.0
Gram	6451	-9953	17.2	-39.4
Rapeseed & Mustard	21916	3640	58.3	14.4
Madhya Pradesh				
Wheat	57490	50373	100.0	100.0
Gram	37310	31897	64.9	63.3
Lentil	33035	28967	57.5	57.5
Rapeseed & Mustard	72116	66144	125.4	131.3

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Annex Table 5.9: Relative Average Gross Returns of *Rabi* Crops with respect to Wheat in Selected States during TE2023-24

Crop/ State	Gross Returns Over (₹/ha)		Relative Gross Returns Over (%)	
	A_2	A_2+FL	A_2	A_2+FL
Maharashtra				
Wheat	29175	18217	100.0	100.0
Gram	33520	26623	114.9	146.1
Punjab				
Wheat	71976	68896	100.0	100.0
Rapeseed & Mustard	58764	49812	81.6	72.3
Rajasthan				
Wheat	74962	52134	100.0	100.0
Barley	77791	55326	103.8	106.1
Gram	39857	24273	53.2	46.6
Rapeseed & Mustard	61097	44712	81.5	85.8
Uttar Pradesh				
Wheat	58661	48731	100.0	100.0
Barley	60024	48521	102.3	99.6
Gram	36463	18966	62.2	38.9
Lentil	37100	28778	63.2	59.1
Rapeseed & Mustard	56447	47326	96.2	97.1
West Bengal				
Wheat	24928	13047	100.0	100.0
Lentil	32657	17386	131.0	133.3
Rapeseed & Mustard	37577	22484	150.7	172.3

*Due to non-availability of CS data for wheat of Assam, CS data of West Bengal for Assam have been taken

Note : All-India CoC, GVO and gross returns of a crop are weighted average of respective CoCs, GVOs and gross returns of projected States

Source: CACP Calculations using CS data



Annex Table 5.10: Comparison of CACP and State Projected Cost of Production (C_2), and Yield and suggested Minimum Support Price by State for Rabi Crops for RMS2026-27

State/ Crop	CACP Projections on the basis of CS data		State Projections		Suggested Minimum Support Price by State (₹/qtl)
	Yield (qtl/ha)	Cost of Production (₹/qtl)	Yield (qtl/ha)	Cost of Production (₹/qtl)	
Wheat					
Bihar	29.68	1894	Cost estimates are not provided		No Suggestion
Chhattisgarh	25.89	1939	16.80	1961	2575
Gujarat	34.11	1936	34.00	2816	4200
Haryana	45.99	1689	47.21	2162	2652
Himachal Pradesh	20.91	2451	CoP not provided		No Suggestion
Jharkhand	25.98	2096	CoP not provided		3253
Madhya Pradesh	39.08	1550	47.42	1494	2600
Maharashtra	29.12	3091	CoP not provided		4525
Punjab	47.43	1565	46.99	2112	3168
Rajasthan	40.20	1944	Cost estimates are not provided		No Suggestion
Uttar Pradesh	36.80	1944	36.90	1731	No Suggestion
West Bengal	33.06	2337	30.29	2807	3450
Barley					
Himachal Pradesh	16.97	3027	CoP not provided		No Suggestion
Rajasthan	38.06	1802	Cost estimates are not provided		No Suggestion
Uttar Pradesh	34.64	1922	30.25	1554	No Suggestion
Gram					
Andhra Pradesh	16.26	5106	15.00	6185	9277
Bihar	13.83	4199	Cost estimates are not provided		No Suggestion
Chhattisgarh	14.13	4358	10.70	3187	5870
Gujarat	14.79	4300	19.56	4376	7400
Haryana	8.83	5534	11.11	6279	7632
Jharkhand	11.75	5149	CoP not provided		5485
Karnataka	9.42	5355	Cost estimates are not provided		No Suggestion
Madhya Pradesh	12.46	4520	15.60	4112	5850
Maharashtra	14.15	5191	CoP not provided		7372
Rajasthan	12.01	4673	Cost estimates are not provided		No Suggestion
Telangana	11.37	5091	Cost estimates are not provided		No Suggestion
Uttar Pradesh	9.24	6037	12.90	3667	No Suggestion
Lentil					
Bihar	10.13	4658	Cost estimates are not provided		No Suggestion
Madhya Pradesh	10.70	4417	13.02	4109	6900
Uttar Pradesh	8.97	6305	10.35	4107	No Suggestion
West Bengal	11.19	5688	Cost estimates are not provided		No Suggestion

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Annex Table 5.10: Comparison of CACP and State Projected Cost of Production (C_p), and Yield and suggested Minimum Support Price by State for *Rabi Crops* for RMS2026-27

State/Crop	CACP Projections on the basis of CS data		State Projections		Suggested Minimum Support Price by State (₹/qtl)
	Yield (qtl/ha)	Cost of Production (₹/qtl)	Yield (qtl/ha)	Cost of Production (₹/qtl)	
Rapeseed & Mustard					
Assam	9.69	5543	14.00	2653	6545
Bihar	10.22	4756	Cost estimates are not provided		No Suggestion
Gujarat	20.46	4321	19.94	4445	7600
Haryana	17.79	4113	18.09	4356	5305
Himachal Pradesh	8.09	6671	CoP not provided		No Suggestion
Jharkhand	8.18	5867	CoP not provided		6179
Madhya Pradesh	16.13	3438	20.52	3534	6200
Punjab	15.26	4844	Cost estimates are not provided		7064
Rajasthan	15.33	4521	Cost estimates are not provided		No Suggestion
Uttar Pradesh	13.93	5264	13.08	3636	No Suggestion
West Bengal	13.23	5559	12.42	6180	7450
Safflower					
Karnataka	5.75	5738	Cost estimates are not provided		No Suggestion

Source: State Governments and CACP Calculations



Annex Table

Annex Table 5.11: Crop-wise Projection of State having small Area or Production or Sample of Rabi Crops for Marketing Season 2026-27

Crop	State	Reasons for Inclusion in Projection	Action and Recommendation
Wheat	Chhattisgarh	Although State area and production shares of wheat at all-India level during TE2024-25 are just 0.44% and 0.18% only, respectively, yet State area and production shares of wheat in State's rabi cereals are more than adequate. Sample size during TE2023-24 is 38 (1.1%).	Projected Projection is made as State having reasonable area and production shares of wheat in State's rabi cereals.
	Himachal Pradesh	Although State area and production shares of wheat at all-India during TE2024-25 are 0.99% and 0.61% only, respectively, yet State area and production shares of wheat in State cereals is significant. Sample size during TE2023-24 is 318 (9.2%).	Projected Projection is made as State having reasonable area and production shares of wheat in State's cereals.
	Jharkhand	Although State area and production shares of wheat at all-India during TE2024-25 are 0.71% and 0.43%, respectively, yet State area and production shares of wheat in State cereals are reasonable. Sample size during TE2023-24 is 48 (1.4%).	Projected Projection is made as State having reasonable area and production shares of wheat in State's cereals.
	West Bengal	Although State area and production shares of wheat at all-India during TE2024-25 are 0.57% and 0.50%, respectively, yet State area and production shares of wheat in State's rabi cereals is adequate. Sample size during TE2023-24 is 46 (1.3%).	Projected Projection is made as State having adequate area and production shares of wheat in State's rabi cereals.
Gram	Bihar	Although State area and production shares of gram at all-India level during TE2024-25 are 0.66% and 0.58%, yet State area and production shares of gram in State pulses are reasonable. Sample size under CS during TE2023-24 is 32 (4.1%).	Projected Projection is made as State having reasonable area and production shares of gram in State's pulses.
	Haryana	Although State area and production shares of gram at all-India level during TE2024-25 are low, yet State area and production shares of gram in State pulses are reasonable. Sample size under CS during TE2023-24 is 16 (2.1%).	Projected Projection is made as State having reasonable area and production shares of gram in State's pulses.
	Jharkhand	State area and production shares of gram at all-India as well as in State pulses during TE2024-25 are reasonably adequate. Sample size under CS during TE2023-24 is 20 (2.5%).	Projected Projection is made as State having reasonable area and production shares of gram in State's as well as in all-India pulses.

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Annex Table 5.11: Crop-wise Projection of State having small Area or Production or Sample of Rabi Crops for Marketing Season 2026-27

Crop	State	Reasons for Inclusion in Projection	Action and Recommendation
Lentil	West Bengal	Area and production shares of lentil at all-India level during TE2024-25 are 9.30% and 8.58% respectively, and State area and production shares of lentil in State pulses are reasonably adequate. Sample size under CS during TE2023-24 is 48 (21.24%).	Projected Projection is made as State having adequate area and production shares of lentil at all-india as well as in State's pulses.
Rapeseed & Mustard	Himachal Pradesh	Although area and production shares of rapeseed & mustard at all-India level during TE2024-25 are low, yet State area and production shares of rapeseed & mustard in State oilseeds are adequate. CS data are available for TE2023-24 with average Sample size 77 (4.7%).	Projected Projection is made as State having adequate area and production shares of rapeseed & mustard in State's oilseeds.
	Jharkhand	State area and production shares of rapeseed & mustard at all-India as well as in State oilseeds during TE2024-25 are reasonably adequate. CS data are available for TE2023-24 with average sample size of 32 (1.98%).	Projected Projection is made as State having adequate area and production shares of rapeseed & mustard at all-india as well as in State's oilseeds.
	Punjab	Although State area and production shares of rapeseed & mustard all-India at all-India level during TE2024-25 are 0.5% and 0.63% respectively, yet State area and production shares of rapeseed & mustard in State oilseeds are reasonably adequate. Sample size under CS during TE2023-24 is 67 (4.13%).	Projected Projection is made as State having reasonably adequate area and production shares of rapeseed & mustard in State's oilseeds.
	West Bengal	Area and production shares of R&M at all-India level and State area and production shares of R&M in State oilseeds are reasonably adequate. Sample size under CS during TE2023-24 is 215 (13.21%).	Projected Projection is made as State having reasonably adequate area and production shares of rapeseed & mustard in State's oilseeds.
Safflower	Karnataka	State area and production shares of safflower at all-India level are reasonably adequate. CS data is available for TE2023-24 with average sample size of 19 (65.12%).	Projected However, Sample size should be increased.

*Note: Percentage area and production shares presented in Table have been derived on average area and production during TE2024-25
Source: ES&E Division, Ministry of Agriculture & Farmers Welfare, and CACP Calculations*

Annex Table





Annex Table

Annex Table 5.12: Crop-wise States not-included in Projection of Rabi Crops for Marketing Season 2026-27

Crop	State	Reasons for not-inclusion in Projection	Action and Recommendation
Wheat	Karnataka	State area and production shares of wheat at all-India during TE2024-25 are 0.47% and 0.17%, respectively, Also State area and production shares of wheat in State's rabi cereals is inadequate. Sample size during TE2023-24 is 46(1.3%).	Not Projected
Gram	Tamil Nadu	State area and production share of gram at all-India level during TE2024-25 are very low, Also area and production shares in State pulses are low with inadequate sample size.	Not Projected
	West Bengal	State area and production shares of gram at all-India level during TE2024-25 are 0.39% and 0.42%, respectively, but State area and production shares of gram in State pulses are reasonably adequate. However data is available for only 2023-24 during TE2023-24.	Not Projected CS data should be collected every year.
Lentil	Jharkhand	State area and production shares of lentil at all-India level during TE2024-25 are 3.84% and 3.28%, respectively. However data is available for only 2023-24 during TE2023-24.	Not Projected CS data should be collected every year.
Rapeseed & Mustard	Chhattisgarh	Although State area and production shares of rapeseed & mustard at all-India level during TE2023-24 are 0.33% and 0.11% respectively, yet State area and production shares of rapeseed & mustard in State oilseeds are reasonably adequate. But CS data are available for 2022-23 and 2023-24 only during TE2023-24, that too, with thin Sample in 2022-23.	Not Projected CS data should be collected every year with increased sample.
	Odisha	State area and production share of R&M at all-India level during TE2024-25 are very low, Also area and production shares in State oilseeds are low.	Not Projected
Safflower	Maharashtra	State area and production shares of safflower at all-India level are reasonably adequate. But CS data is only available for 2023-24 during TE2023-24.	Not Projected CS data should be collected every year with increased sample.

Note: Percentage area and production shares presented in Table have been derived on average area and production during TE2024-25

Source: ES&E Division, Ministry of Agriculture & Farmers Welfare, and CACP Calculations

Annex Table 5.13: Crop-wise Inclusion of States under Comprehensive Scheme

Crop	State	Reasons for inclusion under Comprehensive Scheme	Action and Recommendation
Wheat	Jammu & Kashmir	Although area and production shares of wheat at national level are 0.89% and 0.52%, respectively, yet State area and production shares of wheat in State cereals are very reasonable.	CS data should be collected for the State.
	Uttarakhand	Although State area and production shares of wheat at all-India during TE2024-25 are 0.84% and 0.75%, respectively, yet quantum of State area and production is reasonably adequate.	CS data should be collected for the State.
Barley	Bihar	Although State area and production of barley in State cereals during TE 2024-25 are very low, yet State area and production of barley at all-India level are 1.36% and 0.78%, respectively, which are reasonable adequate.	CS data should be collected for the State.
	Haryana	Although State area and production shares of barley in State cereals during TE2024-25 are very low, yet area and production shares of barley at all-India are 2.07% and 2.52% respectively.	CS data should be collected for the State.
	Madhya Pradesh	Although State area and production shares of barley in State cereals during TE2024-25 are very low, yet area and production shares of barley at all-India are 3.75% and 2.25% respectively.	CS data should be collected for the State.
	Uttarakhand	Although State area and production shares of barley in State cereals during TE2024-25 are small, yet State area and production shares of barley at all-India are 2.94% and 1.35%, which are reasonable. But, there is no collection of CS data.	CS data should be collected for the State.
Lentil	Assam	State area and production shares of Lentil at all-India level during TE2024-25 are 1.55% and 1.23%, respectively, and State area and production shares of Lentil in State pulses are reasonably adequate. But, there is no collection of CS data.	CS data should be collected for the State.
	Odisha	Although State area and production shares of lentil in State pulses during TE2024-25 are small, yet State area and production shares of lentil at all-India level are 0.95% and 0.50%, respectively. The quantum of area and production are reasonable.	CS data should be collected for the State.

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Annex Table



Annex Table

Annex Table 5.13: Crop-wise Inclusion of States under Comprehensive Scheme

Crop	State	Reasons for inclusion under Comprehensive Scheme	Action and Recommendation
Rapeseed & Mustard	Rajasthan	Although State area and production shares in State pulses during TE2024-25 are small, yet State area and production shares of lentil at all-India are reasonable. But, there is no CS data during TE2023-24.	CS data should be collected for the State.
	Uttarakhand	Area and production shares of lentil at all-India level during TE2024-25 are 0.55% and 0.46% respectively, and State area and production shares of lentil in State pulses are reasonably adequate. But, there are no collection of CS data.	CS data should be collected for the State.
	Jammu & Kashmir	Although area and production shares of rapeseed & mustard at all-India level during TE2024-25 are 0.46% and 0.28% respectively, yet State area and production shares of rapeseed & mustard in State oilseeds are reasonably adequate. But, there is no CS under operation within State.	CS data should be collected for the State.
	Tripura	Although State area and production shares of rapeseed & mustard at all-India level during TE2024-25 are 0.08% and 0.04%, respectively, yet State area and production shares of rapeseed & mustard in State oilseeds are reasonably adequate. But, CS is not under operation within State.	CS data should be collected for the State.
Safflower	Uttarakhand	Although State area and production shares of rapeseed & mustard all-India at all-India level during TE2024-25 are 0.23% and 0.15%, respectively, yet State area and production shares of rapeseed & mustard in State oilseeds are reasonably adequate. But, CS data are not collected.	CS data should be collected for the State.
	Andhra Pradesh	State area and production shares of safflower at all-India during TE2024-25 are 4.67% and 3.20%, respectively. But CS data are not available during TE2023-24.	CS data should be collected for the State.
	Telangana	State area and production shares of safflower at all-India level are reasonably adequate. But CS data are not available for TE2023-24.	CS data should be collected for the State.

Note 1: Union Territory of Jammu & Kashmir also Includes Union Territory of Ladakh

2: Percentage area and production shares presented in Table have been derived on average area and production during TE2024-25

Source: ES&E Division, Ministry of Agriculture & Farmers Welfare, and CACP Calculations

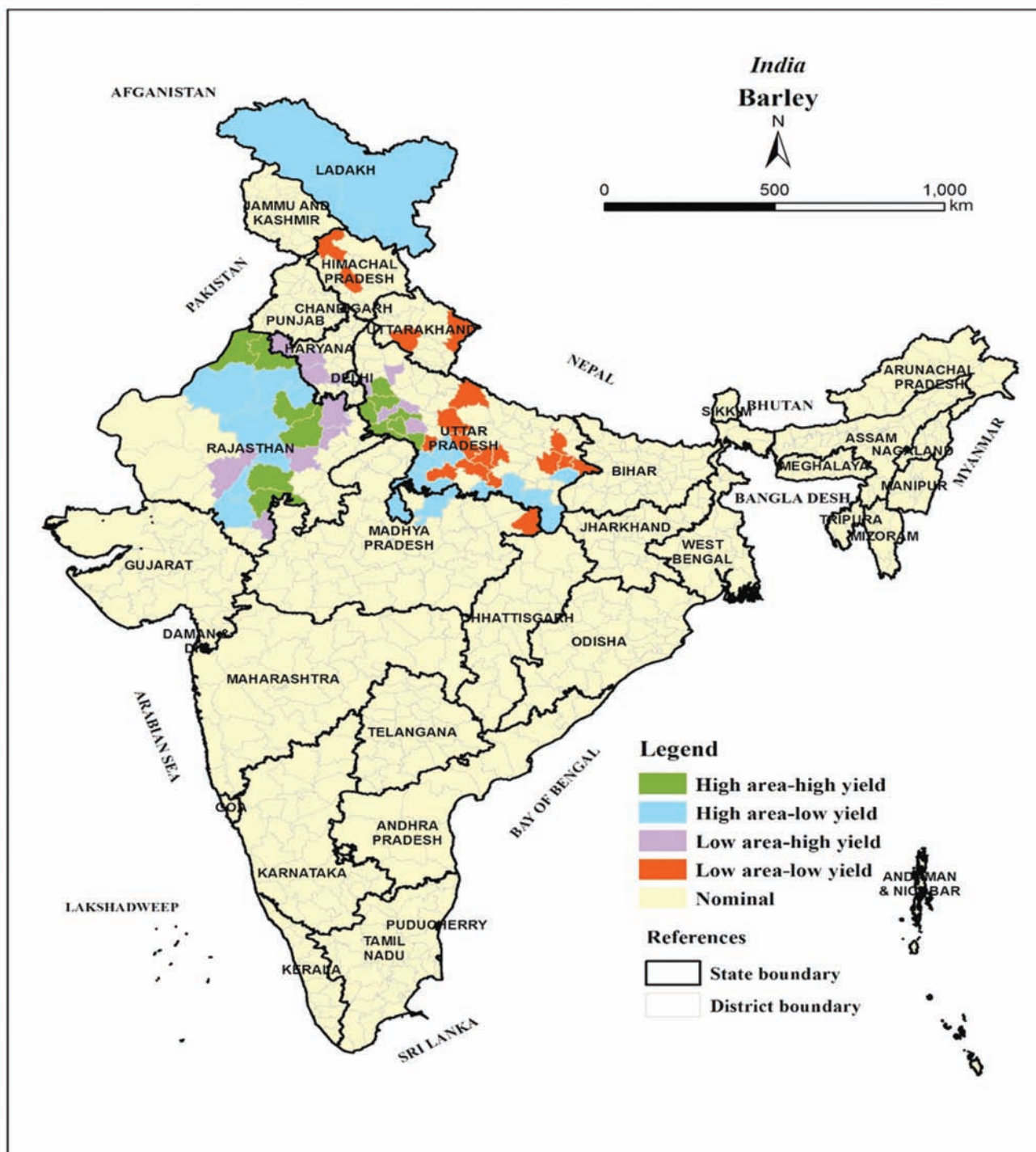


Annex Maps

Annex Table



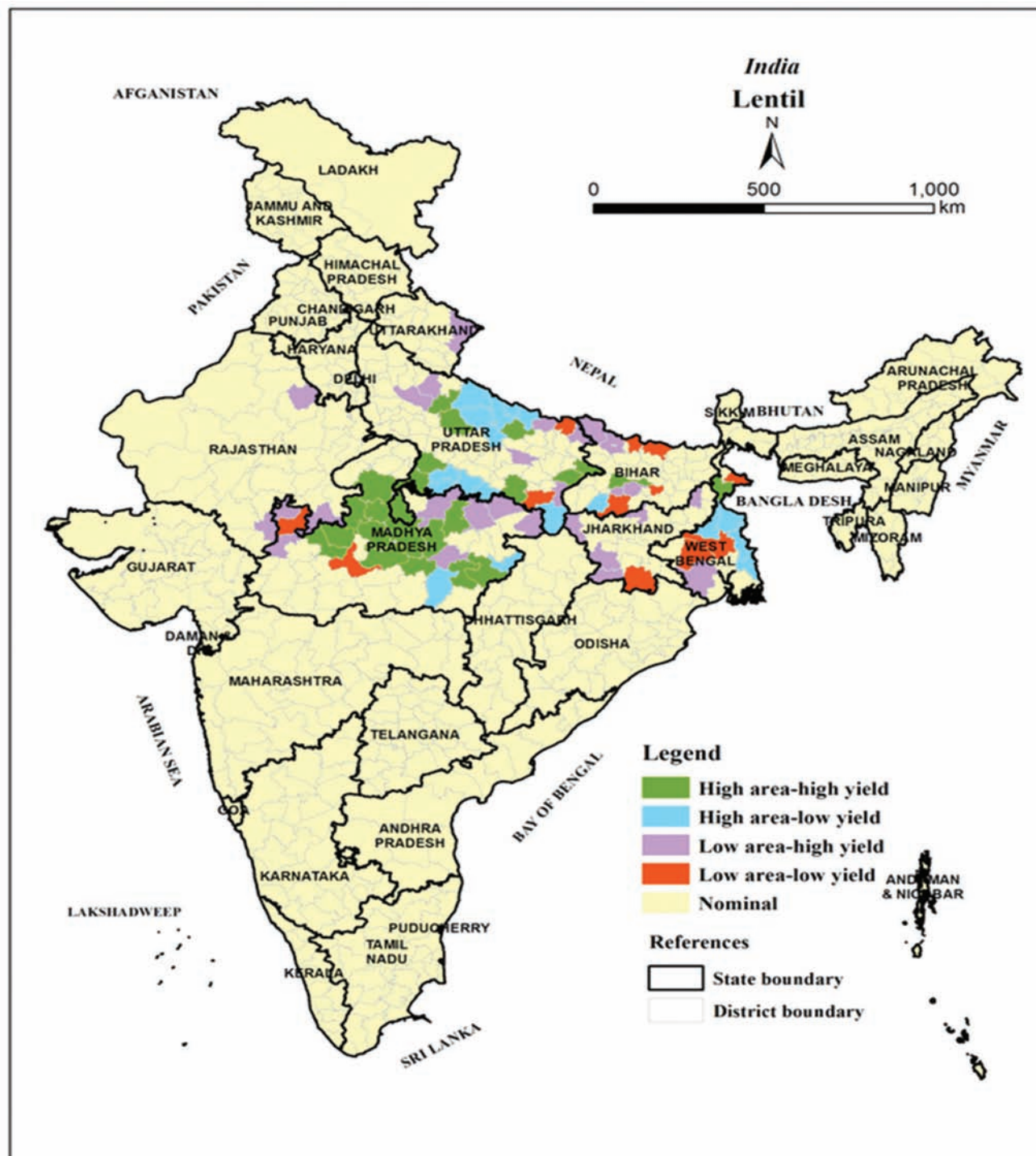
Annex Map 3.1: Classification of Major Barley Producing Districts based on Area Share and Yield



Note: 1. Districts contributing 90% production of respective crops considered for the analysis
 2. All-India Yield and Median of Area Shares of districts in all-India area used as benchmark for categorization
 3. Count of districts under different categories may slightly deviate from Table 3.6 due to formation of some new districts in recent years

Source: 1. Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare
 2. Map drawn by ICAR- National Bureau of Soil Survey and Land Use Planning, Nagpur

Annex Map 3.2: Classification of Major Lentil Producing Districts based on Area Share and Yield



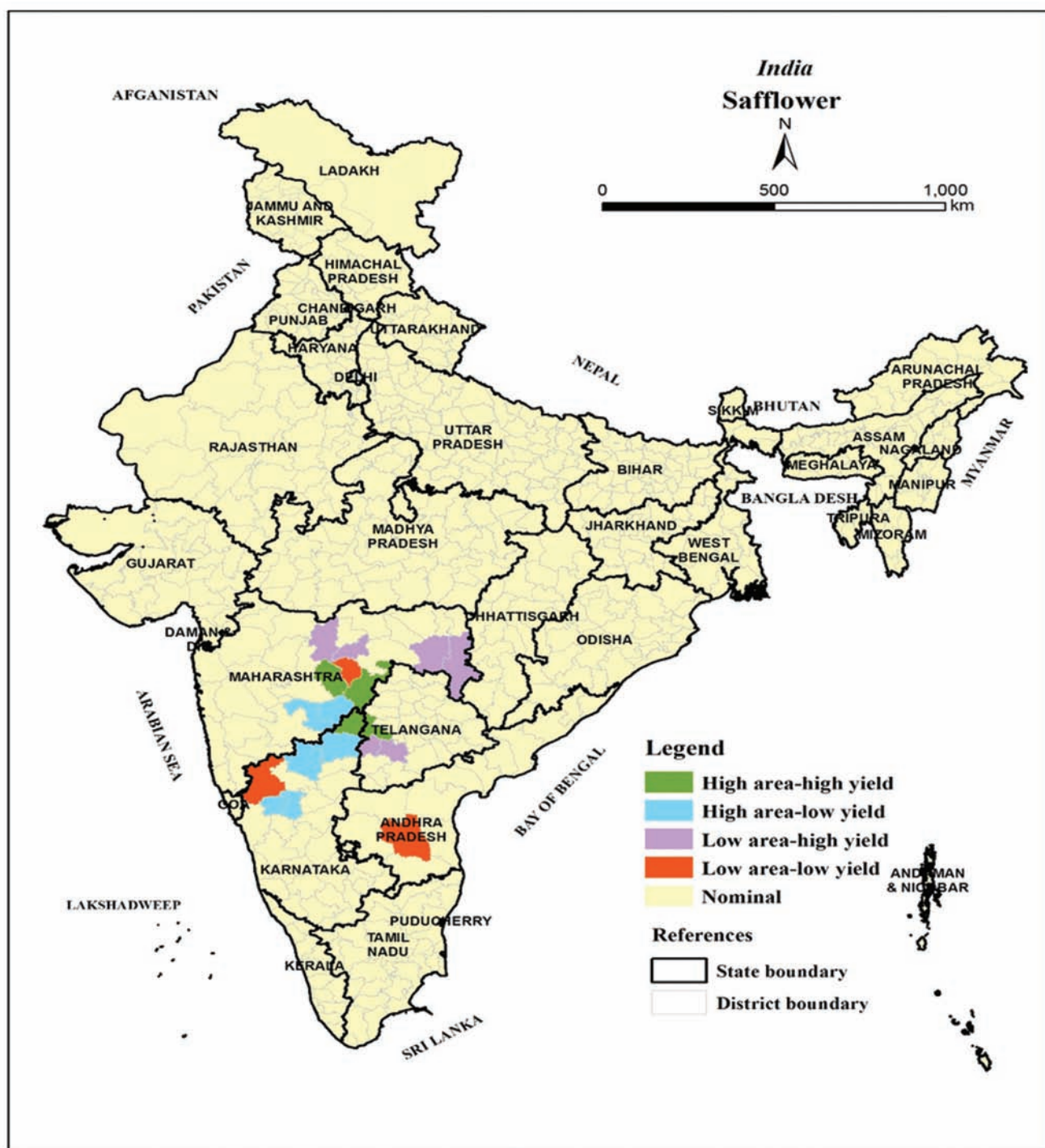
Note: 1. Districts contributing 90% production of respective crops considered for the analysis
2. All-India Yield and Median of Area Shares of districts in all-India area used as benchmark for categorization
3. Count of districts under different categories may slightly deviate from Table 3.7 due to formation of some new districts in recent years

Source: 1. Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare
2. Map drawn by ICAR- National Bureau of Soil Survey and Land Use Planning, Nagpur



Annex Map 3.3: Classification of Major Safflower Producing Districts based on Area Share and Yield

Annex Table



Note: 1. Districts contributing 90% production of respective crops considered for the analysis
 2. All-India Yield and Median of Area Shares of districts in all-India area used as benchmark for categorization
 3. Count of districts under different categories may slightly deviate from Table 3.8 due to formation of some new districts in recent years

Source: 1. Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare
 2. Map drawn by ICAR- National Bureau of Soil Survey and Land Use Planning, Nagpur

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Mrs. Shivani
Mrs. Shraddha Pal

Sr. Translation Officer

Mr. Sant Kumar

Assistant Directors

Mr. Deepak Peeceeyen S
Mr. Vipin Singh
Ms. Parul Singh

Administrative Staff

Mr. Pardeep Kumar
Mr. Pankaj Kumar
Mr. Anil Singh
Mrs. Pooja Budhiraja
Mr. Vikram Singh Bhadauria
Ms. Kamla Manral

Sr. Statistical Officers

Mr. B.K. Barnawal
Mr. Ravi Kumar Sharma
Mr. Ashok Kumar
Mrs. Savitri Ramani



Commission for Agricultural Costs & Prices

Department of Agriculture & Farmers Welfare

Ministry of Agriculture & Farmers Welfare

Government of India, New Delhi

July 2025